

# Emergent Mathematics

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## Contents

|          |                                               |           |
|----------|-----------------------------------------------|-----------|
| <b>1</b> | <b>Fourier Analysis</b>                       | <b>4</b>  |
| 1.1      | Locally Compact Abelian Groups . . . . .      | 4         |
| 1.2      | Pontryagin Duality . . . . .                  | 6         |
| 1.3      | Bessel, Airy, and Pearcey Functions . . . . . | 10        |
| 1.4      | Diffraction Patterns . . . . .                | 12        |
| <b>2</b> | <b>Compact Harmonic Analysis</b>              | <b>12</b> |
| 2.1      | Compact Groups . . . . .                      | 12        |
| 2.2      | Peter–Weyl Decomposition . . . . .            | 14        |
| 2.3      | $SU(2)$ Matrix Coefficients . . . . .         | 17        |
| 2.4      | Angular Spectroscopy . . . . .                | 22        |
| <b>3</b> | <b>Coadjoint Orbits</b>                       | <b>22</b> |
| 3.1      | Hamiltonian Group Actions . . . . .           | 22        |
| 3.2      | Symplectic Reduction . . . . .                | 26        |
| 3.3      | $SU(2)$ Coherent-State Kernels . . . . .      | 29        |
| 3.4      | Spin Spectroscopy . . . . .                   | 30        |
| <b>4</b> | <b>Heisenberg Group</b>                       | <b>30</b> |
| 4.1      | Central Extensions . . . . .                  | 30        |
| 4.2      | Stone–von Neumann Theorem . . . . .           | 33        |
| 4.3      | Hermite and Laguerre Functions . . . . .      | 38        |
| 4.4      | Oscillator Spectroscopy . . . . .             | 39        |
| <b>5</b> | <b>Poincaré Group</b>                         | <b>41</b> |
| 5.1      | Minkowski Isometries . . . . .                | 41        |
| 5.2      | Wigner Classification . . . . .               | 43        |
| 5.3      | Spin-Weighted Harmonics . . . . .             | 44        |
| 5.4      | Relativistic Propagation . . . . .            | 45        |
| <b>6</b> | <b>Potential Scattering</b>                   | <b>46</b> |
| 6.1      | Euclidean Group . . . . .                     | 46        |
| 6.2      | Partial Waves . . . . .                       | 48        |
| 6.3      | Coulomb Functions . . . . .                   | 52        |
| 6.4      | Rutherford Scattering . . . . .               | 52        |
| <b>7</b> | <b>Schwarzschild Spacetime</b>                | <b>53</b> |
| 7.1      | Schwarzschild Isometries . . . . .            | 53        |
| 7.2      | Geodesic Reduction . . . . .                  | 53        |
| 7.3      | Confluent Heun Functions . . . . .            | 54        |
| 7.4      | Black-Hole Ringdown . . . . .                 | 55        |
| <b>8</b> | <b>Determinantal Kernels</b>                  | <b>56</b> |

|           |                                             |            |
|-----------|---------------------------------------------|------------|
| 8.1       | Dyson Symmetry Classes . . . . .            | 56         |
| 8.2       | Determinantal Point Processes . . . . .     | 56         |
| 8.3       | Orthogonal-Polynomial Kernels . . . . .     | 58         |
| 8.4       | KPZ Fluctuations . . . . .                  | 63         |
| <b>9</b>  | <b>Transfer Matrices</b>                    | <b>65</b>  |
| 9.1       | Symmetric Groups . . . . .                  | 65         |
| 9.2       | Transfer Spectra . . . . .                  | 66         |
| 9.3       | Theta Characters . . . . .                  | 68         |
| 9.4       | Lattice Spectroscopy . . . . .              | 70         |
| <b>10</b> | <b><math>SL_2(\mathbb{R})</math></b>        | <b>73</b>  |
| 10.1      | Hyperbolic Isometries . . . . .             | 73         |
| 10.2      | Spherical Transform . . . . .               | 75         |
| 10.3      | Legendre Functions . . . . .                | 75         |
| 10.4      | Hyperbolic Spectroscopy . . . . .           | 80         |
| <b>11</b> | <b><math>SL_2(\mathbb{Z})</math></b>        | <b>82</b>  |
| 11.1      | Modular Group . . . . .                     | 82         |
| 11.2      | Automorphic Spectrum . . . . .              | 82         |
| 11.3      | Modular Functions . . . . .                 | 84         |
| 11.4      | Modular Spectra . . . . .                   | 87         |
| <b>12</b> | <b>Quantum Groups</b>                       | <b>96</b>  |
| 12.1      | Hopf Algebras . . . . .                     | 96         |
| 12.2      | Quantum Peter–Weyl Theory . . . . .         | 98         |
| 12.3      | Basic Hypergeometric Functions . . . . .    | 103        |
| 12.4      | XXZ Spectroscopy . . . . .                  | 107        |
| <b>13</b> | <b>Double Affine Hecke Algebras</b>         | <b>110</b> |
| 13.1      | Double-Affine Braid Groups . . . . .        | 110        |
| 13.2      | Polynomial Representations . . . . .        | 111        |
| 13.3      | Elliptic Hypergeometric Functions . . . . . | 112        |
| 13.4      | Integrable Particle Spectra . . . . .       | 121        |
| <b>14</b> | <b>Algebraic Curves</b>                     | <b>126</b> |
| 14.1      | Legendre Involution . . . . .               | 126        |
| 14.2      | Branched Covers . . . . .                   | 131        |
| 14.3      | Elliptic Integrals . . . . .                | 132        |
| 14.4      | Pendulum Periods . . . . .                  | 134        |
| <b>15</b> | <b>Riemann–Roch</b>                         | <b>134</b> |
| 15.1      | Picard Group . . . . .                      | 134        |
| 15.2      | Sheaf Cohomology . . . . .                  | 135        |
| 15.3      | Theta Functions . . . . .                   | 140        |
| 15.4      | Landau Levels . . . . .                     | 141        |
| <b>16</b> | <b>Affine Grassmannian</b>                  | <b>142</b> |
| 16.1      | Loop Groups . . . . .                       | 142        |
| 16.2      | Perverse-Sheaf Convolution . . . . .        | 143        |
| 16.3      | Satake Characters . . . . .                 | 146        |
| 16.4      | 't Hooft–Line Fusion . . . . .              | 146        |
| <b>17</b> | <b>Finite-Gap Spectral Curves</b>           | <b>147</b> |
| 17.1      | Bloch Monodromy . . . . .                   | 147        |

|           |                                           |            |
|-----------|-------------------------------------------|------------|
| 17.2      | Hyperelliptic Curves . . . . .            | 148        |
| 17.3      | Baker–Akhiezer Functions . . . . .        | 149        |
| 17.4      | Finite-Gap Transport . . . . .            | 152        |
| <b>18</b> | <b>Index Theory</b>                       | <b>154</b> |
| 18.1      | Lattice Translations . . . . .            | 154        |
| 18.2      | K-Theory Classification . . . . .         | 155        |
| 18.3      | Monopole Harmonics . . . . .              | 163        |
| 18.4      | Quantized Hall Transport . . . . .        | 164        |
| <b>19</b> | <b>Cobordism</b>                          | <b>166</b> |
| 19.1      | $SO$ , Spin, and $U$ Structures . . . . . | 166        |
| 19.2      | Pontryagin–Thom Transform . . . . .       | 166        |
| 19.3      | Bordism Genera . . . . .                  | 167        |
| 19.4      | SPT Phases . . . . .                      | 173        |
| <b>20</b> | <b>Fusion Categories</b>                  | <b>177</b> |
| 20.1      | Temperley–Lieb Categories . . . . .       | 177        |
| 20.2      | Fusion Semisimplification . . . . .       | 177        |
| 20.3      | Modular $S$ -Matrix . . . . .             | 181        |
| 20.4      | Anyonic Observables . . . . .             | 182        |

# 1 Fourier Analysis

## 1.1 Locally Compact Abelian Groups

**Exercise 1.1** (Finite-dimensional vector groups, duality, and trace). Let  $k$  be a field. A *vector space* over  $k$  is an abelian group  $V$  with scalar multiplication satisfying

$$a(v + w) = av + aw, \quad (a + b)v = av + bv, \quad (ab)v = a(bv), \quad 1v = v.$$

A map  $T : V \rightarrow W$  is *linear* when  $T(av + bw) = aT(v) + bT(w)$ . Its kernel and image are

$$\ker T = \{v : T(v) = 0\}, \quad \operatorname{im} T = \{T(v) : v \in V\}.$$

The *dual space* of  $V$  is  $V^\vee = \operatorname{Hom}_k(V, k)$ . Its canonical evaluation pairing is

$$\operatorname{ev}_V : V^\vee \times V \longrightarrow k, \quad (\varphi, v) \longmapsto \varphi(v).$$

The *tensor product*  $V \otimes W$  is characterized by a bilinear map  $(v, w) \mapsto v \otimes w$  with the following universal property: every bilinear map  $b : V \times W \rightarrow X$  factors uniquely as

$$V \times W \longrightarrow V \otimes W \xrightarrow{\tilde{b}} X.$$

The evaluation pairing induces a linear map  $\operatorname{ev}_V : V^\vee \otimes V \rightarrow k$ .

A vector space  $V$  is *finite-dimensional* when there is a coevaluation map

$$\operatorname{coev}_V : k \longrightarrow V \otimes V^\vee$$

satisfying the snake identities

$$(\operatorname{id}_V \otimes \operatorname{ev}_V) \circ (\operatorname{coev}_V \otimes \operatorname{id}_V) = \operatorname{id}_V,$$

$$(\operatorname{ev}_V \otimes \operatorname{id}_{V^\vee}) \circ (\operatorname{id}_{V^\vee} \otimes \operatorname{coev}_V) = \operatorname{id}_{V^\vee}.$$

Let  $s_{V, V^\vee} : V \otimes V^\vee \rightarrow V^\vee \otimes V$  exchange the factors. For  $T \in \operatorname{End}(V)$ , define

$$\operatorname{tr}(T) = \operatorname{ev}_V \circ s_{V, V^\vee} \circ (T \otimes \operatorname{id}_{V^\vee}) \circ \operatorname{coev}_V.$$

The *dimension* of  $V$  is the scalar

$$\dim_k(V) = \operatorname{tr}(\operatorname{id}_V) \in k.$$

(a) Let  $c, d : k \rightarrow V \otimes V^\vee$  both satisfy the snake identities with  $\operatorname{ev}_V$ . Apply

$$\operatorname{id}_V \otimes \operatorname{ev}_V \otimes \operatorname{id}_{V^\vee}$$

to  $d \otimes c : k \rightarrow V \otimes V^\vee \otimes V \otimes V^\vee$  and use the snake identity for  $d$  to identify the result with  $c$ . Use the snake identity for  $c$  to identify the same result with  $d$ , and deduce  $c = d$ .

(b) Prove that  $T \mapsto \operatorname{tr}(T)$  is linear and that

$$\operatorname{tr}(ST) = \operatorname{tr}(TS) \quad (S, T \in \operatorname{End}(V)).$$

For a finite-dimensional real vector space  $V$ , equip its additive group with its standard locally compact topology. Prove that every  $p \in V^\vee$  defines a continuous unitary character  $q \mapsto e^{ip(q)}$ , and that every continuous character is of this form.

**Exercise 1.2** (Exterior powers, determinants, and lattice volume). For  $m \geq 0$ , the  $m$ th *exterior power* of a vector space  $V$  is

$$\Lambda^m V = V^{\otimes m} / \langle v_1 \otimes \cdots \otimes v_m : v_i = v_j \text{ for some } i \neq j \rangle,$$

and the image of  $v_1 \otimes \cdots \otimes v_m$  is written  $v_1 \wedge \cdots \wedge v_m$ . An endomorphism  $T$  induces

$$\Lambda^m T(v_1 \wedge \cdots \wedge v_m) = Tv_1 \wedge \cdots \wedge Tv_m.$$

The products

$$\Lambda^r V \otimes \Lambda^s V \longrightarrow \Lambda^{r+s} V, \quad (u_1 \wedge \cdots \wedge u_r) \otimes (v_1 \wedge \cdots \wedge v_s) \longmapsto u_1 \wedge \cdots \wedge u_r \wedge v_1 \wedge \cdots \wedge v_s$$

make  $\Lambda^\bullet V = \bigoplus_{m \geq 0} \Lambda^m V$  the *exterior algebra* of  $V$ .

For a finite-dimensional vector space  $V$ , an integer  $n \in \mathbb{Z}_{\geq 0}$  satisfying

$$\Lambda^n V \neq 0, \quad \Lambda^{n+1} V = 0$$

is its *exterior rank*, written  $\text{rank}(V) = n$ , when it exists and is unique. When  $\text{rank}(V) = n$ , the *determinant* of  $T \in \text{End}(V)$  is the scalar  $\det(T)$  satisfying

$$\Lambda^n T = \det(T) \text{id}_{\Lambda^n V}.$$

Let  $V$  be finite-dimensional.

(a) Prove that the exterior algebra is graded-commutative:

$$\xi \wedge \eta = (-1)^{rs} \eta \wedge \xi, \quad \xi \in \Lambda^r V, \quad \eta \in \Lambda^s V.$$

(b) Prove that the exterior rank exists and is unique, that every  $\Lambda^m V$  is finite-dimensional, and that every endomorphism of  $\Lambda^n V$  is a unique scalar multiple of the identity.

(c) Prove

$$\dim_k(V) = n \cdot 1_k, \quad \dim_k(\Lambda^m V) = \binom{n}{m} 1_k, \quad \Lambda^m V = 0 \quad (m > n).$$

Deduce that  $\dim_k(V)$  determines  $n$  when  $\text{char } k = 0$ , whereas it determines only the residue class of  $n$  modulo  $p$  when  $\text{char } k = p > 0$ .

(d) Prove that the determinant is well-defined. Use functoriality of the highest exterior power to prove

$$\det(ST) = \det(S) \det(T), \quad T \text{ is invertible} \iff \det(T) \neq 0.$$

(e) The coevaluation, trace, and exterior-power constructions extend under scalar extension to the finite free  $k[z]$ -module  $V[z] = V \otimes_k k[z]$ ; in particular,  $\Lambda_{k[z]}^n V[z]$  is free of rank one. Define the determinant of a  $k[z]$ -linear endomorphism through its action on this top exterior power. For  $T \in \text{End}(V)$  prove

$$\det(I + zT) = \sum_{m=0}^n \text{tr}(\Lambda^m T) z^m.$$

Recover  $(1 + z)^n = \sum_m \binom{n}{m} z^m$  by taking  $T = I$ .

For an automorphism  $A$  of a finite-dimensional real vector group, prove that transport of Haar measure multiplies volume by  $|\det A|$ . Deduce the covolume formula for  $A\mathbb{Z}^n \subset \mathbb{R}^n$ .

**Exercise 1.3** (Locally compact abelian groups and translation symmetry). A *locally compact abelian group* is a Hausdorff locally compact topological group whose multiplication is commutative. Put

$$U(1) = \{z \in \mathbb{C} : |z| = 1\}, \quad \mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}.$$

For a finite-dimensional real vector space  $V$ , prove that the topology transported from  $\mathbb{R}^n$  by a linear isomorphism is independent of the isomorphism. Equip its additive group with this topology. Prove that these vector groups,  $\mathbb{Z}^n$ ,  $\mathbb{T}^n$ , and  $U(1)$  are locally compact abelian groups, and construct the explicit isomorphism  $\mathbb{T} \cong U(1)$ . For  $a \in G$ , define the translation  $L_a(x) = a + x$  and the induced action

$$(\lambda_a f)(x) = f(x - a).$$

Prove that translations form an action of  $G$ . Call a linear operator on a translation-stable function space *translation invariant* when it commutes with this action. Prove that finite differences on any abelian group and constant-direction derivatives on a vector group are translation invariant. Show that on  $U(1)$  the functions  $z \mapsto z^n$ ,  $n \in \mathbb{Z}$ , are exactly the continuous characters. Identify their pullbacks to  $\mathbb{T}$  as  $x \mapsto e^{inx}$ .

## 1.2 Pontryagin Duality

**Exercise 1.4** (Hilbert spaces and bounded operators). An *inner product* on a complex vector space is conjugate-linear in its first argument, linear in its second, positive definite, and satisfies  $\langle x, y \rangle = \overline{\langle y, x \rangle}$ . Its norm is  $\|x\| = \sqrt{\langle x, x \rangle}$ . A *Hilbert space* is an inner-product space complete for this norm.

Vectors are *orthogonal* when their inner product is zero. For a subspace  $M \subseteq H$ , define

$$M^\perp = \{x : \langle m, x \rangle = 0 \text{ for every } m \in M\}.$$

A linear map  $T : H \rightarrow K$  between Hilbert spaces is *bounded* when

$$\|T\| = \sup_{\|x\| \leq 1} \|Tx\| < \infty.$$

Write  $B(H, K)$  for the bounded operators and  $B(H) = B(H, H)$ . An *adjoint* of  $T$  is an operator  $T^* : K \rightarrow H$  satisfying

$$\langle x, Ty \rangle = \langle T^*x, y \rangle.$$

An *orthogonal projection* is an operator  $P$  satisfying  $P = P^* = P^2$ . Define the unitary group

$$U(H) = \{U \in B(H) : U^*U = UU^* = I\}$$

and let it act on states by  $\psi \mapsto U\psi$  and on bounded observables by  $T \mapsto UTU^*$ .

- (a) Prove the Cauchy–Schwarz inequality, the projection theorem, and the Riesz representation theorem. Deduce the existence of adjoints and prove

$$\|T\|^2 = \|T^*T\|, \quad (\text{im } T)^\perp = \ker T^*.$$

- (b) For every closed subspace  $M \subseteq H$ , prove  $H = M \oplus M^\perp$  and construct the orthogonal projection  $P_M : H \rightarrow M$ .

- (c) Prove that  $U(H)$  is a group, that its action preserves inner products, adjoints, normality, and orthogonal projections, and that

$$\langle U\psi, (UTU^*)U\psi \rangle = \langle \psi, T\psi \rangle.$$

**Exercise 1.5** ( $L^p$  geometry). For a measure space  $(X, \mathcal{F}, \mu)$  and  $1 \leq p < \infty$ , define

$$L^p(X, \mu) = \left\{ f : X \rightarrow \mathbb{C} \text{ measurable} : \int_X |f|^p d\mu < \infty \right\} / \text{equality a.e.},$$

$$\|f\|_p = \left( \int_X |f|^p d\mu \right)^{1/p}.$$

Let  $L^\infty(X, \mu)$  consist of essentially bounded measurable functions modulo equality almost everywhere, with

$$\|f\|_\infty = \operatorname{ess\,sup}_{x \in X} |f(x)|.$$

Prove Hölder's and Minkowski's inequalities and deduce that  $L^p(X, \mu)$  is complete for  $1 \leq p \leq \infty$ . For  $1 < p < \infty$  with  $p^{-1} + q^{-1} = 1$ , assume that  $(X, \mathcal{F}, \mu)$  is  $\sigma$ -finite and prove that

$$L^q(X, \mu) \longrightarrow L^p(X, \mu)^*, \quad g \longmapsto \left( f \mapsto \int_X \bar{g}f d\mu \right)$$

is an isometric isomorphism.

**Exercise 1.6** (Haar measure, convolution, and regular representations). Prove the Haar theorem: every locally compact group has a nonzero left-invariant regular Borel measure, unique up to a positive scalar. Every locally compact group below is equipped with one such measure  $\mu$ , and Haar measure on a compact group is normalized by  $\mu(G) = 1$ .

Let  $G$  be a locally compact group with left Haar measure  $\mu$ . For  $f, g \in C_c(G)$ , define

$$(f * g)(x) = \int_G f(y)g(y^{-1}x) d\mu(y).$$

The *left regular representation* on  $L^2(G)$  is

$$(\lambda_s \xi)(t) = \xi(s^{-1}t).$$

Identify Haar measure on  $\mathbb{R}^n$ ,  $\mathbb{Z}^n$ , and  $\mathbb{T}^n$ . Prove associativity of convolution, the  $L^1$  convolution inequality, and  $\|f * g\|_2 \leq \|f\|_1 \|g\|_2$ . Prove that  $g \mapsto \lambda_g$  is a strongly continuous unitary representation. For a finite-dimensional representation of a compact group, average an inner product over  $G$  to obtain an invariant one.

**Exercise 1.7** (Pontryagin duality and Abelian Fourier analysis). For a locally compact abelian group  $G$ , its *Pontryagin dual*  $\widehat{G}$  is the group of continuous characters  $\chi : G \rightarrow U(1)$ , with pointwise multiplication and the compact-open topology. The *double-dual map* sends  $x \in G$  to the character  $\chi \mapsto \chi(x)$  of  $\widehat{G}$ .

Identify

$$\widehat{\mathbb{R}^n} \cong \mathbb{R}^n, \quad \widehat{\mathbb{Z}^n} \cong \mathbb{T}^n, \quad \widehat{\mathbb{T}^n} \cong \mathbb{Z}^n,$$

and verify the double-dual map in each case. For a linear subspace  $W \subseteq \mathbb{R}^n$  and a full-rank lattice  $\Gamma \subseteq \mathbb{R}^n$ , compute their annihilators and identify the duals of  $\mathbb{R}^n/W$  and  $\mathbb{R}^n/\Gamma$ .

**Exercise 1.8** (Fourier series). On  $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ , write

$$dm_{\mathbb{T}}(x) = \frac{dx}{2\pi}$$

for normalized Haar measure and define

$$\widehat{f}(k) = \int_{\mathbb{T}} f(x) e^{-ikx} dm_{\mathbb{T}}(x), \quad k \in \mathbb{Z}.$$

The *Fourier series* of  $f$  is  $\sum_{k \in \mathbb{Z}} \widehat{f}(k) e^{ikx}$ .

Prove that  $(e^{ikx})_{k \in \mathbb{Z}}$  is a complete orthonormal family in  $L^2(\mathbb{T})$  and derive

$$f = \sum_{k \in \mathbb{Z}} \widehat{f}(k) e^{ikx}, \quad \|f\|_2^2 = \sum_{k \in \mathbb{Z}} |\widehat{f}(k)|^2.$$

**Exercise 1.9** (Euclidean Fourier inversion and tempered distributions). For a finite-dimensional real Euclidean space  $V$ , extend the Schwartz condition already used on  $\mathbb{R}$  by letting  $\mathcal{S}(V)$  consist of the smooth functions  $f$  satisfying

$$\sup_{q \in V} (1 + \|q\|)^m \|d^r f(q)\| < \infty$$

for all  $m, r \geq 0$ , where the norm on the symmetric covariant tensors is induced by the Euclidean structure. Let  $V$  be an  $n$ -dimensional real Euclidean space with Lebesgue measure  $dq$ . Equip  $V^*$  with its dual Euclidean structure and the resulting Lebesgue measure  $dp$ . For  $f \in \mathcal{S}(V)$ , define

$$(\mathcal{F}_h f)(p) = (2\pi\hbar)^{-n/2} \int_V e^{-ip(q)/\hbar} f(q) dq, \quad \mathcal{F} = \mathcal{F}_1.$$

Prove Fourier inversion on  $\mathcal{S}(V)$  and prove that  $\mathcal{F}_h$  extends uniquely to a unitary operator on  $L^2(V)$ . For  $v \in V$ , prove

$$\mathcal{F}_h(-i\hbar d_v) \mathcal{F}_h^{-1} \widehat{f}(p) = p(v) \widehat{f}(p).$$

A *distribution* on  $V$  is a linear functional  $T$  on  $C_c^\infty(V)$  such that, for every compact  $K \subset V$ , there are  $C_K > 0$  and  $m \geq 0$  satisfying

$$|\langle T, \phi \rangle| \leq C_K \max_{0 \leq r \leq m} \sup_{q \in K} \|d^r \phi(q)\|$$

whenever  $\phi$  is supported in  $K$ . A *tempered distribution* is a linear functional  $T$  on  $\mathcal{S}(V)$  for which there are  $C > 0$  and  $m \geq 0$  satisfying

$$|\langle T, \phi \rangle| \leq C \max_{0 \leq r \leq m} \sup_{q \in V} (1 + \|q\|)^m \|d^r \phi(q)\|.$$

Write  $\mathcal{D}'(V)$  and  $\mathcal{S}'(V)$  for these spaces. For  $v \in V$  and  $q \in V$ , define

$$\langle d_v T, \phi \rangle = -\langle T, d_v \phi \rangle, \quad \langle \delta_q, \phi \rangle = \phi(q),$$

and define the Fourier transform  $\mathcal{F} : \mathcal{S}'(V) \rightarrow \mathcal{S}'(V^*)$  by

$$\langle \mathcal{F}T, \phi \rangle = \langle T, \mathcal{F}\phi \rangle,$$

where the Fourier transform on the test function uses the dual pairing between  $V^*$  and  $V$  with the same normalization. Prove that locally integrable functions define distributions, that distributional derivatives extend ordinary derivatives, and that

$$\mathcal{F}(d_v T) = i p(v) \mathcal{F}T, \quad \mathcal{F}\delta_0 = (2\pi)^{-n/2}.$$

Prove that Fourier inversion extends to  $\mathcal{S}'(V)$ .



**Exercise 1.10** (Fourier uncertainty and Gaussian wave packets). Let  $V$  be a real Euclidean space, let  $v \in V$  be a unit vector, and put

$$(Q_v f)(q) = \langle v, q \rangle f(q), \quad P_v f = -i\hbar d_v f$$

on  $\mathcal{S}(V)$ . Prove

$$[Q_v, P_v] = i\hbar I, \quad \mathcal{F}_\hbar P_v \mathcal{F}_\hbar^{-1} = Q_v.$$

For a normalized  $f \in \mathcal{S}(V)$  define

$$(\Delta_f Q_v)^2 = \|(Q_v - \langle Q_v \rangle_f) f\|^2, \quad (\Delta_f P_v)^2 = \|(P_v - \langle P_v \rangle_f) f\|^2.$$

Apply Cauchy–Schwarz to the two centered vectors and use the imaginary part of their inner product to prove

$$\Delta_f Q_v \Delta_f P_v \geq \frac{\hbar}{2}.$$

For  $V = \mathbb{R}$ , determine the equality condition and prove that every equality state is, up to normalization and a constant phase,

$$f(x) = \exp \left[ -\frac{(x - x_0)^2}{4\sigma^2} + \frac{ip_0 x}{\hbar} \right], \quad \Delta Q = \sigma, \quad \Delta P = \frac{\hbar}{2\sigma}.$$

Evolve this packet with the free Schrödinger Hamiltonian and prove

$$(\Delta Q(t))^2 = \sigma^2 + \frac{\hbar^2 t^2}{4m^2 \sigma^2}.$$

State the preparation, free-propagation, spatial-resolution, and momentum-calibration assumptions under which width measurements test both the uncertainty product and the spreading law.

**Exercise 1.11** (Free-particle dispersion). Substitute the plane wave  $\psi(q, t) = e^{i(k(q) - \omega t)}$ , with  $k \in V^*$ , into the free Schrödinger equation and prove

$$\hbar\omega = \frac{\hbar^2 \|k\|^2}{2m}.$$

Set  $p = \hbar k$  and  $E = \hbar\omega$ . Derive

$$E(p) = \frac{\|p\|^2}{2m}, \quad v_{\text{group}} = \frac{p^\sharp}{m},$$

where  $p^\sharp \in V$  is the Euclidean-dual vector to  $p$ . Deduce the time-of-flight displacement of a wave packet concentrated near  $p_0$  and the de Broglie relation between its measured wavelength and momentum.

**Exercise 1.12** (Lattices and Poisson summation). A *lattice* in  $\mathbb{R}^n$  is a subgroup  $\Gamma = A\mathbb{Z}^n$  for some invertible linear map  $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ . Its covolume is  $\text{vol}(\Gamma) = |\det A|$ , and a fundamental cell is  $F = A[0, 1]^n$ . Its *integral dual* and *reciprocal lattice* are

$$\Gamma^\vee = \{y \in \mathbb{R}^n : y \cdot \gamma \in \mathbb{Z} \text{ for every } \gamma \in \Gamma\}, \quad \Gamma^{\text{rec}} = 2\pi\Gamma^\vee.$$

For  $f \in \mathcal{S}(\mathbb{R}^n)$ , its  $\Gamma$ -periodization is

$$P_\Gamma f(x) = \sum_{\gamma \in \Gamma} f(x + \gamma).$$

Prove that  $P_\Gamma f$  converges with all derivatives and compute its Fourier coefficients. With the Fourier-transform normalization above, prove

$$P_\Gamma f(x) = \frac{(2\pi)^{n/2}}{\text{vol}(\Gamma)} \sum_{\xi \in \Gamma^{\text{rec}}} \mathcal{F}f(\xi) e^{i\xi \cdot x}.$$

Set  $x = 0$  to obtain the Poisson summation formula and compute the transform of a Gaussian in the stated normalization.

### 1.3 Bessel, Airy, and Pearcey Functions

**Exercise 1.13** (Rising factorials and the binomial series). For  $a \in \mathbb{C}$  and  $m \in \mathbb{Z}_{\geq 0}$ , define

$$(a)_0 = 1, \quad (a)_m = a(a+1) \cdots (a+m-1) \quad (m \geq 1).$$

For  $|z| < 1$ , define the *binomial series*

$$B_a(z) = \sum_{m=0}^{\infty} \frac{(a)_m}{m!} z^m.$$

Prove that the binomial series converges for  $|z| < 1$  and satisfies

$$(1-z)B'_a(z) = aB_a(z), \quad B_a(0) = 1.$$

For the branch normalized to equal 1 at  $z = 0$ , deduce

$$B_a(z) = (1-z)^{-a}.$$

For  $a = -n$ ,  $n \in \mathbb{Z}_{\geq 0}$ , prove that the series terminates and, after replacing  $z$  by  $-z$ , recover

$$(1+z)^n = \sum_{m=0}^n \binom{n}{m} z^m.$$

**Exercise 1.14** (Gamma and beta integrals). For  $s, t > 0$ , define

$$\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx, \quad B(s, t) = \int_0^1 x^{s-1} (1-x)^{t-1} dx.$$

Use Tonelli's theorem and the change of variables  $x = ru$ ,  $y = r(1-u)$  to prove

$$B(s, t) = \frac{\Gamma(s)\Gamma(t)}{\Gamma(s+t)}.$$

Use integration by parts to prove  $\Gamma(s+1) = s\Gamma(s)$ , and use the pushforward of Euclidean measure under the norm map to obtain

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \quad \int_{\mathbb{R}^n} e^{-|x|^2/2} dx = (2\pi)^{n/2}.$$

State the uses of Tonelli, Fubini, monotone convergence, and dominated convergence in the derivation.

**Exercise 1.15** (Bessel functions and radial Fourier analysis). For  $\alpha > -1$ , the Bessel function of the first kind is

$$J_\alpha(z) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m + \alpha + 1)} \left(\frac{z}{2}\right)^{2m+\alpha},$$

with a branch of  $z^\alpha$  fixed when  $\alpha \notin \mathbb{Z}$ .

Prove

$$z^2 J_\alpha''(z) + z J_\alpha'(z) + (z^2 - \alpha^2) J_\alpha(z) = 0$$

and the recurrence identities

$$J_{\alpha-1}(z) + J_{\alpha+1}(z) = \frac{2\alpha}{z} J_\alpha(z), \quad J_{\alpha-1}(z) - J_{\alpha+1}(z) = 2J_\alpha'(z).$$

Average the Euclidean Fourier kernel over spheres and express the Fourier transform of a radial Schwartz function on  $\mathbb{R}^n$  as a Hankel transform with kernel  $J_{n/2-1}$ . Derive the corresponding inversion identity.

**Exercise 1.16** (Airy–Pearcey diffraction). For a monochromatic scalar wave with wave number  $k = 2\pi/\lambda$ , define the fold and cusp canonical fields as oscillatory integrals

$$\mathcal{A}_{\text{fold}}(a; k) = \int_{\mathbb{R}}^{\text{osc}} e^{ik(t^3/3 - at)} dt,$$

$$\mathcal{A}_{\text{cusp}}(a, b; k) = \int_{\mathbb{R}}^{\text{osc}} e^{ik(t^4/4 + at^2/2 + bt)} dt.$$

For a phase function, call the image of its degenerate stationary points in parameter space its *caustic*. Prove that the fold caustic is  $a = 0$  and that the cusp caustic is the discriminant curve

$$4a^3 + 27b^2 = 0$$

of the stationary cubic  $t^3 + at + b$ .

Using the Airy function, prove

$$\mathcal{A}_{\text{fold}}(a; k) = 2\pi k^{-1/3} \text{Ai}(-k^{2/3}a).$$

Define the Pearcey function by

$$\text{Pe}(A, B) = \int_{\mathbb{R}}^{\text{osc}} e^{i(t^4/4 + At^2/2 + Bt)} dt$$

and derive

$$\mathcal{A}_{\text{cusp}}(a, b; k) = k^{-1/4} \text{Pe}(k^{1/2}a, k^{3/4}b).$$

Deduce that fold dark fringes occur at  $a = a_n k^{-2/3}$ , where  $\text{Ai}(-a_n) = 0$ , and that the two cusp control scales are  $k^{-1/2}$  and  $k^{-3/4}$ . Predict the normalized intensities  $|\text{Ai}|^2$  and  $|\text{Pe}|^2$  near fold and cusp caustics. State the scalar-wave, short-wavelength, canonical-form, finite-aperture, and detector-resolution assumptions under which measured fringe locations test the discriminant geometry and its wavelength scaling.

## 1.4 Diffraction Patterns

**Exercise 1.17** (Bessel diffraction). Let  $j_{\nu,k}$  be the  $k$ th positive zero of  $J_\nu$ . Separate the wave equation on a circular membrane of radius  $R$  with Dirichlet boundary and prove that its normal-mode frequencies are

$$\omega_{m,k} = \frac{c}{R} j_{|m|,k},$$

with radial factor  $J_{|m|}(j_{|m|,k}r/R)$ . Deduce the nodal circles and angular nodal lines and state how measured resonance ratios determine the ratios of Bessel zeros without knowledge of  $c/R$ .

For a uniformly illuminated circular aperture of radius  $a$ , average the Fraunhofer kernel over the aperture, put  $k = 2\pi/\lambda$ , and derive

$$\frac{I(\theta)}{I(0)} = \left[ \frac{2J_1(ka \sin \theta)}{ka \sin \theta} \right]^2.$$

Prove that the dark-ring angles satisfy  $ka \sin \theta_k = j_{1,k}$  and obtain the small-angle prediction  $\theta_1 \sim j_{1,1}\lambda/(2\pi a)$ . Identify which membrane resonances and diffraction minima measure zeros of the same Bessel functions.

**Exercise 1.18** (Crystal diffraction and Bragg peaks). For a finite motif  $B \subset \mathbb{R}^n$  with scattering weights  $c_b$ , its *structure factor* is

$$S_B(q) = \sum_{b \in B} c_b e^{-iq \cdot b}.$$

Let  $\Gamma_N$  be a finite rectangular portion of  $\Gamma$ . In the single-scattering approximation, prove that the amplitude at momentum transfer  $q$  factors as

$$\mathcal{A}_N(q) = S_B(q) \sum_{\gamma \in \Gamma_N} e^{-iq \cdot \gamma}.$$

Use geometric sums and Poisson summation to show that the normalized intensity concentrates at  $q \in \Gamma^{\text{rec}}$  as the crystal grows. For elastic scattering with incident and outgoing wavevectors of magnitude  $2\pi/\lambda$ , derive the Laue condition and, for planes spaced by  $d$ , the Bragg relation

$$2d \sin \theta = m\lambda.$$

Show how zeros of  $S_B$  produce missing reflections. Compare the peak condition with W. H. Bragg and W. L. Bragg, *Proceedings of the Royal Society A* **88** (1913). Predict the reciprocal-lattice peak locations and motif extinctions, and state the kinematic single-scattering, elastic, coherent, periodic-crystal, finite-size, wavelength-calibration, and angular-resolution assumptions.

## 2 Compact Harmonic Analysis

### 2.1 Compact Groups

**Exercise 2.1** (Compact Lie groups and invariant Laplacians). A *smooth manifold* of dimension  $n$  is a Hausdorff, second-countable space with a maximal atlas of homeomorphisms to open subsets of  $\mathbb{R}^n$  whose transition maps are smooth. A map of smooth manifolds is *smooth* when its chart representatives are smooth; it is a *diffeomorphism* when it has a smooth inverse.

The *tangent space*  $T_p M$  is the vector space of derivations  $v : C^\infty(M) \rightarrow \mathbb{R}$  at  $p$  satisfying

$$v(fg) = f(p)v(g) + g(p)v(f).$$

For a smooth map  $F : M \rightarrow N$ , define

$$dF_p : T_p M \longrightarrow T_{F(p)} N, \quad dF_p(v)(f) = v(f \circ F).$$

Put  $TM = \coprod_{p \in M} T_p M$ . A *smooth vector bundle* is a smooth map  $\pi : E \rightarrow M$  locally isomorphic by fiberwise linear maps to  $U \times W \rightarrow U$  for a finite-dimensional real vector space  $W$ . A *smooth section* of  $E$  is a smooth map  $s : M \rightarrow E$  satisfying  $\pi s = \text{id}_M$ ; write  $\Gamma(E)$  for the vector space of smooth sections of  $E$ . The differentials of chart maps give  $TM$  its smooth vector-bundle structure. A smooth section of  $TM$  is a *vector field*.

A *Lie group* is a topological group with a smooth manifold structure for which multiplication and inversion are smooth. Its Lie algebra  $\mathfrak{g} = T_e G$  has the bracket induced by left-invariant vector fields.

A *Riemannian metric* on an  $n$ -manifold  $M$  is a smooth family of positive-definite inner products  $g_p$  on  $T_p M$ . Its Riemannian density is characterized by

$$|\text{vol}_g|_p(v_1 \wedge \cdots \wedge v_n) = \det(g_p(v_i, v_j))^{1/2},$$

and determines the Riemannian measure  $d\text{vol}_g$ . The gradient is defined by

$$g(\text{grad}_g f, X) = Xf.$$

The divergence is determined by

$$\int_M Xf d\text{vol}_g = - \int_M f \text{div}_g X d\text{vol}_g$$

for compactly supported  $f$ , and the Laplace–Beltrami operator is  $\Delta_g = \text{div}_g \text{grad}_g$ .

An isometry  $F : (M, g) \rightarrow (N, h)$  is a diffeomorphism satisfying  $h_{F(p)}(dF_p v, dF_p w) = g_p(v, w)$ . For a Lie group  $G$  and  $g \in G$ , write  $L_g(h) = gh$ . A Riemannian metric on  $G$  is *left-invariant* when  $L_g$  is an isometry for every  $g$ .

- Prove that an isometry preserves the Riemannian density, gradient, divergence, and Laplace–Beltrami operator.
- Prove that a left-invariant Riemannian metric on a Lie group has left-invariant Riemannian measure.
- For the Euclidean metric, recover Lebesgue measure and  $\Delta = \sum_j \partial_j^2$ .

**Exercise 2.2** (Finite representations and Maschke’s theorem). For a group  $G$ , a representation on a vector space  $V$  is a homomorphism  $\rho : G \rightarrow \text{GL}(V)$ . A subspace is invariant when it is preserved by every  $\rho(g)$ , and a nonzero representation is irreducible when it has no proper nonzero invariant subspace. A *unitary representation* on a Hilbert space  $H$  is a homomorphism to  $U(H)$ ; for a topological group it is required to be strongly continuous.

Let  $G$  be finite, let the characteristic of  $k$  not divide  $|G|$ , and let  $V$  be a finite-dimensional representation. Average an arbitrary projection over  $G$  to construct an invariant complement to every invariant subspace. Deduce that every finite-dimensional representation is a direct sum of irreducible representations.

Let  $H$  be a finite-dimensional complex Hilbert space carrying a unitary representation of  $G$ , and let  $A \in \text{End}(H)$  be self-adjoint and commute with the action. Prove that every spectral subspace of  $A$

is invariant and decomposes into irreducible representations. If  $A_s$  is a norm-continuous family of such operators and a spectral cluster remains isolated, use its spectral projection to prove that the multiplicity of every irreducible type in the cluster is locally constant. Deduce that a change in these multiplicities requires a collision with another spectral cluster.

**Exercise 2.3** (Schur's lemma). An intertwiner  $T : (V, \rho) \rightarrow (W, \sigma)$  between representations satisfies  $T\rho(g) = \sigma(g)T$ .

Prove that a nonzero intertwiner between irreducible complex finite-dimensional representations is an isomorphism and that every endomorphism of an irreducible representation is scalar. Deduce uniqueness of irreducible multiplicities in a decomposition.

**Exercise 2.4** (Characters and finite-group harmonic analysis). The *character* of a finite-dimensional representation  $V$  is the class function  $\chi_V(g) = \text{Tr}(\rho_V(g))$ . For class functions on a finite group, define

$$\langle f, h \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{f(g)} h(g).$$

Use averaging of linear maps and Schur's lemma to prove orthonormality of the irreducible characters. Prove that they form a complete orthonormal family in the class functions and that  $\langle \chi_U, \chi_V \rangle_G = \dim \text{Hom}_G(U, V)$ . Deduce the multiplicity formula and the character table constraints.

**Exercise 2.5** (One-parameter subgroups and the exponential map). Prove existence and uniqueness of the one-parameter subgroup generated by  $X \in \mathfrak{g}$ . For a closed matrix group, prove that it is  $t \mapsto e^{tX}$ , identify the commutator bracket, and compute the first nontrivial terms of the Baker–Campbell–Hausdorff formula.

## 2.2 Peter–Weyl Decomposition

**Exercise 2.6** ( $C^*$ -algebras and spectra). A  $C^*$ -algebra is a complete normed complex algebra  $A$  with a conjugate-linear involution satisfying

$$(ab)^* = b^* a^*, \quad \|ab\| \leq \|a\| \|b\|, \quad \|a^* a\| = \|a\|^2.$$

For a unital  $C^*$ -algebra, define

$$\text{spec}_A(a) = \{\lambda \in \mathbb{C} : a - \lambda 1 \text{ is not invertible}\}.$$

An element is *normal* when  $a^* a = a a^*$ .

Prove that  $\text{spec}_A(a)$  is nonempty and compact and that

$$r(a) = \lim_{n \rightarrow \infty} \|a^n\|^{1/n}.$$

For normal  $a$ , prove  $r(a) = \|a\|$ . For  $A = B(H)$  and  $U \in U(H)$ , prove

$$\text{spec}_{B(H)}(UTU^*) = \text{spec}_{B(H)}(T).$$

**Exercise 2.7** (Commutative Gelfand theory). For a commutative  $C^*$ -algebra  $A$ , a *character* is a nonzero homomorphism  $\chi : A \rightarrow \mathbb{C}$ . Let  $\hat{A}$  be the space of characters with its weak-\* topology. The *Gelfand transform* is

$$\Gamma : A \longrightarrow C_0(\hat{A}), \quad \Gamma(a)(\chi) = \chi(a).$$

Prove the commutative Gelfand theorem [19]: every character is continuous,  $\hat{A}$  is locally compact Hausdorff, and the Gelfand transform is an isometric \*-isomorphism  $A \cong C_0(\hat{A})$ . Deduce that  $A$  is unital exactly when  $\hat{A}$  is compact.

**Exercise 2.8** (Continuous functional calculus). An element  $a$  of a  $C^*$ -algebra is *self-adjoint* when  $a = a^*$  and *positive*, written  $a \geq 0$ , when  $a = b^*b$  for some  $b$ .

For every normal element  $a$  of a unital  $C^*$ -algebra, apply commutative Gelfand theory to  $C^*(1, a)$  and prove that there is a unique unital isometric  $*$ -homomorphism

$$C(\text{spec}(a)) \longrightarrow C^*(1, a), \quad f \longmapsto f(a),$$

sending the coordinate function to  $a$ , and prove the continuous spectral mapping theorem. Deduce that positive elements have unique positive square roots and that  $a \geq 0$  if and only if  $a = a^*$  and  $\text{spec}_A(a) \subseteq [0, \infty)$ .

**Exercise 2.9** (Complex Riesz–Markov representation). For a locally compact Hausdorff space  $X$ , call a linear functional  $\varphi : C_c(X) \rightarrow \mathbb{C}$  *positive* when  $\varphi(f) \geq 0$  for every real-valued  $f \geq 0$ . Prove the positive Riesz–Markov theorem [28]: if such a functional is bounded on functions supported in each compact set, then there is a unique regular Borel measure  $\mu_\varphi$  satisfying  $\varphi(f) = \int_X f d\mu_\varphi$ . Prove that  $\mu_\varphi$  is positive and that its values on compact sets recover the local bounds of  $\varphi$ .

Prove the complex Riesz–Markov theorem [28]: every  $\varphi \in C_0(X)^*$  has a unique finite complex regular measure  $\mu_\varphi$  such that

$$\varphi(f) = \int_X f d\mu_\varphi, \quad \|\varphi\| = |\mu_\varphi|(X).$$

Deduce the positive theorem for functionals bounded in the  $C_0(X)$  norm and characterize positive functionals by positive measures.

**Exercise 2.10** (Spectral distributions of states). For a unital  $C^*$ -algebra, a positive linear functional is one satisfying  $\varphi(a^*a) \geq 0$  for every  $a$ , and a *state* is a positive functional with  $\varphi(1) = 1$ .

Let  $A$  be a unital  $C^*$ -algebra, let  $a = a^* \in A$ , and let  $\varphi$  be a state. Apply the continuous functional calculus and Riesz–Markov theorem to prove that there is a unique Borel probability measure  $\mu_{\varphi, a}$  on  $\text{spec}(a)$  such that

$$\varphi(f(a)) = \int_{\text{spec}(a)} f(\lambda) d\mu_{\varphi, a}(\lambda), \quad f \in C(\text{spec}(a)).$$

Deduce

$$\varphi(a) = \int \lambda d\mu_{\varphi, a}(\lambda), \quad \varphi(a^2) - \varphi(a)^2 = \int (\lambda - \varphi(a))^2 d\mu_{\varphi, a}(\lambda).$$

**Exercise 2.11** ( $C_0(X)$  representations). A *projection-valued measure* on a measurable space  $X$  is a map  $E$  from measurable subsets of  $X$  to orthogonal projections on  $H$  satisfying

$$E(\emptyset) = 0, \quad E(X) = I, \quad E\left(\bigsqcup_{j=1}^{\infty} B_j\right) = \sum_{j=1}^{\infty} E(B_j)$$

for pairwise disjoint measurable sets, where the sum converges in the strong operator topology. The spectral integral  $\int_X f dE$  for bounded measurable  $f$  is defined by uniform approximation from simple functions. It is regular when each scalar measure  $B \mapsto \langle x, E(B)y \rangle$  is a regular Borel measure.

Let  $X$  be locally compact Hausdorff. A representation  $\pi : C_0(X) \rightarrow B(H)$  is *nondegenerate* when  $\overline{\pi(C_0(X))H} = H$ . Prove that every nondegenerate  $*$ -representation determines scalar measures by applying complex Riesz–Markov to  $f \mapsto \langle x, \pi(f)y \rangle$ . Use polarization and multiplicativity to assemble them into a unique regular projection-valued measure  $E_\pi$  satisfying

$$\pi(f) = \int_X f(x) dE_\pi(x), \quad f \in C_0(X).$$

**Exercise 2.12** (Bounded normal spectral theorem). For a normal operator  $T \in B(H)$ , apply the  $C_0(X)$  representation theorem to the continuous functional calculus  $C(\text{spec}(T)) \rightarrow B(H)$  and prove that there is a unique projection-valued measure  $E_T$  on  $\text{spec}(T)$  satisfying

$$T = \int_{\text{spec}(T)} z dE_T(z), \quad f(T) = \int_{\text{spec}(T)} f(z) dE_T(z).$$

Deduce that a normal operator on a finite-dimensional Hilbert space has the intrinsic spectral decomposition

$$H = \bigoplus_{\lambda \in \text{spec}(T)} E_T(\{\lambda\})H, \quad T|_{E_T(\{\lambda\})H} = \lambda I.$$

**Exercise 2.13** (Compact spectral theorem). An operator  $K \in B(H)$  is compact when it maps the closed unit ball to a set with compact closure.

For a compact normal operator  $K$ , prove that every nonzero spectral value is an eigenvalue of finite multiplicity, that the nonzero eigenvalues can accumulate only at zero, and that

$$K = \sum_{\lambda \in \text{spec}(K) \setminus \{0\}} \lambda E_K(\{\lambda\})$$

in operator norm, and prove that the spectral subspaces in the displayed sum are mutually orthogonal.

**Exercise 2.14** (Compact convolution operators). Let  $G$  be compact with normalized Haar measure and  $k \in C(G)$ . Prove that

$$(T_k f)(x) = \int_G k(y^{-1}x) f(y) dy$$

defines a compact operator on  $L^2(G)$  by uniformly approximating its continuous kernel by finite sums of product functions. Prove that  $T_k$  commutes with left translations and that  $k(g^{-1}) = \overline{k(g)}$  makes  $T_k$  self-adjoint. Deduce that every nonzero eigenspace of such a self-adjoint convolution operator is a finite-dimensional invariant subspace of the left regular representation.

**Exercise 2.15** (Density of matrix coefficients). For a unitary representation  $\pi : G \rightarrow U(H_\pi)$ , its matrix coefficients are the functions  $g \mapsto \langle y, \pi(g)x \rangle$ .

Construct continuous convolution kernels that approximate the identity on  $C(G)$ . Apply the compact spectral theorem to their self-adjoint symmetrizations and prove that the linear span of matrix coefficients of finite-dimensional unitary representations is uniformly dense in  $C(G)$  and dense in  $L^2(G)$ .

**Exercise 2.16** (Schur orthogonality). For irreducible finite-dimensional unitary representations  $\pi$  and  $\sigma$ , prove that

$$A \mapsto \int_G \pi(g) A \sigma(g)^* dg$$

is zero when  $\pi \not\cong \sigma$  and is  $A \mapsto \text{tr}(A) I_{H_\pi} / \dim H_\pi$  when  $\pi = \sigma$ . Deduce that the coefficient map identifies  $H_\pi \otimes H_\pi^\vee$ , after multiplication by  $\sqrt{\dim H_\pi}$ , with an isometric subspace of  $L^2(G)$ , and that distinct irreducible coefficient spaces are orthogonal, where

$$C_\pi : H_\pi \otimes H_\pi^\vee \longrightarrow L^2(G), \quad C_\pi(x \otimes \varphi)(g) = \varphi(\pi(g)x).$$

**Exercise 2.17** (Peter–Weyl decomposition). The unitary dual  $\widehat{G}$  is the set of unitary-equivalence classes of irreducible finite-dimensional unitary representations of  $G$ .



Use density and Schur orthogonality to prove

$$L^2(G) \cong \widehat{\bigoplus_{\pi \in \widehat{G}} H_\pi \otimes H_\pi^\vee},$$

with the left regular representation acting as  $I_{H_\pi} \otimes \pi^\vee$  on the coefficient summand  $C_\pi(H_\pi \otimes H_\pi^\vee)$ . Equivalently, after relabelling this summand by its left type, it is a direct sum of  $\dim H_\pi$  copies of  $\pi^\vee$ . For  $f \in L^2(G) \subseteq L^1(G)$ , define  $\widehat{f}(\pi) = \int_G f(g) \pi(g)^* dg$  and prove

$$f = \sum_{\pi \in \widehat{G}} (\dim H_\pi) \operatorname{tr}(\widehat{f}(\pi) \pi(\cdot)), \quad \|f\|_2^2 = \sum_{\pi \in \widehat{G}} (\dim H_\pi) \operatorname{tr}(\widehat{f}(\pi)^* \widehat{f}(\pi)),$$

where the first sum converges in  $L^2(G)$ . Prove uniform convergence on  $C(G)$  for the summability means supplied by the approximate identities in the density exercise.

**Exercise 2.18** (Character inversion). Let  $G$  be compact with normalized Haar measure, let  $f$  be a finite linear combination of irreducible characters, and use the operator-valued Fourier coefficients defined above. If  $f$  is central, use Schur's lemma to write  $\widehat{f}(\pi) = c_\pi I_{d_\pi}$ . Derive

$$f(e) = \sum_{\pi \in \widehat{G}} d_\pi \operatorname{Tr}(\widehat{f}(\pi)) = \sum_{\pi \in \widehat{G}} d_\pi^2 c_\pi$$

and prove that left convolution by  $f$  acts by the scalar  $c_\pi$  on  $C_\pi(H_\pi \otimes H_\pi^\vee)$ , equivalently by  $c_{\pi^\vee}$  on every copy of the left type  $\pi$  in  $L^2(G)$ .

## 2.3 $SU(2)$ Matrix Coefficients

**Exercise 2.19** (Irreducible representations of  $SU(2)$ ). Define

$$SU(2) = \{U \in \operatorname{End}(\mathbb{C}^2) : U^*U = \operatorname{id}, \det U = 1\}.$$

Its Lie algebra is

$$\mathfrak{su}(2) = \{X \in \operatorname{End}(\mathbb{C}^2) : X^* = -X, \operatorname{Tr} X = 0\}.$$

Let the distinguished Pauli endomorphisms satisfy

$$\sigma_a \sigma_b = \delta_{ab} I + i \sum_{c=1}^3 \varepsilon_{abc} \sigma_c, \quad \operatorname{Tr}(\sigma_a \sigma_b) = 2\delta_{ab},$$

put  $e_a = -i\sigma_a/2$ , and, in a smooth unitary representation  $\pi$ , define the self-adjoint angular-momentum operators

$$J_a = i\hbar d\pi(e_a), \quad J_\pm = J_1 \pm iJ_2, \quad J^2 = J_1^2 + J_2^2 + J_3^2.$$

Prove  $[J_a, J_b] = i\hbar \varepsilon_{abc} J_c$  and derive the raising and lowering relations with  $J_3$ .

Let  $SU(2)$  act by conjugation on the real Euclidean space of traceless Hermitian endomorphisms, with inner product  $\langle A, B \rangle = \frac{1}{2} \operatorname{Tr}(AB)$ . Prove that the induced homomorphism

$$SU(2) \longrightarrow SO(3)$$

is surjective with kernel  $\{\pm I\}$ . For a complex vector space  $V$ , set  $\operatorname{Sym}^0(V) = \mathbb{C}$  and, for  $N \geq 1$ , define

$$\operatorname{Sym}^N(V) = V^{\otimes N} / \langle w - \sigma w : w \in V^{\otimes N}, \sigma \in S_N \rangle,$$

where  $S_N$  permutes the tensor factors. For  $j \in \{0, \frac{1}{2}, 1, \dots\}$ , put

$$V_j = \text{Sym}^{2j}(\mathbb{C}^2)$$

with the induced  $SU(2)$  action. Using raising and lowering operators, prove that every finite-dimensional irreducible smooth complex representation is isomorphic to exactly one  $V_j$ , that it has dimension  $2j+1$ , and that it has  $J_3$ -eigenvalues  $m\hbar$  for  $m = -j, -j+1, \dots, j$ . Compute the spectrum of  $J^2 = J_1^2 + J_2^2 + J_3^2$ . For  $j = \frac{1}{2}$ , relate the two  $J_3$  outcomes to the two Stern–Gerlach beams. For a calibrated uniform field in the third direction, define the Zeeman Hamiltonian  $\hat{H}_B = -\gamma B J_3$  and derive its spectral values and spacing.

**Exercise 2.20** (Weyl integration, character formulas, and orbital Fourier transforms). Let

$$t_\phi = \begin{pmatrix} e^{i\phi} & 0 \\ 0 & e^{-i\phi} \end{pmatrix}, \quad 0 \leq \phi \leq \pi,$$

and let  $T \subset SU(2)$  be the resulting maximal torus. Prove that every element of  $SU(2)$  is conjugate to exactly one  $t_\phi$  in this interval. Identify  $SU(2)$  with the unit sphere in the quaternions, disintegrate its normalized Haar measure over conjugacy classes, and prove the rank-one Weyl integration formula

$$\int_{SU(2)} F(g) dg = \frac{2}{\pi} \int_0^\pi F(t_\phi) \sin^2 \phi d\phi$$

for every continuous class function  $F$ .

For the representations  $V_j$  above, compute

$$\chi_j(t_\phi) = \frac{\sin((2j+1)\phi)}{\sin \phi} = U_{2j}(\cos \phi)$$

where the Chebyshev polynomial of the second kind is defined by

$$U_n(\cos \phi) = \frac{\sin((n+1)\phi)}{\sin \phi}$$

with its endpoint values supplied by continuity. Derive  $U_0(x) = 1$ ,  $U_1(x) = 2x$ , and  $U_{n+1}(x) = 2xU_n(x) - U_{n-1}(x)$ , and use the Weyl formula to prove the character orthogonality relations directly. For normalized area measure  $d\mu$  on  $S^2$ , prove

$$\int_{S^2} e^{iar_3} d\mu(r) = \frac{\sin a}{a}$$

and deduce the rank-one Kirillov formula

$$\chi_j(t_{\theta/2}) = (2j+1) \frac{\theta/2}{\sin(\theta/2)} \int_{S^2} e^{i(j+1/2)\theta r_3} d\mu(r).$$

Identify the shift  $j \mapsto j + \frac{1}{2}$  and the Weyl denominator as the two corrections relating the orbital Fourier transform to the exact quantum character.

**Exercise 2.21** (Clebsch–Gordan decomposition). Prove

$$V_{j_1} \otimes V_{j_2} \cong \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V_j.$$

Construct coupled angular-momentum eigenstates from highest-weight vectors and compute the singlet and triplet decomposition of  $V_{1/2} \otimes V_{1/2}$ .

**Exercise 2.22** (Gauss hypergeometric functions). For  $c \notin \mathbb{Z}_{\leq 0}$  the *Gauss hypergeometric function* is

$${}_2F_1\left(\begin{matrix} a, b \\ c \end{matrix}; z\right) = \sum_{m=0}^{\infty} \frac{(a)_m (b)_m}{(c)_m} \frac{z^m}{m!}.$$

It is interpreted as a finite sum when a numerator parameter is a nonpositive integer and otherwise converges for  $|z| < 1$ . The Gauss hypergeometric operator is

$$\mathcal{L}_{a,b,c} = z(1-z) \frac{d^2}{dz^2} + (c - (a+b+1)z) \frac{d}{dz} - ab.$$

Prove that  ${}_2F_1(a, b; c; z)$  is the solution of  $\mathcal{L}_{a,b,c}f = 0$  analytic at  $z = 0$  with  $f(0) = 1$ . Derive Euler's integral

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1} (1-u)^{c-b-1} (1-zu)^{-a} du,$$

and Gauss's summation

$${}_2F_1(a, b; c; 1) = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}.$$

Prove the integral under  $\operatorname{Re} c > \operatorname{Re} b > 0$  and the summation under  $\operatorname{Re}(c-a-b) > 0$ , initially with the gamma factors in their holomorphic domains, and then extend both identities throughout their common meromorphic domains.

**Exercise 2.23** (Kummer, Whittaker, and Laguerre functions). For  $b \notin \mathbb{Z}_{\leq 0}$ , define the Kummer function

$$M(a, b, z) = {}_1F_1(a; b; z) = \sum_{m=0}^{\infty} \frac{(a)_m}{(b)_m} \frac{z^m}{m!}.$$

For  $2\mu + 1 \notin \mathbb{Z}_{\leq 0}$ , define

$$M_{\lambda, \mu}(z) = e^{-z/2} z^{\mu+1/2} M\left(\mu - \lambda + \frac{1}{2}, 2\mu + 1, z\right),$$

with a branch of  $z^{\mu+1/2}$  fixed. The Whittaker function  $W_{\lambda, \mu}$  is the solution of

$$w'' + \left(-\frac{1}{4} + \frac{\lambda}{z} + \frac{1/4 - \mu^2}{z^2}\right) w = 0$$

normalized by  $W_{\lambda, \mu}(z) \sim e^{-z/2} z^{\lambda}$  as  $z \rightarrow +\infty$ .

For  $\alpha > -1$  and  $n \in \mathbb{Z}_{\geq 0}$ , define

$$L_n^{(\alpha)}(x) = \frac{x^{-\alpha} e^x}{n!} \frac{d^n}{dx^n} (e^{-x} x^{n+\alpha}) = \frac{(\alpha+1)_n}{n!} M(-n, \alpha+1, x).$$

Write  $L_n = L_n^{(0)}$ .

Prove locally uniformly in  $z$  that

$$\lim_{B \rightarrow +\infty} {}_2F_1\left(\begin{matrix} a, B \\ c \end{matrix}; \frac{z}{B}\right) = M(a, c, z),$$

and derive Kummer's equation

$$zy'' + (b-z)y' - ay = 0.$$

Apply the substitution  $w(z) = e^{-z/2} z^{\mu+1/2} y(z)$  to obtain the Whittaker equation with  $a = \mu - \lambda + \frac{1}{2}$  and  $b = 2\mu + 1$ .

Derive the equality of the two displayed definitions of  $L_n^{(\alpha)}$ . Use the Rodrigues formula and integration by parts to prove

$$\int_0^\infty L_m^{(\alpha)}(x) L_n^{(\alpha)}(x) x^\alpha e^{-x} dx = \frac{\Gamma(n + \alpha + 1)}{n!} \delta_{mn},$$

and derive

$$(n + 1) L_{n+1}^{(\alpha)}(x) = (2n + \alpha + 1 - x) L_n^{(\alpha)}(x) - (n + \alpha) L_{n-1}^{(\alpha)}(x).$$

Prove

$$\sum_{n \geq 0} L_n^{(\alpha)}(x) t^n = (1 - t)^{-\alpha-1} \exp\left(-\frac{xt}{1-t}\right), \quad |t| < 1,$$

and obtain the corresponding coefficient contour integral around  $t = 0$ .

**Exercise 2.24** (Jacobi polynomials and their spectral equation). For  $\alpha, \beta > -1$ , the *Jacobi polynomial* is

$$P_n^{(\alpha, \beta)}(x) = \frac{(\alpha + 1)_n}{n!} {}_2F_1\left(\begin{matrix} -n, n + \alpha + \beta + 1 \\ \alpha + 1 \end{matrix}; \frac{1-x}{2}\right).$$

Prove

$$\left( (1-x^2) \frac{d^2}{dx^2} + (\beta - \alpha - (\alpha + \beta + 2)x) \frac{d}{dx} \right) P_n^{(\alpha, \beta)} = -n(n + \alpha + \beta + 1) P_n^{(\alpha, \beta)}.$$

Derive Jacobi orthogonality on  $[-1, 1]$  with weight  $(1-x)^\alpha(1+x)^\beta$  and derive the three-term recurrence.

**Exercise 2.25** (Spherical harmonics and atomic orbitals). Identify  $S^2$  with  $SO(3)/SO(2)$  and  $L^2(S^2)$  with the right  $SO(2)$ -invariant subspace of  $L^2(SO(3))$ . Use Peter–Weyl to obtain

$$L^2(S^2) \cong \bigoplus_{\ell=0}^{\infty} V_\ell$$

Choose the normalized  $J_3$ -eigenstates by the Condon–Shortley convention

$$J_- |\ell m\rangle = \hbar \sqrt{(\ell + m)(\ell - m + 1)} |\ell, m - 1\rangle.$$

Define the Wigner matrix coefficients

$$D_{mn}^\ell(g) = \langle \ell m | \pi_\ell(g) | \ell n \rangle.$$

Prove that the functions  $Y_{\ell m} \propto \overline{D_{m0}^\ell}$ ,  $-\ell \leq m \leq \ell$ , form a complete orthogonal family in  $V_\ell$ . For the intrinsic Laplace–Beltrami operator of the unit round metric, derive  $\Delta_{S^2} Y_{\ell m} = -\ell(\ell + 1) Y_{\ell m}$  and completeness of the spherical harmonics. Derive the Jacobi realization

$$Y_{\ell m}(\theta, \phi) = C_{\ell m} e^{im\phi} (\sin \theta)^{|m|} P_{\ell-|m|}^{(|m|, |m|)}(\cos \theta)$$

with  $C_{\ell m}$  fixed by unit norm. On

$$L^2(\mathbb{R}^3, d^3x) \cong L^2((0, \infty), r^2 dr) \otimes L^2(S^2, d\Omega),$$

consider the Coulomb Hamiltonian

$$\hat{H}_C = -\frac{\hbar^2}{2M}\Delta_{\mathbb{R}^3} - \frac{\kappa}{r}, \quad M, \kappa > 0.$$

Separate  $\hat{H}_C$  in spherical coordinates and prove that its radial operator on the  $\ell$ th angular sector is

$$-\frac{\hbar^2}{2M} \left( \frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} - \frac{\ell(\ell+1)}{r^2} \right) - \frac{\kappa}{r}$$

on  $L^2((0, \infty), r^2 dr)$ . Recover  $\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r)Y_{\ell m}(\theta, \phi)$ . Put

$$a_0 = \frac{\hbar^2}{M\kappa}, \quad \beta = \frac{\sqrt{-2ME}}{\hbar}, \quad \rho = 2\beta r$$

for  $E < 0$ , and set  $u(r) = rR(r)$ . Reduce the radial eigenvalue equation to the Whittaker equation with

$$\lambda = \frac{1}{a_0\beta}, \quad \mu = \ell + \frac{1}{2}.$$

Use regularity at the origin and square-integrability at infinity to prove that

$$\ell + 1 - \lambda = -n_r, \quad n_r \in \mathbb{Z}_{\geq 0}.$$

Writing  $n = n_r + \ell + 1$ , deduce

$$E_n = -\frac{M\kappa^2}{2\hbar^2 n^2}, \quad R_{n\ell}(r) \propto e^{-r/(na_0)} \left( \frac{2r}{na_0} \right)^\ell L_{n-\ell-1}^{(2\ell+1)} \left( \frac{2r}{na_0} \right).$$

Prove that the spatial multiplicity of  $E_n$  is  $n^2$  and derive the spectral-line frequencies

$$\omega_{n_i \rightarrow n_f} = \frac{M\kappa^2}{2\hbar^3} \left| \frac{1}{n_f^2} - \frac{1}{n_i^2} \right|.$$

Determine the angular nodes of the  $s$ ,  $p$ , and  $d$  orbitals and relate them to symmetry-enforced zeros in photoelectron angular distributions in the plane-wave final-state approximation. Compare the predicted momentum-space nodal suppression with Weiß et al., *Nature Communications* **6** (2015), and distinguish the orbital-symmetry prediction, plane-wave approximation, measured intensity, and final-state scattering corrections.

**Exercise 2.26** (Clebsch–Gordan and Wigner coefficients). Use the normalized  $J_3$ -eigenstates fixed by the Condon–Shortley ladder convention above. Define

$$|jm\rangle = \sum_{m_1+m_2=m} C_{j_1 m_1, j_2 m_2}^{jm} |j_1 m_1\rangle \otimes |j_2 m_2\rangle,$$

with all coefficients real and with the coefficient of  $|j_1, j_1\rangle \otimes |j_2, j - j_1\rangle$  in  $|jj\rangle$  positive, and set

$$\begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & m_3 \end{pmatrix} = \frac{(-1)^{j_1-j_2-m_3}}{\sqrt{2j+1}} C_{j_1 m_1, j_2 m_2}^{j, -m_3}.$$

Define the Wigner  $6j$  symbol by

$$|(j_1 j_2) j_{12}, j_3; jm\rangle = \sum_{j_{23}} (-1)^{j_1+j_2+j_3+j} \sqrt{(2j_{12}+1)(2j_{23}+1)} \begin{Bmatrix} j_1 & j_2 & j_{12} \\ j_3 & j & j_{23} \end{Bmatrix} |j_1, (j_2 j_3) j_{23}; jm\rangle.$$

For a rotation through angle  $\beta$  about the second axis, set

$$d_{m'm}^j(\beta) = \langle jm' | e^{-i\beta J_2/\hbar} | jm \rangle.$$

Use the  $SU(2)$  differential equation and the value at  $\beta = 0$  to express  $d_{m'm}^j(\beta)$  as a normalized product of powers of  $\sin(\beta/2)$  and  $\cos(\beta/2)$  with a Jacobi polynomial in  $\cos \beta$ . Deduce its orthogonality from Peter–Weyl. Use the Clebsch–Gordan decomposition to derive the linearization formula for a product of two Wigner coefficients and identify its coefficients as products of Clebsch–Gordan coefficients.

## 2.4 Angular Spectroscopy

**Exercise 2.27** (Angular amplitudes). Derive the orthogonality relations for the Clebsch–Gordan coefficients and the triangle and magnetic selection rules for the  $3j$  symbols. Prove the Gaunt identity

$$\int_{S^2} \overline{Y_{\ell_f m_f}} Y_{1q} Y_{\ell_i m_i} d\Omega = (-1)^{m_f} \sqrt{\frac{3(2\ell_i + 1)(2\ell_f + 1)}{4\pi}} \begin{pmatrix} \ell_i & 1 & \ell_f \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell_i & 1 & \ell_f \\ m_i & q & -m_f \end{pmatrix}.$$

Deduce the electric-dipole rules  $\ell_f = \ell_i \pm 1$  and  $m_f = m_i + q$ , and prove that the relative Zeeman-resolved line intensities are the squared absolute values of the second  $3j$  symbol after a common reduced matrix element is removed. Use the  $6j$  identity to derive the corresponding relative intensities when orbital and spin angular momenta are coupled in different orders. For an oriented initial state, express the dipole radiation pattern as the squared absolute value of the resulting spherical-harmonic amplitude and identify its symmetry-enforced dark directions and absent spectral lines. State the rotational-symmetry, electric-dipole, weak-field, polarization-control, oriented-state-preparation, common-reduced-matrix-element, and angular- and line-resolution assumptions under which the relative intensities and selection rules are spectroscopic predictions.

## 3 Coadjoint Orbits

### 3.1 Hamiltonian Group Actions

**Exercise 3.1** (Differential forms and Cartan calculus). The cotangent space at  $p$  is  $T_p^*M = (T_p M)^*$ , and  $T^*M = \coprod_{p \in M} T_p^*M$  has the dual vector-bundle structure.

A *differential  $k$ -form* on  $M$  is a smooth section of  $\Lambda^k T^*M$ ; write  $\Omega^k(M)$  for the space of differential  $k$ -forms. The exterior products on the cotangent spaces define

$$\wedge : \Omega^r(M) \times \Omega^s(M) \longrightarrow \Omega^{r+s}(M).$$

For a smooth map  $F : M \rightarrow N$ , define

$$(F^* \alpha)_p(v_1, \dots, v_k) = \alpha_{F(p)}(dF_p v_1, \dots, dF_p v_k).$$

For a vector field  $X$ , define contraction by

$$(\iota_X \alpha)_p(v_2, \dots, v_k) = \alpha_p(X_p, v_2, \dots, v_k).$$

The *exterior derivative* is the family  $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$  satisfying  $df(X) = Xf$ , the graded Leibniz rule,  $d^2 = 0$ , and  $F^*d = dF^*$ . A *local flow* of  $X$  is a smooth local family  $\Phi_t$  satisfying  $\Phi_0 = \text{id}$ ,  $\Phi_{t+s} = \Phi_t\Phi_s$  whenever both sides are defined, and  $\frac{d}{dt}\big|_{t=0} \Phi_t(p) = X_p$ . Define

$$\mathcal{L}_X\alpha = \frac{d}{dt}\bigg|_{t=0} \Phi_t^*\alpha.$$

Verify that the differentials of chart maps give  $TM$  and  $T^*M$  the stated smooth vector-bundle structures and that  $d(G \circ F) = dG \circ dF$ . Construct  $d$  locally and prove its uniqueness. For contraction  $\iota_X$ , prove Cartan's identity

$$\mathcal{L}_X = d\iota_X + \iota_X d.$$

For  $A \in \Omega^1(M)$ , put  $F = dA$ . Prove that replacing  $A$  by  $A + d\chi$ , with  $\chi \in C^\infty(M)$ , leaves  $F$  unchanged. Use Cartan's identity to derive

$$\mathcal{L}_X A = \iota_X F + d(\iota_X A).$$

**Exercise 3.2** (Local Poincaré lemma). Let  $U$  be a star-shaped open neighborhood of 0 in a finite-dimensional real vector space. For  $\alpha \in \Omega^k(U)$  with  $k > 0$ , define

$$(K\alpha)_x(v_1, \dots, v_{k-1}) = \int_0^1 t^{k-1} \alpha_{tx}(x, v_1, \dots, v_{k-1}) dt.$$

Use Cartan's identity for the radial flow to prove

$$dK + Kd = \text{id}$$

on positive-degree forms. Deduce that every closed form of positive degree on  $U$  is exact, with a primitive vanishing at the origin.

**Exercise 3.3** (Integration of forms and Stokes' theorem). An *orientation* of an  $n$ -manifold is a continuous choice of component of the nonzero elements in  $\Lambda^n T_p^*M$ . The integral  $\int_M \omega$  of a compactly supported top form is defined using orientation-preserving charts and a partition of unity. Prove that the integral is independent of charts and partition of unity. For an oriented manifold with boundary, prove

$$\int_M d\omega = \int_{\partial M} \omega.$$

Deduce the fundamental theorem of calculus, Green's theorem, the divergence theorem, and the classical curl theorem. For time-dependent magnetic two-form  $B$  and electric one-form  $E$ , prove the equivalence of

$$\int_{\partial S} E = -\frac{d}{dt} \int_S B \quad \text{and} \quad dE = -\partial_t B,$$

recovering the observed flux-emf law.

**Exercise 3.4** (de Rham cohomology). A differential form  $\omega$  is *closed* if  $d\omega = 0$  and *exact* if  $\omega = d\eta$  for some form  $\eta$ . Since  $d^2 = 0$ , the *de Rham cohomology* of  $M$  is

$$H_{\text{dR}}^k(M) = \frac{\ker(d : \Omega^k(M) \rightarrow \Omega^{k+1}(M))}{\text{im}(d : \Omega^{k-1}(M) \rightarrow \Omega^k(M))}.$$

Pullback induces maps on de Rham cohomology, and the wedge product makes  $H_{\text{dR}}^*(M)$  a graded-commutative algebra.

Use the local Poincaré lemma and Stokes' theorem to prove that an exact  $k$ -form has zero integral over every closed oriented  $k$ -manifold mapped smoothly into  $M$ . Prove  $H_{\text{dR}}^1(S^1) \cong \mathbb{R}$ , with the isomorphism given by integration.

On  $\mathbb{R}^2 \setminus \{0\}$ , show that

$$\alpha = \frac{-y dx + x dy}{x^2 + y^2}$$

is closed and has integral  $2\pi$  around the unit circle, hence is not exact. Compute  $H_{\text{dR}}^1(\mathbb{R}^2 \setminus \{0\}) = \mathbb{R} \cdot [\alpha]$ . Deduce that a closed electromagnetic field strength has local potentials, while it has a global potential  $F = dA$  exactly when  $[F] = 0$  in de Rham cohomology.

**Exercise 3.5** (Singular homology, de Rham comparison, and integral periods). For a space  $X$ , a *singular  $k$ -simplex* is a continuous map  $\sigma : \Delta^k \rightarrow X$ . Let  $C_k(X; \mathbb{Z})$  be the free abelian group on these simplices and define

$$\partial\sigma = \sum_{j=0}^k (-1)^j \sigma|_{[v_0, \dots, \widehat{v_j}, \dots, v_k]}.$$

Prove  $\partial^2 = 0$  and define

$$H_k(X; \mathbb{Z}) = \ker \partial / \text{im } \partial.$$

For a continuous map  $f : X \rightarrow Y$ , define  $f_{\#}$  by composition with singular simplices and prove that it is a chain map. Construct the prism operator for a homotopy and deduce that homotopic maps induce the same map on homology. For  $A \subset X$ , define the relative chain complex  $C_*(X, A) = C_*(X)/C_*(A)$  and derive the long exact sequence of a pair.

For an abelian group  $A$ , define singular cochains by  $C^k(X; A) = \text{Hom}(C_k(X; \mathbb{Z}), A)$  with coboundary  $\delta\varphi = \varphi\partial$ , and define  $H^k(X; A)$ . For cochains with values in a commutative ring, define the cup product by evaluating the two factors on the front and back faces of a simplex. Prove the graded Leibniz identity for  $\delta$  and deduce the graded cohomology ring.

For coefficients in a field  $F$ , split the cycle and boundary subspaces of  $C_*(X; F)$  and prove the universal-coefficient isomorphism

$$H^k(X; F) \cong \text{Hom}_F(H_k(X; F), F).$$

For a smooth manifold  $M$ , construct compatible smooth approximations of singular simplices relative to their faces and a prism homotopy, thereby proving that smooth singular simplices compute singular homology. Set

$$I(\omega)(\sigma) = \int_{\Delta^k} \sigma^* \omega.$$

Use Stokes' theorem to prove  $I(d\omega) = \delta I(\omega)$  and hence obtain the natural de Rham map

$$I : H_{\text{dR}}^k(M) \longrightarrow H^k(M; \mathbb{R}).$$

Prove the de Rham theorem [14]: this map is an isomorphism of graded rings. Deduce that a closed form is exact exactly when all its periods over smooth singular cycles vanish.

Use the coefficient homomorphism  $\mathbb{Z} \rightarrow \mathbb{R}$  to define an *integral de Rham class* as a class in the image of  $H^k(M; \mathbb{Z}) \rightarrow H^k(M; \mathbb{R})$ . Use the integral-period form of the de Rham theorem to prove that  $[\omega]$  is integral exactly when

$$\int_c \omega \in \mathbb{Z} \quad \text{for every } [c] \in H_k(M; \mathbb{Z}).$$



For a closed oriented  $k$ -manifold  $N$  and a smooth map  $f : N \rightarrow M$ , identify the period over  $f_*[N]$  with  $\int_N f^*\omega$  and prove its invariance under oriented bordism of the map.

**Exercise 3.6** (Cotangent phase space). For the projection  $\pi : T^*Q \rightarrow Q$ , define the tautological one-form by

$$\theta_\alpha(v) = \alpha(d\pi(v)), \quad \alpha \in T^*Q, \quad v \in T_\alpha(T^*Q).$$

The canonical two-form on  $T^*Q$  is  $\omega = d\theta$ .

For a finite-dimensional real vector space  $V$ , construct the canonical identifications

$$TV \cong V \times V, \quad T^*V \cong V \times V^*.$$

Prove that a phase  $S \in C^\infty(Q)$  determines the momentum  $p = dS_q \in T_q^*Q$ . For a diffeomorphism  $F : Q \rightarrow Q$ , construct its cotangent lift and prove that it preserves  $\theta$  and  $d\theta$ . Under  $T^*V \cong V \times V^*$ , prove

$$\omega((q, p), (q', p')) = p(q') - p'(q).$$

**Exercise 3.7** (Symplectic manifolds and Hamiltonian fields). A *symplectic form* on a smooth manifold  $M$  is a closed nondegenerate two-form  $\omega$ . A *symplectic vector space* is a finite-dimensional real vector space equipped with a nondegenerate alternating bilinear form; it carries the corresponding constant symplectic form. For  $f \in C^\infty(M)$ , define

$$\iota_{X_f}\omega = -df, \quad \{f, g\} = -\omega(X_f, X_g).$$

The local flow of  $X_f$  is the *Hamiltonian flow* of  $f$ . A diffeomorphism  $F : (M, \omega_M) \rightarrow (N, \omega_N)$  is a *canonical transformation* when  $F^*\omega_N = \omega_M$ .

Prove that the canonical two-form  $d\theta$  on  $T^*Q$  is nondegenerate and hence symplectic. Prove that the Poisson bracket is antisymmetric, satisfies the Leibniz rule and Jacobi identity, and is preserved by canonical transformations. For the Hamiltonian flow  $\Phi_t$  of  $H$ , prove

$$\frac{d}{dt}f(\Phi_t(x)) = \{f, H\}(\Phi_t(x)).$$

**Exercise 3.8** (Liouville's theorem). On a  $2n$ -dimensional symplectic manifold, prove that Hamiltonian flow preserves  $\omega$ , the volume form  $\omega^n/n!$ , and every density depending only on the Hamiltonian. On a symplectic vector space, prove that the infinitesimal generator of a linear Hamiltonian flow has trace zero and that the flow has determinant one.

**Exercise 3.9** (Moment maps and Noether charges). For a smooth left action of a Lie group  $G$  on  $M$  and  $\xi \in \mathfrak{g}$ , define the fundamental vector field

$$\xi_M(x) = d(g \mapsto gx)_e(\xi).$$

Define

$$\text{Ad}_g = d(h \mapsto ghg^{-1})_e, \quad (\text{Ad}_g^*\ell)(\xi) = \ell(\text{Ad}_{g^{-1}}\xi).$$

An action of  $G$  on  $(M, \omega)$  is *Hamiltonian* when it preserves  $\omega$  and has an equivariant map  $\mu : M \rightarrow \mathfrak{g}^*$  satisfying

$$\iota_{\xi_M}\omega = -d\mu^\xi, \quad \mu^\xi(x) = \langle \mu(x), \xi \rangle, \quad \mu(gx) = \text{Ad}_g^*\mu(x).$$

The map  $\mu$  is the *moment map*.

Prove

$$X_{\mu^\xi} = \xi_M, \quad \{\mu^\xi, \mu^\eta\} = \mu^{[\xi, \eta]}.$$

If a Hamiltonian  $H$  is  $G$ -invariant, prove Noether's conservation law

$$\frac{d}{dt} \mu^\xi(\Phi_t(x)) = 0$$

for its Hamiltonian flow  $\Phi_t$ .

Let  $G$  act on a manifold  $Q$  and lift the action to  $T^*Q$ . Prove that the lifted action preserves the tautological form and is Hamiltonian with

$$\langle \mu(\alpha_q), \xi \rangle = \alpha_q(\xi_Q(q)).$$

For the distinguished Euclidean configuration space  $Q = \mathbb{R}^3$ , identify the moment maps for translations and rotations with linear momentum  $p$  and angular momentum  $q \times p$ . Deduce conservation of  $p$  for a translation-invariant Hamiltonian and conservation of  $q \times p$  for

$$H(q, p) = \frac{\|p\|^2}{2m} + V(\|q\|).$$

### 3.2 Symplectic Reduction

**Exercise 3.10** (Symplectic reduction). Let a compact Lie group  $G$  act on  $(M, \omega)$  with equivariant moment map  $\mu : M \rightarrow \mathfrak{g}^*$ . Suppose 0 is a regular value, meaning  $d\mu_x$  is surjective for every  $x \in \mu^{-1}(0)$ , and suppose  $G$  acts freely on  $\mu^{-1}(0)$ . Prove [51] that

$$M//G = \mu^{-1}(0)/G$$

is a smooth manifold and the quotient map  $\pi : \mu^{-1}(0) \rightarrow M//G$  is a submersion.

For the inclusion  $i : \mu^{-1}(0) \hookrightarrow M$ , prove that  $i^*\omega$  is invariant and vanishes on the tangent directions to the  $G$ -orbits. Prove that there is a unique symplectic form  $\omega_{\text{red}}$  such that

$$\pi^*\omega_{\text{red}} = i^*\omega, \quad \dim(M//G) = \dim M - 2 \dim G.$$

If  $H$  is  $G$ -invariant, prove that its restriction descends to  $H_{\text{red}}$  and that the Hamiltonian flow of  $H$  on  $\mu^{-1}(0)$  projects to the Hamiltonian flow of  $H_{\text{red}}$ . Write the reduced equations governing motion at the imposed value of the conserved Noether charges.

**Exercise 3.11** (Coadjoint orbits and the Kirillov form). For  $\ell \in \mathfrak{g}^*$ , define

$$\mathcal{O}_\ell = \text{Ad}^*(G)\ell, \quad G_\ell = \{g \in G : \text{Ad}_g^*\ell = \ell\},$$

and

$$\mathfrak{g}_\ell = \{X \in \mathfrak{g} : \ell([X, Y]) = 0 \text{ for every } Y \in \mathfrak{g}\}.$$

For  $X \in \mathfrak{g}$ , set

$$(\text{ad}_X^* \ell)(Y) = -\ell([X, Y]).$$

On  $T_\ell \mathcal{O}_\ell$ , define

$$\omega_\ell(\text{ad}_X^* \ell, \text{ad}_Y^* \ell) = \ell([X, Y]).$$

For  $\eta = \text{Ad}_g^* \ell$ , transport  $\omega_\ell$  by the coadjoint action to define  $\omega_\eta$ .

Prove

$$T_\ell \mathcal{O}_\ell = \{\text{ad}_X^* \ell : X \in \mathfrak{g}\} \cong \mathfrak{g}/\mathfrak{g}_\ell.$$

Prove that the Kirillov form is well-defined, alternating, nondegenerate, and  $G$ -invariant. Prove the cyclic identity

$$\ell([X, [Y, Z]]) + \ell([Y, [Z, X]]) + \ell([Z, [X, Y]]) = 0$$

and prove directly that it is the cyclic identity satisfied by the transported field  $\omega$  on  $\mathcal{O}_\ell$ . Prove that

$$\mu_{\mathcal{O}} : \mathcal{O}_\ell \longrightarrow \mathfrak{g}^*, \quad \mu_{\mathcal{O}}(\eta) = -\eta,$$

is an equivariant moment map with the sign conventions above.

For the distinguished group

$$SO(3) = \{R \in \text{End}(\mathbb{R}^3) : R^* R = \text{id}, \det R = 1\},$$

use the Euclidean inner product and orientation to identify  $\mathfrak{so}(3)$  and its dual with  $\mathbb{R}^3$ , with bracket  $[X, Y] = X \times Y$ . Prove that  $\mathcal{O}_\ell$  is the sphere of radius  $L = \|\ell\|$ , that

$$\omega_\ell(X \times \ell, Y \times \ell) = \ell \cdot (X \times Y),$$

and that the Kirillov form is  $L^{-1}$  times the oriented Euclidean area form on this sphere. Deduce

$$\int_{\mathcal{O}_\ell} \omega = 4\pi L.$$

**Exercise 3.12** (Connections, curvature, and holonomy). A *connection* on a vector bundle  $E \rightarrow M$  is an  $\mathbb{R}$ -bilinear map  $\nabla : \Gamma(TM) \times \Gamma(E) \rightarrow \Gamma(E)$ , linear over smooth functions in the first variable and satisfying  $\nabla_X(fs) = X(f)s + f\nabla_X s$ . Along a curve  $\gamma$  it induces a covariant derivative  $D/dt$ ; its solutions  $Ds/dt = 0$  define *parallel transport* between endpoint fibers.

The *curvature* of  $\nabla$  is the  $\text{End}(E)$ -valued two-form

$$R^\nabla(X, Y)s = \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X, Y]} s.$$

The *holonomy group* at  $p$  consists of the automorphisms of  $E_p$  obtained by parallel transport around loops based at  $p$ .

Prove existence, uniqueness, composition, and inverse properties of parallel transport. Prove that curvature is tensorial and gives the leading failure of transport around an infinitesimal parallelogram to return a vector to itself. Deduce that a flat connection has path-independent transport on every simply connected coordinate domain.

**Exercise 3.13** (Principal connections and gauge covariance). A *principal  $G$ -bundle* is a smooth map  $P \rightarrow M$  with a free right  $G$ -action, transitive on each fiber, and local  $G$ -equivariant trivializations. For a homomorphism  $\rho : G \rightarrow \text{GL}(W)$ , its associated vector bundle is  $P \times_G W = (P \times W)/((pg, w) \sim (p, \rho(g)w))$ . A *principal connection* is an equivariant  $\mathfrak{g}$ -valued one-form  $A$  on  $P$  reproducing fundamental vertical fields. It induces connections on every associated bundle, and its *curvature* is  $F_A = dA + \frac{1}{2}[A \wedge A]$ . A *gauge transformation* is a  $G$ -equivariant bundle automorphism covering  $\text{id}_M$ .

Prove  $d_A F_A = 0$ . For a gauge transformation  $\Phi$ , prove  $F_{\Phi^* A} = \Phi^* F_A$  and that holonomy changes by conjugacy. Prove that a flat connection has trivial holonomy on contractible loops and is gauge-equivalent to a product connection on a connected base exactly when its holonomy homomorphism is trivial. For  $G = U(1)$ , compute the holonomy around a generator of the fundamental group and exhibit the obstruction to removing a flat connection globally.

**Exercise 3.14** (Integral curvature and circle cocycles). A *good cover* of a manifold is an open cover  $\{U_i\}_{i \in I}$  for which every nonempty finite intersection is contractible. A *smooth complex vector bundle* is a smooth vector bundle locally modeled on a finite-dimensional complex vector space by fiberwise complex-linear product maps; it is a *complex line bundle* when its complex rank is one. A *Hermitian vector bundle* carries a smoothly varying positive-definite Hermitian form, and a connection on it is *unitary* when it preserves that form. Let  $F \in \Omega^2(M; i\mathbb{R})$  be closed and choose, on a good cover,

$$dA_i = F|_{U_i}.$$

On  $U_i \cap U_j$ , choose  $f_{ij} : U_i \cap U_j \rightarrow i\mathbb{R}$  with  $A_j - A_i = df_{ij}$  and  $f_{ji} = -f_{ij}$ . Prove that

$$c_{ijk} = f_{ij} + f_{jk} + f_{ki}$$

is constant on every nonempty triple intersection.

Prove the following good-cover form of the integral de Rham theorem [14]:

$$\frac{1}{2\pi i} \int_{\Sigma} h^* F \in \mathbb{Z} \quad \text{for every smooth } h : \Sigma \rightarrow M$$

from a closed oriented surface if and only if the  $f_{ij}$  may be chosen so that  $c_{ijk} \in 2\pi i\mathbb{Z}$ . Under this condition put

$$g_{ij} = e^{f_{ij}}.$$

Prove the cocycle identities for the  $g_{ij}$ , construct the associated Hermitian line bundle, and show that the local operators  $d + A_i$  define a unitary connection of curvature  $F$ . Conversely, compare local unitary frames of a Hermitian line bundle with connection and recover the integral period condition from its transition functions.

**Exercise 3.15** (Prequantum line bundles on coadjoint orbits). A *prequantum line bundle* over  $(M, \omega)$  is a Hermitian line bundle  $L \rightarrow M$  with a unitary connection  $\nabla$  satisfying

$$F_{\nabla} = -\frac{i}{\hbar} \omega.$$

Prove that a prequantum line bundle exists if and only if

$$\frac{1}{2\pi\hbar} \int_{\Sigma} f^* \omega \in \mathbb{Z}$$

for every smooth map  $f : \Sigma \rightarrow M$  from a closed oriented surface. Apply the circle-cocycle construction above to  $F = -i\omega/\hbar$ , identify its local connection forms with symplectic potentials, and prove that

$$A_i \longmapsto A_i + d\phi_i, \quad f_{ij} \longmapsto f_{ij} + \phi_j - \phi_i$$

produces an isomorphic prequantum line bundle with connection. Show that two prequantum line bundles with the same curvature differ by a flat Hermitian line bundle, so curvature alone need not determine the prequantization up to isomorphism. For the  $SO(3)$  coadjoint orbit of radius  $L$ , prove from the Kirillov period that prequantization exists exactly when  $2L/\hbar$  is an integer.

**Exercise 3.16** (Kostant–Souriau operators). For  $f \in C^\infty(M)$ , define

$$\mathcal{Q}_{\text{pre}}(f) = -i\hbar \nabla_{X_f} + f.$$

Use the curvature identity for  $\nabla$  and the convention for Hamiltonian vector fields to prove

$$[\mathcal{Q}_{\text{pre}}(f), \mathcal{Q}_{\text{pre}}(g)] = i\hbar \mathcal{Q}_{\text{pre}}(\{f, g\}).$$

For a Hamiltonian  $G$ -action with moment map  $\mu$ , define

$$d\pi_{\text{pre}}(\xi) = \frac{1}{i\hbar} \mathcal{Q}_{\text{pre}}(\mu^\xi).$$

Prove

$$[d\pi_{\text{pre}}(\xi), d\pi_{\text{pre}}(\eta)] = d\pi_{\text{pre}}([\xi, \eta]),$$

and call  $d\pi_{\text{pre}}$  the prequantum Lie-algebra representation on sections of  $L$ .

### 3.3 $SU(2)$ Coherent-State Kernels

**Exercise 3.17** (Coadjoint-orbit harmonic analysis). (a) Let  $\mathcal{O}_L$  be the coadjoint sphere of radius  $L > 0$  with its Kirillov form, and regard it as an  $SU(2)$  orbit through the spin covering. Put

$$N = \frac{2L}{\hbar}.$$

Assume  $N \in \mathbb{Z}_{>0}$  and prove [70] that  $\mathcal{O}_L$  has the  $SU(2)$ -invariant complex structure  $J$  compatible with the Kirillov form and positive orbit metric; thus  $J^2 = -\text{id}$  and  $\omega(-, J-)$  is positive definite. Let  $T^{0,1}\mathcal{O}_L$  be the  $-i$ -eigenspace of the complex-linear extension of  $J$ . Call a section  $s$  of the prequantum line bundle *polarized* when  $\nabla_X s = 0$  for every  $X \in T^{0,1}\mathcal{O}_L$ . Identify the prequantum bundle with  $\mathcal{O}(N) \rightarrow \mathbb{P}^1$ , solve the polarization equation in the two affine charts, and prove [70] that the polarized sections form the  $SU(2)$  representation

$$V_j = \text{Sym}^N(\mathbb{C}^2), \quad j = \frac{N}{2}.$$

Call this polarized-section space the *geometric quantization* of  $\mathcal{O}_L$ . The action on  $V_j$  is induced from the tensor-power action on  $(\mathbb{C}^2)^{\otimes N}$ . Put  $u_\theta = t_{\theta/2}$  and use  $\chi_j(u_\theta) = U_{2j}(\cos(\theta/2))$  to identify the quantized orbit with the spin- $j$  character.

- (b) Write  $\rho_j$  for the action on  $V_j$ . Let  $P_j$  be the orthogonal projection onto its highest-weight line, let  $d\mu$  be the invariant probability measure on  $\mathcal{O}_L \cong SU(2)/U(1)$ , and define

$$Q_{gU(1)} = \rho_j(g)P_j\rho_j(g)^*.$$

Prove that  $Q$  is well defined on the orbit and use Schur's lemma to derive the coherent-state resolution and transform identities

$$(2j+1) \int_{\mathcal{O}_L} Q_x d\mu(x) = I, \quad \psi = (2j+1) \int_{\mathcal{O}_L} Q_x \psi d\mu(x),$$

$$(2j+1) \int_{\mathcal{O}_L} \|Q_x \psi\|^2 d\mu(x) = \|\psi\|^2.$$

If  $n_x, n_y$  are the corresponding unit orbit directions, prove that the reproducing kernel is

$$\text{Tr}(Q_x Q_y) = \left( \frac{1 + n_x \cdot n_y}{2} \right)^{2j}.$$

### 3.4 Spin Spectroscopy

**Exercise 3.18** (Coadjoint-orbit spin spectroscopy). Normalize the infinitesimal generators by  $[J_a, J_b] = i\hbar\epsilon_{abc}J_c$ . Prove that  $J_3$  has values  $m\hbar$  and that a  $2\pi$  rotation acts by  $(-1)^{2j}$ , whereas a  $4\pi$  rotation acts trivially. For the Zeeman Hamiltonian  $\hat{H}_B = -\gamma BJ_3$ , derive

$$E_m = -\gamma Bm\hbar, \quad m = -j, -j+1, \dots, j.$$

Use the Weyl character to derive the normalized coherent rotation amplitude

$$\mathcal{A}_j(t) = \frac{1}{2j+1} \text{Tr} e^{-it\hat{H}_B/\hbar} = \frac{U_{2j}(\cos(\gamma Bt/2))}{2j+1},$$

and determine its exact destructive-interference times from the zeros of  $U_{2j}$ . Deduce the  $2j+1$  resolved Stern–Gerlach outcomes, the Zeeman spacing  $|\gamma|B\hbar$ , and the  $4\pi$  restoration measured by spin interferometry, stating the field-gradient, weak-coupling, beam-resolution, and coherence assumptions separately.

## 4 Heisenberg Group

### 4.1 Central Extensions

**Exercise 4.1** (Weyl multipliers). A *normalized multiplier* on a locally compact group  $G$  is a continuous map  $\sigma : G \times G \rightarrow \mathbb{T}$  satisfying

$$\sigma(x, e) = \sigma(e, x) = 1, \quad \sigma(x, y)\sigma(xy, z) = \sigma(y, z)\sigma(x, yz).$$

Let  $(E, \omega)$  be a symplectic vector space. For  $E = V \oplus V^*$ , the canonical symplectic form used below is

$$\omega((q, p), (q', p')) = p(q') - p'(q).$$

For  $\lambda \in \mathbb{R}$ , define

$$\sigma_{\lambda\omega}(z, z') = e^{i\lambda\omega(z, z')/2}.$$

Prove that  $\sigma_{\lambda\omega}$  is a normalized multiplier for every  $\lambda \in \mathbb{R}$ .

**Exercise 4.2** (Commutator bicharacter). For a multiplier  $\sigma$  on an abelian group  $A$ , define

$$c_\sigma(x, y) = \sigma(x, y)\overline{\sigma(y, x)}.$$

Multipliers  $\sigma$  and  $\sigma'$  are cohomologous when there is a continuous map  $b : A \rightarrow \mathbb{T}$ ,  $b(e) = 1$ , with

$$\sigma'(x, y) = b(x)b(y)\overline{b(x+y)}\sigma(x, y).$$

For  $E = V \oplus V^*$ , call  $\sigma_{\omega/\hbar}$  the Weyl multiplier.

Prove that  $c_\sigma$  is an alternating bicharacter and depends only on the cohomology class of  $\sigma$ . For the Weyl multiplier on  $V \oplus V^*$ , prove

$$c_{\sigma_{\omega/\hbar}}((q, p), (q', p')) = e^{i(p(q') - p'(q))/\hbar}.$$

Prove

$$c_{\sigma_{\lambda\omega}}(z, z') = e^{i\lambda\omega(z, z')}.$$

**Exercise 4.3** (Heisenberg groups and central characters). A  $\sigma$ -projective unitary representation of a group  $E$  is a strongly continuous map  $U : E \rightarrow U(H)$  satisfying  $U_z U_{z'} = \sigma(z, z') U_{z+z'}$  and  $U_0 = I$ . For a symplectic vector space  $(E, \omega)$ , its *Heisenberg group* is

$$\mathbb{H}(E, \omega) = E \times \mathbb{R}$$

with multiplication

$$(z, s)(z', s') = (z + z', s + s' + \tfrac{1}{2}\omega(z, z')).$$

Write  $\mathbb{H}_n = \mathbb{H}(\mathbb{R}^n \oplus (\mathbb{R}^n)^*, \omega)$  for the canonical symplectic form on  $\mathbb{R}^n \oplus (\mathbb{R}^n)^*$ . Its Lie algebra is  $\mathfrak{h}(E, \omega) = E \oplus \mathbb{R}Z$  with

$$[(z, sZ), (z', s'Z)] = \omega(z, z')Z.$$

Prove that  $\mathbb{H}(E, \omega)$  is a locally compact group with center  $\{0\} \times \mathbb{R}$ . Prove that projective representations of  $E$  with multiplier  $e^{i\lambda\omega/2}$  correspond to representations of  $\mathbb{H}(E, \omega)$  with central character  $e^{i\lambda s}$ .

**Exercise 4.4** (Heisenberg coadjoint orbits and Schrödinger polarization). For a connected simply connected nilpotent group  $G$  and  $\ell \in \mathfrak{g}^*$ , a *polarization* is a Lie subalgebra  $\mathfrak{p} \subseteq \mathfrak{g}$  satisfying

$$\ell([\mathfrak{p}, \mathfrak{p}]) = 0, \quad \dim \mathfrak{p} = \frac{\dim \mathfrak{g} + \dim \mathfrak{g}_\ell}{2}.$$

For  $P = \exp(\mathfrak{p})$ , set  $\chi_\ell(\exp X) = e^{i\ell(X)}$ . The induced representation  $\text{Ind}_P^G \chi_\ell$  acts by left translation on the  $L^2$ -sections over  $G/P$  satisfying  $F(gp) = \chi_\ell(p)^{-1}F(g)$ .

Identify

$$\mathfrak{h}(E, \omega)^* = E^* \oplus \mathbb{R}Z^*.$$

For  $(\xi, \lambda) \in E^* \oplus \mathbb{R}Z^*$ , prove

$$\text{Ad}_{(z, s)}^*(\xi, \lambda) = (\xi - \lambda\omega(z, \cdot), \lambda).$$

Deduce that the orbits with  $\lambda = 0$  are points and that, for  $\lambda \neq 0$ ,

$$\mathcal{O}_\lambda = \{(\xi, \lambda) : \xi \in E^*\}.$$

Prove that

$$E \longrightarrow \mathcal{O}_\lambda, \quad z \longmapsto (-\lambda\omega(z, \cdot), \lambda),$$

pulls the Kirillov form back to  $\lambda\omega$ .

For  $E = V \oplus V^*$  and  $\mathfrak{p} = V^* \oplus \mathbb{R}Z$ , prove that  $\mathfrak{p}$  is a polarization at  $(0, \lambda)$ . Identify  $\mathbb{H}(E, \omega)/\exp(\mathfrak{p})$  with  $V$  and compute the induced action to obtain

$$(\pi_\lambda(q, p, s)\psi)(x) = e^{i\lambda(s+p(x-q/2))}\psi(x-q).$$

Prove that this representation has central character  $e^{i\lambda s}$ . For  $\lambda = 0$ , recover every character of  $E$  from a point orbit.

**Exercise 4.5** (Moyal algebra of a Heisenberg orbit). Let  $(E, \omega)$  be a symplectic vector space, let  $\Pi : E^* \rightarrow E$  be the inverse of  $v \mapsto \omega(v, -)$ , and define

$$\{f, g\} = dg(\Pi(df)), \quad \Pi(k, \ell) = \ell(\Pi(k)).$$

For  $\lambda \neq 0$ , use the orbit map  $E \rightarrow \mathcal{O}_\lambda$  to identify  $\omega$  with the pullback of  $\lambda^{-1}$  times the Kirillov form. A *differential formal deformation* of  $C^\infty(E)$  is an associative  $\mathbb{C}[[\hbar]]$ -bilinear product on  $C^\infty(E)[[\hbar]]$  of the form

$$f \star g = fg + \sum_{r \geq 1} \hbar^r C_r(f, g),$$

where the  $C_r$  are bidifferential operators, 1 is a unit, and

$$C_1(f, g) - C_1(g, f) = i\{f, g\}.$$

Let  $D_\Pi$  be the constant bidifferential operator characterized on decomposable functions by

$$D_\Pi(f \otimes g)(x, y) = d_y g(\Pi(dx f)).$$

Define

$$f \star g = m \circ \exp\left(\frac{i\hbar}{2} D_\Pi\right)(f \otimes g),$$

where  $m$  restricts a function on  $E \times E$  to the diagonal. Prove associativity from the commutation of the induced constant bidifferential operators on  $E^3$ , and prove

$$f \star g = fg + \frac{i\hbar}{2} \{f, g\} + O(\hbar^2).$$

For  $k, \ell \in E^*$  put  $e_k(x) = e^{ik(x)}$  and derive

$$e_k \star e_\ell = e^{-i\hbar \Pi(k, \ell)/2} e_{k+\ell}.$$

Check the multiplier identity directly and prove that every affine symplectic transformation of  $E$  preserves  $\star$ . Call this intrinsic formal algebra the *Moyal algebra* of the Heisenberg orbit.

**Exercise 4.6** (Canonical commutation and Weyl relations). For  $\hbar > 0$ ,  $\ell \in V^*$ , and  $v \in V$ , define on  $\mathcal{S}(V)$

$$Q_\ell \psi(x) = \ell(x) \psi(x), \quad P_v \psi(x) = -i\hbar d\psi_x(v).$$

For

$$(T_q \psi)(x) = \psi(x - q), \quad (M_p^h \psi)(x) = e^{ip(x)/\hbar} \psi(x),$$

prove that the corresponding one-parameter groups have infinitesimal generators  $-iP_v/\hbar$  and  $iQ_\ell/\hbar$ , respectively, and prove

$$M_p^h T_q = e^{ip(q)/\hbar} T_q M_p^h.$$

For

$$W_\hbar(q, p) = e^{-ip(q)/(2\hbar)} M_p^h T_q,$$

prove

$$W_\hbar(z) W_\hbar(z') = e^{i\omega(z, z')/(2\hbar)} W_\hbar(z + z')$$

and identify  $W_\hbar(z)$  with  $\pi_{1/\hbar}(z, 0)$  under the canonical set-theoretic section  $z \mapsto (z, 0)$ . Prove that this section is not a homomorphism and that its failure to preserve multiplication is exactly the displayed multiplier. Differentiate the Weyl relations on  $\mathcal{S}(V)$  to prove

$$[Q_\ell, P_v] = i\hbar \ell(v) I.$$



## 4.2 Stone–von Neumann Theorem

**Exercise 4.7** (Unbounded self-adjoint operators and the Cayley transform). An unbounded operator is a linear map  $A : \mathcal{D}(A) \rightarrow H$  on a dense domain. Its adjoint has domain

$$\mathcal{D}(A^*) = \{y \in H : x \mapsto \langle y, Ax \rangle \text{ is bounded on } \mathcal{D}(A)\}.$$

For  $y \in \mathcal{D}(A^*)$ , the vector  $A^*y$  is determined by  $\langle A^*y, x \rangle = \langle y, Ax \rangle$  for  $x \in \mathcal{D}(A)$ . The operator is *self-adjoint* when  $A = A^*$ , including equality of domains.

For a self-adjoint operator  $A$ , prove that  $A \pm iI$  are bijective from  $\mathcal{D}(A)$  to  $H$  with bounded inverses and that the Cayley transform

$$C_A = (A - iI)(A + iI)^{-1}$$

is unitary with  $\ker(I - C_A) = 0$ . Prove conversely that every unitary  $C$  with  $\ker(I - C) = 0$  determines a self-adjoint operator through the inverse Cayley transform.

**Exercise 4.8** (Unbounded spectral theorem). Apply the bounded normal spectral theorem to  $C_A$  and transport its projection-valued measure by the inverse Cayley map. Prove that there is a unique projection-valued measure  $E_A$  on  $\mathbb{R}$  satisfying

$$A = \int_{\mathbb{R}} \lambda dE_A(\lambda), \quad \mathcal{D}(A) = \left\{ x : \int_{\mathbb{R}} \lambda^2 d\langle x, E_A(\lambda)x \rangle < \infty \right\}.$$

**Exercise 4.9** (Closed quadratic forms and Friedrichs realizations). Let  $H$  be a Hilbert space. A densely defined Hermitian form is a map  $q : \mathcal{Q}(q) \times \mathcal{Q}(q) \rightarrow \mathbb{C}$  on a dense linear subspace, linear in the second variable, conjugate-linear in the first, and satisfying  $q(x, y) = \overline{q(y, x)}$ . It is *bounded below* by  $c \in \mathbb{R}$  when

$$q(x, x) \geq c\|x\|^2 \quad (x \in \mathcal{Q}(q)).$$

For such a form put

$$\|x\|_q^2 = q(x, x) + (1 - c)\|x\|^2.$$

Call  $q$  *closed* when  $\mathcal{Q}(q)$  is complete in this norm, and *closable* when every sequence  $x_n \in \mathcal{Q}(q)$  satisfying  $x_n \rightarrow 0$  in  $H$  and  $\|x_n - x_m\|_q \rightarrow 0$  also satisfies  $\|x_n\|_q \rightarrow 0$ .

Prove the representation theorem for closed semibounded forms [57]: there is a unique self-adjoint operator  $A_q$  bounded below by  $c$  such that

$$\mathcal{Q}(q) = \mathcal{D}((A_q - cI)^{1/2}), \quad q(x, y) = \langle (A_q - cI)^{1/2}x, (A_q - cI)^{1/2}y \rangle + c\langle x, y \rangle,$$

and  $x \in \mathcal{D}(A_q)$  exactly when there is  $z \in H$  satisfying  $q(y, x) = \langle y, z \rangle$  for every  $y \in \mathcal{Q}(q)$ , in which case  $A_q x = z$ . Prove uniqueness from the latter characterization.

Let  $A_0$  be a densely defined symmetric operator bounded below by  $c$ , and put  $q_0(x, y) = \langle x, A_0 y \rangle$  on  $\mathcal{D}(A_0)$ . Prove directly from symmetry, the lower bound, and density that  $q_0$  is closable [57]. Define the *Friedrichs realization* of  $A_0$  to be the operator associated to its closed form  $\bar{q}_0$ . Prove that it extends  $A_0$ , is self-adjoint, and preserves the lower bound. For  $-d^2/dx^2$  on  $C_c^\infty(0, \ell)$ , define  $H_0^1(0, \ell)$  as the completion of  $C_c^\infty(0, \ell)$  in the norm

$$\|f\|_{H^1}^2 = \int_0^\ell (|f|^2 + |f'|^2) dx.$$

Integrate by parts to identify this as the closed form domain and derive the Dirichlet spectral values  $\pi^2 n^2 / \ell^2$ . State how changing the interval length changes these spectral lines.

**Exercise 4.10** (Resolvents, compact resolvent, and the min–max principle). For a closed operator  $A$ , define its resolvent set by

$$\rho(A) = \{z \in \mathbb{C} : A - zI : \mathcal{D}(A) \rightarrow H \text{ is bijective with bounded inverse}\}$$

and put  $R_A(z) = (A - zI)^{-1}$ . For a self-adjoint  $A$ , use the spectral measure to prove

$$R_A(z) = \int_{\mathbb{R}} \frac{1}{\lambda - z} dE_A(\lambda), \quad \|R_A(z)\| \leq \frac{1}{|\operatorname{Im} z|} \quad (z \notin \mathbb{R}),$$

and derive the resolvent identity

$$R_A(z) - R_A(w) = (z - w)R_A(z)R_A(w).$$

If a positively oriented contour  $\Gamma$  in  $\rho(A)$  encloses an isolated compact part  $\Sigma$  of the spectrum, prove

$$E_A(\Sigma) = -\frac{1}{2\pi i} \int_{\Gamma} R_A(z) dz.$$

Call  $A$  an operator with *compact resolvent* when  $R_A(z)$  is compact for one  $z \in \rho(A)$ . Prove from the resolvent identity that it is then compact for every  $z \in \rho(A)$ . If  $H$  is infinite-dimensional and  $A$  is self-adjoint, bounded below, and has compact resolvent, prove that its spectrum consists of eigenvalues of finite multiplicity

$$\lambda_1 \leq \lambda_2 \leq \cdots, \quad \lambda_n \longrightarrow +\infty,$$

listed with multiplicity. For a nonzero subspace  $L \subset \mathcal{D}(A)$  put

$$\mathcal{R}_A(L) = \sup_{x \in L \setminus \{0\}} \frac{\langle x, Ax \rangle}{\langle x, x \rangle}.$$

Use the spectral projections, without choosing eigenvectors, to prove the min–max identity

$$\lambda_n = \inf_{\dim L=n} \mathcal{R}_A(L).$$

For a bounded self-adjoint perturbation  $B$ , apply the same identity to prove

$$|\lambda_n(A + B) - \lambda_n(A)| \leq \|B\|.$$

For a closed semibounded form  $q$  whose associated operator has compact resolvent, extend the argument to prove the form-domain identity

$$\lambda_n(A_q) = \inf_{\substack{L \subset \mathcal{Q}(q) \\ \dim L=n}} \sup_{x \in L \setminus \{0\}} \frac{q(x, x)}{\langle x, x \rangle}.$$

For two such forms  $q_1, q_2$  with a common form domain, deduce

$$q_1(x, x) \leq q_2(x, x) \quad \text{for every } x \implies \lambda_n(A_{q_1}) \leq \lambda_n(A_{q_2}).$$

State how a calibrated operator-norm bound on an uncontrolled perturbation gives an error bar for every predicted discrete spectral line.

**Exercise 4.11** (Stone's theorem). Use the measurable functional calculus of  $E_A$  to prove that  $e^{-itA}$  is a strongly continuous unitary group with generator  $-iA$ . Conversely, let  $(U_t)_{t \in \mathbb{R}}$  be a strongly continuous unitary group and define

$$\mathcal{D}(A) = \left\{ x : \lim_{t \rightarrow 0} i \frac{U_t x - x}{t} \text{ exists} \right\}, \quad Ax = \lim_{t \rightarrow 0} i \frac{U_t x - x}{t}.$$

Prove that  $\mathcal{D}(A)$  is dense and that  $A$  is symmetric. Use the strong integrals

$$i \int_0^\infty e^{-t} U_t dt, \quad -i \int_{-\infty}^0 e^t U_t dt$$

to prove that  $A - iI$  and  $A + iI$  are surjective. Deduce that  $A$  is self-adjoint, prove  $U_t = e^{-itA}$ , and prove uniqueness of  $A$ .

**Exercise 4.12** (Irreducibility of the Schrödinger representation). Prove that an operator on  $L^2(V)$  commuting with every translation and every modulation is scalar: first use the Fourier transform to identify the commutant of translations, and then impose commutation with modulations. Deduce that  $\pi_\lambda$  is irreducible for  $\lambda \neq 0$ .

**Exercise 4.13** (Stone–von Neumann theorem). Let  $U$  be a strongly continuous irreducible unitary representation of  $\mathbb{H}(V \oplus V^*, \omega)$  with central character  $U(0, s) = e^{i\lambda s} I$ , where  $\lambda \neq 0$ . Integrate Schwartz functions on  $V \oplus V^*$  against the canonical section of  $U$ . Verify in the Schrödinger model that Fourier inversion sends such an integrated operator with Schwartz symbol to an integral operator with Schwartz kernel. Prove the Stone–von Neumann theorem [27]: every such  $U$  is unitarily equivalent to  $\pi_\lambda$ . Deduce that the nonzero central character  $\lambda$  determines the irreducible representation up to unitary equivalence.

**Exercise 4.14** (Trace and Hilbert–Schmidt classes). For  $x, y \in H$ , let

$$\theta_{x,y}(\xi) = \langle y, \xi \rangle x.$$

The *trace class*  $S_1(H)$  consists of operators admitting a representation

$$A = \sum_{j=1}^{\infty} \theta_{x_j, y_j}, \quad \sum_{j=1}^{\infty} \|x_j\| \|y_j\| < \infty,$$

where the series converges in operator norm, with norm the infimum of the displayed sums. Define

$$\text{Tr}(A) = \sum_{j=1}^{\infty} \langle y_j, x_j \rangle.$$

The *Hilbert–Schmidt class* is

$$S_2(H) = \{A \in B(H) : A^* A \in S_1(H)\}, \quad \|A\|_2^2 = \text{Tr}(A^* A).$$

Prove that  $S_1(H)$ ,  $\text{Tr}$ , and  $S_2(H)$  are well-defined and that every trace-class operator is compact. For a positive compact operator  $A$ , use its spectral projections to prove

$$A \in S_1(H) \iff \sum_j \lambda_j(A) < \infty, \quad \text{Tr}(A) = \sum_j \lambda_j(A),$$

where the nonzero spectral values are repeated with multiplicity. Prove that  $\langle A, B \rangle_{S_2} = \text{Tr}(A^* B)$  makes  $S_2(H)$  a Hilbert space and that the pairing  $(A, T) \mapsto \text{Tr}(AT)$  identifies  $S_1(H)^*$  isometrically with  $B(H)$ .

**Exercise 4.15** (Coefficient and Fourier–Wigner transforms). Let  $V$  be an  $n$ -dimensional real vector space with Lebesgue measure  $dq$ , let  $\hbar > 0$ , and equip  $V^*$  with the dual measure  $dp$  for which  $(2\pi\hbar)^{-n/2} \int_V e^{-ip(q)/\hbar} f(q) dq$  is unitary on  $L^2$ . Use the translation, modulation, and Weyl operators  $T_q, M_p^h, W_h(q, p)$  already normalized by the Weyl relations. The *Fourier–Wigner transform* is

$$\mathcal{W}_h : S_1(L^2(V)) \longrightarrow C_b(V \oplus V^*), \quad \mathcal{W}_h(A)(z) = (2\pi\hbar)^{-n/2} \text{Tr}(AW_h(z)).$$

Prove

$$W_h(z)^* = W_h(-z).$$

For  $A, B \in S_1(L^2(V)) \cap S_2(L^2(V))$ , prove the Fourier–Wigner orthogonality identity

$$\int_{V \oplus V^*} \overline{\mathcal{W}_h(A)(z)} \mathcal{W}_h(B)(z) dz = \text{Tr}(A^*B).$$

Deduce that  $\mathcal{W}_h$  extends uniquely to a unitary map

$$S_2(L^2(V)) \longrightarrow L^2(V \oplus V^*)$$

and, whenever  $\mathcal{W}_h(A) \in L^1(V \oplus V^*)$ , prove

$$A = (2\pi\hbar)^{-n/2} \int_{V \oplus V^*} \mathcal{W}_h(A)(z) W_h(z)^* dz$$

in the weak operator topology.

**Exercise 4.16** (Schrödinger character). For  $V = \mathbb{R}^n$ , let  $f \in C^\infty(\mathbb{H}_n)$  satisfy

$$(q, p, s) \longmapsto f(\exp(q, p, s)) \in \mathcal{S}(\mathbb{R}^{2n+1})$$

under the canonical identification  $\mathfrak{h}_n = \mathbb{R}^n \oplus (\mathbb{R}^n)^* \oplus \mathbb{R}$ . With Haar measure  $dq dp ds$ , put

$$\pi_\lambda(f) = \int_{\mathbb{H}_n} f(g) \pi_\lambda(g) dg.$$

Prove

$$\text{Tr } \pi_\lambda(f) = (2\pi)^n |\lambda|^{-n} \int_{\mathbb{R}} f(0, 0, s) e^{i\lambda s} ds.$$

Derive this identity directly from the Schrödinger kernel and identify the factor  $|\lambda|^{-n}$  with the symplectic-volume scaling of  $\mathcal{O}_\lambda$ .

**Exercise 4.17** (Heisenberg Plancherel and inversion). For  $f \in L^1(\mathbb{H}_n) \cap L^2(\mathbb{H}_n)$ , use Haar measure  $dq dp ds$  and the scalar Fourier transform  $\hat{h}(\lambda) = \int_{\mathbb{R}} h(s) e^{-i\lambda s} ds = (2\pi)^{1/2} (\mathcal{F}h)(\lambda)$ , so its inverse carries  $(2\pi)^{-1}$ , and define

$$\hat{f}(\lambda) = \int_{\mathbb{H}_n} f(g) \pi_\lambda(g)^* dg.$$

Compute the integral kernel of  $\hat{f}(\lambda)$  and apply Euclidean Plancherel in the central and phase-space variables. Deduce the Plancherel measure  $(2\pi)^{-(n+1)} |\lambda|^n d\lambda$  and prove

$$\|f\|_2^2 = (2\pi)^{-(n+1)} \int_{\mathbb{R} \setminus \{0\}} \|\hat{f}(\lambda)\|_{\text{HS}}^2 |\lambda|^n d\lambda.$$

Prove the inversion formula and the distributional orthogonality of Weyl operators. Prove that representations with trivial central character have zero Plancherel measure.

**Exercise 4.18** (Weyl quantization and Wigner distributions). For  $a \in \mathcal{S}(V \oplus V^*)$ , define  $\text{Op}_h^W(a) : \mathcal{S}(V) \rightarrow \mathcal{S}(V)$  by

$$(\text{Op}_h^W(a)\psi)(x) = \frac{1}{(2\pi\hbar)^n} \int_{V \times V^*} e^{ip(x-y)/\hbar} a\left(\frac{x+y}{2}, p\right) \psi(y) dy dp.$$

For polynomial symbols, use the same formula as an oscillatory integral on  $\mathcal{S}(V)$ .

For  $\psi, \varphi \in \mathcal{S}(V)$ , define

$$W_h(\psi, \varphi)(q, p) = \frac{1}{(2\pi\hbar)^n} \int_V e^{-ip(y)/\hbar} \overline{\psi(q - y/2)} \varphi(q + y/2) dy.$$

Write  $W_h[\psi] = W_h(\psi, \psi)$ .

For  $\ell \in V^*$  and  $v \in V$ , prove

$$\text{Op}_h^W((q, p) \mapsto \ell(q)) = Q_\ell, \quad \text{Op}_h^W((q, p) \mapsto p(v)) = P_v.$$

Prove

$$\text{Op}_h^W(a)^* = \text{Op}_h^W(\bar{a}), \quad \|\text{Op}_h^W(a)\|_{\text{HS}} = (2\pi\hbar)^{-n/2} \|a\|_2,$$

and

$$\langle \psi, \text{Op}_h^W(a)\varphi \rangle = \int_{V \oplus V^*} a(q, p) W_h(\psi, \varphi)(q, p) dq dp.$$

Prove that for  $a, b \in \mathcal{S}(V \oplus V^*)$  there is a unique  $a \star_h b \in \mathcal{S}(V \oplus V^*)$  satisfying

$$\text{Op}_h^W(a \star_h b) = \text{Op}_h^W(a) \text{Op}_h^W(b).$$

Derive the oscillatory-integral formula for  $\star_h$  and prove

$$a \star_h b = ab + \frac{i\hbar}{2} \{a, b\} + O(\hbar^2), \quad \frac{a \star_h b - b \star_h a}{i\hbar} = \{a, b\} + O(\hbar^2)$$

in the Schwartz-symbol topology. Prove that the complete asymptotic expansion agrees coefficient by coefficient with the Moyal product. First prove that every coefficient of that product preserves Schwartz functions, so for  $E = V \oplus V^*$  it restricts to a formal product on  $\mathcal{S}(V \oplus V^*)[[\hbar]]$ . For  $k \in (V \oplus V^*)^*$  put  $e_k(z) = e^{ik(z)}$ . Extend Weyl quantization to these oscillatory symbols and prove that  $\text{Op}_h^W(e_k)$  is unitary and that

$$\text{Op}_h^W(e_k) \text{Op}_h^W(e_\ell) = e^{-i\hbar \Pi(k, \ell)/2} \text{Op}_h^W(e_{k+\ell}).$$

Deduce the operator realization of the Moyal plane-wave identity.

**Exercise 4.19** (Groenewold–van Hove obstruction). Specialize to  $V = \mathbb{R}$  and write  $Q = Q_{\text{id}_\mathbb{R}}$  and  $P = P_1$ . Prove that the polynomials of degree at most two form a Lie subalgebra for the Poisson bracket and that Weyl quantization is an exact Lie-algebra quantization on this subalgebra. Quantize  $q^2 p^2$  using

$$q^2 p^2 = \frac{1}{9} \{q^3, p^3\} = \frac{1}{3} \{q^2 p, q p^2\}$$

and prove that there is no linear quantization of the polynomial Poisson algebra satisfying

$$\mathcal{Q}(1) = I, \quad \mathcal{Q}(q) = Q, \quad \mathcal{Q}(p) = P, \quad \mathcal{Q}(q^m) = Q^m, \quad \mathcal{Q}(p^m) = P^m,$$

together with

$$\mathcal{Q}(\{f, g\}) = (i\hbar)^{-1} [\mathcal{Q}(f), \mathcal{Q}(g)]$$

on a common invariant domain.

**Exercise 4.20** (Classical and quantum evolution). Let  $H \in C^\infty(V \oplus V^*, \mathbb{R})$  be subquadratic, meaning that every derivative of order at least two is bounded. Define its Hamiltonian field by

$$da(X_H) = \{a, H\}$$

for every smooth  $a$ , let  $\Phi_t$  be the flow of  $X_H$ , and let  $\text{Op}_h^W(H)$  denote its self-adjoint realization. Define

$$U_h(t) = e^{-it \text{Op}_h^W(H)/\hbar}.$$

For  $a \in \mathcal{S}(V \oplus V^*)$ , define

$$a_t = a \circ \Phi_t, \quad A_h(t) = U_h(t)^* \text{Op}_h^W(a) U_h(t).$$

Prove

$$\frac{d}{dt} a_t = \{a_t, H\}, \quad \frac{d}{dt} A_h(t) = \frac{i}{\hbar} [\text{Op}_h^W(H), A_h(t)].$$

Prove [38] the semiclassical operator estimate that the Moyal commutator remainder obtained from symbols with the displayed derivative bounds has operator norm  $O(\hbar)$ , uniformly for  $|t| \leq T$ , after division by  $i\hbar$ . Apply Duhamel's formula to prove, for every  $T > 0$ , Egorov's theorem

$$\sup_{|t| \leq T} \|A_h(t) - \text{Op}_h^W(a \circ \Phi_t)\| = O_T(\hbar).$$

Prove exact equality for quadratic  $H$ . Derive Ehrenfest's equations for the expectations of  $Q_\ell$  and  $P_v$  for every  $\ell \in V^*$  and  $v \in V$ .

**Exercise 4.21** (Linear symplectic covariance). Let  $S$  be a linear symplectic automorphism of  $V \oplus V^*$ . Construct unitary implementers for cotangent lifts of linear automorphisms of  $V$ , for symplectic shears defined by quadratic forms, and, after fixing a positive inner product  $b : V \rightarrow V^*$ , for the symplectic exchange  $(q, p) \mapsto (-b^{-1}p, bq)$  implemented by the  $b$ -Fourier transform. Prove for these generators that

$$\mu(S) \text{Op}_h^W(a) \mu(S)^* = \text{Op}_h^W(a \circ S^{-1}), \quad W_h[\mu(S)\psi] = W_h[\psi] \circ S^{-1}.$$

Deduce the covariance for their products and prove that each implementer is determined up to phase. Prove the metaplectic lifting theorem [27]: the multiplier of these implementers has a canonical double cover on which the projective representation becomes unitary, and its two lifts of a fixed symplectic map differ by sign. Use the Groenewold–van Hove obstruction to show why exact covariance cannot extend to every nonlinear canonical transformation.

### 4.3 Hermite and Laguerre Functions

**Exercise 4.22** (Hermite polynomials and functions). Define the physicists' Hermite polynomials by

$$e^{2xt-t^2} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}.$$

The normalized *Hermite functions* are

$$h_n(x) = \frac{e^{-x^2/2} H_n(x)}{(2^n n! \sqrt{\pi})^{1/2}}.$$

Use the generating function and the Gaussian integral to prove that  $(h_n)_{n \geq 0}$  is an orthonormal family in  $L^2(\mathbb{R})$ . Derive the Hermite differential equation and recurrence relations.

**Exercise 4.23** (Mehler kernel). For  $|r| < 1$ , use the Hermite generating function and Gaussian integration to derive Mehler's identity

$$\sum_{n=0}^{\infty} r^n h_n(x) h_n(y) = \frac{1}{\sqrt{\pi(1-r^2)}} \exp\left(-\frac{(1+r^2)(x^2+y^2)-4rxy}{2(1-r^2)}\right).$$

**Exercise 4.24** (Hermite completeness). Let  $K_r$  be the integral operator with the Mehler kernel for  $0 < r < 1$ . Use its explicit Gaussian form to prove

$$K_r f \longrightarrow f \quad \text{in } L^2(\mathbb{R}) \quad (r \uparrow 1).$$

Prove that the image of  $K_r$  lies in the closed linear span of the Hermite functions and deduce that  $(h_n)_{n \geq 0}$  is complete in  $L^2(\mathbb{R})$ .

**Exercise 4.25** (Wigner–Laguerre identity). Use the generalized Laguerre polynomials and their orthogonality. For

$$H_{\text{cl}}(q, p) = \frac{p^2}{2m} + \frac{m\omega^2 q^2}{2}, \quad \psi_n(q) = \left(\frac{m\omega}{\hbar}\right)^{1/4} h_n\left(\sqrt{\frac{m\omega}{\hbar}} q\right),$$

prove

$$W_{\hbar}[\psi_n](q, p) = \frac{(-1)^n}{\pi \hbar} e^{-2H_{\text{cl}}(q, p)/(\hbar\omega)} L_n\left(\frac{4H_{\text{cl}}(q, p)}{\hbar\omega}\right).$$

Verify the position and momentum marginals and identify the zeros and sign changes of the Wigner distribution from those of  $L_n$ .

#### 4.4 Oscillator Spectroscopy

**Exercise 4.26** (Spectral lines and time evolution). Let  $A$  be self-adjoint, let  $B \in B(H)$ , and set  $B(t) = e^{itA/\hbar} B e^{-itA/\hbar}$ . For eigenvalues  $\lambda, \mu$  of  $A$ , write  $P_{\lambda} = E_A(\{\lambda\})$  and prove

$$P_{\mu} B(t) P_{\lambda} = e^{it(\mu-\lambda)/\hbar} P_{\mu} B P_{\lambda}.$$

Use the functional calculus to prove the factorized weak identity

$$B(t) = \left(\int_{\mathbb{R}} e^{it\mu/\hbar} dE_A(\mu)\right) B \left(\int_{\mathbb{R}} e^{-it\lambda/\hbar} dE_A(\lambda)\right).$$

When  $A$  has pure-point spectrum, deduce in the weak operator topology that

$$B(t) = \sum_{\lambda, \mu} e^{it(\mu-\lambda)/\hbar} P_{\mu} B P_{\lambda}.$$

For an initial state  $\psi$ , define the line strength

$$I_{\mu\lambda}(\psi) = \|P_{\mu} B P_{\lambda} \psi\|^2.$$

Prove that a line at  $(\mu - \lambda)/\hbar$  occurs only when this quantity is nonzero, and identify vanishing spectral blocks as selection rules.

**Exercise 4.27** (Oscillator spectroscopy). The *Schwartz space*  $\mathcal{S}(\mathbb{R})$  consists of the smooth functions  $\psi$  satisfying

$$\sup_{x \in \mathbb{R}} (1 + |x|)^m |\psi^{(r)}(x)| < \infty$$

for all  $m, r \geq 0$ . On  $\mathcal{S}(\mathbb{R})$  let

$$(Q\psi)(x) = x\psi(x), \quad (P\psi)(x) = -i\hbar\psi'(x).$$

For

$$\hat{H} = -\frac{\hbar^2}{2M} \frac{d^2}{dx^2} + \frac{M\omega^2 x^2}{2},$$

use the Hermite functions to prove that

$$\psi_n(x) = \left(\frac{M\omega}{\hbar}\right)^{1/4} h_n\left(\sqrt{\frac{M\omega}{\hbar}}x\right), \quad E_n = \hbar\omega\left(n + \frac{1}{2}\right)$$

give the complete spectral decomposition of  $\hat{H}$ . Construct the creation and annihilation operators and compute

$$\langle \psi_k, Q\psi_n \rangle = \sqrt{\frac{\hbar}{2M\omega}} (\sqrt{n+1} \delta_{k,n+1} + \sqrt{n} \delta_{k,n-1}).$$

Deduce the spectral-line prediction  $|E_{n\pm 1} - E_n|/\hbar = \omega$  and the relative line strengths for a linearly coupled oscillator. For  $\alpha \in \mathbb{C}$ , prove convergence and normalization of

$$\psi_\alpha = e^{-|\alpha|^2/2} \sum_{n \geq 0} \frac{\alpha^n}{\sqrt{n!}} \psi_n,$$

show that its number statistics are Poisson with mean  $|\alpha|^2$ , and derive its sinusoidal position expectation. Analytically continue Mehler's identity with the usual  $t - i0$  prescription to obtain the oscillator propagator for  $t \notin (\pi/\omega)\mathbb{Z}$ ,

$$K_t(x, y) = \sqrt{\frac{M\omega}{2\pi i \hbar \sin(\omega t)}} \exp\left[\frac{iM\omega}{2\hbar \sin(\omega t)} ((x^2 + y^2) \cos(\omega t) - 2xy)\right].$$

Determine the Maslov phase obtained by continuation across each caustic. Prove

$$e^{-2\pi i \hat{H}/(\hbar\omega)} = -I, \quad e^{-4\pi i \hat{H}/(\hbar\omega)} = I,$$

and distinguish the projective and observable revival period  $2\pi/\omega$  from the exact operator period  $4\pi/\omega$ . Use these formulas to predict the spectral spacing, coherent oscillation frequency, occupation-number distribution, and revival period measured in an oscillator experiment.

**Exercise 4.28** (Laguerre tomography). For  $F \in \mathcal{S}(\mathbb{R}^2)$ ,  $n_\theta = (\cos \theta, \sin \theta)$ , and  $n_\theta^\perp = (-\sin \theta, \cos \theta)$ , define the Radon transform by

$$(\mathcal{R}F)(\theta, s) = \int_{\mathbb{R}} F(sn_\theta + tn_\theta^\perp) dt.$$

With

$$\hat{h}(k) = \int_{\mathbb{R}} h(s) e^{-iks} ds, \quad \hat{F}(\xi) = \int_{\mathbb{R}^2} F(z) e^{-i\xi \cdot z} dz,$$



which are the same unnormalized transforms and hence differ from the unitary transform  $\mathcal{F}$  by  $(2\pi)^{1/2}$  and  $2\pi$  respectively, prove the Fourier-slice identity

$$\widehat{\mathcal{R}F}(\theta, k) = \widehat{F}(kn_\theta)$$

and the inversion formula

$$F(z) = \frac{1}{(2\pi)^2} \int_0^\pi \int_{\mathbb{R}} |k| \widehat{\mathcal{R}F}(\theta, k) e^{ikn_\theta \cdot z} dk d\theta.$$

For the dimensionless quadrature

$$X_\theta = \sqrt{\frac{m\omega}{\hbar}} Q \cos \theta + \frac{P}{\sqrt{m\hbar\omega}} \sin \theta,$$

prove that its probability density in a state  $\psi$  is the Radon transform of  $W_\hbar[\psi]$  along the corresponding phase-space lines. Apply the Fourier-slice inversion formula to reconstruct the Wigner distribution from the family of quadrature densities. For the oscillator state  $\psi_n$ , use the Wigner–Laguerre identity to prove that the reconstructed radial profile is

$$\frac{(-1)^n}{\pi\hbar} e^{-2H_{\text{cl}}/(\hbar\omega)} L_n\left(\frac{4H_{\text{cl}}}{\hbar\omega}\right),$$

while every quadrature density is a scaled  $e^{-x^2} H_n(x)^2$ . Deduce the predicted number of quadrature nodes, the radial zero circles determined by  $L_n$ , and the sign at the phase-space origin. State how homodyne quadrature data distinguish adjacent number states and detect Wigner negativity after inverse Radon reconstruction. State the harmonic-oscillator, number-state-preparation, quadrature-phase, detection-efficiency, inverse-reconstruction, sampling, and spatial-resolution assumptions under which these nodes, zero circles, and signs are observable.

## 5 Poincaré Group

### 5.1 Minkowski Isometries

**Exercise 5.1** (Minkowski momentum orbits and relativistic dispersion). A *time-oriented Minkowski space* is a real four-dimensional vector space  $M$  with a symmetric nondegenerate form  $g$  of signature  $(1, 3)$  and a chosen component of the cone  $\{v : g(v, v) > 0\}$ . Let  $SO^+(M, g)$  be the group preserving  $g$ , the orientation, and the time orientation. Define the proper orthochronous Poincaré group by

$$\mathcal{P}^+(M, g) = M \rtimes SO^+(M, g), \quad (a, A)(b, B) = (a + Ab, AB).$$

Prove that  $(a, A)$  acts on the affine space underlying  $M$  by  $x \mapsto a + Ax$ , that this identifies  $\mathcal{P}^+(M, g)$  with its orientation- and time-orientation-preserving affine isometries, and that the induced action on momentum covectors factors through  $SO^+(M, g)$ . Use  $g$  to identify  $M^\vee$  with  $M$ . For  $m \geq 0$ , define the positive mass shell

$$\mathcal{O}_m^+ = \{p \in M^\vee : g^{-1}(p, p) = m^2 c^2, p^\sharp \text{ is future-pointing}\},$$

with the zero covector removed when  $m = 0$ .

Let  $u$  be a future-pointing observer with  $g(u, u) = c^2$ . Define

$$E_u(p) = g(p^\sharp, u), \quad \mathbf{p}_u = p^\sharp - \frac{E_u(p)}{c^2} u, \quad |\mathbf{p}_u|^2 = -g(\mathbf{p}_u, \mathbf{p}_u).$$

Prove

$$E_u(p)^2 = c^2 |\mathbf{p}_u|^2 + m^2 c^4$$

and show that  $SO^+(M, g)$  acts transitively on each  $\mathcal{O}_m^+$ . For  $m > 0$ , prove that the stabilizer of a point is the rotation group of its negative-definite orthogonal complement and hence is isomorphic to  $SO(3)$ . For  $m = 0$ , show that the stabilizer preserves the Euclidean plane  $p^\perp/\mathbb{R}p$  and is isomorphic to  $SO(2) \ltimes \mathbb{R}^2$ .

Put  $Q(p) = g^{-1}(p, p)$ . For a Lorentz-invariant volume form  $dV$  on  $M^\vee$ , choose along the shell a vector field  $N$  satisfying  $dQ(N) = 1$ . Define

$$(\iota_N dV)(v_1, v_2, v_3) = dV(N, v_1, v_2, v_3)$$

and restrict this three-form to the shell. Prove that the restriction is independent of  $N$  and constructs an invariant measure  $d\mu_m$ , unique up to scale. Show that in the observer decomposition it is proportional to

$$\frac{d^3 \mathbf{p}_u}{E_u(\mathbf{p})}.$$

Deduce that the mass-shell relation and this measure are independent of the observer used to express them.

**Exercise 5.2** (The Lorentz spin cover and finite-dimensional fields). Let  $S$  be a two-dimensional complex vector space equipped with a nonzero alternating form  $\epsilon : \Lambda^2 S \rightarrow \mathbb{C}$ . Let  $\text{Herm}(S)$  be the fixed real subspace of  $S \otimes \bar{S}$  under conjugation followed by exchange of the factors, and use  $\epsilon \otimes \bar{\epsilon}$  to define its determinant. Prove that

$$q(X) = \det X$$

has signature  $(1, 3)$ , with the positive-definite forms selecting a time orientation. For  $A \in SL(S)$ , define

$$X \mapsto (A \otimes \bar{A})X.$$

Prove that this action preserves  $q$  and the chosen orientations and induces an exact sequence

$$1 \longrightarrow \{\pm I\} \longrightarrow SL(S) \longrightarrow SO^+(\text{Herm}(S), q) \longrightarrow 1.$$

Prove [31] the finite-dimensional Lorentz-representation classification: every irreducible finite-dimensional complex representation of  $SL(S)$ , regarded as the spin cover of the real Lorentz group, is uniquely of the form

$$\text{Sym}^{2j_L} S \otimes \text{Sym}^{2j_R} \bar{S}, \quad j_L, j_R \in \frac{1}{2}\mathbb{Z}_{\geq 0}.$$

Identify the scalar, left- and right-handed Weyl-spinor, Lorentz-vector, and self-dual and anti-self-dual two-form representations. Restrict to the massive little group  $SU(2) \subset SL(S)$  and derive

$$(j_L, j_R)|_{SU(2)} \cong \bigoplus_{j=|j_L-j_R|}^{j_L+j_R} V_j.$$

Prove that no nontrivial representation in the displayed family is unitary for the full Lorentz group, and distinguish these covariant field representations from the unitary one-particle representations constructed below.

## 5.2 Wigner Classification

**Exercise 5.3** (Wigner induction and the Poincaré particle classification). Let

$$\tilde{\mathcal{P}} = M \rtimes SL(S)$$

be the spin Poincaré group. Fix a Lorentz orbit  $\mathcal{O} \subset M^\vee$ , a point  $p_0 \in \mathcal{O}$ , its little group  $K \subset SL(S)$ , and a unitary representation  $\sigma : K \rightarrow U(V_\sigma)$ . Define the associated bundle

$$\mathcal{E}_\sigma = SL(S) \times_K V_\sigma \longrightarrow \mathcal{O}$$

as the quotient by  $(A, v) \sim (Ak, \sigma(k)^{-1}v)$ , with  $[A, v]$  lying over  $Ap_0$ . Define the Hilbert space  $L^2(\mathcal{O}, \mathcal{E}_\sigma; d\mu)$  of its square-integrable sections. For  $(a, A) \in \tilde{\mathcal{P}}$ , define intrinsically

$$[U(a, A)\psi](p) = e^{-ip(a)/\hbar} A \cdot \psi(A^{-1}p),$$

where  $A \cdot$  is the induced map between the associated fibers. Prove that this is a strongly continuous unitary representation and that the translation spectral measure is supported on  $\mathcal{O}$ .

Call a unitary Poincaré representation *positive-energy* when its translation spectral measure is supported in the closed future cone. Prove the Mackey–Wigner imprimitivity theorem [50, 68]: every irreducible positive-energy unitary representation whose translation spectrum is supported on one orbit is obtained by this construction, and two such representations are equivalent exactly when their orbit and little-group representation data are equivalent. Deduce that a massive particle is labelled by  $m > 0$  and an irreducible  $SU(2)$  representation  $V_j$ , so it has  $2j + 1$  rest-frame polarizations. For a massless orbit, show that little-group representations trivial on the translation subgroup are labelled by helicity  $h \in \frac{1}{2}\mathbb{Z}$ , while nontrivial translation action produces the continuous-spin family and an additional positive scale.

On the common invariant space of smooth vectors, let  $P$  be the translation generator and let  $J$  be the Lorentz bivector generator. Use the isomorphism

$$\Lambda^2 M \longrightarrow \mathfrak{so}(M, g), \quad u \wedge v \longmapsto [w \mapsto g(v, w)u - g(u, w)v]$$

to fix the normalization of  $J$ . The chosen orientation and  $g$  determine the Hodge operator by  $\alpha \wedge * \beta = \langle \alpha, \beta \rangle_g \text{vol}_g$ . Define the Pauli–Lubanski covector by

$$W = *(P \wedge J).$$

With this sign convention for the Minkowski form, prove

$$P^2 = m^2 c^2 I, \quad W^2 = -m^2 c^2 \hbar^2 j(j+1)I$$

in the massive representation and

$$P^2 = 0, \quad W = \hbar P$$

in a finite-helicity massless representation. Explain why this classification constrains masses, spins, helicities, and polarization counts but does not determine the numerical masses or interactions.

**Exercise 5.4** (Covariant wave equations and helicity polarizations). Use units  $c = 1$  in this exercise. Let  $\square$  be the constant-coefficient d’Alembertian whose symbol is  $-g^{-1}(p, p)$ . For  $\hat{\phi} \in C_c^\infty(\mathcal{O}_m^+)$ , put

$$\phi(x) = \int_{\mathcal{O}_m^+} e^{-ip(x)/\hbar} \hat{\phi}(p) d\mu_m(p).$$

Prove

$$(\square + m^2/\hbar^2)\phi = 0$$

and show that completion in the mass-shell norm realizes the positive energy spin-zero Wigner representation.

Let  $\text{Cl}(M, g)$  be the Clifford algebra defined by

$$vw + wv = 2g(v, w)I.$$

A *complex spin module* is a four-dimensional  $\mathbb{Z}/2\mathbb{Z}$ -graded irreducible module  $\Sigma = \Sigma^+ \oplus \Sigma^-$  for the complexified Clifford algebra in which multiplication by  $M$  exchanges the two graded summands. Let  $\mathcal{D}$  be Clifford multiplication followed by differentiation. Prove  $\mathcal{D}^2 = \square$  and show that

$$(i\hbar\mathcal{D} - m)\psi = 0$$

implies the Klein–Gordon equation and that its positive-frequency solution space realizes the massive spin-1/2 mass shell. At  $m = 0$ , prove that the two chiral positive-frequency solution spaces carry helicities  $\pm\frac{1}{2}$ .

Put  $M_{\mathbb{C}} = M \otimes_{\mathbb{R}} \mathbb{C}$ . For a null covector  $p$ , define the complex transverse-polarization space

$$\mathcal{T}_p = \{\varepsilon \in M_{\mathbb{C}}^{\vee} : g^{-1}(p, \varepsilon) = 0\}/\mathbb{C}p.$$

Interpret  $\varepsilon$  as the amplitude of a plane-wave vector potential and the quotient by  $\mathbb{C}p$  as its gauge equivalence. Show that rotations in the massless little group split  $\mathcal{T}_p$  into helicities  $+1$  and  $-1$ , giving the two plane-wave polarizations of the source-free Maxwell equations. Show that the transverse traceless symmetric square has precisely the two helicity lines  $+2$  and  $-2$ . State the gauge, positive-energy, and finite-helicity assumptions required for these polarization counts, without invoking the spin–statistics relation.

### 5.3 Spin-Weighted Harmonics

**Exercise 5.5** (Spin-weighted spherical harmonics and radiation patterns). Write  $S^2 = SU(2)/U(1)$ . The *spin-weight- $s$  line bundle* is the bundle associated to the character of  $U(1)$  under which rotation of an oriented tangent frame through  $\psi$  acts by  $e^{-is\psi}$ . Write it as

$$\mathcal{L}_s = SU(2) \times_{U(1)} \mathbb{C}_s \longrightarrow S^2.$$

For integers  $\ell \geq |s|$  and  $-\ell \leq m \leq \ell$ , define

$${}_sY_{\ell m}(\theta, \phi) = (-1)^s \sqrt{\frac{2\ell+1}{4\pi}} \overline{D_{m, -s}^{\ell}}(R_z(\phi)R_y(\theta)),$$

using the Wigner coefficients in the Condon–Shortley convention. Verify the frame-equivariance law, prove

$$\int_{S^2} {}_sY_{\ell m} \overline{{}_sY_{\ell' m'}} d\Omega = \delta_{\ell\ell'} \delta_{mm'},$$

and prove completeness in  $L^2(S^2, \mathcal{L}_s)$ . Use the Jacobi-polynomial formula for the Wigner  $d$ -functions to express every  ${}_sY_{\ell m}$  as a phase times powers of  $\sin(\theta/2)$  and  $\cos(\theta/2)$  times a Jacobi polynomial.

Deduce that helicity- $h$  radiation has no angular multipoles with  $\ell < |h|$ . In the spin-weight  $-2$  convention, derive

$$_{-2}Y_{2,2} = \sqrt{\frac{5}{64\pi}}(1 + \cos\theta)^2 e^{2i\phi}, \quad _{-2}Y_{2,-2} = \sqrt{\frac{5}{64\pi}}(1 - \cos\theta)^2 e^{-2i\phi}.$$

For a prepared pure quadrupole helicity mode, deduce the normalized angular intensity

$$\frac{I_{\pm}(\theta)}{I_{\pm}^{\max}} = \left( \frac{1 \pm \cos\theta}{2} \right)^4$$

and its symmetry-enforced dark direction. State the linearized massless-spin-2, far-field, single-mode, polarization-frame, and angular-resolution assumptions under which this pattern and the absence of monopole and dipole spin-2 radiation are tested.

## 5.4 Relativistic Propagation

**Exercise 5.6** (Bessel mass-shell kernels and relativistic signal propagation). Use units  $c = \hbar = 1$ . Fix a future-pointing observer and use its orthogonal complement, with the negative Minkowski form, as the Euclidean space in which  $x, p \in \mathbb{R}^3$ . Put  $E_p = \sqrt{|p|^2 + m^2}$ , and let  $\Theta$  be the Heaviside function. Define the retarded Klein–Gordon kernel by

$$G_{\text{ret}}(t, x) = \Theta(t) \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} \frac{\sin(E_p t)}{E_p} e^{ip \cdot x}.$$

Using the Fourier transform of tempered distributions and the radial Fourier–Bessel transform, prove

$$(\square + m^2)G_{\text{ret}} = \delta_{(0,0)}$$

and, for  $r = |x|$  and  $\tau = \sqrt{t^2 - r^2}$ ,

$$G_{\text{ret}}(t, r) = \Theta(t) \left[ \frac{\delta(t^2 - r^2)}{2\pi} - \Theta(t^2 - r^2) \frac{m J_1(m\tau)}{4\pi\tau} \right].$$

Deduce exact vanishing outside the light cone, a light-cone impulse, and the massive interior tail. Show that the tail kernel has proper-time zeros  $m\tau = j_{1,n}$ , where  $j_{1,n}$  are the positive zeros of  $J_1$ .

For  $\text{Re } z > 0$ , define the modified Bessel function by

$$K_\nu(z) = \int_0^\infty e^{-z \cosh u} \cosh(\nu u) du.$$

For the positive-frequency mass-shell kernel

$$\Delta_+(t, x) = \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} e^{-iE_p t + ip \cdot x},$$

prove that at spacelike separation  $\rho = \sqrt{r^2 - t^2} > 0$ ,

$$\Delta_+(t, r) = \frac{m}{4\pi^2 \rho} K_1(m\rho).$$

Derive

$$K_1(z) \sim \sqrt{\frac{\pi}{2z}} e^{-z} \quad (z \rightarrow +\infty)$$

by Laplace's method. Restore  $c$  and  $\hbar$  to identify the Compton length  $\hbar/(mc)$  and the invariant tail-node lengths  $j_{1,n}\hbar/(mc)$ . Distinguish this nonzero spacelike positive-frequency correlation from the causal response represented by  $G_{\text{ret}}$ .

For a wave packet concentrated near spatial momentum  $p_0$ , derive

$$v_{\text{group}} = \frac{c^2 p_0}{\sqrt{c^2 |p_0|^2 + m^2 c^4}}, \quad \Delta t \sim \frac{L m^2 c}{2 |p_0|^2} \quad (|p_0| \gg mc)$$

for its delay relative to a light signal over distance  $L$ . State the flat-spacetime, free-field, point-impulse, wave-packet, and detector-smearing assumptions under which the causal front, Bessel tail nodes, exponential spacelike decay, and time-of-flight delay are physical predictions.

## 6 Potential Scattering

### 6.1 Euclidean Group

**Exercise 6.1** (Outgoing Green kernels and the Lippmann–Schwinger equation). Let  $\mu > 0$  be a reduced mass, let  $k > 0$ , put

$$E = \frac{\hbar^2 k^2}{2\mu}, \quad H = -\frac{\hbar^2}{2\mu} \Delta + V,$$

and initially take  $V$  to be a real Schwartz function on  $\mathbb{R}^3$ . For an incident direction  $\omega \in S^2$ , call a solution of  $(H - E)\psi = 0$  *outgoing* when

$$\psi(x) = e^{ik\omega \cdot x} + f(\hat{x}, \omega; k) \frac{e^{ik|x|}}{|x|} + o(|x|^{-1}), \quad \hat{x} = \frac{x}{|x|},$$

and call  $f$  its *scattering amplitude*. For outgoing direction  $\omega'$ , define the scattering angle by  $\cos \theta = \omega' \cdot \omega$ .

Define the Euclidean group and its unitary action by

$$E(3) = \mathbb{R}^3 \rtimes SO(3), \quad (U_{a,R}\phi)(x) = \phi(R^{-1}(x - a)).$$

Prove that the free Hamiltonian commutes with this action, that  $H$  commutes with every rotation when  $V$  is radial, and that translations are broken unless  $V$  is constant. For radial  $V$ , prove

$$f(R\omega', R\omega; k) = f(\omega', \omega; k).$$

Prove in  $\mathcal{D}'(\mathbb{R}^3)$  that

$$G_k^+(x) = \frac{e^{ik|x|}}{4\pi|x|}, \quad (-\Delta - k^2)G_k^+ = \delta_0,$$

and verify the Sommerfeld condition  $(\partial_r - ik)G_k^+ = o(r^{-1})$ . Derive the Lippmann–Schwinger equation

$$\psi(x) = e^{ik\omega \cdot x} - \frac{2\mu}{\hbar^2} \int_{\mathbb{R}^3} G_k^+(x - y) V(y) \psi(y) dy.$$

First take  $V$  compactly supported and use

$$G_k^+(x - y) = \frac{e^{ik|x|}}{4\pi|x|} e^{-ik\hat{x} \cdot y} + O(|x|^{-2})$$

for bounded  $y$  to prove

$$f(\omega', \omega; k) = -\frac{\mu}{2\pi\hbar^2} \int_{\mathbb{R}^3} e^{-ik\omega' \cdot y} V(y) \psi(y) dy.$$

Extend the formula to Schwartz  $V$  by controlling the tails with its decay.

**Exercise 6.2** (Probability current, Born series, and screened-Coulomb counts). For the probability current

$$j_\psi = \frac{\hbar}{\mu} \text{Im}(\bar{\psi} \nabla \psi),$$

divide the outgoing radial flux through  $r^2 d\Omega$  by the incident flux and derive

$$\frac{d\sigma}{d\Omega} = |f(\omega', \omega; k)|^2.$$

Replace  $\psi$  in the amplitude by the incident plane wave to obtain the first Born term

$$f_B(\omega', \omega; k) = -\frac{\mu}{2\pi\hbar^2} \int_{\mathbb{R}^3} e^{-iQ \cdot y} V(y) dy, \quad Q = k(\omega' - \omega).$$

On any Banach space  $X$  containing the incident plane wave and on which

$$K_k \phi = \frac{2\mu}{\hbar^2} G_k^+ * (V\phi)$$

is bounded, prove that  $\|K_k\|_{X \rightarrow X} < 1$  implies the convergent Born series

$$\psi = \sum_{n \geq 0} (-K_k)^n e^{ik\omega \cdot x}.$$

Writing  $q = \|K_k\|_{X \rightarrow X}$ , prove the geometric remainder bounds

$$\|\psi - e^{ik\omega \cdot x}\|_X \leq \frac{q}{1-q} \|e^{ik\omega \cdot x}\|_X,$$

$$\|\psi - (I - K_k)e^{ik\omega \cdot x}\|_X \leq \frac{q^2}{1-q} \|e^{ik\omega \cdot x}\|_X,$$

and explain which bound controls the first and second Born approximations to the amplitude when the amplitude integral is a bounded functional on  $X$ . For the screened Coulomb potential

$$V_\alpha(x) = g \frac{e^{-\alpha|x|}}{|x|}, \quad \alpha > 0,$$

justify the same calculation by approximation, prove

$$\int_{\mathbb{R}^3} e^{-iQ \cdot x} \frac{e^{-\alpha|x|}}{|x|} dx = \frac{4\pi}{\|Q\|^2 + \alpha^2}, \quad \|Q\|^2 = 4k^2 \sin^2 \frac{\theta}{2},$$

and derive

$$\frac{d\sigma_B}{d\Omega} = \frac{4\mu^2 g^2}{\hbar^4 (4k^2 \sin^2(\theta/2) + \alpha^2)^2}.$$

State the weak-coupling, single-scattering, wave-packet, far-field, and angular resolution assumptions under which this differential cross section predicts detector counts.

## 6.2 Partial Waves

**Exercise 6.3** (Scattering operators and elastic unitarity). For a real radial potential supported in  $\{x : \|x\| \leq R\}$ , define the on-shell scattering operator  $S(k)$  on angular wave packets by sending the incoming far-field coefficient to the outgoing one, with the free propagation phases normalized so that  $V = 0$  gives  $S(k) = I$ . Use conservation of probability current to prove that  $S(k)$  is unitary, and use rotation covariance together with the multiplicity-free decomposition

$$L^2(S^2) = \widehat{\bigoplus_{\ell \geq 0} \mathcal{H}_\ell}$$

where  $\mathcal{H}_\ell$  is the degree- $\ell$  spherical-harmonic summand, to prove

$$S(k)|_{\mathcal{H}_\ell} = S_\ell(k)I, \quad S_\ell(k) = e^{2i\delta_\ell(k)}$$

for real phase shifts  $\delta_\ell$ , determined modulo  $\pi$ .

**Exercise 6.4** (Partial waves, Legendre amplitudes, and the optical theorem). Put

$$j_\ell(z) = \sqrt{\frac{\pi}{2z}} J_{\ell+1/2}(z).$$

Project the Euclidean Fourier kernel onto the spherical-harmonic summands and derive the distinguished expansion

$$e^{ikr \cos \theta} = \sum_{\ell \geq 0} i^\ell (2\ell + 1) j_\ell(kr) P_\ell(\cos \theta),$$

where  $P_\ell = P_\ell^{(0,0)}$  is the Legendre polynomial. Separate the radial equation and show that the regular radial function  $u_\ell$  satisfies

$$u_\ell''(r) + \left( k^2 - \frac{\ell(\ell+1)}{r^2} - \frac{2\mu}{\hbar^2} V(r) \right) u_\ell(r) = 0,$$

with

$$u_\ell(r) \sim \sin \left( kr - \frac{\ell\pi}{2} + \delta_\ell(k) \right) \quad (r \rightarrow \infty).$$

Match incoming and outgoing spherical waves to derive

$$f(\theta; k) = \frac{1}{2ik} \sum_{\ell \geq 0} (2\ell + 1) (S_\ell(k) - 1) P_\ell(\cos \theta) = \frac{1}{k} \sum_{\ell \geq 0} (2\ell + 1) e^{i\delta_\ell} \sin \delta_\ell P_\ell(\cos \theta).$$

Use Legendre orthogonality and  $P_\ell(1) = 1$  to prove

$$\sigma_{\text{el}}(k) = \frac{4\pi}{k^2} \sum_{\ell \geq 0} (2\ell + 1) \sin^2 \delta_\ell(k), \quad \sigma_{\text{el}}(k) = \frac{4\pi}{k} \text{Im } f(0; k),$$

and obtain the partial-wave unitarity bound

$$\sigma_\ell(k) \leq \frac{4\pi}{k^2} (2\ell + 1).$$

Identify exactly where reality of the potential and absence of inelastic channels enter the proof.



**Exercise 6.5** (Hard-sphere phase shifts and effective-range scattering). Define the spherical Neumann function intrinsically by

$$n_\ell(z) = -(-z)^\ell \left( \frac{1}{z} \frac{d}{dz} \right)^\ell \frac{\cos z}{z}.$$

Prove that it satisfies the spherical Bessel equation, that its Wronskian with  $j_\ell$  is  $z^{-2}$ , and that  $n_\ell(z) \sim -(2\ell - 1)!! z^{-\ell-1}$  as  $z \downarrow 0$ . For scattering from an impenetrable sphere of radius  $a$ , impose the Dirichlet condition at  $r = a$  and write the exterior radial factor as

$$\cos \delta_\ell j_\ell(kr) - \sin \delta_\ell n_\ell(kr).$$

Prove

$$\tan \delta_\ell(k) = \frac{j_\ell(ka)}{n_\ell(ka)}.$$

Use the Bessel series to show that  $\delta_0 = -ka$  modulo  $\pi$  and  $\delta_\ell = O((ka)^{2\ell+1})$  for  $\ell > 0$ . Deduce the low-energy predictions

$$f(\theta; k) \longrightarrow -a, \quad \sigma_{\text{el}}(k) \longrightarrow 4\pi a^2 \quad (ka \downarrow 0).$$

For a short-range  $s$ -wave phase shift, define the effective-range parameters by

$$k \cot \delta_0(k) = -\frac{1}{a_s} + \frac{1}{2} r_e k^2 + O(k^4).$$

For the hard sphere, use  $\delta_0 = -ka$  to prove

$$a_s = a, \quad r_e = \frac{2a}{3}.$$

State how low-energy measurements at two or more momenta distinguish the scattering length from the effective range.

**Exercise 6.6** (Square-well thresholds and Breit–Wigner resonances). For the attractive square well

$$V(r) = \begin{cases} -V_0, & 0 < r < a, \\ 0, & r > a, \end{cases} \quad V_0 > 0,$$

put  $K = (k^2 + 2\mu V_0/\hbar^2)^{1/2}$  and match the logarithmic derivatives of the regular  $s$ -wave solutions to prove

$$K \cot(Ka) = k \cot(ka + \delta_0(k)).$$

Define the scattering length by  $\delta_0(k) = -ka_s + o(k)$  and derive

$$a_s = a - \frac{\tan(\gamma a)}{\gamma}, \quad \gamma = \frac{\sqrt{2\mu V_0}}{\hbar}.$$

Show that  $a_s$  diverges when  $\gamma a = (n + \frac{1}{2})\pi$ ,  $n \in \mathbb{Z}_{\geq 0}$ , and identify these values with the appearance of an  $s$ -wave state at zero binding energy.

If an elastic phase shift has a simple resonance at  $E_R$  with

$$\cot \delta_\ell(E) = \frac{2(E_R - E)}{\Gamma} + O\left(\frac{(E - E_R)^2}{\Gamma^2}\right), \quad \Gamma > 0,$$

derive the Breit–Wigner line shape

$$\sigma_\ell(E) = \frac{4\pi}{k^2} (2\ell + 1) \frac{\Gamma^2/4}{(E - E_R)^2 + \Gamma^2/4}$$

to leading order, prove that its full width at half maximum is  $\Gamma$ , and show that the Wigner delay  $2\hbar d\delta_\ell/dE$  equals  $4\hbar/\Gamma$  at the resonance. State the hard-core or square-well model, elastic-channel, energy-resolution, and finite-range assumptions needed to infer a radius, scattering length, resonance energy, or width from measured angular distributions.

**Exercise 6.7** (Jost functions, Levinson’s theorem, and bound-state counting). Let  $V$  be a real radial potential supported in  $[0, R]$  and suppose  $rV(r)$  is integrable. For angular momentum  $\ell$ , let  $\varphi_\ell(k, r)$  be the regular solution of the radial equation, normalized by  $\varphi_\ell(k, r) \sim r^{\ell+1}$  at the origin, and let  $f_\ell(k, r)$  be the outgoing solution normalized by

$$f_\ell(k, r) \sim e^{ikr} \quad (r \rightarrow \infty).$$

Define the Jost function by the  $r$ -independent Wronskian

$$F_\ell(k) = W(f_\ell(k, -), \varphi_\ell(k, -)).$$

Prove from constancy of the Wronskian and reality of  $V$  that, for  $k > 0$ ,

$$F_\ell(-k) = \overline{F_\ell(k)}.$$

With the conventional partial-wave asymptotic  $\sin(kr - \ell\pi/2 + \delta_\ell)$ , include the free centrifugal phase and prove

$$S_\ell(k) = e^{2i\delta_\ell(k)} = (-1)^\ell \frac{F_\ell(-k)}{F_\ell(k)}.$$

Prove that zeros  $k = i\kappa$ ,  $\kappa > 0$ , are exactly the square-integrable bound states of energy  $-\hbar^2\kappa^2/(2\mu)$ . Choose the continuous phase shift with  $\delta_\ell(k) \rightarrow 0$  as  $k \rightarrow \infty$ . Prove the Jost-function argument-principle form of Levinson’s theorem [53] and prove

$$\delta_\ell(0) - \delta_\ell(\infty) = n_\ell\pi$$

when there is no zero-energy resonance, where  $n_\ell$  is the number of bound states in the  $\ell$ th channel. In the  $s$ -wave channel derive from a small semicircle around  $k = 0$  the threshold formula

$$\delta_0(0) - \delta_0(\infty) = \left(n_0 + \frac{1}{2}\right)\pi$$

when there is a zero-energy resonance.

For the square well above and away from its threshold values, prove

$$n_0 = \left\lfloor \frac{\gamma a}{\pi} + \frac{1}{2} \right\rfloor$$

and verify that each divergence of the scattering length increases the continuous value of  $\delta_0(0)$  by  $\pi$ . State the finite-range, elasticity, zero-energy extrapolation, and phase-unwrapping assumptions under which scattering data determine an integer bound-state count.

**Exercise 6.8** (Marchenko reconstruction and the one-soliton potential). For the Schrödinger operator  $L = -d^2/dx^2 + V$  with a real potential on the line satisfying  $\int_{\mathbb{R}} (1 + |x|)|V(x)| dx < \infty$ , let  $f_+(k, x)$  and  $f_-(k, x)$  be the Jost solutions normalized by

$$f_+(k, x)e^{-ikx} \rightarrow 1 \quad (x \rightarrow +\infty), \quad f_-(k, x)e^{ikx} \rightarrow 1 \quad (x \rightarrow -\infty).$$

For real  $k \neq 0$ , define the right-incident reflection and transmission coefficients by the asymptotics

$$\psi_R(k, x) \sim e^{-ikx} + R(k)e^{ikx} \quad (x \rightarrow +\infty), \quad \psi_R(k, x) \sim T(k)e^{-ikx} \quad (x \rightarrow -\infty).$$

The bound states are the poles  $i\kappa_j$ ,  $\kappa_j > 0$ , of the meromorphic transmission coefficient. Normalize their real eigenfunctions so that

$$\|\psi_j\|_2 = 1, \quad \psi_j(x) \sim c_j^{1/2} e^{-\kappa_j x} \quad (x \rightarrow +\infty),$$

and call  $c_j > 0$  the right norming constants. Define

$$\mathcal{F}(s) = \frac{1}{2\pi} \int_{\mathbb{R}} R(k) e^{iks} dk + \sum_j c_j e^{-\kappa_j s}.$$

The Marchenko kernel is the solution of

$$K(x, y) + \mathcal{F}(x + y) + \int_x^\infty K(x, z) \mathcal{F}(z + y) dz = 0, \quad y \geq x,$$

and the reconstructed potential is

$$V(x) = -2 \frac{d}{dx} K(x, x).$$

Assume  $R(-k) = \overline{R(k)}$ ,  $\kappa_j, c_j > 0$ , and that  $\mathcal{F}$  is real and absolutely continuous with  $\mathcal{F}, \mathcal{F}' \in L^1([a, \infty))$  for every  $a$ . On  $L^2([x, \infty))$  define

$$(H_x h)(y) = \int_x^\infty \mathcal{F}(y + z) h(z) dz,$$

and assume  $I + H_x$  is strictly positive for every  $x$ . Prove the Marchenko existence and uniqueness theorem [22]: the integral equation has a unique solution, its diagonal defines a real potential with the stated weighted integrability, and its right scattering data are exactly  $(R, \{\kappa_j, c_j\})$ . Prove that two potentials with the same data therefore coincide.

In the reflectionless one-bound-state case put

$$R(k) = 0, \quad \mathcal{F}(s) = 2\kappa e^{2\kappa x_0} e^{-\kappa s}.$$

Solve the integral equation and prove

$$K(x, y) = -\frac{2\kappa e^{2\kappa x_0} e^{-\kappa(x+y)}}{1 + e^{-2\kappa(x-x_0)}}, \quad V(x) = -2\kappa^2 \operatorname{sech}^2(\kappa(x - x_0)).$$

Compute its transmission coefficient

$$T(k) = \frac{k + i\kappa}{k - i\kappa}$$

and recover the bound-state pole from transmission data. Let the norming constant evolve by  $c(t) = c(0)e^{8\kappa^3 t}$  and show that  $u(x, t) = -V(x, t)$  solves the Korteweg–de Vries equation

$$u_t + 6uu_x + u_{xxx} = 0$$

and is the soliton

$$u(x, t) = 2\kappa^2 \operatorname{sech}^2(\kappa(x - x_0 - 4\kappa^2 t)).$$

State the one-dimensional, reflectionless, full-line, coherent-scattering, and weakly dispersive assumptions under which the measured pole and pulse speed test the inverse-scattering reconstruction.

### 6.3 Coulomb Functions

**Exercise 6.9** (Coulomb radial functions and phase shifts). For the repulsive Coulomb potential  $V(r) = \kappa/r$ , with  $\kappa > 0$ , put

$$\rho = kr, \quad \eta = \frac{\mu\kappa}{\hbar^2 k}, \quad C_\ell(\eta) = \frac{2^\ell e^{-\pi\eta/2} |\Gamma(\ell + 1 + i\eta)|}{(2\ell + 1)!}.$$

Verify that the regular radial solution is

$$F_\ell(\eta, \rho) = C_\ell(\eta) \rho^{\ell+1} e^{-i\rho} M(\ell + 1 - i\eta, 2\ell + 2, 2i\rho).$$

Derive from the Kummer series and the beta integral that

$$M(a, b, z) = \frac{\Gamma(b)}{\Gamma(a)\Gamma(b-a)} \int_0^1 e^{zt} t^{a-1} (1-t)^{b-a-1} dt$$

when  $\text{Re } b > \text{Re } a > 0$ . Analyze its two endpoints for the displayed Coulomb parameters and prove

$$F_\ell(\eta, \rho) = \sin\left(\rho - \eta \log(2\rho) - \frac{\ell\pi}{2} + \sigma_\ell(\eta)\right) + o(1), \quad \sigma_\ell(\eta) = \arg \Gamma(\ell + 1 + i\eta),$$

where the argument is chosen continuously from  $\sigma_\ell(0) = 0$ . Deduce the Coulomb partial-wave factor

$$S_\ell^C = e^{2i\sigma_\ell} = \frac{\Gamma(\ell + 1 + i\eta)}{\Gamma(\ell + 1 - i\eta)}.$$

### 6.4 Rutherford Scattering

**Exercise 6.10** (Coulomb amplitude and Rutherford detector counts). Because the Coulomb potential is not short-range, use the logarithmically modified radial phase above and define its nonforward amplitude by Abel regularization of the partial-wave expression.

For  $\text{Re } s > 0$ , use the hypergeometric form of  $P_\ell$  to prove

$$\int_{-1}^1 (1-x)^{s-1} P_\ell(x) dx = 2^s (-1)^\ell \frac{\Gamma(s)^2}{\Gamma(s-\ell)\Gamma(s+\ell+1)}.$$

Continue in  $s$  and use the Legendre generating function to obtain, for  $-1 < x < 1$ , the Abel-summed identity

$$\lim_{r \uparrow 1} \sum_{\ell \geq 0} (2\ell + 1)(S_\ell^C - 1) r^\ell P_\ell(x) = -2i\eta 2^{i\eta} \frac{\Gamma(1+i\eta)}{\Gamma(1-i\eta)} (1-x)^{-1-i\eta}.$$

With  $x = \cos \theta$ , derive the exact Coulomb amplitude away from the forward direction,

$$f_C(\theta) = -\frac{\eta}{2k \sin^2(\theta/2)} \frac{\Gamma(1+i\eta)}{\Gamma(1-i\eta)} \exp\left(-i\eta \log \sin^2 \frac{\theta}{2}\right),$$

and hence the Rutherford law

$$\frac{d\sigma_C}{d\Omega} = \left( \frac{\kappa}{4E \sin^2(\theta/2)} \right)^2.$$

For incident particle rate  $\dot{N}_{\text{in}}$ , target areal density  $n_T$ , acquisition time  $T$ , detector efficiency  $\epsilon(\theta)$ , and a narrow solid-angle bin  $\Delta\Omega$  away from  $\theta = 0$ , derive

$$N(\theta; \Delta\Omega) \simeq \dot{N}_{\text{in}} n_T T \epsilon(\theta) \frac{d\sigma_C}{d\Omega} \Delta\Omega.$$

State separately the assumptions of nonrelativistic spinless two-body motion, a thin target, pointlike unscreened charges, single scattering, negligible inelasticity, far-field detection, and sufficiently fine angular bins. State how screening, finite charge radius, recoil calibration, multiple scattering, and detector acceptance alter the comparison with measured counts.

## 7 Schwarzschild Spacetime

### 7.1 Schwarzschild Isometries

**Exercise 7.1** (Schwarzschild geometry and gravitational redshift). A *Lorentzian metric* on a four-manifold is a smooth nondegenerate symmetric form of signature  $(+, -, -, -)$ . Call a tangent vector timelike, null, or spacelike according as its squared length is positive, zero, or negative. Prove the fundamental theorem of pseudo-Riemannian geometry: there is a unique torsion-free metric-compatible connection  $\nabla$ , characterized by the Koszul formula. Define geodesics by  $\nabla_{\dot{\gamma}}\dot{\gamma} = 0$ , curvature by

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z,$$

and  $\text{Ric}(X, Y) = \text{Tr}(Z \mapsto R(Z, X)Y)$ . Let

$$r_s = \frac{2G_N M}{c^2}, \quad f(r) = 1 - \frac{r_s}{r},$$

and on  $(r_s, \infty) \times \mathbb{R} \times S^2$  consider

$$g = f(r)c^2 dt^2 - f(r)^{-1} dr^2 - r^2 g_{S^2}.$$

Prove Birkhoff's theorem [67]: every spherically symmetric Lorentzian vacuum solution with  $\text{Ric} = 0$  is locally isometric to a member of this family. Compute its Ricci tensor and verify  $\text{Ric} = 0$ . Compute the invariant

$$R_{abcd}R^{abcd} = \frac{48G_N^2 M^2}{c^4 r^6}$$

and use it to distinguish the curvature singularity at  $r = 0$  from the coordinate singularity at  $r = r_s$ .

Prove that time translation preserves  $g$ . For its timelike Killing field  $\partial_t$ , prove that  $g(k, \partial_t)$  is conserved along a null geodesic with tangent  $k$ . Compare it with the frequency measured by static observers to derive

$$\frac{\nu_{\text{rec}}}{\nu_{\text{emit}}} = \sqrt{\frac{f(r_{\text{emit}})}{f(r_{\text{rec}})}}.$$

In particular, for reception at infinity predict  $\nu_{\infty}/\nu_r = \sqrt{1 - r_s/r}$ . State the static-emitter, static-receiver, geometric-optics, negligible-backreaction, and spectral-resolution assumptions under which a clock or spectral-line measurement tests the formula.

### 7.2 Geodesic Reduction

**Exercise 7.2** (Weierstrass elliptic functions and Schwarzschild light rays). Restrict a Schwarzschild null geodesic to an equatorial great circle. Let  $\mathcal{E} = f(r)c^2 \dot{t}$  and  $L = r^2 \dot{\phi}$  be its conserved energy and angular momentum per affine parameter, put  $b = cL/\mathcal{E}$ , and set  $u = 1/r$ . Derive

$$\left(\frac{du}{d\phi}\right)^2 = \frac{1}{b^2} - u^2 + r_s u^3, \quad \frac{d^2 u}{d\phi^2} + u = \frac{3}{2} r_s u^2.$$

For  $g_2^3 - 27g_3^2 \neq 0$ , prove the Weierstrass uniformization theorem: there is a lattice  $\Lambda \subset \mathbb{C}$ , unique up to homothety with the prescribed invariants, and a unique even  $\Lambda$ -periodic meromorphic function satisfying

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3, \quad \wp(z) = z^{-2} + O(z^2),$$

and this function gives the universal covering of the nonsingular cubic. For a nonradial orbit whose cubic is nonsingular, prove from uniqueness of solutions on that cubic that it is given by

$$u(\phi) = \frac{1/3 + 4\wp(\phi - \phi_0; g_2, g_3)}{r_s}, \quad g_2 = \frac{1}{12}, \quad g_3 = \frac{1}{216} - \frac{r_s^2}{16b^2}.$$

Show that its nontrivial finite-impact-parameter discriminant degeneration occurs at

$$r_{\text{ph}} = \frac{3}{2}r_s = \frac{3G_{\text{N}}M}{c^2}, \quad b_c = \frac{3\sqrt{3}G_{\text{N}}M}{c^2},$$

the unstable circular photon orbit and critical capture impact parameter. Expand the orbit equation for  $r_s/b \ll 1$  to derive the deflection angle

$$\delta\phi = \frac{4G_{\text{N}}M}{bc^2} + O\left(\frac{G_{\text{N}}^2M^2}{b^2c^4}\right).$$

Predict the far-field shadow angular radius  $b_c/D$  for an observer at distance  $D$ . State the vacuum, nonrotating, geometric-optics, thin-lens, far-observer, and source-model assumptions separately from the bending and shadow predictions.

### 7.3 Confluent Heun Functions

**Exercise 7.3** (Regge–Wheeler equations and confluent-Heun scattering). Define the tortoise coordinate by

$$r_* = r + r_s \log\left(\frac{r}{r_s} - 1\right).$$

For integer spin  $s \in \{0, 1, 2\}$  and  $\ell \geq s$ , insert the scalar, Maxwell, and linearized Einstein equations into the spin-weighted spherical-harmonic decomposition and prove the separation theorem [58, 16]: after expansion in modes with time dependence  $e^{-i\omega t}$  and the corresponding spherical harmonics, the radial master field satisfies

$$\frac{d^2\psi}{dr_*^2} + \left(\frac{\omega^2}{c^2} - V_{\ell,s}(r)\right)\psi = 0,$$

$$V_{\ell,s}(r) = f(r) \left(\frac{\ell(\ell+1)}{r^2} + (1-s^2)\frac{r_s}{r^3}\right).$$

Show that  $r = r_s$  and  $r = \infty$  are respectively regular and irregular singular endpoints. After putting  $z = r/r_s$  and factoring the ingoing or outgoing exponential and power behavior, transform the radial equation to

$$y'' + \left(\alpha + \frac{\beta+1}{z} + \frac{\gamma+1}{z-1}\right)y' + \left(\frac{\mu}{z} + \frac{\nu}{z-1}\right)y = 0.$$

When the exponent difference is nonintegral, define  $H_0$  and  $H_1$  to be the Frobenius solutions with exponent zero normalized by  $H_0(0) = H_1(1) = 1$ ; in resonant cases use their analytic limits. Call these the local confluent Heun functions. Substitute their power series into the equation to derive

the three-term coefficient recurrences and their radii of convergence. Determine  $\alpha, \beta, \gamma, \mu, \nu$  from  $\omega r_s/c$ ,  $\ell$ , and  $s$  for the chosen asymptotic factors.

Define the horizon-ingoing and infinity-outgoing solutions by

$$\psi_{\text{in}} \sim e^{-i\omega r_*/c} \quad (r_* \rightarrow -\infty), \quad \psi_{\text{out}} \sim e^{+i\omega r_*/c} \quad (r_* \rightarrow +\infty),$$

and define the connection determinant

$$D_{\ell,s}(\omega) = W(\psi_{\text{in}}, \psi_{\text{out}}).$$

Use constancy of the Wronskian to express reflection and transmission amplitudes as confluent-Heun connection coefficients and prove flux conservation for real  $\omega$ .

## 7.4 Black-Hole Ringdown

**Exercise 7.4** (Black-hole resonances, photon spheres, and ringdown). Define a Schwarzschild quasinormal frequency to be a zero of the confluent-Heun connection determinant  $D_{\ell,s}(\omega)$  in the lower half-plane. Show that it is equivalently a mode ingoing at the horizon and outgoing at infinity, and that its time dependence is a damped oscillation.

For  $\ell \rightarrow \infty$ , locate the maximum of the Regge–Wheeler potential at the photon sphere. Prove the leading parabolic-barrier WKB connection formula: if  $V$  has a nondegenerate maximum  $V_0$  and  $V_0'' = d^2V/dr_*^2 < 0$  there, an outgoing resonance satisfies

$$\frac{\omega^2/c^2 - V_0}{\sqrt{-2V_0''}} = -i \left( n + \frac{1}{2} \right) + O(\ell^{-1})$$

in the eikonal scaling. Apply it with the Langer replacement  $\ell(\ell+1) \sim (\ell + \frac{1}{2})^2$ , to derive

$$\omega_{\ell n} = \Omega_{\text{ph}} \left( \ell + \frac{1}{2} \right) - i\lambda_{\text{ph}} \left( n + \frac{1}{2} \right) + O(\Omega_{\text{ph}}\ell^{-1}),$$

$$\Omega_{\text{ph}} = \lambda_{\text{ph}} = \frac{c^3}{3\sqrt{3}G_{\text{N}}M}, \quad n = 0, 1, 2, \dots$$

Relate  $\Omega_{\text{ph}}$  to the critical impact parameter by  $\Omega_{\text{ph}} = c/b_c$ . If the source has cosmological redshift  $z_{\text{cos}}$  and angular-diameter distance  $D_A$ , predict the observed ringdown frequency, e-folding time, and shadow angular radius

$$f_{\ell n}^{\text{obs}} = \frac{\text{Re } \omega_{\ell n}}{2\pi(1 + z_{\text{cos}})}, \quad \tau_{\ell n}^{\text{obs}} = \frac{1 + z_{\text{cos}}}{|\text{Im } \omega_{\ell n}|}, \quad \theta_{\text{sh}} = \frac{b_c}{D_A},$$

all from the same photon-sphere geometry. State the isolated, nonrotating, asymptotically flat, linear-perturbation, eikonal, detector-frame, cosmological-redshift, and mode-identification assumptions under which shadow and ringdown measurements test these linked predictions. For the axial gravitational fundamental mode  $(\ell, n) = (2, 0)$ , compute

$$\frac{GM\omega}{c^3} \approx 0.4811 - 0.0962i$$

from the eikonal formula. Numerically continue the two local Heun solutions, locate the corresponding zero of  $D_{2,2}$ , and compare with

$$\frac{GM\omega}{c^3} \approx 0.3737 - 0.0890i.$$

Record that the leading formula overestimates this low-multipole real part by about 29% and is a controlled prediction only as  $\ell \rightarrow \infty$ .

## 8 Determinantal Kernels

### 8.1 Dyson Symmetry Classes

**Exercise 8.1** (Dyson ensembles and unitary orbit Jacobians). For  $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{H}$ , put  $\beta_D = \dim_{\mathbb{R}} \mathbb{F} \in \{1, 2, 4\}$ . The Gaussian Dyson ensemble on the self-adjoint operators on  $\mathbb{F}^N$  has density

$$d\mathbb{P}_{N,\beta_D}(H) = C_{N,\beta_D} \exp\left[-\frac{\beta_D N}{4\sigma^2} \text{Tr}(H^2)\right] dH$$

for the invariant Hilbert–Schmidt volume. For  $\lambda = (\lambda_1, \dots, \lambda_N)$  define

$$\Delta(\lambda) = \det[\lambda_i^{j-1}]_{i,j=1}^N.$$

Use the exterior-power determinant to prove

$$\Delta(\lambda) = \prod_{i < j} (\lambda_j - \lambda_i).$$

Compute the Jacobian of the unitary conjugation-orbit map on the regular spectral locus, including the stabilizer and permutation factors, and derive

$$p_{N,2}(\lambda) = \frac{1}{Z_{N,2}} e^{-N \sum_i \lambda_i^2 / (2\sigma^2)} |\Delta(\lambda)|^2.$$

**Exercise 8.2** (Dyson level repulsion). For the real and quaternionic classes, show that each off-diagonal spectral direction has real multiplicity  $\beta_D$  and deduce

$$p_{N,\beta_D}(\lambda) = \frac{1}{Z_{N,\beta_D}} e^{-\beta_D N \sum_i \lambda_i^2 / (4\sigma^2)} |\Delta(\lambda)|^{\beta_D}.$$

For  $2 \times 2$  GOE and GUE, separate the trace from the traceless operator, rescale the gap to unit mean, and obtain

$$P_1(s) = \frac{\pi}{2} s e^{-\pi s^2/4}, \quad P_2(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}.$$

Identify the exponent  $\beta_D$  as the small-gap repulsion prediction and state the symmetry-sector separation required in spectral data.

### 8.2 Determinantal Point Processes

**Exercise 8.3** (Scattering projections and fermionic correlations). Let

$$H_R = -\frac{\hbar^2}{2\mu} \Delta + V$$

be the Dirichlet realization of a real Schrödinger operator on the ball  $B_R \subset \mathbb{R}^3$ . Fix a Fermi energy  $E_F$  outside its spectrum and let

$$P_R = \mathbf{1}_{(-\infty, E_F]}(H_R)$$

have rank  $N$ . Suppose its integral kernel  $K_R$  is continuous. Prove intrinsically from  $P_R^2 = P_R = P_R^*$  that

$$\int_{B_R} K_R(x, z) K_R(z, y) dz = K_R(x, y), \quad \int_{B_R} K_R(x, x) dx = N,$$



and that every matrix  $[K_R(x_i, x_j)]$  is positive semidefinite.

Let the symmetric group  $S_N$  act on the  $N$ -fold Hilbert tensor product by permuting tensor factors, and define its alternating sector by the sign character. Prove that  $\Lambda^N$  in  $P_R$  is the one-dimensional filled-fermion state space and that the position density of its unit vector, independently of its phase, is

$$p_R(x_1, \dots, x_N) = \frac{1}{N!} \det[K_R(x_i, x_j)]_{i,j=1}^N.$$

Use the projection identity and determinant expansion to prove that  $p_R$  is a probability density and that, for  $1 \leq k \leq N$ ,

$$\frac{N!}{(N-k)!} \int_{B_R^{N-k}} p_R(x_1, \dots, x_N) dx_{k+1} \cdots dx_N = \det[K_R(x_i, x_j)]_{i,j=1}^k.$$

Assume now that  $V$  is radial and supported in a fixed ball. In angular momentum channel  $\ell$ , use the exterior asymptotic

$$u_\ell(r; k) \sim \sin\left(kr - \frac{\ell\pi}{2} + \delta_\ell(k)\right)$$

and the Dirichlet condition at  $r = R$  to derive

$$kR - \frac{\ell\pi}{2} + \delta_\ell(k) = n\pi + o(1) \quad (R \rightarrow \infty).$$

Where  $\delta_\ell$  is differentiable and no level crosses the chosen energy window, deduce that the shift of the radial density of states from its free value is  $\delta'_\ell(k)/\pi$ . Include the multiplicity  $2\ell + 1$  and state the finite-volume, elastic, threshold, angular-momentum cutoff, and detector-resolution assumptions under which fermion counts below  $E_F$  test the phase-shift prediction.

**Exercise 8.4** (Fredholm determinants and trace identities). Let  $A$  be an arbitrary complex trace-class operator on a Hilbert space. Its exterior powers are trace class and satisfy  $|\text{Tr}(\Lambda^n A)| \leq \|A\|_1^n/n!$ . Define

$$\det(I + zA) = \sum_{n=0}^{\infty} z^n \text{Tr}(\Lambda^n A).$$

Prove absolute local uniform convergence and, by finite-rank approximation, the trace-norm continuity estimate

$$|\det(I + A) - \det(I + B)| \leq \|A - B\|_1 \exp(1 + \|A\|_1 + \|B\|_1).$$

Prove the trace-class spectral determinant theorem: if  $\lambda_j(A)$  are the nonzero spectral values with algebraic multiplicity, then

$$\det(I + zA) = \prod_j (1 + z\lambda_j(A)), \quad \text{Tr}(A^m) = \sum_j \lambda_j(A)^m.$$

Deduce

$$\det(I - zA) = \exp\left(-\sum_{m \geq 1} \frac{z^m}{m} \text{Tr}(A^m)\right) \quad (|z| < \|A\|^{-1}).$$

Prove that the determinant is entire and identify its zeros. For a trace-class integral operator on  $L^2(X, \mu)$  whose kernel admits the standard Fredholm expansion, derive

$$\det(I + zA) = 1 + \sum_{n \geq 1} \frac{z^n}{n!} \int_{X^n} \det[A(x_i, x_j)]_{i,j=1}^n d\mu^{\otimes n}.$$

**Exercise 8.5** (Point processes and determinantal statistics). Let  $X$  be a locally compact second-countable Hausdorff space and let  $\mu$  be a Radon measure on  $X$ . A simple point process is a random locally finite counting measure  $\Xi = \sum_i \delta_{x_i}$  with distinct points almost surely. When its  $k$ th factorial moment measure is absolutely continuous with respect to  $\mu^{\otimes k}$ , its density  $\rho_k$  is defined, for every nonnegative compactly supported measurable function  $f$ , by

$$\mathbb{E} \sum_{i_1, \dots, i_k}^{\neq} f(x_{i_1}, \dots, x_{i_k}) = \int_{X^k} f \rho_k d\mu^{\otimes k}.$$

A *determinantal point process* with measurable Hermitian kernel  $K$  is one for which

$$\rho_k(x_1, \dots, x_k) = \det[K(x_i, x_j)]_{i,j=1}^k.$$

For the existence and counting formulas below, the integral operator defined by  $K$  is a locally trace-class positive contraction on  $L^2(X, \mu)$ .

For a relatively compact measurable set  $B$ , put  $N_B = \Xi(B)$  and  $K_B = \mathbf{1}_B K \mathbf{1}_B$ . Use factorial moments and the Fredholm series to prove

$$\mathbb{E}[z^{N_B}] = \det(I + (z - 1)K_B), \quad \mathbb{P}(N_B = 0) = \det(I - K_B).$$

Random variables are independent when their joint law is the product of their marginal laws. Prove that  $N_B$  is distributed as a sum of independent Bernoulli variables whose parameters are the eigenvalues of  $K_B$ .

### 8.3 Orthogonal-Polynomial Kernels

**Exercise 8.6** (Unitary polynomial ensembles and Christoffel–Darboux kernels). Let  $\mu$  be a Radon measure on  $\mathbb{R}$  such that  $w(x)d\mu(x)$  has infinite support and finite moments, and let  $P_j$  be its monic orthogonal polynomials, normalized by

$$\int P_j(x)P_k(x)w(x)d\mu(x) = H_j\delta_{jk}, \quad H_j > 0.$$

The associated unitary polynomial ensemble has density, with respect to  $\mu^{\otimes N}$ ,

$$\mathfrak{p}_N(x_1, \dots, x_N) = \frac{1}{N! \prod_{j=0}^{N-1} H_j} \Delta(x)^2 \prod_{i=1}^N w(x_i)$$

and kernel

$$K_N(x, y) = \sqrt{w(x)w(y)} \sum_{j=0}^{N-1} \frac{P_j(x)P_j(y)}{H_j}.$$

Prove Andréief's identity

$$\int_{X^N} \det[f_i(x_j)] \det[g_i(x_j)] \prod_j d\mu(x_j) = N! \det \left[ \int_X f_i g_j d\mu \right].$$

Deduce that the unitary polynomial ensemble is determinantal with kernel  $K_N$ . Derive from the monic three-term recurrence the Christoffel–Darboux formula

$$K_N(x, y) = \sqrt{w(x)w(y)} \frac{P_N(x)P_{N-1}(y) - P_{N-1}(x)P_N(y)}{H_{N-1}(x - y)} \quad (x \neq y),$$

with the diagonal value obtained by continuity, and prove

$$\int K_N(x, z) K_N(z, y) d\mu(z) = K_N(x, y), \quad \int K_N(x, x) d\mu(x) = N.$$

**Exercise 8.7** (Simple and coalescing saddle estimates). Let

$$I_N(u) = \frac{1}{2\pi i} \int_{\mathcal{C}_N(u)} e^{N\Phi_u(z)} a_u(z) dz,$$

where  $u$  ranges over a compact set, the phases and amplitudes are holomorphic on fixed neighborhoods of the contours, and the contour lengths grow at most polynomially in  $N$ . Assume that the phases, amplitudes, contour parametrizations, and all derivatives used below are uniformly bounded on these neighborhoods.

Suppose first that  $\mathcal{C}_N(u)$  passes through finitely many nondegenerate critical points  $z_j(u)$  along steepest directions. Assume that the critical points remain uniformly separated, that  $|\Phi_u''(z_j(u))|$  is bounded below by a positive constant, and that, outside fixed neighborhoods of these points,

$$\operatorname{Re} \Phi_u(z) \leq \max_j \operatorname{Re} \Phi_u(z_j(u)) - \delta$$

for one  $\delta > 0$  independent of  $u$ . Rescale each fixed local neighborhood by  $z - z_j = N^{-1/2}w$ , use Taylor's theorem and Gaussian domination there, and use the displayed gap on the complement to derive the sum of the Gaussian saddle contributions with absolute remainder  $O(N^{-3/2}e^{NM(u)})$ , where  $M(u) = \max_j \operatorname{Re} \Phi_u(z_j(u))$ . Deduce an  $O(N^{-1})$  relative error whenever the normalized leading saddle sum is bounded away from zero, uniformly in  $u$ .

Suppose instead that two critical points coalesce at  $u = u_*$  and that a local analytic variable  $w$  has been exhibited for which

$$\Phi_u(z) = c(u) + \frac{w^3}{3} - s(u)w, \quad s(u_*) = 0,$$

with the variable, its inverse, and their first three derivatives uniformly bounded. Assume that the transformed amplitude and its first two derivatives are uniformly bounded, that the local contour deforms to the Airy rays without crossing a singularity, and that the complement satisfies a uniform real-part gap and contains no other contributing saddle. Prove, uniformly for bounded  $N^{2/3}s(u)$ , that  $I_N(u)$  is the corresponding linear combination of

$$N^{-1/3} \operatorname{Ai}(N^{2/3}s(u)), \quad N^{-2/3} \operatorname{Ai}'(N^{2/3}s(u)),$$

with a uniform  $O(N^{-1})$  remainder after the common exponential factor is removed. Prove exponentially weighted bounds when  $N^{2/3}s(u)$  lies on the positive real half-line. State precisely how the contour gap and amplitude bounds make both conclusions uniform.

**Exercise 8.8** (Hermite steepest descent). Write  $\psi_j^H$  for the normalized oscillator Hermite functions. From the generating function derive

$$H_j(x) = \frac{j!}{2\pi i} \oint e^{2xz - z^2} z^{-j-1} dz$$

and the resulting contour integral for  $\psi_j^H$ . After the GUE scaling used below, show that the critical-point equation is quadratic, has two simple conjugate roots for  $E \in (-2\sigma, 2\sigma)$ , and has a

double root at  $E = 2\sigma$ . Exhibit the local Gaussian coordinates in the bulk and the cubic Airy coordinate at the edge. Apply the simple- and coalescing-saddle estimates to establish uniform Plancherel–Rotach formulas on compact bulk windows and uniform exponentially weighted Airy formulas on edge half-lines, including the same estimates for the neighboring indices  $j = N - 1, N$  needed in the Christoffel–Darboux kernel.

**Exercise 8.9** (GUE bulk statistics). For  $E \in (-2\sigma, 2\sigma)$ , define

$$\rho_{\text{sc}}(E) = \frac{1}{2\pi\sigma^2} \sqrt{4\sigma^2 - E^2}, \quad K_{\text{sin}}(u, v) = \frac{\sin \pi(u - v)}{\pi(u - v)}.$$

Define

$$\phi_{j,N}(x) = \left(\frac{N}{2\sigma^2}\right)^{1/4} \psi_j^{\text{H}}\left(\sqrt{\frac{N}{2\sigma^2}}x\right), \quad K_N^{\text{GUE}}(x, y) = \sum_{j=0}^{N-1} \phi_{j,N}(x) \phi_{j,N}(y).$$

Insert the bulk Hermite asymptotics into the Christoffel–Darboux identity and prove

$$\frac{1}{N\rho_{\text{sc}}(E)} K_N^{\text{GUE}}\left(E + \frac{u}{N\rho_{\text{sc}}(E)}, E + \frac{v}{N\rho_{\text{sc}}(E)}\right) \longrightarrow K_{\text{sin}}(u, v)$$

locally uniformly. For the limiting sine process prove

$$\text{Var } N_{[0,L]} = \int_0^L K_{\text{sin}}(u, u) du - \int_0^L \int_0^L |K_{\text{sin}}(u, v)|^2 du dv = \frac{1}{\pi^2} \log L + O(1).$$

State the unfolding, symmetry-sector separation, and finite-window assumptions required when testing the spacing and number-variance predictions with resonance or Hamiltonian spectra.

**Exercise 8.10** (GUE edge statistics). Define

$$K_{\text{Ai}}(u, v) = \frac{\text{Ai}(u) \text{Ai}'(v) - \text{Ai}'(u) \text{Ai}(v)}{u - v}, \quad F_2(s) = \det(I - K_{\text{Ai}})_{L^2(s, \infty)},$$

with diagonal values defined by continuity.

Insert the turning-point asymptotics into the Christoffel–Darboux identity and prove

$$\frac{\sigma}{N^{2/3}} K_N^{\text{GUE}}\left(2\sigma + \frac{\sigma u}{N^{2/3}}, 2\sigma + \frac{\sigma v}{N^{2/3}}\right) \longrightarrow K_{\text{Ai}}(u, v).$$

Use the exponentially weighted bounds to pass to the Fredholm series and deduce the GUE largest-eigenvalue law

$$\mathbb{P}\left(\frac{N^{2/3}}{\sigma}(\lambda_{\max} - 2\sigma) \leq s\right) \longrightarrow F_2(s).$$

State the centering, scale calibration, symmetry class, and finite-size assumptions required when testing this prediction with extreme resonance or Hamiltonian levels.

**Exercise 8.11** (Pfaffian kernels and the orthogonal and symplectic edge laws). For a skew-symmetric  $2n \times 2n$  matrix  $A$ , define

$$\text{Pf}(A) = \frac{1}{2^n n!} \sum_{\sigma \in S_{2n}} \text{sgn}(\sigma) \prod_{j=1}^n A_{\sigma(2j-1), \sigma(2j)}.$$

Prove  $\text{Pf}(A)^2 = \det A$ . Let  $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  and let  $K(x, y)$  be a  $2 \times 2$  matrix kernel satisfying  $K(y, x) = -K(x, y)^\top$ . A simple point process is *Pfaffian with kernel  $K$*  when

$$\rho_k(x_1, \dots, x_k) = \text{Pf}[K(x_i, x_j)]_{i,j=1}^k.$$

Define the Fredholm Pfaffian by its correlation expansion and prove, first for finite-rank kernels and then by trace-norm approximation,

$$\text{Pf}(J - K)^2 = \det(I + JK), \quad \mathbb{P}(N_D = 0) = \text{Pf}(J - K_D).$$

Normalize the Gaussian orthogonal, unitary, and symplectic ensembles so that their limiting spectral support is  $[-2\sigma, 2\sigma]$ . Derive the Pfaffian correlation formulas and prove the soft-edge limits for the orthogonal and symplectic ensembles [65]: their scaled matrix kernels are built from  $\text{Ai}$ ,  $\text{Ai}'$ , and integrals of  $\text{Ai}$  and converge in trace norm to kernels  $K_1^{\text{Ai}}$  and  $K_4^{\text{Ai}}$ . Define

$$F_1(s) = \text{Pf}(J - K_{1,s}^{\text{Ai}}), \quad F_4(s) = \text{Pf}(J - K_{4,s}^{\text{Ai}}).$$

If  $q$  is the real pole-free solution

$$q'' = sq + 2q^3, \quad q(s) \sim \text{Ai}(s) \quad (s \rightarrow +\infty),$$

prove the Tracy–Widom identities [65]

$$F_1(s)^2 = F_2(s) \exp\left(-\int_s^\infty q(x) dx\right),$$

$$F_4(s/\sqrt{2})^2 = F_2(s) \cosh^2\left(\frac{1}{2} \int_s^\infty q(x) dx\right)$$

and prove that the three symmetry classes have distinct edge laws despite the same semicircle edge. State the antiunitary-symmetry, Kramers-pairing, centering, scale-calibration, and finite-size assumptions under which extreme-level measurements distinguish  $F_1$ ,  $F_2$ , and  $F_4$ .

**Exercise 8.12** (Partitions and symmetric polynomials). An *integer partition* is a finite nonincreasing sequence  $\lambda = (\lambda_1, \dots, \lambda_\ell)$  of positive integers, with the empty sequence allowed. Put

$$|\lambda| = \sum_i \lambda_i, \quad \ell(\lambda) = \ell,$$

and define its Young diagram to have  $\lambda_i$  left-justified boxes in row  $i$ .

For  $N \geq 1$ , let

$$\Lambda_N = \mathbb{C}[x_1, \dots, x_N]^{S_N}.$$

Define

$$e_r = \sum_{1 \leq i_1 < \dots < i_r \leq N} x_{i_1} \cdots x_{i_r}, \quad h_r = \sum_{1 \leq i_1 \leq \dots \leq i_r \leq N} x_{i_1} \cdots x_{i_r},$$

$$p_r = \sum_{i=1}^N x_i^r, \quad e_0 = h_0 = 1.$$

Set  $e_r = 0$  for  $r > N$  and  $h_r = 0$  for  $r < 0$ . Prove that every element of  $\Lambda_N$  is uniquely a polynomial in  $e_1, \dots, e_N$ . Prove

$$\sum_{r=0}^N e_r t^r = \prod_{i=1}^N (1 + x_i t), \quad \sum_{r \geq 0} h_r t^r = \prod_{i=1}^N \frac{1}{1 - x_i t},$$

and deduce

$$\left( \sum_{r \geq 0} h_r t^r \right) \left( \sum_{r=0}^N (-1)^r e_r t^r \right) = 1, \quad r h_r = \sum_{j=1}^r p_j h_{r-j}.$$

**Exercise 8.13** (Semistandard tableaux and Schur polynomials). For a partition  $\lambda$  of length at most  $N$ , a *semistandard Young tableau* of shape  $\lambda$  is a filling  $T$  of its Young diagram by entries in  $\{1, \dots, N\}$  that is weakly increasing along rows and strictly increasing down columns. If  $m_i(T)$  is the number of entries equal to  $i$ , define

$$s_\lambda(x_1, \dots, x_N) = \sum_T \prod_i x_i^{m_i(T)}.$$

Encode  $h_r$  by lattice paths directed east and south whose east steps at height  $i$  have weight  $x_i$ . For sources  $A_i = (i - \lambda_i, N)$  and sinks  $B_j = (j, 1)$ , expand  $\det[h_{\lambda_i - i + j}]_{i,j=1}^N$  as a signed sum of path families and pair every intersecting family with the family obtained by exchanging the two tails after its first intersection. Show that the remaining nonintersecting families correspond weight-preservingly to semistandard tableaux by reading each path's east-step heights from right to left, and deduce

$$s_\lambda(x_1, \dots, x_N) = \det[h_{\lambda_i - i + j}]_{i,j=1}^N.$$

Call this the *Schur polynomial*. Derive the alternant formula

$$s_\lambda(x) = \frac{\det[x_i^{\lambda_j + N - j}]_{i,j=1}^N}{\det[x_i^{N - j}]_{i,j=1}^N}$$

and prove  $s_{(r)} = h_r$ .

**Exercise 8.14** (RSK, last-passage values, and the finite Schur measure). To insert an integer  $x$  into a semistandard Young tableau, replace the first entry strictly larger than  $x$  in the first row and insert the replaced entry in the next row; append when no such entry exists. For  $W = (W_{ij}) \in M_{m,n}(\mathbb{Z}_{\geq 0})$ , form the lexicographically ordered two-line word containing  $W_{ij}$  copies of  $(i, j)$ , insert the lower letters, and record the upper letters in the created boxes. The resulting pair  $(P, Q)$  of tableaux of common shape  $\lambda$  is the RSK image of  $W$ .

For nonnegative weights  $W = (W_{ij})$ , define

$$G(m, n) = \max_{\pi: (1,1) \nearrow (m,n)} \sum_{(i,j) \in \pi} W_{ij}.$$

For parameters  $\alpha_i, \gamma_j \geq 0$  with  $\alpha_i \gamma_j < 1$ , the geometric model has independent weights

$$\mathbb{P}(W_{ij} = k) = (1 - \alpha_i \gamma_j)(\alpha_i \gamma_j)^k, \quad k \in \mathbb{Z}_{\geq 0}.$$

Prove that row insertion is reversible and hence that RSK is a bijection. Show that the length of the first row of its shape is the maximal weight of an up-right path:

$$\lambda_1 = G(m, n).$$

Use the contents of the two RSK tableaux and the geometric weight law to derive

$$\mathbb{P}(\lambda) = \prod_{i,j} (1 - \alpha_i \gamma_j) s_\lambda(\alpha) s_\lambda(\gamma)$$

and the Cauchy identity

$$\sum_\lambda s_\lambda(\alpha) s_\lambda(\gamma) = \prod_{i,j} (1 - \alpha_i \gamma_j)^{-1}.$$

**Exercise 8.15** (Interlacing partitions and Schur paths). For partitions  $\mu$  and  $\lambda$ , write  $\mu \prec \lambda$  when

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \cdots.$$

Prove the branching formula

$$s_\lambda(\alpha_1, \dots, \alpha_m) = \sum_{\emptyset = \lambda^{(0)} \prec \dots \prec \lambda^{(m)} = \lambda} \prod_{k=1}^m \alpha_k^{|\lambda^{(k)}| - |\lambda^{(k-1)}|}.$$

Encode each interlacing chain by nonintersecting paths on the integer lattice. For a finite directed acyclic graph with compatible sources  $u_i$  and sinks  $v_i$ , let  $Z(u_i, v_j)$  be the weighted sum of paths from  $u_i$  to  $v_j$ . Use the first intersection and tail exchange as a sign-reversing involution to prove

$$\sum_{\substack{(\pi_1, \dots, \pi_r) \\ \text{vertex-disjoint}}} \prod_i \text{wt}(\pi_i) = \det[Z(u_i, v_j)]_{i,j=1}^r.$$

Apply this identity to the interlacing-chain encoding of  $s_\lambda$ .

**Exercise 8.16** (Finite Schur kernels). For finite alphabets, set

$$\Phi(z) = \frac{\prod_j (1 - \gamma_j/z)}{\prod_i (1 - \alpha_i z)}.$$

Choose concentric contours with

$$\max_j \gamma_j < |w| < |z| < \min_i \alpha_i^{-1}.$$

For  $r, s \in \mathbb{Z} + \frac{1}{2}$ , define

$$K_{\alpha, \gamma}(r, s) = \frac{1}{(2\pi i)^2} \oint \oint \frac{\Phi(z)}{\Phi(w)} \frac{1}{z - w} z^{-r-1/2} w^{s-1/2} dz dw.$$

Encode  $\lambda$  by

$$\mathcal{X}(\lambda) = \{\lambda_i - i + \frac{1}{2} : i \geq 1\}.$$

Glue the two nonintersecting path ensembles associated with  $s_\lambda(\alpha)$  and  $s_\lambda(\gamma)$  along their common boundary  $\lambda$ . Apply the Andréief identity successively to prove that the occupation correlations of the glued ensemble are determinants. Evaluate the resulting one-particle propagators by coefficient extraction and derive the kernel  $K_{\alpha, \gamma}$ . Deduce the exact finite Fredholm formula

$$\mathbb{P}(G(m, n) \leq L) = \det(I - K_{\alpha, \gamma})_{\ell^2(\{L + \frac{1}{2}, L + \frac{3}{2}, \dots\})}.$$

State how repeated last-passage or growth experiments with the specified geometric weights test this distribution without invoking an asymptotic limit.

## 8.4 KPZ Fluctuations

**Exercise 8.17** (Exponential last-passage percolation). Let  $E_{ij}$  be independent rate-one exponential random variables and define

$$G_{\text{exp}}(m, n) = \max_{\pi: (1,1) \nearrow (m,n)} \sum_{(i,j) \in \pi} E_{ij}.$$

For random variables  $Y_\varepsilon, Y$ , write  $Y_\varepsilon \implies Y$  when their laws converge against bounded continuous test functions. Set  $\alpha_i = \gamma_j = e^{-\varepsilon/2}$  in the  $n \times n$  geometric model and prove that  $\varepsilon W_{ij}$  converges in distribution to a rate-one exponential variable as  $\varepsilon \downarrow 0$ . Take this limit in the finite Schur Fredholm formula and identify the resulting finite- $n$  kernel with the Laguerre Christoffel–Darboux kernel.

**Exercise 8.18** (Laguerre edge limit). Use the generating function to derive

$$L_n^{(\alpha)}(x) = \frac{1}{2\pi i} \oint \frac{\exp(-xz/(1-z))}{(1-z)^{\alpha+1} z^{n+1}} dz.$$

For the Laguerre kernel, write its scaled phase explicitly and locate the double saddle at the upper spectral edge. Exhibit the cubic coordinate and contours satisfying the gap hypotheses of the coalescing-saddle exercise. Apply that exercise to derive convergence of the rescaled Laguerre kernel to  $K_{\text{Ai}}$  with uniform exponentially weighted bounds, and use those bounds to pass termwise to the Fredholm series.

**Exercise 8.19** (Last-passage fluctuations). Combine the exponential last-passage determinant with the Laguerre edge limit to prove

$$\frac{G_{\text{exp}}(n, n) - 4n}{2^{4/3} n^{1/3}} \implies \chi_2, \quad \mathbb{P}(\chi_2 \leq s) = F_2(s).$$

Deduce the  $n^{1/3}$  height-fluctuation scale. Balance this scale against the quadratic curvature of the deterministic limit shape to predict the transverse  $n^{2/3}$  scale. State the independent-weight, boundary, centering, scale-calibration, and finite-size assumptions required when the Tracy–Widom prediction is compared with interface-height or directed transport data.

**Exercise 8.20** (Extended Airy kernels, the Airy process, and KPZ scaling). Let

$$\mathcal{H}_{\text{Ai}} = -\frac{d^2}{dx^2} + x$$

be the self-adjoint Airy operator on  $L^2(\mathbb{R})$  defined by its closed quadratic form. Verify in the generalized spectral sense that

$$\mathcal{H}_{\text{Ai}} \text{Ai}(x + \lambda) = -\lambda \text{Ai}(x + \lambda).$$

For times  $s, t \in \mathbb{R}$ , define the extended Airy kernel by

$$K_{\text{Ai}}^{\text{ext}}(s, x; t, y) = \begin{cases} \int_0^\infty e^{-\lambda(s-t)} \text{Ai}(x + \lambda) \text{Ai}(y + \lambda) d\lambda, & s \geq t, \\ -\int_{-\infty}^0 e^{-\lambda(s-t)} \text{Ai}(x + \lambda) \text{Ai}(y + \lambda) d\lambda, & s < t. \end{cases}$$

For  $t_1 < \dots < t_m$  and thresholds  $a_1, \dots, a_m$ , let  $\chi$  act on  $\bigoplus_{j=1}^m L^2(\mathbb{R})$  by multiplication by  $\mathbf{1}_{(a_j, \infty)}$  in the  $j$ th summand. Prove the Airy-process existence theorem [55]: the following Fredholm determinants are consistent probability distributions and determine a stationary process with continuous paths, denoted  $\mathcal{A}_2$ :

$$\mathbb{P}(\mathcal{A}_2(t_j) \leq a_j, 1 \leq j \leq m) = \det(I - \chi K_{\text{Ai}}^{\text{ext}} \chi).$$

Use the spectral semigroup identity for  $\mathcal{H}_{\text{Ai}}$  to verify their marginal consistency, and recover  $\mathbb{P}(\mathcal{A}_2(t) \leq a) = F_2(a)$ .



For exponential last-passage percolation, put

$$a_N(u) = \lfloor 2^{2/3} u N^{2/3} \rfloor.$$

Prove the multipoint PNG/last-passage edge theorem in its exponential specialization [55] to obtain, in finite-dimensional distributions,

$$\frac{G_{\text{exp}}(N + a_N(u), N - a_N(u)) - 4N}{2^{4/3} N^{1/3}} \implies \mathcal{A}_2(u) - u^2.$$

Deduce the KPZ correlation-length exponent  $2/3$ , fluctuation exponent  $1/3$ , the Tracy–Widom one-point law, and the nontrivial spatial correlation function of the centered process. Express the corresponding droplet-growth prediction with nonuniversal velocity, fluctuation, and transverse-length calibration constants. State the curved-geometry, short-range-noise, growth-time, spatial-window, and finite-resolution assumptions under which interface measurements test the 1:2:3 scaling and Airy-process correlations.

## 9 Transfer Matrices

### 9.1 Symmetric Groups

**Exercise 9.1** (Determinantal path transfer operators). Let  $X_0, \dots, X_L$  be finite configuration sets. For  $1 \leq s \leq L$ , let

$$A_s : \mathbb{C}^{X_{s-1}} \longrightarrow \mathbb{C}^{X_s}, \quad (A_s f)(y) = \sum_{x \in X_{s-1}} a_s(y, x) f(x),$$

where  $a_s(y, x) \geq 0$  is the weight of a directed edge from  $x$  to  $y$ . Put  $\mathsf{T} = A_L \cdots A_1$ . Prove that

$$\langle \delta_y, \mathsf{T} \delta_x \rangle = \sum_{\pi: x \rightarrow y} \prod_{e \in \pi} \text{wt}(e).$$

For  $r \geq 1$ , let  $S_r$  act on every  $r$ -fold tensor power by permuting factors. Prove that  $A_s^{\otimes r}$  intertwines these actions, that the sign-isotypic sector is the exterior power, and that the induced transfer operator on that sector is  $\Lambda^r \mathsf{T}$ . For ordered configurations  $x = (x_1, \dots, x_r)$  and  $y = (y_1, \dots, y_r)$ , prove intrinsically from the exterior-power construction that

$$\langle \delta_{y_1} \wedge \cdots \wedge \delta_{y_r}, (\Lambda^r \mathsf{T})(\delta_{x_1} \wedge \cdots \wedge \delta_{x_r}) \rangle = \det \left[ \langle \delta_{y_i}, \mathsf{T} \delta_{x_j} \rangle \right]_{i,j=1}^r.$$

When the layered graph is planar and the boundary order prevents a vertex-disjoint family from permuting its endpoints, apply the nonintersecting-path determinant to identify this matrix coefficient with the positive partition function of  $r$  vertex-disjoint paths. Deduce that determinantal path kernels are matrix coefficients of fermionic transfer operators.

For vertex weights  $W_{ij} \geq 0$  on an  $m \times n$  rectangle and  $\beta > 0$ , define the directed-polymer partition function

$$Z_\beta(i, j) = \sum_{\pi: (1,1) \nearrow (i,j)} \exp \left( \beta \sum_{(a,b) \in \pi} W_{ab} \right).$$

Realize  $Z_\beta(m, n)$  as a matrix coefficient of the product of the anti-diagonal transfer operators. With missing boundary values equal to zero, prove

$$Z_\beta(i, j) = e^{\beta W_{ij}} (Z_\beta(i-1, j) + Z_\beta(i, j-1)), \quad Z_\beta(1, 1) = e^{\beta W_{11}}.$$

If  $G(m, n)$  is the last-passage value, prove

$$G(m, n) \leq \frac{1}{\beta} \log Z_\beta(m, n) \leq G(m, n) + \frac{1}{\beta} \log \binom{m+n-2}{m-1},$$

and hence recover last-passage percolation as the zero-temperature limit of a transfer-operator model. For independent rate-one exponential weights, combine this limit with the last-passage fluctuation law to obtain

$$\lim_{n \rightarrow \infty} \lim_{\beta \rightarrow \infty} \mathbb{P} \left( \frac{\beta^{-1} \log Z_\beta(n, n) - 4n}{2^{4/3} n^{1/3}} \leq s \right) = F_2(s).$$

State the directed-path, independent-weight, zero-temperature-before-large-size, boundary, and scale-calibration assumptions under which polymer free-energy measurements test this prediction.

## 9.2 Transfer Spectra

**Exercise 9.2** (Transfer spectra and correlation lengths). Let  $X$  be a finite configuration space and let  $T : \mathbb{C}^X \rightarrow \mathbb{C}^X$  be a self-adjoint transfer operator satisfying

$$\langle \delta_y, T \delta_x \rangle > 0 \quad (x, y \in X).$$

Prove [59] the finite Perron–Frobenius statement that a strictly positive matrix has a positive simple spectral radius and that every other spectral value has strictly smaller modulus. Apply it to  $T$ , denote its spectral radius by  $\lambda_0$ , and choose its normalized positive eigenvector  $u_0$ . Write the remaining distinct spectral values as  $\lambda_1, \lambda_2, \dots$ , ordered so that

$$\lambda_0 > |\lambda_1| \geq |\lambda_2| \geq \dots, \quad T = \sum_j \lambda_j P_j.$$

For configuration observables  $A$  and  $B$ , acting by multiplication, define their separation- $n$  cylinder correlation by

$$\langle A_0 B_n \rangle_M = \frac{\text{Tr}(A T^n B T^{M-n})}{\text{Tr}(T^M)}, \quad 0 \leq n \leq M.$$

For fixed  $n$ , take  $M \rightarrow \infty$  and prove

$$\langle A_0 B_n \rangle = \lambda_0^{-n} \langle u_0, A T^n B u_0 \rangle, \quad \langle A \rangle = \langle u_0, A u_0 \rangle.$$

Deduce the connected-correlation expansion

$$C_{AB}(n) = \sum_{j \geq 1} \left( \frac{\lambda_j}{\lambda_0} \right)^n \langle u_0, A P_j B u_0 \rangle.$$

If the  $j = 1$  coefficient is nonzero, prove that the exponential correlation length in lattice units is

$$\xi^{-1} = \log \left| \frac{\lambda_0}{\lambda_1} \right|.$$

For a self-adjoint observable  $A$  whose connected correlation is dominated by  $C_{AA}(n) = c r^{|n|}$ ,  $|r| < 1$ , compute the static structure factor

$$S_A(k) = \sum_{n \in \mathbb{Z}} C_{AA}(n) e^{-ikn} = c \frac{1 - r^2}{1 - 2r \cos k + r^2}.$$

Locate its peak for  $r > 0$  and  $r < 0$ , determine its width for  $|r| \uparrow 1$ , and state the translation-invariance, equilibrium, diagonalizability, observable-overlap, finite-size, and detector-resolution assumptions under which scattering measurements determine  $\xi$  and the subleading transfer eigenvalue.

**Exercise 9.3** (The planar Ising model and Kramers–Wannier duality). Let  $G = (V, E)$  be a finite connected graph with a fixed planar embedding. For couplings  $K_e > 0$ , define

$$Z_G^f(K) = \sum_{s: V \rightarrow \{-1, 1\}} \exp \left( \sum_{e=uv} K_e s_u s_v \right).$$

Let  $G^*$  be the planar dual, fix the spin of its outer-face vertex to  $+1$ , and denote the corresponding partition function by  $Z_{G^*}^+(K^*)$ . Let  $\mathcal{E}(G)$  be the family of edge subsets in which every vertex has even degree. The dual couplings are determined by

$$e^{-2K_{e^*}} = \tanh K_e.$$

Expand each edge factor as

$$e^{K_e s_u s_v} = \cosh K_e (1 + s_u s_v \tanh K_e)$$

and prove

$$Z_G^f(K) = 2^{|V|} \prod_{e \in E} \cosh K_e \sum_{\gamma \in \mathcal{E}(G)} \prod_{e \in \gamma} \tanh K_e.$$

Identify  $\mathcal{E}(G)$  with the domain walls of dual spin configurations whose outer-face spin is fixed, and deduce the exact planar duality

$$Z_G^f(K) = 2^{|V|} \prod_{e \in E} (\cosh K_e e^{-K_{e^*}}) Z_{G^*}^+(K^*).$$

For horizontal and vertical square-lattice couplings  $K_h, K_v$ , show that the duality interchanges the directions and that its fixed locus is

$$\sinh(2K_h) \sinh(2K_v) = 1.$$

Assuming the infinite-volume ferromagnetic model has a single transition and that duality interchanges its high- and low-temperature phases, deduce in the isotropic case

$$K_c = \frac{1}{2} \log(1 + \sqrt{2}).$$

State separately the exact finite-graph identity, the infinite-volume assumptions, and the conclusion about the critical temperature.

**Exercise 9.4** (Ising star–triangle transformation). For  $K_1, K_2, K_3 > 0$  and boundary spins  $s_1, s_2, s_3$ , prove that there are unique  $A > 0$  and real  $L_{12}, L_{23}, L_{31}$  satisfying

$$\sum_{s_0 = \pm 1} e^{K_1 s_0 s_1 + K_2 s_0 s_2 + K_3 s_0 s_3} = A e^{L_{12} s_1 s_2 + L_{23} s_2 s_3 + L_{31} s_3 s_1}.$$

Put

$$\begin{aligned} w_0 &= 2 \cosh(K_1 + K_2 + K_3), & w_1 &= 2 \cosh(-K_1 + K_2 + K_3), \\ w_2 &= 2 \cosh(K_1 - K_2 + K_3), & w_3 &= 2 \cosh(K_1 + K_2 - K_3). \end{aligned}$$

Derive

$$A = (w_0 w_1 w_2 w_3)^{1/4},$$

$$e^{4L_{12}} = \frac{w_0 w_3}{w_1 w_2}, \quad e^{4L_{23}} = \frac{w_0 w_1}{w_2 w_3}, \quad e^{4L_{31}} = \frac{w_0 w_2}{w_1 w_3}.$$

Show that replacing every star of one sublattice of the honeycomb model by a triangle changes the partition function only by the product of the local factors  $A$ . In the isotropic case derive

$$e^{4L} = \frac{\cosh(3K)}{\cosh K}.$$

Combine this transformation with planar duality and, under the single-transition hypothesis, obtain

$$K_c^{\text{tri}} = \frac{1}{4} \log 3, \quad K_c^{\text{hex}} = \frac{1}{2} \log(2 + \sqrt{3}).$$

Identify the coupling normalization and lattice geometry needed when these critical temperatures are compared with an Ising magnet or a programmable spin array.

### 9.3 Theta Characters

**Exercise 9.5** (Ising transfer matrices and free fermions). Let  $\mathcal{S}_N = \{-1, 1\}^{\mathbb{Z}/N\mathbb{Z}}$  and  $K_h, K_v > 0$ . For row configurations  $s, t \in \mathcal{S}_N$  define

$$T_N^{\text{Is}}(K_h, K_v)_{s,t} = \exp \left[ \frac{K_h}{2} \sum_i (s_i s_{i+1} + t_i t_{i+1}) + K_v \sum_i s_i t_i \right].$$

Let  $R_N$  be cyclic row translation. On  $\mathcal{H}_N = \ell^2(\mathcal{S}_N)$  let

$$(Z_j f)(s) = s_j f(s), \quad (X_j f)(s) = f(s^{(j)}), \quad Y_j = iX_j Z_j,$$

where  $s^{(j)}$  is obtained by reversing  $s_j$ . Put

$$P = \prod_{j=1}^N X_j, \quad c_{2j-1} = \left( \prod_{\ell < j} X_\ell \right) Z_j, \quad c_{2j} = \left( \prod_{\ell < j} X_\ell \right) Y_j.$$

Let  $K_v^* > 0$  satisfy  $e^{-2K_v^*} = \tanh K_v$ , and set

$$\mathcal{K}_+ = \left\{ \frac{(2m+1)\pi}{N} : 0 \leq m < N \right\}, \quad \mathcal{K}_- = \left\{ \frac{2m\pi}{N} : 0 \leq m < N \right\}.$$

Write  $a_x$  and  $a_t$  for the spatial and transfer-step lattice spacings and  $L = Na_x$  for the circumference.

(a) Verify

$$R_N T_N^{\text{Is}}(K_h, K_v) = T_N^{\text{Is}}(K_h, K_v) R_N.$$

(b) Prove

$$T_N^{\text{Is}}(K_h, K_v) = (2 \sinh(2K_v))^{N/2} e^{\frac{K_h}{2} \sum_j Z_j Z_{j+1}} e^{K_v^* \sum_j X_j} e^{\frac{K_h}{2} \sum_j Z_j Z_{j+1}}.$$

(c) Verify the Clifford relations

$$c_r c_s + c_s c_r = 2\delta_{rs} I, \quad X_j = i c_{2j-1} c_{2j}, \quad Z_j Z_{j+1} = i c_{2j} c_{2j+1} \quad (j < N),$$

and

$$Z_N Z_1 = -i P c_{2N} c_1.$$

- (d) On the  $P = +1$  and  $P = -1$  sectors use, respectively, the distinguished Fourier modes with momenta in  $\mathcal{K}_+$  and  $\mathcal{K}_-$  to diagonalize the induced action on the Clifford generators. Derive

$$\cosh \gamma(k) = \cosh(2K_h) \cosh(2K_v^*) - \sinh(2K_h) \sinh(2K_v^*) \cos k, \quad \gamma(k) \geq 0,$$

and prove that the transfer eigenvalues are

$$\Lambda = (2 \sinh(2K_v))^N \exp \left( \frac{1}{2} \sum_{k \in \mathcal{K}_\pm} \epsilon_k \gamma(k) \right), \quad \epsilon_k \in \{-1, 1\},$$

with the fermion-parity condition determined by the chosen  $P$  sector.

- (e) At  $K_h = K_v = K_c$  prove

$$\gamma(k) = 2 \operatorname{arsinh} |\sin(k/2)| = |k| + O(|k|^3).$$

Apply Euler–Maclaurin summation to the filled negative-energy modes and derive the chiral vacuum correction  $-1/48$ . Define  $c = 1/2$  by writing this correction as  $-c/24$ . Let  $q$  be the formal variable recording chiral excitation grade. After subtracting the sector ground energy and measuring gaps in units  $2\pi(a_x/a_t)/L$ , use the parity projections to obtain the chiral scaling products

$$\begin{aligned} \chi_0(q) &= \frac{q^{-1/48}}{2} \left[ \prod_{m \geq 0} (1 + q^{m+1/2}) + \prod_{m \geq 0} (1 - q^{m+1/2}) \right], \\ \chi_{1/2}(q) &= \frac{q^{-1/48}}{2} \left[ \prod_{m \geq 0} (1 + q^{m+1/2}) - \prod_{m \geq 0} (1 - q^{m+1/2}) \right], \\ \chi_{1/16}(q) &= q^{1/24} \prod_{m \geq 1} (1 + q^m). \end{aligned}$$

**Exercise 9.6** (Theta-character transfer spectra). Let  $\Lambda_0(L) > 0$  be the Perron eigenvalue of  $T_N^{\text{Is}}(K_c, K_c)$  and let  $\Lambda_{a,r,\bar{r}}(L)$  denote its low-lying symmetry-resolved eigenvalues. Call  $e_\infty$  the nonuniversal bulk energy per unit length and  $v_{\text{crit}}$  the calibrated critical propagation speed. Define the left and right central charges  $c, \bar{c}$  as the coefficients of the universal vacuum-energy correction, define  $h_a, \bar{h}_a$  as the lowest left and right scaling weights in sector  $a$ , and let  $r, \bar{r} \geq 0$  denote the excitation grades recorded by the corresponding character coefficients. Define

$$E_{a,r,\bar{r}}(L) = -\frac{1}{a_t} \log |\Lambda_{a,r,\bar{r}}(L)|.$$

Use the critical Clifford spectrum and Euler–Maclaurin summation to prove the finite-size asymptotic

$$E_{a,r,\bar{r}}(L) = e_\infty L + \frac{2\pi v_{\text{crit}}}{L} \left( h_a + \bar{h}_a + r + \bar{r} - \frac{c + \bar{c}}{24} \right) + o(L^{-1}).$$

If  $R_N$  acts on the same state by  $e^{i\mathfrak{p}a_x}$ , prove

$$\mathfrak{p} = \frac{2\pi}{L} (h_a - \bar{h}_a + r - \bar{r}) \pmod{2\pi/a_x}.$$

Prove that  $v_{\text{crit}} = a_x/a_t$  for the displayed isotropic transfer normalization. Use the three Ising character products to prove that the coefficient of  $q^{h_a+r-c/24} \bar{q}^{\bar{h}_a+\bar{r}-\bar{c}/24}$  gives the predicted multiplicity at these energy and momentum values. Express every excitation gap directly as

$$E_j(L) - E_0(L) = -\frac{1}{a_t} \log \left| \frac{\Lambda_j(L)}{\Lambda_0(L)} \right|.$$

Compute the first spectral gaps and degeneracies in the three Ising sectors. Deduce the universal gap ratios and the  $L^{-1}$  Casimir-energy correction. State how finite-size transfer eigenvalues determine  $c$ , the primary weights, and  $v_{\text{crit}}$ , including the symmetry-sector resolution, anisotropy corrections, and irrelevant-operator corrections required for the comparison.

## 9.4 Lattice Spectroscopy

**Exercise 9.7** (Renormalization, Ising scaling fields, and critical exponents). For a transfer operator on a circle of circumference  $L$ , normalize by its Perron eigenvalue and define

$$H_L = -\frac{1}{a_t} \log(T_L/\Lambda_0(L)).$$

At the critical Ising point, use the character sectors above to identify the primary scaling dimensions

$$x_1 = 0, \quad x_\sigma = \frac{1}{8}, \quad x_\varepsilon = 1.$$

Define a scale transformation by  $L \mapsto L/b$  followed by the energy rescaling that keeps the critical velocity fixed. Call a coupling  $g_i$  a *scaling field* of dimension  $x_i$  when it is an eigen-coordinate for the linearized transformation with law

$$g_i \mapsto b^{y_i} g_i, \quad y_i = 2 - x_i.$$

Prove that the thermal perturbation has  $y_t = 1$  and the magnetic perturbation has  $y_h = 15/8$ .

Put  $t = (T - T_c)/T_c$  and let  $h$  be the dimensionless magnetic field in a fixed microscopic normalization. Define the critical exponents by

$$\xi(t, 0) \asymp |t|^{-\nu}, \quad \langle \sigma(0)\sigma(r) \rangle_c \asymp r^{-\eta} \quad (t = h = 0),$$

$$M(t, 0) \asymp (-t)^{\beta_{\text{mag}}}, \quad \chi(t, 0) \asymp |t|^{-\gamma}, \quad M(0, h) \asymp |h|^{1/\delta}, \quad C(t, 0) \asymp |t|^{-\alpha},$$

where the two-dimensional correlation law has exponent  $d - 2 + \eta = \eta$ . Assume the scaling hypothesis for the singular free-energy density,

$$f_{\text{sing}}(t, h) = b^{-2} f_{\text{sing}}(b^{y_t} t, b^{y_h} h),$$

and derive

$$\nu = 1, \quad \beta_{\text{mag}} = \frac{1}{8}, \quad \gamma = \frac{7}{4}, \quad \delta = 15, \quad \alpha = 0.$$

Separately use the critical two-point scaling law  $2x_\sigma = d - 2 + \eta$  to derive  $\eta = 1/4$ ; do not infer it from the free-energy homogeneity law alone. Show that a perturbation with  $x_i > 2$  gives corrections to finite-size levels that decay as a power of  $L$ , whereas  $x_i < 2$  opens a scale set by  $|g_i|^{-1/y_i}$ . State the locality, single-fixed-point, scaling-window, field-calibration, thermodynamic-limit, and finite-size assumptions under which magnetization, susceptibility, correlation-length, and heat-capacity measurements test these exponent relations.

**Exercise 9.8** (Izergin determinant and domain-wall ice entropy). Let  $V = \mathbb{C}e_+ \oplus \mathbb{C}e_-$  be the two-state edge space. For an additive spectral parameter  $u$ , put

$$R(u) = \begin{pmatrix} \sin(u + \eta) & 0 & 0 & 0 \\ 0 & \sin u & \sin \eta & 0 \\ 0 & \sin \eta & \sin u & 0 \\ 0 & 0 & 0 & \sin(u + \eta) \end{pmatrix}$$

relative to the ordered weight lines  $e_+ \otimes e_+$ ,  $e_+ \otimes e_-$ ,  $e_- \otimes e_+$ , and  $e_- \otimes e_-$ . Interpret its six nonzero matrix coefficients as the six arrow configurations satisfying the ice rule. Verify from the sine addition formula that

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v).$$

For horizontal rapidities  $x = (x_1, \dots, x_N)$ , vertical rapidities  $y = (y_1, \dots, y_N)$ , and crossing parameter  $\eta$ , write

$$a(x_i, y_j) = \sin(x_i - y_j + \eta), \quad b(x_i, y_j) = \sin(x_i - y_j), \quad c = \sin \eta.$$

Impose domain-wall boundary arrows, with the top and bottom arrows pointing inward and the left and right arrows pointing outward, and let  $Z_N^{\text{DW}}(x; y)$  be the sum of products of these local weights over all allowed arrow configurations.

Use the Yang–Baxter equation to prove that  $Z_N^{\text{DW}}$  is separately symmetric in the  $x_i$  and in the  $y_j$ . At  $x_N = y_N$ , show that the vertex at their intersection has weight  $c$ , that its row and column freeze, and that

$$Z_N^{\text{DW}}(x; y) \Big|_{x_N=y_N} = c \prod_{i=1}^{N-1} a(x_i, y_N) \prod_{j=1}^{N-1} a(x_N, y_j) Z_{N-1}^{\text{DW}}(\hat{x}_N; \hat{y}_N).$$

Here a hat means that the indicated rapidity is omitted. With  $X_i = e^{2ix_i}$  and  $Y_j = e^{2iy_j}$ , prove that  $X_i^{(N-1)/2} Z_N^{\text{DW}}$  is a polynomial of degree  $N-1$  in  $X_i$  and that the analogous statement holds for every  $Y_j$ . Prove the Korepin interpolation lemma by induction and polynomial interpolation [45]: separate symmetry, these degree bounds, the displayed freezing recursion, and  $Z_1^{\text{DW}} = c$  uniquely determine the domain-wall partition function.

Verify that the determinant below has these properties and deduce the Izergin identity

$$Z_N^{\text{DW}}(x; y) = \frac{\prod_{i,j=1}^N a(x_i, y_j) b(x_i, y_j)}{\prod_{1 \leq i < j \leq N} \sin(x_i - x_j) \sin(y_j - y_i)} \det \left[ \frac{c}{a(x_i, y_j) b(x_i, y_j)} \right]_{i,j=1}^N.$$

Take the homogeneous ice limit

$$x_i - y_j = \eta = \frac{\pi}{3}, \quad a = b = c = \frac{\sqrt{3}}{2},$$

and divide every local weight by  $\sqrt{3}/2$ . Let  $A_N$  be the resulting unit-weight partition function. When every  $x_i = x$  and every  $y_j = y$ , put  $z = x - y$ ,  $a(z) = \sin(z + \eta)$ ,  $b(z) = \sin z$ , and write the unnormalized partition function as  $Z_N^{\text{hom}}(z; \eta)$ . Take the confluent limit of the Izergin determinant and prove

$$Z_N^{\text{hom}}(z; \eta) = \frac{[a(z)b(z)]^{N^2}}{\prod_{j=0}^{N-1} (j!)^2} \det \left[ \frac{d^{i+j}}{dz^{i+j}} \frac{c}{a(z)b(z)} \right]_{i,j=0}^{N-1}.$$

Prove [46] the homogeneous determinant evaluation obtained from its Desnanot–Jacobi recurrence at  $z = \eta = \pi/3$ :

$$\frac{A_N}{A_{N-1}} = \frac{(3N-2)!(N-1)!}{(2N-1)!(2N-2)!}, \quad N \geq 2, \quad A_1 = 1,$$

and deduce

$$A_N = \prod_{j=0}^{N-1} \frac{(3j+1)!}{(N+j)!}.$$

Compute  $A_N$  for  $1 \leq N \leq 5$  and use Stirling's formula to derive

$$\lim_{N \rightarrow \infty} \frac{1}{N^2} \log A_N = \log \frac{3\sqrt{3}}{4}.$$

Deduce the exact finite-size state count and the residual configurational entropy per vertex

$$s_{\text{DW}} = k_B \log \frac{3\sqrt{3}}{4}$$

for a unit-weight domain-wall ice sample. State the domain-wall preparation, equal-weight calibration, equilibration, and finite-size assumptions under which configuration counts or free-energy measurements test the determinant and product identities.

**Exercise 9.9** (Harper monodromy and  $SL_2(\mathbb{R})$  band geometry). Let  $\mathcal{E} = V \oplus V^*$  be a symplectic phase space with inverse Poisson tensor  $\Pi$ . Choose  $k, \ell \in \mathcal{E}^*$  with

$$-\hbar \Pi(k, \ell) = 2\pi\alpha$$

and let  $U$  and  $V$  be the Weyl quantizations of  $e_k$  and  $e_\ell$ . Use the Moyal plane-wave identity to derive

$$UV = e^{2\pi i \alpha} VU.$$

Interpret  $\alpha$  as the magnetic flux through one lattice cell in units of the flux quantum, and define the magnetic nearest-neighbor Hamiltonian

$$H_\alpha = U + U^* + V + V^*.$$

Pass from the abstract Weyl relation to its independent magnetic-lattice representation on  $\ell^2(\mathbb{Z})$ ; this is not the Schrödinger representation on  $L^2(\mathbb{R})$ . With transverse phase  $\theta$ , take

$$(U\psi)_m = \psi_{m+1}, \quad (V\psi)_m = e^{i(2\pi\alpha m + \theta)} \psi_m$$

and derive the Harper operator

$$(H_{\alpha, \theta} \psi)_m = \psi_{m+1} + \psi_{m-1} + 2 \cos(2\pi\alpha m + \theta) \psi_m.$$

Prove that the commutator phase is the holonomy around one lattice cell. On an  $N_x \times N_y$  periodic sample, show that  $N_x N_y \alpha \notin \mathbb{Z}$  obstructs consistent torus boundary phases, whereas integral total flux makes their corner cocycle trivial. Prove that nonzero integral total flux still obstructs a single globally periodic vector potential.

Rewrite  $H_{\alpha, \theta} \psi = E\psi$  as

$$\begin{pmatrix} \psi_{m+1} \\ \psi_m \end{pmatrix} = A_m(E, \theta) \begin{pmatrix} \psi_m \\ \psi_{m-1} \end{pmatrix}, \quad A_m(E, \theta) = \begin{pmatrix} E - 2 \cos(2\pi\alpha m + \theta) & -1 \\ 1 & 0 \end{pmatrix}.$$

For coprime integers  $p, q$  with  $q > 0$ , put  $\alpha = p/q$  and define the  $q$ -step monodromy and discriminant by

$$M_{p/q}(E, \theta) = A_{q-1}(E, \theta) \cdots A_0(E, \theta), \quad \Delta_{p/q}(E, \theta) = \text{Tr } M_{p/q}(E, \theta).$$



Interpret  $M_{p/q}$  as the one-dimensional transfer matrix through one magnetic unit cell and contrast its spectral trace with the row-to-row partition-function traces above. Define

$$SL_2(\mathbb{R}) = \{A \in \text{End}(\mathbb{R}^2) : \det A = 1\}.$$

Prove that every  $A_m(E, \theta)$  and  $M_{p/q}(E, \theta)$  belongs to this group. For  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ , prove that

$$A \cdot z = \frac{az + b}{cz + d}$$

preserves the upper half-plane. Show that the eigenvalues of a monodromy matrix satisfy

$$\lambda^2 - \Delta_{p/q}(E, \theta)\lambda + 1 = 0.$$

Call the monodromy elliptic when  $|\Delta_{p/q}| < 2$ , parabolic when  $|\Delta_{p/q}| = 2$  and  $M_{p/q} \neq \pm I$ , and hyperbolic when  $|\Delta_{p/q}| > 2$ ; treat  $M_{p/q} = \pm I$  as the central limiting case. Prove that the Bloch condition  $\psi_{m+q} = e^{iq\kappa}\psi_m$  is equivalent to

$$\Delta_{p/q}(E, \theta) = 2 \cos(q\kappa),$$

that band interiors have elliptic monodromy, open band edges have parabolic monodromy, closed gaps have central monodromy, and open gaps have hyperbolic monodromy with growth exponent

$$\gamma(E, \theta) = \text{arcosh} \left( \frac{|\Delta_{p/q}(E, \theta)|}{2} \right).$$

For any chosen transfer period  $q$  at zero flux, use

$$T_q(\cos t) = \cos(qt)$$

to derive

$$\Delta_{0,q}(E, \theta) = 2T_q \left( \frac{E - 2 \cos \theta}{2} \right).$$

At half a flux quantum, compute the two bands

$$E_{\pm}(\kappa, \theta) = \pm 2 \sqrt{\cos^2 \kappa + \cos^2 \theta}.$$

Deduce the flux-dependent band edges, touching energy, and exponential attenuation per period in every gap. State the nearest-neighbor, uniform-flux, periodicity, contact-matching, coherence, and finite-size assumptions under which microwave, photonic, or electronic lattice spectroscopy tests the Chebyshev discriminant, half-flux splitting, and hyperbolic attenuation exponent.

## 10 $SL_2(\mathbb{R})$

### 10.1 Hyperbolic Isometries

**Exercise 10.1** (The hyperbolic plane and rank-one decompositions). Let

$$G = SL_2(\mathbb{R}), \quad K = SO(2),$$

write

$$k_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

and let  $G$  act on the upper half-plane by fractional linear transformations. Define

$$a_r = \begin{pmatrix} e^{r/2} & 0 \\ 0 & e^{-r/2} \end{pmatrix}, \quad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix},$$

with  $r, x \in \mathbb{R}$ . Give the upper half-plane the metric  $ds^2 = (dx^2 + dy^2)/y^2$  and fix its curvature radius to be one.

Use the polar and QR decompositions of an element of  $SL_2(\mathbb{R})$  to prove

$$K \times \mathbb{R} \times \mathbb{R} \longrightarrow G, \quad (k, r, x) \longmapsto ka_r n_x$$

is a diffeomorphism and

$$G = K\{a_r : r \geq 0\}K.$$

Prove that  $K$  is the stabilizer of  $i$  and that the orbit map identifies  $G/K$  isometrically with the hyperbolic plane. Identify the two-point Weyl group and prove that a  $K$ -bi-invariant function is determined by its value on  $a_r$ ,  $r \geq 0$ .

Derive geodesic polar coordinates about  $i$ ,

$$ds^2 = dr^2 + \sinh^2 r d\theta^2, \quad d\text{vol}_{\mathbb{H}^2} = \sinh r dr d\theta.$$

Normalize  $dk$  to have total mass one and derive the Iwasawa and radial integration formulas

$$\begin{aligned} \int_G F(g) dg &= \int_K \int_{\mathbb{R}} \int_{\mathbb{R}} F(ka_r n_x) e^r dx dr dk, \\ \int_{\mathbb{H}^2} f(z) d\text{vol}(z) &= 2\pi \int_0^\infty f(a_r K) \sinh r dr \end{aligned}$$

for radial  $f$ .

**Exercise 10.2** (Hyperbolic Gauss–Bonnet and Weyl asymptotics). For  $R > 0$ , let  $\mathbb{H}_R^2$  be the upper half-plane with the metric

$$ds_R^2 = R^2 \frac{dx^2 + dy^2}{y^2},$$

of constant curvature  $-R^{-2}$ . Let  $\Gamma$  be a torsion-free cocompact subgroup of  $PSL_2(\mathbb{R}) = SL_2(\mathbb{R})/\{\pm I\}$  and let  $S = \Gamma \backslash \mathbb{H}_R^2$  be a closed oriented surface of genus  $g$ .

For a geodesic triangle  $D$  with interior angles  $\alpha, \beta, \gamma$ , use the Levi–Civita connection one-form, Stokes' theorem, and the exterior turns at the vertices to prove

$$\text{Area}(D) = R^2(\pi - \alpha - \beta - \gamma).$$

Choose a geodesic triangulation of  $S$ , sum this identity over its faces, and use

$$3F = 2E, \quad V - E + F = \chi(S) = 2 - 2g$$

to derive

$$\int_S K dA = 2\pi\chi(S), \quad \text{Area}(S) = 4\pi(g - 1)R^2.$$

Let

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots$$

be the spectral values of the positive Laplacian  $-\Delta_S$ , repeated with multiplicity, and put

$$N_S(\Lambda) = \#\{j : \lambda_j \leq \Lambda\}.$$

Prove the two-dimensional Weyl law [38]

$$N_S(\Lambda) = \frac{\text{Area}(S)}{4\pi} \Lambda + o(\Lambda)$$

to prove

$$N_S(\Lambda) = (g-1)R^2\Lambda + o(\Lambda).$$

For a membrane or wave circuit with propagation speed  $c_s$  and modal frequencies  $\omega_j = c_s\sqrt{\lambda_j}$ , derive

$$\#\{j : \omega_j \leq \Omega\} = (g-1)\frac{R^2\Omega^2}{c_s^2} + o(\Omega^2).$$

State the compact-quotient, constant-curvature, boundary-free, high-frequency, propagation-speed calibration, loss, and mode-resolution assumptions under which the measured mode count tests both the topological area normalization and Weyl asymptotics.

## 10.2 Spherical Transform

**Exercise 10.3** (Spherical convolution and the rank-one Gelfand pair). For  $K$ -bi-invariant functions  $f_1, f_2 \in C_c(G)$ , define

$$(f_1 * f_2)(g) = \int_G f_1(h)f_2(h^{-1}g) dh.$$

Use the Cartan decomposition to prove that  $g$  and  $g^{-1}$  belong to the same double coset  $KgK$ . Apply the anti-involution  $g \mapsto g^{-1}$  to prove

$$f_1 * f_2 = f_2 * f_1.$$

Thus  $(G, K)$  is a *Gelfand pair*.

A normalized continuous  $K$ -bi-invariant function  $\varphi$  is *spherical* when

$$\int_K \varphi(gkh) dk = \varphi(g)\varphi(h).$$

Show that integration against  $\varphi$  is a character of the commutative convolution algebra, and prove the converse for characters represented by continuous functions. Deduce that spherical functions are the distinguished special functions diagonalizing radial convolution.

## 10.3 Legendre Functions

**Exercise 10.4** (Poisson kernels and rank-one spherical functions). For the sign convention

$$\Delta_{\mathbb{H}^2} = y^2(\partial_x^2 + \partial_y^2),$$

derive its radial part

$$\Delta_{\text{rad}} = \frac{d^2}{dr^2} + \coth r \frac{d}{dr}.$$

For  $\lambda \in \mathbb{R}$ , define

$$\varphi_\lambda(r) = \frac{1}{2\pi} \int_0^{2\pi} (\cosh r - \sinh r \cos \theta)^{-1/2-i\lambda} d\theta.$$

For  $z > 1$  and  $\nu \in \mathbb{C}$ , define the Legendre function of the first kind by analytic continuation of

$$P_\nu(z) = {}_2F_1\left(\begin{matrix} -\nu, \nu+1 \\ 1 \end{matrix}; \frac{1-z}{2}\right), \quad P_\nu(1) = 1,$$

along the real interval  $[1, \infty)$ . Interpret the integrand as a complex power of the hyperbolic Poisson kernel. Prove that  $\varphi_\lambda$  is real-valued, even in  $\lambda$ , normalized at  $r = 0$ , and satisfies the spherical product formula. Differentiate under the integral to prove

$$\Delta_{\mathbb{H}^2} \varphi_\lambda = -(\lambda^2 + \frac{1}{4})\varphi_\lambda, \quad \varphi_\lambda(eK) = 1.$$

Regarded as a  $K$ -bi-invariant function on  $G$  and written in the radial coordinate  $a_r K$ , it is determined by

$$\varphi_\lambda''(r) + \coth r \varphi_\lambda'(r) = -(\lambda^2 + \frac{1}{4})\varphi_\lambda(r), \quad \varphi_\lambda(0) = 1, \quad \varphi_\lambda'(0) = 0.$$

Show that the normalized radial eigenfunction satisfying

$$\Delta \varphi_\lambda = -(\lambda^2 + \frac{1}{4})\varphi_\lambda, \quad \varphi_\lambda(0) = 1,$$

is the conical Legendre function

$$\varphi_\lambda(r) = P_{-1/2+i\lambda}(\cosh r).$$

**Exercise 10.5** (Finite-radius conical orthogonality). For real  $\lambda, \mu$ , derive the Wronskian identity

$$(\mu^2 - \lambda^2) \int_0^R \varphi_\lambda(r) \varphi_\mu(r) \sinh r \, dr = \left[ \sinh r (\varphi_\mu \varphi_\lambda' - \varphi_\lambda \varphi_\mu') \right]_0^R.$$

Impose the Dirichlet condition  $\varphi_\lambda(R) = 0$ . Use the radial Sturm–Liouville operator to prove that its positive roots  $\lambda_j$  are real and simple and that the corresponding conical functions are orthogonal in  $L^2((0, R), \sinh r \, dr)$ . Compare the large- $r$  equation with a constant-coefficient equation and show how the discrete parameters become continuous as  $R \rightarrow \infty$ .

**Exercise 10.6** (Abel transforms and Euclidean Fourier reduction). For a smooth compactly supported radial function  $f$  on  $\mathbb{H}^2$ , define its Abel transform by

$$(\mathcal{A}f)(t) = e^{t/2} \int_{\mathbb{R}} f(a_t n_x K) \, dx.$$

Use the Iwasawa integration formula and the Poisson integral for  $\varphi_\lambda$  to prove

$$\tilde{f}(\lambda) := \int_{\mathbb{H}^2} f(z) \varphi_\lambda(z) \, d\text{vol}(z) = \int_{\mathbb{R}} (\mathcal{A}f)(t) e^{-i\lambda t} \, dt.$$

Prove that  $\mathcal{A}$  intertwines radial convolution on  $G/K$  with ordinary convolution on  $\mathbb{R}$ . Deduce injectivity of the spherical transform from Euclidean Fourier inversion and recover the commutativity of radial convolution.

For  $f$  supported in a geodesic disk of radius  $R$ , determine the support and parity properties of  $\mathcal{A}f$ . Derive the rank-one formulas

$$(\mathcal{A}f)(t) = \sqrt{2} \int_{|t|}^{\infty} \frac{f(r) \sinh r}{\sqrt{\cosh r - \cosh t}} \, dr,$$

$$f(r) = -\frac{1}{\pi\sqrt{2}} \int_r^\infty \frac{(\mathcal{A}f)'(t)}{\sqrt{\cosh t - \cosh r}} dt,$$

and use them to express  $f(r)$  in terms of the inverse Euclidean Fourier transform of  $\tilde{f}$ .

**Exercise 10.7** (Harish–Chandra normalization). For real  $\lambda \neq 0$ , define  $c(\lambda)$  by the asymptotic normalization

$$\varphi_\lambda(r) = c(\lambda)e^{(i\lambda-1/2)r} + c(-\lambda)e^{(-i\lambda-1/2)r} + o(e^{-r/2}).$$

Apply endpoint asymptotics to the Poisson integral and evaluate the resulting beta integrals to derive

$$c(\lambda) = \frac{\Gamma(i\lambda)}{\sqrt{\pi}\Gamma(\frac{1}{2} + i\lambda)}, \quad |c(\lambda)|^{-2} = \pi\lambda \tanh(\pi\lambda).$$

**Exercise 10.8** (Mehler–Fock inversion). For the spherical transform defined by the Abel transform, use radial integration to obtain

$$\tilde{f}(\lambda) = 2\pi \int_0^\infty f(r)\varphi_\lambda(r) \sinh r \, dr.$$

For  $\Phi \in C_c^\infty((0, \infty)^2)$ , define the diagonal distribution by

$$\iint \Phi(\lambda, \mu) \delta(\lambda - \mu) \, d\lambda \, d\mu = \int_0^\infty \Phi(\lambda, \lambda) \, d\lambda.$$

Interpret the following identity after pairing both sides with such test functions: truncate the radial integral at  $R$ , use the Harish–Chandra asymptotics in the Wronskian identity, and apply the distributional limit of  $\sin(R(\lambda - \mu))/(\lambda - \mu)$  to prove

$$2\pi \int_0^\infty \varphi_\lambda(r)\varphi_\mu(r) \sinh r \, dr = \frac{2\pi}{\lambda \tanh(\pi\lambda)} \delta(\lambda - \mu).$$

Deduce the Mehler–Fock inversion and Plancherel formulas

$$f(r) = \frac{1}{2\pi^2} \int_0^\infty \tilde{f}(\lambda) P_{-1/2+i\lambda}(\cosh r) |c(\lambda)|^{-2} \, d\lambda,$$

$$\|f\|_2^2 = \frac{1}{2\pi^2} \int_0^\infty |\tilde{f}(\lambda)|^2 |c(\lambda)|^{-2} \, d\lambda.$$

**Exercise 10.9** (Principal and complementary series). Put  $M = \{\pm I\}$  and  $P = MAN$ , where  $A = \{a_r : r \in \mathbb{R}\}$  and  $N = \{n_x : x \in \mathbb{R}\}$ . For  $\epsilon \in \{0, 1\}$  and  $\lambda \in \mathbb{R}$ , let  $\mathcal{H}_{\lambda, \epsilon}$  be the measurable functions  $F : G \rightarrow \mathbb{C}$  satisfying

$$F(g(-I)^j a_r n_x) = (-1)^{\epsilon j} e^{-(1/2+i\lambda)r} F(g), \quad \int_K |F(k)|^2 \, dk < \infty,$$

with functions identified almost everywhere on  $K$ . Define

$$(\pi_{\lambda, \epsilon}(g_0)F)(g) = F(g_0^{-1}g).$$

Use the Iwasawa integration formula to prove that this is a unitary representation. In the compact picture, prove

$$\mathcal{H}_{\lambda, \epsilon} \cong \{f \in L^2(K) : f(k_{\theta+\pi}) = (-1)^\epsilon f(k_\theta)\}$$

and that its distinguished complete orthogonal family of  $K$ -characters is

$$e^{i\ell\theta}, \quad \ell \equiv \epsilon \pmod{2}.$$

For  $\epsilon = 0$ , identify the constant  $K$ -type and prove that its normalized matrix coefficient on  $a_r$  is the spherical function

$$\langle 1, \pi_{\lambda,0}(a_r)1 \rangle = \varphi_\lambda(r) = P_{-1/2+i\lambda}(\cosh r).$$

Let  $B$  be the invariant bilinear form on  $\mathfrak{sl}_2(\mathbb{R})$  whose induced metric on  $G/K$  has curvature  $-1$ . The inverse pairing gives a canonical tensor  $C_B \in \mathfrak{g} \otimes \mathfrak{g}$ ; define the *Casimir element*  $\Omega_B \in U(\mathfrak{g})$  as its image under multiplication. Prove that it is central and normalize its sign so that the induced operator on  $K$ -fixed functions is  $\Delta_{\mathbb{H}^2}$ . Show that it acts by  $-(\lambda^2 + 1/4)$ .

For  $0 < \sigma < 1/2$ , replace  $i\lambda$  in the covariance law by  $\sigma$ . Put  $w = S$  and, in its initial convergence region, define the standard intertwining operator by

$$(A_\sigma F)(g) = \int_{\mathbb{R}} F(gwn_x) dx.$$

Prove the Knapp–Stein theorem [43]:  $A_\sigma$  has a meromorphic continuation to the induced representations, and the form  $\langle F, A_\sigma F' \rangle_{L^2(K)}$  is positive and invariant on the even compact picture for  $0 < \sigma < 1/2$ . Deduce the complementary-series representations, compute their Casimir value  $\sigma^2 - 1/4$ , and explain why their  $K$ -fixed matrix coefficients are the analytic continuations  $P_{-1/2+\sigma}(\cosh r)$  of the conical functions. Show that the parameter  $-\sigma$  gives an equivalent representation through  $A_\sigma$ .

**Exercise 10.10** (Holomorphic discrete series). Put

$$C = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}.$$

Use the Cayley transform  $g \mapsto CgC^{-1}$  to conjugate  $G$  to

$$SU(1,1) = \left\{ h = \begin{pmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{pmatrix} : |\alpha|^2 - |\beta|^2 = 1 \right\},$$

acting on the unit disk by

$$h \cdot z = \frac{\alpha z + \beta}{\bar{\beta} z + \bar{\alpha}}, \quad j(h, z) = \bar{\beta} z + \bar{\alpha}.$$

Verify

$$Ck_\theta C^{-1} = \begin{pmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{pmatrix},$$

which fixes the signs of the  $K$ -weights below. Prove the cocycle identity for  $j$ . For an integer  $n \geq 2$ , let

$$\mathcal{A}_n^2(\mathbb{D}) = \left\{ f \text{ holomorphic on } \mathbb{D} : \frac{n-1}{\pi} \int_{\mathbb{D}} |f(z)|^2 (1 - |z|^2)^{n-2} dx dy < \infty \right\}$$

and define

$$(D_n^+(h)f)(z) = j(h^{-1}, z)^{-n} f(h^{-1} \cdot z).$$

Derive

$$1 - |h \cdot z|^2 = \frac{1 - |z|^2}{|j(h, z)|^2}, \quad dA(h \cdot z) = |j(h, z)|^{-4} dA(z),$$

and prove that  $D_n^+$  is unitary. Show that the monomials form a complete orthogonal system and compute

$$\|z^m\|_n^2 = \frac{m!(n-1)!}{(m+n-1)!}.$$

Determine their  $K$ -characters and obtain the one-sided weights  $n, n+2, n+4, \dots$  for the displayed  $k_\theta$  and left-action conventions. Construct  $D_n^-$  on the antiholomorphic weighted Bergman space and obtain the opposite weights.

For the normalized monomial  $v_m$  and the radial element  $a_r$ , expand the automorphy factor by the binomial series and use the beta integral to show that

$$\langle v_m, D_n^+(a_r)v_m \rangle = (\cosh(r/2))^{-n} {}_2F_1\left(-m, n+m; 1; \tanh^2(r/2)\right).$$

Verify the formula at  $m = 0$  and identify the terminating hypergeometric factor as a Jacobi polynomial. Prove the irreducibility of these two representations [43] and call them the holomorphic and antiholomorphic discrete series.

**Exercise 10.11** (The unitary dual and group Plancherel decomposition). An irreducible unitary representation of  $G$  is an irreducible continuous homomorphism from  $G$  to the unitary operators on a Hilbert space, and two such representations are equivalent when they are conjugate by a unitary map. Prove the classification of the unitary dual of  $SL_2(\mathbb{R})$  [43]: it consists of the principal series  $\pi_{\lambda, \epsilon}$ , the even complementary series, the holomorphic and antiholomorphic discrete series  $D_n^\pm$ , their limit representations, and the trivial representation, with the equivalences and reducible endpoint parameters removed. Record the central character,  $K$ -types, and Casimir value of every family, and identify exactly which families have a nonzero  $K$ -fixed vector. Here the *central character* is the scalar by which  $-I$  acts, whose existence follows from Schur's lemma.

For  $f \in C_c^\infty(G)$  define

$$\widehat{f}(\pi) = \int_G f(g) \pi(g^{-1}) dg.$$

For a square-integrable irreducible representation  $\pi$ , define its *formal degree*  $d_\pi$  by the coefficient orthogonality identity

$$\int_G \langle x, \pi(g)y \rangle \overline{\langle x', \pi(g)y' \rangle} dg = d_\pi^{-1} \langle x, x' \rangle \langle y', y \rangle.$$

Prove the group Plancherel theorem [43] in the normalization  $\int_K dk = 1$ :  $L^2(G)$  decomposes into Hilbert–Schmidt operator spaces over the two principal series and the discrete series, while the complementary series, limit representations, and trivial representation have Plancherel measure zero. The two continuous densities are, up to the common Haar-normalization constant,

$$\lambda \tanh(\pi\lambda) d\lambda, \quad \lambda \coth(\pi\lambda) d\lambda,$$

for even and odd parity, and the formal degree of  $D_n^\pm$  is proportional to  $n-1$ . Write the resulting Plancherel norm and inversion formula for  $f(e)$ .

If a source transforms by the  $K$ -character  $e^{i\ell_{\text{in}}\theta}$  and a detector by  $e^{i\ell_{\text{out}}\theta}$ , prove that only representations containing both  $K$ -types contribute to the resolved matrix-coefficient response. Deduce the parity selection rule for the principal continuum and the one-sided selection rules for the two discrete-series towers.

**Exercise 10.12** (Associated Legendre functions and hyperbolic-disk modes). For  $m \in \mathbb{Z}_{\geq 0}$ ,  $z > 1$ , and  $\nu \in \mathbb{C}$ , define

$$P_\nu^{-m}(z) = \frac{1}{m!} \left( \frac{z-1}{z+1} \right)^{m/2} {}_2F_1\left( -\nu, \nu+1; \frac{1-z}{2} \right).$$

In geodesic polar coordinates, derive the full Laplace–Beltrami operator

$$\Delta_{\mathbb{H}^2} = \frac{\partial^2}{\partial r^2} + \coth r \frac{\partial}{\partial r} + \frac{1}{\sinh^2 r} \frac{\partial^2}{\partial \theta^2}.$$

For the intrinsic  $SO(2)$  character  $e^{i\ell\theta}$ , derive the radial equation

$$R'' + \coth r R' - \frac{\ell^2}{\sinh^2 r} R = -(\lambda^2 + \frac{1}{4})R.$$

Show that the solution regular at the origin is proportional to

$$P_{-1/2+i\lambda}^{-|\ell|}(\cosh r)$$

and determine its normalization from  $R(r) \sim r^{|\ell|}$  as  $r \downarrow 0$ .

For the geodesic disk  $D_R$ , prove that its Dirichlet eigenmodes are

$$e^{i\ell\theta} P_{-1/2+i\lambda_{n\ell}}^{-|\ell|}(\cosh r), \quad P_{-1/2+i\lambda_{n\ell}}^{-|\ell|}(\cosh R) = 0.$$

Establish orthogonality for distinct pairs  $(n, \ell)$  and identify the  $\ell = 0$  family with the conical modes in Mehler–Fock theory.

## 10.4 Hyperbolic Spectroscopy

**Exercise 10.13** (Harish–Chandra wave kernels and geodesic phase fronts). Let  $u_{tt} = v^2 \Delta_{\mathbb{H}^2} u$  have radial initial data  $u(0, r) = f(r)$  and  $u_t(0, r) = g(r)$  in  $C_c^\infty$ . Put  $\omega_\lambda = v(\lambda^2 + 1/4)^{1/2}$ . Use Mehler–Fock inversion to prove

$$u(t, r) = \frac{1}{2\pi^2} \int_0^\infty \left( \cos(t\omega_\lambda) \tilde{f}(\lambda) + \frac{\sin(t\omega_\lambda)}{\omega_\lambda} \tilde{g}(\lambda) \right) P_{-1/2+i\lambda}(\cosh r) |c(\lambda)|^{-2} d\lambda.$$

For a point source at  $z_0$ , take  $f = 0$ ,  $g = \delta_{z_0}$ , use  $G$ -invariance to replace  $r$  by the hyperbolic distance  $d(z, z_0)$ , and derive the impulse kernel

$$K_t(d) = \frac{1}{2\pi^2} \int_0^\infty \frac{\sin(t\omega_\lambda)}{\omega_\lambda} P_{-1/2+i\lambda}(\cosh d) |c(\lambda)|^{-2} d\lambda.$$

Prove [62] finite propagation speed for the wave equation and the one-dimensional stationary-phase estimate for a nondegenerate critical point, including its uniform form for compact parameter sets. Apply them to show that the leading phase fronts are the geodesic circles  $d(z, z_0) = vt$ . Derive the high-frequency spectral density from  $|c(\lambda)|^{-2} = \pi \lambda \tanh(\pi \lambda)$  and state the interior tail present for the two-dimensional wave equation.

**Exercise 10.14** (Hyperbolic-drum spectra and circuit phase fronts). Let  $D_R \subset \mathbb{H}^2$  be the geodesic disk of radius  $R$  in curvature  $-1$ , impose Dirichlet boundary conditions, and let the wave speed be  $v$ . For  $n \geq 1$  and  $\ell \in \mathbb{Z}$ , prove that the normal modes are indexed by the positive solutions of

$$P_{-1/2+i\lambda_{n\ell}}^{-|\ell|}(\cosh R) = 0$$

and have angular frequencies

$$\omega_{n\ell} = v \sqrt{\lambda_{n\ell}^2 + \frac{1}{4}}.$$



For  $\ell = 0$ , use the  $c$ -function asymptotics to derive the large- $R$  quantization law

$$\lambda_{n0}R + \arg c(\lambda_{n0}) = \left(n - \frac{1}{2}\right)\pi + o(1),$$

with the indexing convention stated explicitly. Use the associated Legendre equations to compare the  $(n, \ell)$  branches and deduce the curvature-dependent reordering of radial and angular modes. Order the pairs  $(n, \ell)$  by increasing continuum eigenvalue, with multiplicity, and write the resulting parameter as  $\lambda_j$ .

For the  $\{3, 7\}$  tessellation used in Lenggenhager et al., *Nature Communications* **13** (2022), let  $Q_R$  be the graph Laplacian in their sign convention, put  $L_R = -Q_R \geq 0$ , and write  $\Lambda_j(L_R)$  for its eigenvalues. The experiment uses curvature  $-4$ , geodesic edge length  $a = 0.545275$ , and disk coordinate radius  $h = \tanh a$ . At an interior vertex  $p$ , define

$$(Q_R f)(p) = \sum_{q \sim p} (f(q) - f(p)).$$

For the seven tangent displacements  $v_j$  to neighboring vertices, treat  $a$  as the long-wavelength expansion parameter, use the continuum normalization  $\|v_j\| = a$ , and verify the isotropic moment identities

$$\sum_{j=1}^7 v_j = 0, \quad \sum_{j=1}^7 v_j \otimes v_j = \frac{7}{2}a^2 I.$$

Apply the covariant Taylor expansion of  $f(\exp_p v_j)$  to prove the long-wavelength calibration

$$\Lambda_j(L_R) = \frac{7}{4}a^2 \nu_j^{(-4)} + O(a^3), \quad \nu_j^{(-4)} = 4\nu_j^{(-1)},$$

where  $\nu_j^{(-1)} = \lambda_j^2 + \frac{1}{4}$  is the curvature- $-1$  continuum eigenvalue. Prove that the metric and geodesic-radius normalizations satisfy  $g_{-1} = 4g_{-4}$  and  $R_{-1} = 2R_{-4}$ .

For circuit capacitance  $C$  and grounding inductance  $L$ , derive from the grounded circuit Laplacian

$$J(\omega) = -i\omega C Q_R + \frac{1}{i\omega L} I$$

that the measured circuit resonance satisfies

$$\omega_j^{\text{circ}} = \frac{1}{\sqrt{LC \Lambda_j(L_R)}}, \quad \nu_j^{(-1)} \sim \frac{\Lambda_j(L_R)}{7a^2} = \frac{1}{28\pi^2 a^2 LC (f_j^{\text{circ}})^2}.$$

Use this calibration to compare the inferred parameters  $\lambda_{n\ell}$  and measured voltage eigenmodes with the associated Legendre zeros and their predicted angular-momentum reordering. For the time-resolved experiment, plot the measured phase against  $d_{\mathbb{H}^2}(z, z_0)$  and against Euclidean distance, and compare the result with the Harish-Chandra impulse kernel and its geodesic phase fronts. State separately the constant-curvature, long-wavelength, Dirichlet-boundary, circuit-calibration, weak-loss, mode-identification, time-resolution, and finite-size assumptions, and quantify errors caused by boundary shape, loss, finite frequency resolution, and discretization.

## 11 $SL_2(\mathbb{Z})$

### 11.1 Modular Group

**Exercise 11.1** (The modular group and its fundamental domain). Let  $SL_2(\mathbb{Z})$  act on the upper half-plane by

$$\gamma\tau = \frac{a\tau + b}{c\tau + d}, \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

and put

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Use the Euclidean algorithm on the first column of a matrix in  $SL_2(\mathbb{Z})$  to prove that  $S$  and  $T$  generate the group. Determine the kernel of the action and prove

$$PSL_2(\mathbb{Z}) = SL_2(\mathbb{Z})/\{\pm I\} \cong \langle s, t \mid s^2 = (st)^3 = 1 \rangle.$$

Let

$$\mathcal{F} = \{\tau \in \mathbb{H} : |\operatorname{Re} \tau| \leq \frac{1}{2}, |\tau| \geq 1\}.$$

Show that every orbit meets  $\mathcal{F}$  by first maximizing  $\operatorname{Im}(\gamma\tau)$  and then translating by a power of  $T$ . Prove that two interior points of  $\mathcal{F}$  are never equivalent, identify the side pairings, and determine the stabilizers of  $i$  and  $\rho = e^{2\pi i/3}$ . Prove that the only cusp orbit is represented by infinity and compute from the hyperbolic area form

$$dA = \frac{dx dy}{y^2}$$

that  $\operatorname{Area}(PSL_2(\mathbb{Z}) \backslash \mathbb{H}) = \pi/3$ .

### 11.2 Automorphic Spectrum

**Exercise 11.2** (Maass forms, Eisenstein series, and automorphic Plancherel theory). Let  $\Gamma = PSL_2(\mathbb{Z})$  act on the upper half-plane and give  $\Gamma \backslash \mathbb{H}$  the invariant measure  $d\mu = y^{-2} dx dy$ . Put

$$\mathcal{L}_{\mathbb{H}} = -\Delta_{\mathbb{H}^2} = -y^2(\partial_x^2 + \partial_y^2).$$

A *Maass cusp form* with spectral parameter  $r \in \mathbb{R}_{\geq 0} \cup i(0, \frac{1}{2})$  is a smooth  $\Gamma$ -invariant square-integrable function  $u$  satisfying

$$\mathcal{L}_{\mathbb{H}} u = (\tfrac{1}{4} + r^2)u, \quad \int_0^1 u(x + iy) dx = 0.$$

For  $z > 0$ , symmetrize the integral representation of the modified Bessel function in  $u$  and substitute  $t = e^u$  to obtain

$$K_{\nu}(z) = \frac{1}{2} \int_0^{\infty} t^{\nu-1} \exp\left[-\frac{z}{2}(t + t^{-1})\right] dt.$$

Derive its differential equation, use invariance under  $x \mapsto x + 1$  to separate the distinguished Fourier characters, and prove that every Maass cusp form has an expansion

$$u(x + iy) = \sum_{n \neq 0} a_n \sqrt{y} K_{ir}(2\pi|n|y) e^{2\pi i n x}.$$

For  $\operatorname{Re} s > 1$ , define

$$E(z, s) = \frac{1}{2} \sum_{\substack{(c,d) \in \mathbb{Z}^2 \\ (c,d)=1}} \frac{y^s}{|cz + d|^{2s}}.$$

Prove absolute convergence,  $\Gamma$ -invariance, and

$$\mathcal{L}_{\mathbb{H}} E(z, s) = s(1 - s)E(z, s).$$

Use Poisson summation to derive its Fourier expansion and identify the nonconstant terms as multiples of  $\sqrt{y}K_{s-1/2}(2\pi|n|y)e^{2\pi inx}$ .

Prove the automorphic spectral theorem [39]:  $E(z, \frac{1}{2} + ir)$  supplies the continuous spectrum and the constant function together with the Maass cusp forms supplies the discrete spectrum of  $L^2(\Gamma \backslash \mathbb{H})$ . Write the resulting Plancherel and inversion formulas with the Eisenstein measure  $dr/(4\pi)$ . Deduce that the cusp profiles of discrete resonant modes have exponentially decaying  $K$ -Bessel harmonics and state the cusp truncation, boundary, loss, and mode-resolution assumptions required to test them in a hyperbolic resonator.

**Exercise 11.3** (Selberg trace formula and hyperbolic length spectroscopy). For a primitive hyperbolic conjugacy class  $[\gamma]$  in  $\Gamma$ , define its closed-geodesic length  $\ell_\gamma > 0$  by

$$|\operatorname{Tr} \gamma| = 2 \cosh(\ell_\gamma/2).$$

Let  $h$  be even, holomorphic in a strip, and rapidly decreasing there, and put

$$g(u) = \frac{1}{2\pi} \int_{\mathbb{R}} h(r) e^{-iru} dr.$$

Prove the Selberg trace formula for the modular surface [39]: after subtracting the continuous Eisenstein contribution, its regularized spectral trace is the sum of the identity, elliptic, parabolic, and hyperbolic terms, with hyperbolic term

$$\sum_{[\gamma]_{\text{prim}}} \sum_{m \geq 1} \frac{\ell_\gamma}{2 \sinh(m\ell_\gamma/2)} g(m\ell_\gamma).$$

The identity contribution is

$$\frac{\operatorname{Area}(\Gamma \backslash \mathbb{H})}{4\pi} \int_{\mathbb{R}} r \tanh(\pi r) h(r) dr.$$

Recover the Weyl area term from it. Choose a family of functions  $g$  concentrated near a positive length  $L$  and prove that the oscillatory remainder in the smoothed spectral density detects the primitive lengths and their repetitions. Conversely, choose  $h$  concentrated near a spectral parameter and determine the resolution with which a finite length window can recover it.

For an idealized wave or circuit realization of a compactly truncated modular surface, express the measured resonance parameters as  $\lambda_j = \frac{1}{4} + r_j^2$  and state separately the curvature calibration, cusp truncation, boundary correction, unfolding, finite spectral window, and loss assumptions under which the length peaks inferred from resonance data may be compared with the closed-geodesic lengths.

### 11.3 Modular Functions

**Exercise 11.4** (Lattice theta series). Let  $L \subset \mathbb{R}^d$  be a full Euclidean lattice and define

$$L^\vee = \{y \in \mathbb{R}^d : \langle x, y \rangle \in \mathbb{Z} \text{ for every } x \in L\}.$$

Call  $L$  integral when  $L \subseteq L^\vee$ , unimodular when  $L = L^\vee$ , and even when  $\|x\|^2 \in 2\mathbb{Z}$  for every  $x \in L$ . For  $\tau$  in the upper half-plane and  $q = e^{2\pi i\tau}$ , its theta series is

$$\Theta_L(\tau) = \sum_{x \in L} q^{\|x\|^2/2}.$$

When  $L$  is even, its coefficient of  $q^n$  counts lattice vectors of squared length  $2n$ .

Apply Poisson summation to a Gaussian and prove

$$\Theta_L(-1/\tau) = \frac{(-i\tau)^{d/2}}{\text{vol}(L)} \Theta_{L^\vee}(\tau),$$

using the branch positive on the positive imaginary axis. Show that evenness gives  $\Theta_L(\tau+1) = \Theta_L(\tau)$ . Explain why a general lattice theta series transforms with its dual rather than being a scalar modular form by itself.

**Exercise 11.5** (Modular forms, zeta functions, and Eisenstein series). A *modular form of integral weight  $k$*  for  $SL_2(\mathbb{Z})$  is a holomorphic function  $f$  on the upper half-plane satisfying

$$f(\gamma\tau) = (c\tau + d)^k f(\tau)$$

and holomorphic at the cusp, meaning that its Fourier expansion  $f(\tau) = \sum_{n \geq 0} a_n q^n$  has no negative powers. It is a *cusp form* if  $a_0 = 0$ . Write  $M_k$  for the vector space of modular forms of weight  $k$ .

For  $\text{Re } s > 1$ , define the Riemann zeta function by

$$\zeta(s) = \sum_{n \geq 1} n^{-s}.$$

For a  $d$ -dimensional Euclidean lattice  $L$ , with  $d \geq 1$ , and  $\text{Re } s > d/2$ , define its Epstein zeta function by

$$Z_L(s) = \sum_{x \in L \setminus \{0\}} \|x\|^{-2s}.$$

For  $m \geq 2$  and  $\tau$  in the upper half-plane, define

$$G_{2m}(\tau) = \sum_{(c,d) \in \mathbb{Z}^2 \setminus \{(0,0)\}} \frac{1}{(c\tau + d)^{2m}}, \quad E_{2m}(\tau) = \frac{G_{2m}(\tau)}{2\zeta(2m)}.$$

(a) Put  $\theta_L(t) = \Theta_L(it)$ . For  $\text{Re } s > d/2$ , prove

$$\pi^{-s} \Gamma(s) Z_L(s) = \int_0^\infty (\theta_L(t) - 1) t^{s-1} dt.$$

Split the integral at 1 and use Poisson duality to prove the meromorphic continuation

$$\begin{aligned} \pi^{-s} \Gamma(s) Z_L(s) &= \int_1^\infty (\theta_L(t) - 1) t^{s-1} dt \\ &\quad + \frac{1}{\text{vol}(L)} \int_1^\infty (\theta_{L^\vee}(t) - 1) t^{d/2-s-1} dt + \frac{1}{\text{vol}(L)(s - d/2)} - \frac{1}{s}. \end{aligned}$$

Deduce

$$\pi^{-s}\Gamma(s)Z_L(s) = \frac{1}{\text{vol}(L)}\pi^{-(d/2-s)}\Gamma(d/2-s)Z_{L^\vee}(d/2-s),$$

$Z_L(0) = -1$ , and

$$\text{res}_{s=d/2} Z_L(s) = \frac{\pi^{d/2}}{\text{vol}(L)\Gamma(d/2)}.$$

(b) For  $L = \mathbb{Z}$ , use  $Z_{\mathbb{Z}}(s) = 2\zeta(2s)$  to derive the functional equation of  $\zeta$ . Use Fourier series and Parseval to prove

$$\zeta(2) = \frac{\pi^2}{6}, \quad \zeta(4) = \frac{\pi^4}{90}, \quad \zeta(6) = \frac{\pi^6}{945},$$

and then deduce

$$\zeta(-1) = -\frac{1}{12}, \quad \zeta(-3) = \frac{1}{120}.$$

**Exercise 11.6** (Holomorphic Eisenstein series). Prove absolute local uniform convergence of  $G_{2m}$  and prove

$$G_{2m}\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^{2m}G_{2m}(\tau)$$

for every  $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$ . Using Poisson summation in one lattice direction, derive

$$G_{2m}(\tau) = 2\zeta(2m) + \frac{2(2\pi i)^{2m}}{(2m-1)!} \sum_{n \geq 1} \sigma_{2m-1}(n)q^n, \quad \sigma_r(n) = \sum_{d|n} d^r.$$

Deduce that  $E_{2m}$  is a modular form of weight  $2m$  and that

$$E_4(\tau) = 1 + 240 \sum_{n \geq 1} \sigma_3(n)q^n, \quad E_6(\tau) = 1 - 504 \sum_{n \geq 1} \sigma_5(n)q^n.$$

**Exercise 11.7** (Hecke operators, eigenforms, and Euler products). For  $f(\tau) = \sum_{m \geq 0} a_m q^m$  of modular weight  $k$ , define

$$(T_n f)(\tau) = n^{k-1} \sum_{\substack{ad=n \\ d>0}} d^{-k} \sum_{b \bmod d} f\left(\frac{a\tau + b}{d}\right).$$

Prove that  $T_n$  preserves modular forms and cusp forms and derive

$$[q^m]T_n f = \sum_{d|(m,n)} d^{k-1} a_{mn/d^2}.$$

Use this formula to prove

$$T_m T_n = \sum_{d|(m,n)} d^{k-1} T_{mn/d^2}$$

and hence that the Hecke operators commute. Define the Petersson inner product on cusp forms and prove that every  $T_n$  is self-adjoint.

If  $f$  is a normalized simultaneous eigenform, prove that its eigenvalue under  $T_n$  is  $a_n$ , that

$$a_m a_n = \sum_{d|(m,n)} d^{k-1} a_{mn/d^2}, \quad a_{p^{r+1}} = a_p a_{p^r} - p^{k-1} a_{p^{r-1}},$$

and that

$$L(f, s) = \sum_{n \geq 1} \frac{a_n}{n^s} = \prod_p \left( 1 - a_p p^{-s} + p^{k-1-2s} \right)^{-1}$$

in its half-plane of absolute convergence. Apply this to every normalized eigenform available at this stage.

For an arithmetic hyperbolic resonator on which a finite family of Hecke correspondences is implemented as commuting calibrated operators, prove that joint Laplace–Hecke measurements resolve common spectral subspaces and predict the displayed multiplicative relations among their response coefficients. State the arithmetic-quotient, correspondence implementation, normalization, and finite-resolution assumptions.

**Exercise 11.8** (Theta and eta functions). Set

$$\begin{aligned} \vartheta_2(\tau) &= \sum_{n \in \mathbb{Z}} q^{(n+1/2)^2/2}, \\ \vartheta_3(\tau) &= \sum_{n \in \mathbb{Z}} q^{n^2/2}, & \eta(\tau) &= q^{1/24} \prod_{m \geq 1} (1 - q^m). \\ \vartheta_4(\tau) &= \sum_{n \in \mathbb{Z}} (-1)^n q^{n^2/2}, \end{aligned}$$

Use Poisson summation on shifted and phase-twisted Gaussians to derive

$$\vartheta_2(-1/\tau) = (-i\tau)^{1/2} \vartheta_4(\tau), \quad \vartheta_3(-1/\tau) = (-i\tau)^{1/2} \vartheta_3(\tau), \quad \vartheta_4(-1/\tau) = (-i\tau)^{1/2} \vartheta_2(\tau).$$

Use the branch positive for  $\tau$  on the positive imaginary axis. Prove directly from the series that

$$\vartheta_2(\tau + 1) = e^{\pi i/4} \vartheta_2(\tau), \quad \vartheta_3(\tau + 1) = \vartheta_4(\tau), \quad \vartheta_4(\tau + 1) = \vartheta_3(\tau).$$

Write these  $S$ - and  $T$ -laws as a vector-valued transformation law of weight  $\frac{1}{2}$  for  $(\vartheta_2, \vartheta_3, \vartheta_4)^T$  and prove that it is not an instance of a scalar integral-weight transformation law.

**Exercise 11.9** (Jacobi triple product and eta transformations). Establish the Jacobi triple product, first as an identity of formal Laurent series,

$$\sum_{n \in \mathbb{Z}} z^n q^{n^2/2} = \prod_{m \geq 1} (1 - q^m)(1 + zq^{m-1/2})(1 + z^{-1}q^{m-1/2}),$$

and use specializations to recover product forms for the theta functions. Deduce

$$2\eta(\tau)^3 = \vartheta_2(\tau)\vartheta_3(\tau)\vartheta_4(\tau).$$

Use the theta transformations and the branch positive on the positive imaginary axis to derive

$$\eta(\tau + 1) = e^{\pi i/12} \eta(\tau), \quad \eta(-1/\tau) = (-i\tau)^{1/2} \eta(\tau).$$

Prove that these laws give  $\eta$  weight  $\frac{1}{2}$  with a nontrivial scalar multiplier, rather than an integral-weight law with trivial multiplier. Explain how the two sides encode charge sectors and independent oscillator modes, respectively.

**Exercise 11.10** (Modular functions, the discriminant, and the  $J$ -function). Put  $PSL_2(\mathbb{R}) = SL_2(\mathbb{R})/\{\pm I\}$  and define  $PSL_2(\mathbb{Z})$  similarly. Let  $\Gamma \subset PSL_2(\mathbb{R})$  be commensurable with  $PSL_2(\mathbb{Z})$  and suppose that the cusp at infinity has width one. A *modular function* for  $\Gamma$  is a  $\Gamma$ -invariant function meromorphic on the upper half-plane and at the cusps. Write  $X_\Gamma$  for the compactified quotient. A *normalized Hauptmodul* is a modular function inducing an isomorphism  $X_\Gamma \cong \mathbb{P}^1(\mathbb{C})$  and having Fourier expansion

$$q^{-1} + O(q)$$

at infinity.

Define

$$\Delta(\tau) = \eta(\tau)^{24}, \quad j(\tau) = \frac{E_4(\tau)^3}{\Delta(\tau)}, \quad J(\tau) = j(\tau) - 744 = \sum_{n \geq -1} c(n)q^n.$$

Let  $\mathbb{H}$  be the upper half-plane, let  $\rho = e^{2\pi i/3}$ , and let  $\mathcal{F}$  be the standard fundamental domain for  $PSL_2(\mathbb{Z})$ . Apply the argument principle to a truncated copy of  $\mathcal{F}$ , pair its boundary arcs by  $S$  and  $T$ , and prove that every nonzero modular form  $f$  of weight  $k$  satisfies

$$v_\infty(f) + \frac{1}{2}v_i(f) + \frac{1}{3}v_\rho(f) + \sum_{\substack{[z] \in PSL_2(\mathbb{Z}) \backslash \mathbb{H} \\ [z] \neq [i], [\rho]}} v_z(f) = \frac{k}{12}.$$

Use the eta transformation laws and product to show that  $\Delta$  is a weight-12 cusp form with a simple zero at infinity and no zero in the upper half-plane. Deduce

$$M_{12} = \mathbb{C}E_4^3 \oplus \mathbb{C}\Delta, \quad \Delta = \frac{E_4^3 - E_6^2}{1728}.$$

Deduce that  $S_{12} = \mathbb{C}\Delta$ , so the Hecke operators make  $\Delta = \sum_{n \geq 1} \tau(n)q^n$  a normalized simultaneous eigenform. Use the Hecke relations to compute  $\tau(mn)$  for coprime  $m, n$  and the recurrence for  $\tau(p^r)$ . Show that  $j$  has one simple pole on  $X_{PSL_2(\mathbb{Z})}$  and hence induces an isomorphism with  $\mathbb{P}^1(\mathbb{C})$ . Expand the products and prove that

$$J(\tau) = q^{-1} + 196884q + 21493760q^2 + 864299970q^3 + O(q^4),$$

so  $J$  is the normalized Hauptmodul for  $PSL_2(\mathbb{Z})$ .

## 11.4 Modular Spectra

**Exercise 11.11** (Leech theta series and its first diffraction shell). If an integral lattice  $L$  is used as a crystal lattice, its physical reciprocal lattice is  $L^{\text{rec}} = 2\pi L^\vee$ .

Call a vector of squared length 2 in an integral Euclidean lattice a *root*. For an even integral lattice  $L$ , define its discriminant group and quadratic form by

$$A_L = L^\vee/L, \quad q_L(x + L) = \frac{\langle x, x \rangle}{2} \pmod{\mathbb{Z}}.$$

Prove that even overlattices of  $L$  correspond to isotropic subgroups of  $(A_L, q_L)$ . Call two unimodular lattices  $p$ -neighbors when their intersection has index  $p$  in each, and construct their common index- $p$  sublattice from an isotropic line in  $L/pL$ . Use these constructions to develop the neighbor and glue-code analysis for rank-24 even unimodular lattices and prove the rootless case of the Niemeier

classification [20]: there is a unique isometry class of even unimodular Euclidean lattices of rank 24 having no roots. Call its members *Leech lattices*, and fix one representative  $\Lambda$ . Use Poisson duality to prove that  $\Theta_\Lambda$  is a modular form of weight 12. Use the absence of roots and the description of  $M_{12}$  to derive

$$\Theta_\Lambda = E_4^3 - 720\Delta = 1 + 196560q^2 + O(q^3).$$

For the flat torus  $T_\Lambda = \mathbb{R}^{24}/\Lambda$ , let  $K_t$  be its heat kernel. Prove

$$\int_{T_\Lambda} K_t(x, x) dx = \sum_{\xi \in \Lambda^*} e^{-4\pi^2 t \|\xi\|^2} = \Theta_\Lambda(4\pi i t),$$

and deduce that its first nonconstant heat-trace term is  $196560e^{-16\pi^2 t}$ . For the idealized 24-dimensional crystal with lattice  $\Lambda$  and a one-point motif, deduce that the first nonzero reciprocal-lattice shell has radius  $4\pi$  in the physical reciprocal convention and consists of 196560 Bragg peaks. State how a nontrivial motif changes their intensities without changing the shell radius.

**Exercise 11.12** (Theta heat kernels, spectral zeta functions, and Casimir pressure). For a massless scalar field between parallel Dirichlet plates of area  $A$  and separation  $a$ , neglect edge effects and take the normal-mode frequencies to be

$$\omega_{n,k} = c \sqrt{|k|^2 + \left(\frac{\pi n}{a}\right)^2}, \quad n \geq 1, \quad k \in \mathbb{R}^2.$$

Define the normal-direction theta sum and the heat trace per unit area by

$$\Theta_a(t) = \sum_{n \in \mathbb{Z}} e^{-\pi^2 t n^2 / a^2},$$

$$\frac{K_D(t)}{A} = \sum_{n \geq 1} \int_{\mathbb{R}^2} \frac{d^2 k}{(2\pi)^2} e^{-t(|k|^2 + \pi^2 n^2 / a^2)} = \frac{\Theta_a(t) - 1}{8\pi t}.$$

(a) Apply the Jacobi theta transformation to prove

$$\Theta_a(t) = \frac{a}{\sqrt{\pi t}} \sum_{m \in \mathbb{Z}} e^{-a^2 m^2 / t}$$

and hence

$$\frac{K_D(t)}{A} = \frac{a}{8\pi^{3/2} t^{3/2}} - \frac{1}{8\pi t} + \frac{a}{4\pi^{3/2} t^{3/2}} \sum_{m \geq 1} e^{-a^2 m^2 / t}.$$

Identify the first term as the infinite-volume contribution, the second as plate-local, and the remaining image sum as the interaction term.

(b) For  $\mu > 0$  and  $\text{Re } s$  sufficiently large, define the regulated vacuum energy density by

$$\frac{\mathcal{E}_D(s)}{A} = \frac{\hbar c}{2} \mu^{2s} \sum_{n \geq 1} \int_{\mathbb{R}^2} \frac{d^2 k}{(2\pi)^2} \left( |k|^2 + \frac{\pi^2 n^2}{a^2} \right)^{1/2-s}.$$

Use the Mellin identity

$$\lambda^{1/2-s} = \frac{1}{\Gamma(s - \frac{1}{2})} \int_0^\infty t^{s-3/2} e^{-t\lambda} dt$$



to express  $\mathcal{E}_D(s)$  through the heat trace. Remove the bulk and plate-local terms identified above and evaluate the image integrals to continue the interaction energy as

$$\frac{\mathcal{E}_D^{\text{int}}(s)}{A} = \frac{\hbar c \mu^{2s} a^{2s-3}}{8\pi^{3/2}} \frac{\Gamma(2-s)}{\Gamma(s-\frac{1}{2})} \zeta(4-2s).$$

Evaluate at  $s = 0$  and prove

$$\frac{\mathcal{E}_D}{A} = -\frac{\pi^2 \hbar c}{1440 a^3}.$$

- (c) For the ideal electromagnetic field, account for the two polarizations and deduce

$$\frac{\mathcal{E}_{\text{EM}}}{A} = -\frac{\pi^2 \hbar c}{720 a^3}, \quad P = -\frac{\partial(\mathcal{E}_{\text{EM}}/A)}{\partial a} = -\frac{\pi^2 \hbar c}{240 a^4}.$$

State how a force measurement tests the  $a^{-4}$  law and its coefficient, separating the ideal prediction from finite conductivity, temperature, roughness, alignment, and edge corrections.

**Exercise 11.13** (Bosonic and fermionic trace generating functions). For a finite-dimensional complex vector space  $V$ , an endomorphism  $T$  induces  $\text{Sym}^m T$  and  $\Lambda^m T$ . Define

$$Z_B(T; z) = \sum_{m \geq 0} \text{tr}(\text{Sym}^m T) z^m, \quad Z_F(T; z) = \sum_{m \geq 0} \text{tr}(\Lambda^m T) z^m.$$

Let  $p(n)$  be the number of integer partitions of  $n$ , with  $p(0) = 1$ .

- (a) Let  $V$  be a finite-dimensional complex vector space and  $T \in \text{End}(V)$ . Using the determinant defined by the action on the highest exterior power, prove

$$Z_B(T; z) = \frac{1}{\det(I - zT)}, \quad Z_F(T; z) = \det(I + zT).$$

The first identity holds for  $|z|$  below the reciprocal spectral radius; the second is a polynomial identity. Prove them first for a direct sum of generalized spectral subspaces and then by polynomial continuation, without choosing vectors in those subspaces. If  $\lambda_1, \dots, \lambda_N$  are the eigenvalues counted with algebraic multiplicity, prove, with the empty products interpreted as 1,

$$\text{tr}(\text{Sym}^m T) = \sum_{1 \leq i_1 \leq \dots \leq i_m \leq N} \lambda_{i_1} \cdots \lambda_{i_m}, \quad \text{tr}(\Lambda^m T) = \sum_{1 \leq i_1 < \dots < i_m \leq N} \lambda_{i_1} \cdots \lambda_{i_m}.$$

- (b) Let the one-particle Hamiltonian have one-dimensional spectral subspaces at energies  $j\hbar\omega$ ,  $j \geq 1$ . Apply the identities to the finite spectral truncation and then pass coefficientwise to the limit to derive

$$\prod_{j \geq 1} \frac{1}{1 - q^j} = \sum_{n \geq 0} p(n) q^n, \quad \prod_{j \geq 1} (1 + q^j) = \sum_{n \geq 0} p_{\text{dist}}(n) q^n,$$

where  $p_{\text{dist}}(n)$  counts partitions into distinct parts. Construct the multiplicity map separating powers of two and prove

$$\prod_{j \geq 1} (1 + q^j) = \prod_{j \geq 1} \frac{1}{1 - q^{2j-1}}.$$

Interpret the coefficient of  $q^n$  as the exact degeneracy at excitation energy  $n\hbar\omega$  for the corresponding bosonic or fermionic system.

(c) For the bosonic system put  $q = e^{-\beta_{\text{th}}\hbar\omega}$ , where  $\beta_{\text{th}} = (k_B T)^{-1}$ , and define

$$Z(\beta_{\text{th}}) = \prod_{j \geq 1} \frac{1}{1 - e^{-\beta_{\text{th}} j \hbar \omega}}, \quad \Psi(\beta_{\text{th}}) = \log Z(\beta_{\text{th}}).$$

Define

$$U = -\frac{d\Psi}{d\beta_{\text{th}}}.$$

For

$$\mathcal{S} = k_B(\Psi + \beta_{\text{th}}U), \quad C = k_B\beta_{\text{th}}^2 \frac{d^2\Psi}{d\beta_{\text{th}}^2},$$

prove  $d\mathcal{S} = k_B\beta_{\text{th}} dU$  as  $\beta_{\text{th}}$  varies and express  $d^2\Psi/d\beta_{\text{th}}^2$  as the energy variance. Derive

$$U = \hbar\omega \sum_{j \geq 1} \frac{j}{e^{\beta_{\text{th}} j \hbar \omega} - 1}, \quad C = k_B(\beta_{\text{th}}\hbar\omega)^2 \sum_{j \geq 1} \frac{j^2 e^{\beta_{\text{th}} j \hbar \omega}}{(e^{\beta_{\text{th}} j \hbar \omega} - 1)^2}.$$

(d) State how resolved spectroscopy tests the integer-spaced levels and the partition degeneracies, while equilibrium calorimetry tests the displayed energy and heat-capacity curves. Separate the independent-mode, harmonic, thermal-equilibrium, finite-resolution, and high-energy-cutoff assumptions from these predictions.

**Exercise 11.14** (Virasoro symmetry and chiral finite-size levels). The Virasoro algebra has generators  $L_n$ ,  $n \in \mathbb{Z}$ , and a central generator  $C$ , with

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{1}{12}(m^3 - m)\delta_{m+n,0}C, \quad [C, L_n] = 0.$$

A module has central charge  $c$  when  $C$  acts by  $cI$ . A highest-weight vector  $|h\rangle$  satisfies  $L_0|h\rangle = h|h\rangle$  and  $L_n|h\rangle = 0$  for  $n > 0$ . For a chiral system on a circle of circumference  $\ell$  and propagation speed  $u$ , define the finite-size Hamiltonian on such a module by

$$H_{\text{cyl}} = \frac{2\pi\hbar u}{\ell}(L_0 - c/24).$$

Order the negative Virasoro modes by their indices. Use the commutation relations to rewrite every word as an ordered word. Check every three-letter overlap and use the Jacobi identity to prove that the rewriting is confluent. Filter by word length and deduce that the associated graded algebra is the symmetric algebra on the negative modes. Conclude that the ordered monomials in the negative modes are linearly independent in the universal highest-weight module. Prove that  $L_{-n_1} \cdots L_{-n_r}|h\rangle$  has excitation energy  $2\pi\hbar u(n_1 + \cdots + n_r)/\ell$  above the highest-weight state and identify its exact degeneracy with a partition count before null-vector relations are imposed.

**Exercise 11.15** (Verma and vacuum characters). For  $\tau$  in the upper half-plane put  $q = e^{2\pi i\tau}$ . If  $H_a$  is a positive-energy module with finite-dimensional  $L_0$  spectral subspaces, define

$$\chi_a(\tau) = \text{Tr}_{H_a} q^{L_0 - c/24}$$

when the displayed operator is trace class; the same expression denotes the formal graded trace otherwise.

Use the ordered descendants to prove that the Verma-module character is

$$\chi_{c,h}^{\text{Verma}}(\tau) = q^{h-c/24} \prod_{m \geq 1} (1 - q^m)^{-1}.$$

For the vacuum impose  $L_{-1}|0\rangle = 0$  and derive the corresponding generic vacuum product.

**Exercise 11.16** (Null-vector quotients). Let a singular vector of degree  $r$  in a Verma module generate a submodule isomorphic to the Verma module of highest weight  $h + r$ . Prove that the quotient character is

$$\chi_{c,h}^{\text{Verma}} - \chi_{c,h+r}^{\text{Verma}}.$$

For several singular vectors, prove that their submodules must be combined by inclusion–exclusion, and show that independent subtraction is valid only when the intersections of their descendant submodules have been accounted for.

**Exercise 11.17** (Unitary modular character systems and Ising characters). A finite character vector  $\chi$  is a *unitary modular character system* when

$$\chi(-1/\tau) = S\chi(\tau), \quad \chi(\tau+1) = T\chi(\tau)$$

for unitary matrices  $S, T$ . For a matrix  $M$  with nonnegative integral entries define

$$Z_M(\tau, \bar{\tau}) = \sum_{a,b} \overline{\chi_a(\tau)} M_{ab} \chi_b(\tau).$$

It is *modular invariant* when

$$S^*MS = M, \quad T^*MT = M.$$

For the Ising character system with  $c = 1/2$  and weights  $0, 1/2, 1/16$ , choose the square roots whose  $q$ -expansions have the displayed positive leading coefficients and put

$$\begin{aligned} \chi_0 &= \frac{1}{2} \left( \sqrt{\frac{\vartheta_3}{\eta}} + \sqrt{\frac{\vartheta_4}{\eta}} \right), \\ \chi_{1/2} &= \frac{1}{2} \left( \sqrt{\frac{\vartheta_3}{\eta}} - \sqrt{\frac{\vartheta_4}{\eta}} \right), \\ \chi_{1/16} &= \frac{1}{\sqrt{2}} \sqrt{\frac{\vartheta_2}{\eta}}. \end{aligned}$$

Use the Jacobi triple product to prove

$$\begin{aligned} \sqrt{\frac{\vartheta_3}{\eta}} &= q^{-1/48} \prod_{m \geq 0} (1 + q^{m+1/2}), & \sqrt{\frac{\vartheta_4}{\eta}} &= q^{-1/48} \prod_{m \geq 0} (1 - q^{m+1/2}), \\ \frac{1}{\sqrt{2}} \sqrt{\frac{\vartheta_2}{\eta}} &= q^{1/24} \prod_{m \geq 1} (1 + q^m). \end{aligned}$$

Identify the three characters with the fermionic products obtained from the critical Ising transfer spectrum. Use the theta and eta transformation laws to derive

$$S = \frac{1}{2} \begin{pmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ \sqrt{2} & -\sqrt{2} & 0 \end{pmatrix}, \quad T_{aa} = e^{2\pi i(h_a - c/24)}.$$

Expand each character as  $q^{h-c/24}$  times a power series, identify its coefficients with exact descendant counts, and verify the modular invariance of  $\sum_a |\chi_a|^2$ .

**Exercise 11.18** (Eta asymptotics and bosonic thermodynamics). Use the theta and eta transformation laws to prove

$$\prod_{j \geq 1} (1 - e^{-tj})^{-1} \sim \sqrt{\frac{t}{2\pi}} \exp\left(\frac{\pi^2}{6t}\right) \quad (t \downarrow 0).$$

For  $t = \beta_{\text{th}} \hbar \omega$ , differentiate the transformed eta-product identity to strengthen this to

$$\Psi(\beta_{\text{th}}) = \frac{\pi^2}{6t} + \frac{1}{2} \log \frac{t}{2\pi} - \frac{t}{24} + O(e^{-4\pi^2/t})$$

and derive

$$U = \hbar \omega \left( \frac{\pi^2}{6t^2} - \frac{1}{2t} + \frac{1}{24} + o(1) \right),$$

$$C = k_B \left( \frac{\pi^2}{3t} - \frac{1}{2} + o(1) \right), \quad \mathcal{S} = k_B \left( \frac{\pi^2}{3t} + \frac{1}{2} \log \frac{t}{2\pi} - \frac{1}{2} + o(1) \right).$$

**Exercise 11.19** (Partition saddle). Insert this estimate into Cauchy's coefficient integral, locate its positive real saddle, prove that its equation is

$$n = \frac{U}{\hbar \omega},$$

and identify the leading saddle exponent with the Legendre transform  $\mathcal{S}/k_B$ . Expand the phase through quadratic order and derive the major-arc contribution

$$\frac{1}{4n\sqrt{3}} \exp\left(\pi \sqrt{\frac{2n}{3}}\right).$$

**Exercise 11.20** (Partition minor arcs). Let  $r_n$  be the positive saddle radius and use a Farey dissection of the circle  $|q| = r_n$  whose principal major arc contains  $q = r_n$ . Bound  $\eta(\tau)^{-1}$  on every nonprincipal Farey arc using its modular transformation and prove the Hardy–Ramanujan minor-arc estimate [32]: for some  $\delta > 0$ , the coefficient integral over the complement of the principal major arc is

$$O\left(\exp\left[\left(\pi \sqrt{\frac{2}{3}} - \delta\right) \sqrt{n}\right]\right).$$

Combine this estimate with the major-arc calculation to deduce

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi \sqrt{\frac{2n}{3}}\right).$$

**Exercise 11.21** (Ising state counts). For the diagonal Ising system, define nonnegative integers  $d_a(N)$  by

$$\chi_a(\tau) = q^{h_a - c/24} \sum_{N \geq 0} d_a(N) q^N$$

and define the resolved two-sided degeneracy

$$\rho_{\text{Ising}}(N, \bar{N}) = \sum_{a \in \{0, 1/2, 1/16\}} d_a(N) d_a(\bar{N}).$$

Use the explicit theta-product formulas and the  $S$ -transformation to control each character in a complex neighborhood of the positive real saddle. Prove the uniform minor-arc estimate that, for each Ising character and some  $\delta > 0$ , the complementary coefficient integral is

$$O\left(\exp\left[2\pi\sqrt{\frac{cN}{6}} - \delta\sqrt{N}\right]\right).$$

Combine it with the saddle contribution and prove, as  $N, \bar{N} \rightarrow \infty$  with  $N/\bar{N}$  in a fixed compact subset of  $(0, \infty)$ ,

$$\log \rho_{\text{Ising}}(N, \bar{N}) \sim 2\pi\sqrt{\frac{cN}{6}} + 2\pi\sqrt{\frac{c\bar{N}}{6}}, \quad c = \frac{1}{2}.$$

Derive the resulting asymptotic number of states below a specified high energy on a circle and the corresponding entropy. State the circumference, propagation speed, energy window, vacuum assumption, sector resolution, saddle-point regime, and finite-size limitations required when comparing these predictions with measured or transfer-matrix spectra.

**Exercise 11.22** (Lattice Fock characters). For an even positive-definite lattice  $L$  of rank  $d$ , let  $\varepsilon : L \times L \rightarrow \{\pm 1\}$  satisfy

$$\varepsilon(\lambda, 0) = \varepsilon(0, \lambda) = 1, \quad \varepsilon(\lambda, \mu)\varepsilon(\lambda + \mu, \nu) = \varepsilon(\mu, \nu)\varepsilon(\lambda, \mu + \nu),$$

$$\frac{\varepsilon(\lambda, \mu)}{\varepsilon(\mu, \lambda)} = (-1)^{\langle \lambda, \mu \rangle}.$$

Let  $\mathbb{C}_\varepsilon[L]$  have symbols  $e^\lambda$ ,  $\lambda \in L$ , with  $e^\lambda e^\mu = \varepsilon(\lambda, \mu)e^{\lambda+\mu}$ , and define

$$\mathcal{F}_L = \text{Sym}\left(\bigoplus_{n \geq 1} (L \otimes_{\mathbb{Z}} \mathbb{C})t^{-n}\right) \otimes \mathbb{C}_\varepsilon[L],$$

where  $\text{Sym}(W) = \bigoplus_{r \geq 0} \text{Sym}^r(W)$ . Give  $(L \otimes_{\mathbb{Z}} \mathbb{C})t^{-n}$  degree  $n$ , give the element  $e^\lambda$  of the twisted group algebra degree  $\|\lambda\|^2/2$ , and let  $L_0$  act by total degree.

The standard lattice vertex-operator construction equips  $\mathcal{F}_L$  with a central-charge- $d$  vertex operator algebra structure.

(a) Use the symmetric-power trace formula and the definition of the lattice theta series to prove

$$\text{Tr}_{\mathcal{F}_L} q^{L_0 - d/24} = \frac{\Theta_L(\tau)}{\eta(\tau)^d}.$$

(b) For the Leech lattice, use its theta series to derive

$$\text{Tr}_{\mathcal{F}_\Lambda} q^{L_0 - 1} = \frac{\Theta_\Lambda}{\eta^{24}} = J + 24.$$

(c) Show directly from the graded construction that the coefficient 24 counts the degree-one oscillator states and that

$$196884 = 196560 + 324, \quad 324 = 24 + \dim \text{Sym}^2(\Lambda \otimes_{\mathbb{Z}} \mathbb{C}),$$

splits the degree-two states into nonzero lattice momenta and oscillator states.

**Exercise 11.23** (McKay–Thompson series and monstrous moonshine). Let  $V = \bigoplus_{m \geq 0} V_m$  be a positive-energy vertex operator algebra with finite-dimensional graded pieces and central charge  $c$ , and let  $L_0|_{V_m} = m \text{ id}$ . Let  $G$  act by grading-preserving automorphisms of  $V$ . For  $g \in G$ , define its *McKay–Thompson series* by

$$T_g(\tau) = \text{Tr}_V \left( g q^{L_0 - c/24} \right).$$

Let  $\mathbb{M}$  denote the Monster sporadic simple group.

Carry out the Frenkel–Lepowsky–Meurman construction and prove Borcherds’s monstrous moonshine theorem [13]. If  $\theta$  lifts  $\lambda \mapsto -\lambda$  to the Leech-lattice vertex operator algebra and  $V_\Lambda^{T,+}$  is the  $+1$  eigenspace in its  $\theta$ -twisted module, the orbifold

$$V^\natural = V_\Lambda^+ \oplus V_\Lambda^{T,+}$$

is a central-charge-24 vertex operator algebra with

$$\text{Aut}(V^\natural) = \mathbb{M}, \quad T_1(\tau) = J(\tau),$$

and, for every  $g \in \mathbb{M}$ , the series  $T_g$  is the normalized Hauptmodul for the Conway–Norton genus-zero group  $\Gamma_g$ . Deduce

$$\dim V_0^\natural = 1, \quad \dim V_1^\natural = 0, \quad \dim V_2^\natural = 196884, \quad \dim V_3^\natural = 21493760.$$

Construct the indicated irreducible Monster representations  $\mathbf{d}$ , compute the required character inner products from the conjugacy classes, and verify the first decompositions

$$V_2^\natural \cong \mathbf{1} \oplus \mathbf{196883}, \quad V_3^\natural \cong \mathbf{1} \oplus \mathbf{196883} \oplus \mathbf{21296876}.$$

With  $p = e^{2\pi i \sigma}$ , use Borcherds’s denominator identity

$$p^{-1} \prod_{m>0, n \in \mathbb{Z}} (1 - p^m q^n)^{c(mn)} = J(\sigma) - J(\tau)$$

as a formal product in the region  $|p| < |q| < 1$ . Compare coefficients through order  $p$  and derive the first replication identity

$$\frac{1}{2} \left( J(\tau)^2 - \sum_{n \in \mathbb{Z}} c(n) q^{2n} \right) - \sum_{n \in \mathbb{Z}} c(2n) q^n = 196884.$$

**Exercise 11.24** (Monster-resolved finite-size spectrum). Let  $\chi_\rho$  be the character of an irreducible Monster representation  $\rho$ . Use finite-group character orthogonality to prove that its multiplicity in  $V_m^\natural$  is

$$\mu_\rho(m) = \frac{1}{|\mathbb{M}|} \sum_{g \in \mathbb{M}} \overline{\chi_\rho(g)} [q^{m-1}] T_g(\tau).$$

Deduce that the symmetry-resolved series

$$Z_\rho(\tau) = \frac{\dim(\rho)}{|\mathbb{M}|} \sum_{g \in \mathbb{M}} \overline{\chi_\rho(g)} T_g(\tau)$$

has coefficient  $\dim(\rho) \mu_\rho(m)$  at  $q^{m-1}$ . Model an exactly Monster-symmetric chiral edge of circumference  $\ell$  and propagation speed  $u$  by

$$H = \frac{2\pi \hbar u}{\ell} (L_0 - 1).$$

Derive the excitation energies

$$E_m - E_0 = \frac{2\pi\hbar u}{\ell} m$$

and predict a missing level at  $2\pi\hbar u/\ell$ , followed by levels of degeneracy 196884 and 21493760 at  $4\pi\hbar u/\ell$  and  $6\pi\hbar u/\ell$ . Express the symmetry-resolved degeneracy at level  $m$  as  $\dim(\rho)\mu_\rho(m)$ . State the assumptions of exact Monster symmetry, linear chiral dispersion, periodic boundary conditions, access to the twined sectors, and the low-energy finite-size regime. Explain why a purely chiral central-charge-24 theory must be realized as an anomalous edge or accompanied by a compensating opposite-chirality sector, and distinguish the derived spectral prediction from evidence for a particular physical realization.

**Exercise 11.25** (Affine  $SU(2)$  characters and WZW spectra). Fix an integer  $k \geq 1$ . For integers  $K \geq 2$  and  $m \in \mathbb{Z}$ , put

$$\Theta_{m,K}(\tau, z) = \sum_{n \in \mathbb{Z}} q^{(2Kn+m)^2/(4K)} e^{2\pi i(2Kn+m)z}, \quad q = e^{2\pi i\tau},$$

so that  $\Theta_{m+2K,K} = \Theta_{m,K}$ . For  $0 \leq j \leq k$ , define

$$\chi_j^{(k)}(\tau) = \lim_{z \rightarrow 0} \frac{\Theta_{j+1,k+2}(\tau, z) - \Theta_{-j-1,k+2}(\tau, z)}{\Theta_{1,2}(\tau, z) - \Theta_{-1,2}(\tau, z)}.$$

Prove the Weyl–Kac character formula and the Kac–Peterson modular transformation [40]: these are the specialized integrable affine  $\widehat{\mathfrak{sl}}_2$  characters at level  $k$ , with

$$c = \frac{3k}{k+2}, \quad h_j = \frac{j(j+2)}{4(k+2)}, \quad \chi_j^{(k)}(\tau) = q^{h_j - c/24} (j+1 + O(q)),$$

and

$$\chi_j^{(k)}(-1/\tau) = \sum_{\ell=0}^k S_{j\ell} \chi_\ell^{(k)}(\tau), \quad S_{j\ell} = \sqrt{\frac{2}{k+2}} \sin\left(\frac{(j+1)(\ell+1)\pi}{k+2}\right).$$

Prove the sine orthogonality identity  $S^*S = I$ . For  $a \in \mathbb{C}^\times$  with  $a \neq \pm 1$  and  $n \geq 0$ , define the *quantum integer*

$$[n]_a = \frac{a^n - a^{-n}}{a - a^{-1}}.$$

Put  $\zeta = e^{\pi i/(k+2)}$  and prove

$$\frac{S_{j0}}{S_{00}} = [j+1]_\zeta, \quad [2]_\zeta [j+1]_\zeta = [j]_\zeta + [j+2]_\zeta, \quad [k+2]_\zeta = 0.$$

For the diagonal  $SU(2)_k$  Wess–Zumino–Witten model on a circle of circumference  $L$  and propagation speed  $u$ , take

$$Z_k(\tau, \bar{\tau}) = \sum_{j=0}^k |\chi_j^{(k)}(\tau)|^2, \quad H = \frac{2\pi\hbar u}{L} \left( L_0 + \bar{L}_0 - \frac{c}{12} \right).$$

Derive the primary excitation gaps and their ground-grade multiplicities,

$$\Delta E_j = \frac{\pi\hbar u}{L} \frac{j(j+2)}{k+2}, \quad g_j = (j+1)^2, \quad 0 \leq j \leq k.$$

Predict these sector-resolved gaps and multiplicities for finite-size spectroscopy. Show that the modular  $S$ -transform exchanges the thermal and spatial cycles of the torus partition function. State the assumptions of exact affine  $SU(2)_k$  symmetry, the diagonal modular invariant, linear left- and right-moving dispersion, periodic boundary conditions, sector-resolved preparation, and the low-energy finite-size regime under which spectroscopy tests the theta-quotient characters, the displayed gaps, and the root-of-unity quantum integers.

## 12 Quantum Groups

### 12.1 Hopf Algebras

**Exercise 12.1** (Hopf algebras and  $U_v(\mathfrak{sl}_2)$ ). A *bialgebra* is a unital associative algebra  $H$  with unital algebra homomorphisms

$$\Delta : H \rightarrow H \otimes H, \quad \varepsilon : H \rightarrow \mathbb{C}$$

satisfying coassociativity and the counit identities. A *Hopf algebra* is a bialgebra with a linear antipode  $S : H \rightarrow H$  satisfying

$$m(S \otimes \text{id})\Delta = \eta\varepsilon = m(\text{id} \otimes S)\Delta.$$

Fix a quantum-group parameter  $0 < v < 1$ . For  $n \geq 1$ , prove

$$[2]_v[n]_v = [n-1]_v + [n+1]_v, \quad \lim_{v \rightarrow 1} [n]_v = n,$$

and identify the root-of-unity values in the affine  $SU(2)_k$  character system as the specialization  $v = \zeta$  of the same Laurent polynomials.

For  $0 < v < 1$ ,  $U_v(\mathfrak{sl}_2)$  is generated by  $E, F, K, K^{-1}$  with

$$KEK^{-1} = v^2E, \quad KFK^{-1} = v^{-2}F, \quad [E, F] = \frac{K - K^{-1}}{v - v^{-1}},$$

and

$$\Delta(K) = K \otimes K, \quad \Delta(E) = E \otimes 1 + K \otimes E, \quad \Delta(F) = F \otimes K^{-1} + 1 \otimes F.$$

Its counit, antipode, and compact real-form involution are

$$\varepsilon(E) = \varepsilon(F) = 0, \quad \varepsilon(K) = 1, \quad S(E) = -K^{-1}E, \quad S(F) = -FK, \quad S(K) = K^{-1},$$

$$K^* = K, \quad E^* = KF, \quad F^* = EK^{-1}.$$

Verify that the coproduct preserves the defining relations and that the displayed counit, antipode, and involution make  $U_v(\mathfrak{sl}_2)$  a Hopf  $*$ -algebra. In the  $\hbar$ -adic completion put

$$v = e^{-\hbar}, \quad K = v^H = e^{-\hbar H}.$$

Prove that

$$[H, E] = 2E, \quad [H, F] = -2F, \quad \frac{K - K^{-1}}{v - v^{-1}} = H + O(\hbar^2),$$

and that the relations and coproduct tend to those of  $U(\mathfrak{sl}_2)$  as  $\hbar \rightarrow 0$ . Define

$$\delta(x) = \lim_{\hbar \rightarrow 0} \frac{\Delta(x) - \Delta^{\text{op}}(x)}{\hbar}$$

on the generators of  $\mathfrak{sl}_2$  and prove

$$\delta(H) = 0, \quad \delta(E) = E \otimes H - H \otimes E, \quad \delta(F) = F \otimes H - H \otimes F.$$

Verify directly that  $\delta$  is a 1-cocycle for the adjoint action and that its transpose satisfies the Jacobi identity.



**Exercise 12.2** (Poisson–Lie groups and classical Yang–Baxter equations). A *Poisson–Lie group* is a Lie group  $G$  with a Poisson structure for which multiplication  $G \times G \rightarrow G$  is Poisson. A *Lie bialgebra* is a Lie algebra  $\mathfrak{g}$  with a linear map

$$\delta : \mathfrak{g} \rightarrow \mathfrak{g} \wedge \mathfrak{g}$$

that is a 1-cocycle for the adjoint action and whose transpose defines a Lie bracket on  $\mathfrak{g}^*$ . It is *coboundary* when

$$\delta(x) = [x \otimes 1 + 1 \otimes x, r]$$

for some  $r \in \mathfrak{g} \otimes \mathfrak{g}$ .

Define

$$\text{CYB}(r) = [r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}].$$

Differentiate a Poisson–Lie structure at the identity and prove that it defines a Lie bialgebra. For  $\mathfrak{sl}_2$ , put

$$r = E \otimes F - F \otimes E$$

and prove that its coboundary is the semiclassical cobracket of  $U_v(\mathfrak{sl}_2)$  displayed above. Prove that the co-Jacobi identity is equivalent to  $\text{CYB}(r)$  being invariant under the diagonal adjoint action. Under the additional triangular hypothesis  $\text{CYB}(r) = 0$ , recover the classical Yang–Baxter equation; state the modified classical Yang–Baxter equation as  $\text{CYB}(r) = \Omega$  for an invariant tensor  $\Omega$  and distinguish the case  $\Omega \neq 0$ .

**Exercise 12.3** (Gaussian binomial coefficients). For an indeterminate  $u$ ,  $n \geq 0$ , and  $0 \leq r \leq n$ , define

$$\begin{bmatrix} n \\ r \end{bmatrix}_u = \prod_{j=1}^r \frac{1 - u^{n-r+j}}{1 - u^j},$$

with value 1 for  $r = 0$  and value 0 outside this range.

Derive

$$\begin{bmatrix} n \\ r \end{bmatrix}_u = \begin{bmatrix} n-1 \\ r \end{bmatrix}_u + u^{n-r} \begin{bmatrix} n-1 \\ r-1 \end{bmatrix}_u$$

and prove

$$\prod_{j=0}^{n-1} (1 + tu^j) = \sum_{r=0}^n u^{r(r-1)/2} \begin{bmatrix} n \\ r \end{bmatrix}_u t^r.$$

For  $YX = uXY$ , prove the noncommutative identity

$$(X + Y)^n = \sum_{r=0}^n \begin{bmatrix} n \\ r \end{bmatrix}_u X^r Y^{n-r}.$$

Set  $u = v^2$  and apply it to  $X = E \otimes 1$  and  $Y = K \otimes E$  in  $U_v(\mathfrak{sl}_2)$  to obtain an explicit formula for  $\Delta(E^n)$ . Show that the limit  $v \rightarrow 1$  recovers the ordinary binomial theorem and the undeformed coproduct.

**Exercise 12.4** (Gaussian-binomial state counts). Apply the Gaussian binomial theorem. Prove that the coefficient of  $u^m$  in  $\begin{bmatrix} n \\ r \end{bmatrix}_u$  counts partitions of  $m$  whose diagrams fit inside an  $r \times (n-r)$  rectangle. For  $n$  fermionic levels with energies  $j\hbar\omega$ ,  $0 \leq j < n$ , prove that the coefficient of  $t^r u^{m+r(r-1)/2}$  in

$$\prod_{j=0}^{n-1} (1 + tu^j)$$

is the exact number of  $r$ -particle states at excitation energy  $m\hbar\omega$  above the filled lowest configuration, and identify it with the coefficient of  $u^m$  in  $[n]_u$ .

**Exercise 12.5** (Compact quantum groups and  $SU_v(2)$ ). A *compact quantum group*  $\mathbb{G}$  is a unital  $C^*$ -algebra  $C(\mathbb{G})$  with a unital  $*$ -homomorphism

$$\Delta : C(\mathbb{G}) \longrightarrow C(\mathbb{G}) \otimes_{\min} C(\mathbb{G})$$

where  $\otimes_{\min}$  is the closure of the algebraic tensor product in  $B(H \otimes K)$  for faithful representations on  $H$  and  $K$ . The coproduct satisfies

$$(\Delta \otimes \text{id})\Delta = (\text{id} \otimes \Delta)\Delta$$

and

$$\overline{\Delta(C(\mathbb{G}))(1 \otimes C(\mathbb{G}))} = C(\mathbb{G}) \otimes_{\min} C(\mathbb{G}) = \overline{\Delta(C(\mathbb{G}))(C(\mathbb{G}) \otimes 1)}.$$

Here a state on a unital  $C^*$ -algebra is a linear functional  $h$  with  $h(a^*a) \geq 0$  and  $h(1) = 1$ . A *Haar state* is a state  $h$  satisfying

$$(h \otimes \text{id})\Delta(a) = h(a)1 = (\text{id} \otimes h)\Delta(a).$$

Let  $\text{Pol}(SU_v(2))$  be the universal unital  $*$ -algebra generated by  $\alpha, \gamma$  so that

$$u = \begin{pmatrix} \alpha & -v\gamma^* \\ \gamma & \alpha^* \end{pmatrix}$$

is unitary. Equivalently, its relations are

$$\begin{aligned} \alpha^*\alpha + \gamma^*\gamma &= 1, & \alpha\alpha^* + v^2\gamma\gamma^* &= 1, & \gamma^*\gamma &= \gamma\gamma^*, \\ \alpha\gamma &= v\gamma\alpha, & \alpha\gamma^* &= v\gamma^*\alpha, \end{aligned}$$

Define  $\Delta(u_{ij}) = \sum_k u_{ik} \otimes u_{kj}$ , and let  $C(SU_v(2))$  be the universal  $C^*$ -completion.

Prove that  $\Delta$  is a coassociative  $*$ -homomorphism satisfying the cancellation conditions. On  $\ell^2(\mathbb{Z}_{\geq 0})$ , define for  $\xi \in \mathbb{T}$

$$\pi_\xi(\gamma)e_n = \xi v^n e_n, \quad \pi_\xi(\alpha)e_n = (1 - v^{2n})^{1/2} e_{n-1}, \quad e_{-1} = 0.$$

Verify the relations and prove that

$$h(a) = (1 - v^2) \sum_{n \geq 0} v^{2n} \int_{\mathbb{T}} \langle e_n, \pi_\xi(a)e_n \rangle dm_{\mathbb{T}}(\xi)$$

is the Haar state.

## 12.2 Quantum Peter–Weyl Theory

**Exercise 12.6** (Quantum-group coefficient pairings). Call a finite-dimensional  $U_v(\mathfrak{sl}_2)$ -module  $V$  *type-1* when  $K$  is diagonalizable and

$$V = \bigoplus_{m \in \mathbb{Z}} V[m], \quad K|_{V[m]} = v^m \text{id}_{V[m]}.$$

On the distinguished Hermitian space  $V_1 = \mathbb{C}e_+ \oplus \mathbb{C}e_-$  define the fundamental type-1 module by

$$Ke_{\pm} = v^{\pm 1}e_{\pm}, \quad Ee_- = v^{-1/2}e_+, \quad Ee_+ = 0, \quad Fe_+ = v^{1/2}e_-, \quad Fe_- = 0.$$

For a finite-dimensional type-1 module  $V$ ,  $\xi \in V$ , and  $\varphi \in V^\vee$ , define

$$c_{\varphi, \xi}(x) = \varphi(x\xi) \quad (x \in U_v(\mathfrak{sl}_2)).$$

Let  $\mathcal{O}_v$  be the Hopf algebra generated by these coefficient functionals. Its pairing with  $U_v(\mathfrak{sl}_2)$  is

$$\langle x, c_{\varphi, \xi} \rangle = c_{\varphi, \xi}(x),$$

with multiplication and coproduct characterized by

$$\langle x, ab \rangle = \langle \Delta x, a \otimes b \rangle, \quad \langle xy, a \rangle = \langle x \otimes y, \Delta a \rangle.$$

Prove that the coefficient map of a tensor product is the product of the coefficient maps, that the coefficient map of a dual module is obtained from the antipode, and that the displayed pairing is a nondegenerate Hopf pairing. Apply it to the fundamental module and prove that sending its coefficient corepresentation to the displayed matrix  $u$  extends to a Hopf  $*$ -isomorphism

$$\mathcal{O}_v \longrightarrow \text{Pol}(SU_v(2)).$$

**Exercise 12.7** ( $SU_v(2)$  Peter–Weyl decomposition). For a finite-dimensional unitary corepresentation  $U$ , a coefficient is  $u_\omega = (\omega \otimes \text{id})(U)$  for  $\omega \in B(H_U)^*$ . The algebra  $\text{Pol}(SU_v(2))$  is the unital  $*$ -subalgebra generated by all such coefficients.

Construct the irreducible type-1 unitary  $U_v(\mathfrak{sl}_2)$ -modules  $V_n$  from their highest-weight lines and form their coefficient corepresentations of  $\text{Pol}(SU_v(2))$  using the Hopf pairing. Prove the compact-quantum-group Peter–Weyl theorem [71]: irreducible coefficient spaces are orthogonal for the Haar state and their algebraic direct sum is  $\text{Pol}(SU_v(2))$ . Use the explicit Haar state to compute the orthogonality constants and deduce

$$\text{Pol}(SU_v(2)) = \bigoplus_{n \geq 0} C(V_n)$$

algebraically and as a Hilbert direct sum in the Haar GNS space.

**Exercise 12.8** ( $C^*$ -tensor categories and corepresentations). A finite-dimensional unitary corepresentation of a compact quantum group  $\mathbb{G}$  on  $H_U$  is a unitary  $U \in B(H_U) \otimes C(\mathbb{G})$  with  $(\text{id} \otimes \Delta)(U) = U_{12}U_{13}$ .

A  $\mathbb{C}$ -linear category  $\mathcal{C}$  consists of objects, complex vector spaces  $\text{Hom}_{\mathcal{C}}(X, Y)$ , bilinear associative compositions

$$\text{Hom}_{\mathcal{C}}(Y, Z) \otimes \text{Hom}_{\mathcal{C}}(X, Y) \longrightarrow \text{Hom}_{\mathcal{C}}(X, Z), \quad (g, f) \longmapsto gf,$$

and identity morphisms  $\text{id}_X$ . A  $\mathbb{C}$ -linear tensor category also has a bilinear tensor product on objects and morphisms, a tensor unit  $\mathbf{1}$ , and isomorphisms

$$a_{X,Y,Z} : (X \otimes Y) \otimes Z \longrightarrow X \otimes (Y \otimes Z), \quad l_X : \mathbf{1} \otimes X \longrightarrow X, \quad r_X : X \otimes \mathbf{1} \longrightarrow X.$$

The tensor product satisfies

$$(f'f) \otimes (g'g) = (f' \otimes g')(f \otimes g), \quad \text{id}_X \otimes \text{id}_Y = \text{id}_{X \otimes Y},$$

and the structure maps commute with morphisms; in particular,

$$a_{X',Y',Z'}((f \otimes g) \otimes h) = (f \otimes (g \otimes h))a_{X,Y,Z}.$$

They satisfy the pentagon and triangle identities

$$\begin{aligned} a_{W,X,Y \otimes Z} a_{W \otimes X,Y,Z} &= (\text{id}_W \otimes a_{X,Y,Z}) a_{W,X \otimes Y,Z} (a_{W,X,Y} \otimes \text{id}_Z), \\ (\text{id}_X \otimes l_Y) a_{X,1,Y} &= r_X \otimes \text{id}_Y. \end{aligned}$$

A  $C^*$ -category has Banach morphism spaces and conjugate-linear maps  $f \mapsto f^*$  with

$$\begin{aligned} f^* : Y &\longrightarrow X, & (f^*)^* &= f, & (gf)^* &= f^* g^*, \\ \|gf\| &\leq \|g\| \|f\|, & \|f^* f\| &= \|f\|^2. \end{aligned}$$

A  $C^*$ -tensor category is a  $\mathbb{C}$ -linear tensor category with this structure, with

$$(f \otimes g)^* = f^* \otimes g^*, \quad a^* = a^{-1}, \quad l^* = l^{-1}, \quad r^* = r^{-1}.$$

Associators and unit maps are suppressed in subsequent tensor composites.

For a compact quantum group  $\mathbb{G}$ , let  $\text{Rep}(\mathbb{G})$  have finite-dimensional unitary corepresentations as objects and

$$\text{Hom}_{\mathbb{G}}(U, V) = \{A : H_U \rightarrow H_V : (A \otimes 1)U = V(A \otimes 1)\}$$

as morphism spaces. Composition and adjoints are those of Hilbert-space operators. Define

$$U \boxtimes V = U_{13} V_{23} \quad \text{on } H_U \otimes H_V,$$

and take the trivial corepresentation on  $\mathbb{C}$  as tensor unit, with the Hilbert-space associator and unit maps.

- (a) Prove that the adjoint of an intertwiner is an intertwiner, that  $U \boxtimes V$  is a unitary corepresentation, and that tensor products of intertwiners are intertwiners.
- (b) Verify the pentagon, triangle, and  $C^*$ -identities and deduce that  $\text{Rep}(\mathbb{G})$  is a  $C^*$ -tensor category.
- (c) For  $C(\mathbb{G}) = \mathbb{C}$ , identify it with  $\text{Hilb}_{\text{fd}}$ , the category of finite-dimensional Hilbert spaces and all linear maps.

**Exercise 12.9** (Duals and conjugates). Let  $\mathcal{C}$  be a  $C^*$ -tensor category. A *left dual* of  $V$  is an object  $V^\vee$  with morphisms

$$\text{ev}_V : V^\vee \otimes V \longrightarrow \mathbf{1}, \quad \text{coev}_V : \mathbf{1} \longrightarrow V \otimes V^\vee$$

satisfying

$$\begin{aligned} (\text{id}_V \otimes \text{ev}_V)(\text{coev}_V \otimes \text{id}_V) &= \text{id}_V, \\ (\text{ev}_V \otimes \text{id}_{V^\vee})(\text{id}_{V^\vee} \otimes \text{coev}_V) &= \text{id}_{V^\vee}. \end{aligned}$$

A *right dual*  ${}^\vee V$  has maps

$$\widetilde{\text{ev}}_V : V \otimes {}^\vee V \longrightarrow \mathbf{1}, \quad \widetilde{\text{coev}}_V : \mathbf{1} \longrightarrow {}^\vee V \otimes V$$

satisfying

$$(\widetilde{\text{ev}}_V \otimes \text{id}_V)(\text{id}_V \otimes \widetilde{\text{coev}}_V) = \text{id}_V,$$

$$(\mathrm{id}_V \otimes \widetilde{\mathrm{ev}}_V)(\widetilde{\mathrm{coev}}_V \otimes \mathrm{id}_V) = \mathrm{id}_V.$$

In a  $C^*$ -tensor category, a *conjugate* of an object  $V$  is an object  $\bar{V}$  with morphisms

$$R_V : \mathbf{1} \longrightarrow \bar{V} \otimes V, \quad \bar{R}_V : \mathbf{1} \longrightarrow V \otimes \bar{V}$$

satisfying

$$(\bar{R}_V^* \otimes \mathrm{id}_V)(\mathrm{id}_V \otimes R_V) = \mathrm{id}_V, \quad (R_V^* \otimes \mathrm{id}_{\bar{V}})(\mathrm{id}_{\bar{V}} \otimes \bar{R}_V) = \mathrm{id}_{\bar{V}}.$$

Call a  $C^*$ -tensor category *rigid* when every object has a conjugate. Its dual maps are

$$\mathrm{ev}_V = R_V^*, \quad \mathrm{coev}_V = \bar{R}_V, \quad \widetilde{\mathrm{ev}}_V = \bar{R}_V^*, \quad \widetilde{\mathrm{coev}}_V = R_V.$$

Prove directly from the conjugate equations that these evaluation and coevaluation maps satisfy the left and right snake identities.

**Exercise 12.10** (Standard categorical traces). Let  $\mathcal{C}$  be a rigid  $C^*$ -tensor category with simple tensor unit, so  $\mathrm{End}(\mathbf{1}) = \mathbb{C}$ . A solution of the conjugate equations is *standard* when

$$R_V^*(\mathrm{id}_{\bar{V}} \otimes T)R_V = \bar{R}_V^*(T \otimes \mathrm{id}_{\bar{V}})\bar{R}_V \quad (T \in \mathrm{End}(V)).$$

For a standard solution, define

$$\mathrm{tr}_{\mathcal{C}}(T) = R_V^*(\mathrm{id}_{\bar{V}} \otimes T)R_V, \quad \dim_{\mathcal{C}}(V) = \mathrm{tr}_{\mathcal{C}}(\mathrm{id}_V).$$

For  $\mathcal{C} = \mathrm{Rep}(SU_v(2))$ , write  $\mathrm{tr}_v$  and  $\dim_v$ .

- (a) Let two standard solutions of the conjugate equations be given for  $V$ . Use the snake identities and standardness to construct the unitary isomorphism between their conjugate objects and prove that they define the same trace.
- (b) For  $S : V \rightarrow W$  and  $T : W \rightarrow V$ , prove

$$\mathrm{tr}_{\mathcal{C}}(TS) = \mathrm{tr}_{\mathcal{C}}(ST), \quad \mathrm{tr}_{\mathcal{C}}(T^*T) \geq 0.$$

**Exercise 12.11** (Categorical dimensions). Let  $\mathcal{C}$  be a  $C^*$ -tensor category with simple tensor unit and standard conjugate solutions. An orthogonal direct sum  $X = \bigoplus_{i=1}^r X_i$  is specified by morphisms  $u_i : X_i \rightarrow X$  satisfying

$$u_i^* u_j = \delta_{ij} \mathrm{id}_{X_i}, \quad \sum_{i=1}^r u_i u_i^* = \mathrm{id}_X.$$

- (a) Construct standard solutions for tensor products, conjugates, and existing orthogonal direct sums and prove

$$\dim_{\mathcal{C}}(V \otimes W) = \dim_{\mathcal{C}}(V) \dim_{\mathcal{C}}(W), \quad \dim_{\mathcal{C}}(\bar{V}) = \dim_{\mathcal{C}}(V),$$

$$\dim_{\mathcal{C}}(V \oplus W) = \dim_{\mathcal{C}}(V) + \dim_{\mathcal{C}}(W).$$

- (b) For  $\mathcal{C} = \mathrm{Hilb}_{\mathrm{fd}}$  and a finite-dimensional Hilbert space  $V$ , let  $j_V : \bar{V} \rightarrow V^\vee$  be the Riesz isomorphism  $j_V(\bar{v}) = \langle v, - \rangle$ . From the coevaluation and symmetry define

$$\bar{R}_V = (\mathrm{id}_V \otimes j_V^{-1}) \mathrm{coev}_V : \mathbb{C} \longrightarrow V \otimes \bar{V}, \quad R_V = s_{V, \bar{V}} \bar{R}_V : \mathbb{C} \longrightarrow \bar{V} \otimes V.$$

Prove the conjugate equations and standardness. Prove without choosing a basis that

$$\mathrm{tr}_{\mathrm{Hilb}_{\mathrm{fd}}}(T) = \mathrm{tr}(T), \quad \dim_{\mathrm{Hilb}_{\mathrm{fd}}}(V) = \dim_{\mathbb{C}}(V).$$

**Exercise 12.12** ( $SU_v(2)$  fusion and quantum dimensions). Let  $V_n$  be the irreducible type-1  $U_v(\mathfrak{sl}_2)$ -module of highest weight  $n$ . Construct its conjugate equations and prove

$$V_m \otimes V_n \cong \bigoplus_{k=0}^{\min(m,n)} V_{m+n-2k}.$$

Normalize the conjugate solutions by the compact involution and compute

$$\dim_v(V_n) = [n+1]_v.$$

Prove that finite orthogonal direct sums of the  $V_n$  are closed under tensor products and conjugates and form a rigid  $C^*$ -tensor category. Prove

$$\begin{aligned} \dim_{\mathbb{C}}(V_1) &= 2, & \dim_v(V_1) &= v + v^{-1}, \\ \dim_{\mathbb{C}}(V_n) &= n+1, & \lim_{v \rightarrow 1} \dim_v(V_n) &= n+1, \\ \dim_v(V_n) &> n+1 & (0 < v < 1, n \geq 1). \end{aligned}$$

Use  $V_1 \otimes V_n \cong V_{n+1} \oplus V_{n-1}$  to derive

$$d_{n+1} = (v + v^{-1})d_n - d_{n-1}, \quad d_0 = 1, \quad d_1 = v + v^{-1},$$

where  $d_n = \dim_v(V_n)$ . In terms of the Chebyshev polynomial  $U_n$ , prove

$$\dim_v(V_n) = U_n\left(\frac{v + v^{-1}}{2}\right) = \frac{v^{n+1} - v^{-(n+1)}}{v - v^{-1}}.$$

**Exercise 12.13** (Tannaka–Krein reconstruction of compact quantum symmetry). Let  $\mathcal{C}$  be a rigid  $C^*$ -tensor category with a chosen small skeleton and let  $F : \mathcal{C} \rightarrow \text{Hilb}_{\text{fd}}$  be a faithful unitary tensor functor. Form the coend

$$\text{Pol}(\mathcal{C}, F) = \left( \bigoplus_{U \in \mathcal{C}} F(U)^\vee \otimes F(U) \right) / \sim,$$

where

$$(\varphi \circ F(T)) \otimes v \sim \varphi \otimes F(T)v \quad (T : U \rightarrow V).$$

Let  $q_U : F(U)^\vee \otimes F(U) \rightarrow \text{Pol}(\mathcal{C}, F)$  be the resulting coefficient map and write  $[\varphi \otimes v]_U = q_U(\varphi \otimes v)$ . Use the tensor structure of  $F$  to define multiplication by

$$[\varphi \otimes v]_U [\psi \otimes w]_V = [(\varphi \otimes \psi) \otimes (v \otimes w)]_{U \otimes V}.$$

Define the counit by evaluation and define the coproduct intrinsically by inserting the coevaluation of  $F(U)$  between  $\varphi$  and  $v$ :

$$\Delta q_U = (q_U \otimes q_U)(\text{id}_{F(U)^\vee} \otimes \text{coev}_{F(U)} \otimes \text{id}_{F(U)}).$$

Prove that these maps are well defined and give a bialgebra. Use the chosen conjugates and the Hilbert-space adjoint to construct the antipode and involution and verify the Hopf  $*$ -identities.

Prove the compact Tannaka–Krein reconstruction theorem [71]: the universal  $C^*$ -completion determines a compact quantum group  $\mathbb{G}_F$ , and  $F$  induces a unitary tensor equivalence

$$\mathcal{C} \simeq \text{Rep}(\mathbb{G}_F).$$

Apply the construction to the forgetful functor on  $\text{Rep}(SU_v(2))$ , identify its coefficient Hopf  $*$ -algebra with  $\text{Pol}(SU_v(2))$ , and prove that a unitary monoidal natural isomorphism between two fiber functors induces an isomorphism of the reconstructed compact quantum groups.

### 12.3 Basic Hypergeometric Functions

**Exercise 12.14** (Basic hypergeometric series and summations). For  $m \in \mathbb{N}$ , define

$$(a; Q)_m = \prod_{j=0}^{m-1} (1 - aQ^j), \quad (a; Q)_\infty = \prod_{j=0}^{\infty} (1 - aQ^j),$$

$$(a_1, \dots, a_r; Q)_m = \prod_{i=1}^r (a_i; Q)_m, \quad (a_1, \dots, a_r; Q)_\infty = \prod_{i=1}^r (a_i; Q)_\infty.$$

The *basic hypergeometric series* is

$${}_r\phi_s \left( \begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; Q, z \right) = \sum_{m=0}^{\infty} \frac{(a_1, \dots, a_r; Q)_m}{(b_1, \dots, b_s; Q)_m} \left( (-1)^m Q^{m(m-1)/2} \right)^{1+s-r} z^m.$$

It is terminating when one numerator parameter is  $Q^{-n}$ .

Prove the  $Q$ -Gauss identity

$${}_2\phi_1 \left( \begin{matrix} a, b \\ c \end{matrix}; Q, \frac{c}{ab} \right) = \frac{(c/a, c/b; Q)_\infty}{(c, c/(ab); Q)_\infty}, \quad \left| \frac{c}{ab} \right| < 1,$$

and the terminating  $Q$ -Pfaff–Saalschütz identity

$${}_3\phi_2 \left( \begin{matrix} a, b, Q^{-n} \\ c, abQ^{1-n}/c \end{matrix}; Q, Q \right) = \frac{(c/a, c/b; Q)_n}{(c, c/(ab); Q)_n}$$

by comparing coefficients after multiplication by the denominator  $Q$ -products in the first identity, and by showing that the two sides of the second identity satisfy the same recurrence in  $n$  and have the same initial value.

**Exercise 12.15** (Quantum coupling coefficients and XXZ symmetry). Use the distinguished weight-line realization  $V_1 = \mathbb{C}e_+ \oplus \mathbb{C}e_-$  of the fundamental type-1 module and the distinguished Pauli endomorphisms  $\sigma^x, \sigma^y, \sigma^z$ . Let  $\sigma_j^a$  act as  $\sigma^a$  on the  $j$ th tensor factor and as the identity on the others. On  $V_1^{\otimes L}$  let

$$H_{\text{XXZ}} = J \sum_{j=1}^{L-1} \left( \sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y + \frac{v + v^{-1}}{2} (\sigma_j^z \sigma_{j+1}^z - I) \right. \\ \left. + \frac{v - v^{-1}}{2} (\sigma_j^z - \sigma_{j+1}^z) \right).$$

For  $J \in \mathbb{R}$  and  $0 < v < 1$ , prove that this open-chain Hamiltonian is self-adjoint and commutes with the iterated coproduct action of  $U_v(\mathfrak{sl}_2)$ .

**Exercise 12.16** (Dual  $Q$ -Hahn coupling coefficients). For  $N \in \mathbb{Z}_{\geq 0}$  and parameters for which no denominator vanishes, define the dual  $Q$ -Hahn polynomial by

$$R_n(\mu(x); \gamma, \delta, N \mid Q) = {}_3\phi_2 \left( \begin{matrix} Q^{-n}, Q^{-x}, \gamma\delta Q^{x+1} \\ \gamma Q, Q^{-N} \end{matrix}; Q, Q \right), \quad \mu(x) = Q^{-x} + \gamma\delta Q^{x+1}.$$

For irreducible type-1 modules  $V_a, V_b, V_c$ , let

$$C_{ab}^c : V_c \longrightarrow V_a \otimes V_b$$

be the intertwining isometry normalized by its positive highest-weight coefficient. Its matrix coefficients between the distinguished weight lines are the  $v$ -Clebsch–Gordan coefficients.

Decompose two coupled irreducible spin sectors by the projections  $C_{ab}^c (C_{ab}^c)^*$  and derive the orthogonality relations for their  $v$ -Clebsch–Gordan coefficients. Prove that their weight dependence satisfies the dual  $Q$ -Hahn recurrence with  $Q = v^2$  and normalized amplitudes with the displayed terminating  ${}_3\phi_2$ . Determine  $N, \gamma, \delta$  from the endpoint weights and coupling labels.

**Exercise 12.17** (XXZ transition strengths). Let the weak drive be an irreducible tensor operator  $A_\mu^{(s)}$  for the  $U_v(\mathfrak{sl}_2)$  action. Prove the  $v$ -Wigner–Eckart factorization of its matrix elements into a reduced matrix element and a  $v$ -Clebsch–Gordan coefficient. Derive the relative transition strengths at fixed reduced matrix element as squared  $v$ -Clebsch–Gordan coefficients. State the chain length, open boundary terms, resolved weights, energy scale  $J$ , and nondegeneracy assumptions under which the measured line positions and strengths determine  $v$ .

**Exercise 12.18** (Universal  $R$ -matrices and braided tensor categories). A *universal  $R$ -matrix* for a Hopf algebra  $H$  is an invertible element  $\mathcal{R}$  in a completion of  $H \otimes H$  on which the finite-dimensional tensor-product representations under consideration extend, satisfying

$$\mathcal{R}\Delta(x)\mathcal{R}^{-1} = \Delta^{\text{op}}(x), \quad (\Delta \otimes \text{id})(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{23}, \quad (\text{id} \otimes \Delta)(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{12}.$$

Prove that the  $\hbar$ -adic completion of the displayed  $U_v(\mathfrak{sl}_2)$  is quasitriangular with

$$\mathcal{R}_v = v^{H \otimes H/2} \sum_{n \geq 0} v^{n(n-1)/2} \frac{(v - v^{-1})^n}{[n]_v!} F^n \otimes E^n, \quad [n]_v! = \prod_{j=1}^n [j]_v,$$

with products interpreted in the displayed order. Verify the series on finite-dimensional type-1 modules and the three quasitriangular identities on the generators. On  $V_1 \otimes V_1 = V_2 \oplus V_0$ , prove

$$\check{\mathcal{R}}_v := \tau(\pi_{V_1} \otimes \pi_{V_1})(\mathcal{R}_v) = v^{1/2}P_2 - v^{-3/2}P_0, \quad T = v^{1/2}\check{\mathcal{R}}_v = vP_2 - v^{-1}P_0.$$

For finite-dimensional modules  $V, W$ , define

$$c_{V,W} = \tau \circ (\pi_V \otimes \pi_W)(\mathcal{R}_v) : V \otimes W \longrightarrow W \otimes V.$$

A *braiding* on a  $\mathbb{C}$ -linear tensor category  $\mathcal{C}$  is a family of isomorphisms

$$c_{X,Y} : X \otimes Y \longrightarrow Y \otimes X$$

satisfying, for  $f : X \rightarrow X'$  and  $g : Y \rightarrow Y'$ ,

$$(g \otimes f)c_{X,Y} = c_{X',Y'}(f \otimes g),$$

and the two hexagon identities

$$c_{X,Y \otimes Z} = (\text{id}_Y \otimes c_{X,Z})(c_{X,Y} \otimes \text{id}_Z),$$

$$c_{X \otimes Y, Z} = (c_{X,Z} \otimes \text{id}_Y)(\text{id}_X \otimes c_{Y,Z}).$$

The category is *braided* when it has a braiding.

Prove that  $c_{V,W}$  is a module intertwiner satisfying naturality and that the two coproduct identities for  $\mathcal{R}$  give the two hexagon identities, so that the finite-dimensional module category is braided. Derive

$$\mathcal{R}_{12}\mathcal{R}_{13}\mathcal{R}_{23} = \mathcal{R}_{23}\mathcal{R}_{13}\mathcal{R}_{12}$$



and, on  $V^{\otimes 3}$ ,

$$c_{12}c_{23}c_{12} = c_{23}c_{12}c_{23}.$$

For  $r \geq 2$ , define the braid group by

$$B_r = \langle b_1, \dots, b_{r-1} \mid b_i b_{i+1} b_i = b_{i+1} b_i b_{i+1}, \quad b_i b_j = b_j b_i \ (|i - j| > 1) \rangle.$$

Prove that  $b_j \mapsto c_{j,j+1}$  defines braid-group representations on tensor powers of every finite-dimensional type-1 module.

**Exercise 12.19** (Ribbon categories and ribbon elements). Let  $\mathcal{C}$  be a braided rigid  $\mathbb{C}$ -linear tensor category. A *twist* is a family of isomorphisms  $\theta_X : X \rightarrow X$  satisfying

$$\theta_Y f = f \theta_X \quad (f : X \rightarrow Y)$$

and

$$\theta_1 = \text{id}_1, \quad \theta_{X \otimes Y} = c_{Y,X} c_{X,Y} (\theta_X \otimes \theta_Y).$$

The category is *ribbon* when

$$\theta_{X^\vee} = (\theta_X)^\vee$$

for every object  $X$ .

For the universal  $R$ -matrix, set

$$u = m(S \otimes \text{id})(\mathcal{R}_{21}).$$

Let  $\nu_R \in H$  be central and invertible and satisfy

$$\nu_R^2 = uS(u), \quad S(\nu_R) = \nu_R, \quad \varepsilon(\nu_R) = 1, \quad \Delta(\nu_R) = (\mathcal{R}_{21}\mathcal{R})^{-1}(\nu_R \otimes \nu_R).$$

Prove that

$$\theta_V = \pi_V(\nu_R^{-1})$$

defines a ribbon twist on the category of finite-dimensional  $H$ -modules. Derive invariance of the quantum trace under framed ribbon isotopy.

**Exercise 12.20** (Kauffman bracket skein algebra). Let  $\Sigma$  be an oriented surface and let  $\mathcal{L}(\Sigma)$  be the set of framed-isotopy classes of unoriented links in  $\Sigma \times [0, 1]$ , including the empty link. For  $A \in \mathbb{C}^\times$ , define  $K_A(\Sigma)$  as the complex span of  $\mathcal{L}(\Sigma)$  modulo the following local relations. If  $L_\times, L_0, L_\infty$  agree outside a disk and differ inside it by a crossing and its two smoothings, impose

$$[L_\times] = A[L_0] + A^{-1}[L_\infty].$$

If  $\bigcirc$  is a contractible component disjoint from  $L$ , impose

$$[L \sqcup \bigcirc] = -(A^2 + A^{-2})[L].$$

Define multiplication by stacking links in the interval direction.

Prove that the relations are preserved by framed Reidemeister moves and that stacking makes  $K_A(\Sigma)$  an associative algebra with the empty link as identity. For two simple closed curves meeting once, resolve their crossing to express both multiplication orders in terms of the two smoothings and compute their commutator.

The quantum-group and skein parameters are related by  $A = v^{1/2}$  up to the chosen framing sign. At  $A = -1$ , let  $\rho : \pi_1(\Sigma) \rightarrow SL_2(\mathbb{C})$  and assign to an unoriented loop  $\gamma$  the function

$$[\gamma] \mapsto -W_\gamma(\rho) = -\text{tr } \rho(\gamma),$$

extending multiplicatively over disjoint components. Use Cayley–Hamilton to prove

$$W_\alpha W_\beta = W_{\alpha\beta} + W_{\alpha\beta^{-1}}$$

and show that the assignment respects both skein relations. Deduce that the classical specialization is commutative and maps to the algebra of Wilson-trace functions on flat  $SL_2$  connections.

For  $\Sigma = T^2$ , put

$$x = W_{1,0}, \quad y = W_{0,1}, \quad z = W_{1,1}.$$

Use successive skein resolutions and the Euclidean algorithm to express every  $W_{p,q}$  as a polynomial in  $x, y, z$ , reduced modulo

$$x^2 + y^2 + z^2 - xyz - 4 = 0.$$

Use the commutator computed above to prove that  $K_A(T^2)$  is noncommutative for generic  $A$  and that its specialization at  $A = -1$  maps to the torus Wilson-trace algebra.

**Exercise 12.21** (Quantum recoupling coefficients). For  $N \in \mathbb{Z}_{\geq 0}$  and the terminating specialization  $\gamma = Q^{-N-1}$ , define the  $Q$ -Racah polynomial by

$$R_n(\mu(x); \alpha, \beta, \gamma, \delta \mid Q) = {}_4\phi_3 \left( \begin{matrix} Q^{-n}, \alpha\beta Q^{n+1}, Q^{-x}, \gamma\delta Q^{x+1} \\ \alpha Q, \beta\delta Q, \gamma Q \end{matrix}; Q, Q \right), \quad \mu(x) = Q^{-x} + \gamma\delta Q^{x+1}.$$

The matrix coefficients of the recoupling map

$$(V_a \otimes V_b) \otimes V_c \longrightarrow V_a \otimes (V_b \otimes V_c)$$

between fixed intermediate irreducible summands are the  $v$ -6j coefficients.

Compute the change between the two coupling orders of  $V_a \otimes V_b \otimes V_c$  and express its entries as terminating  ${}_4\phi_3$  series with base  $Q = v^2$ . Identify them with  $Q$ -Racah polynomials and derive their orthogonality from unitarity of the recoupling map for the compact real form. Prove the pentagon identity by comparing the five coupling orders of four factors. For three resolved XXZ spin sectors, derive the recoupling probabilities as squared  $v$ -6j coefficients and state the unitarity and spectral-resolution assumptions required for this probability interpretation.

**Exercise 12.22** (Type-A Hecke actions). For  $v \neq 0$ , the Hecke algebra  $\mathcal{H}_v(S_r)$  is generated by  $T_1, \dots, T_{r-1}$  with

$$\begin{aligned} T_i T_{i+1} T_i &= T_{i+1} T_i T_{i+1}, & T_i T_j &= T_j T_i \quad (|i - j| > 1), \\ (T_i - v)(T_i + v^{-1}) &= 0. \end{aligned}$$

Let  $V_1$  be the fundamental  $U_v(\mathfrak{sl}_2)$ -module. On  $V_1 \otimes V_1 \cong V_2 \oplus V_0$ , let  $P_2, P_0$  be the invariant orthogonal projections, and define

$$T = vP_2 - v^{-1}P_0, \quad \rho_r(T_i) = \text{id}^{\otimes(i-1)} \otimes T \otimes \text{id}^{\otimes(r-i-1)}.$$

Prove that the coproduct action of  $U_v(\mathfrak{sl}_2)$  commutes with  $\rho_r(T_i)$ . Derive the braid relation from the  $Q$ -Racah recoupling identity and the Hecke relation from the two displayed spectral projections. Put  $e_i = (v - T_i)/(v + v^{-1})$  and prove that the action factors through the Temperley–Lieb relations

$$e_i^2 = e_i, \quad e_i e_{i\pm 1} e_i = (v + v^{-1})^{-2} e_i.$$

**Exercise 12.23** (Hecke Poincaré identity). Put  $\tau = v^2$ . For  $w \in S_r$ , let  $\ell(w)$  be the least number of adjacent transpositions in an expression for  $w$ , and let  $T_w = T_{i_1} \cdots T_{i_{\ell(w)}}$  for any reduced expression  $w = s_{i_1} \cdots s_{i_{\ell(w)}}$ . Prove from the braid relations that  $T_w$  is well-defined and that

$$\sum_{w \in S_r} \tau^{\ell(w)} = \prod_{j=1}^r \frac{1 - \tau^j}{1 - \tau}.$$

Prove that

$$e_+ = \frac{\sum_{w \in S_r} v^{\ell(w)} T_w}{\sum_{w \in S_r} v^{2\ell(w)}}, \quad e_- = \frac{\sum_{w \in S_r} (-v^{-1})^{\ell(w)} T_w}{\sum_{w \in S_r} v^{-2\ell(w)}}$$

are idempotents satisfying  $T_i e_+ = v e_+$  and  $T_i e_- = -v^{-1} e_-$ . Under the fundamental Hecke action, identify their images with the quantum symmetric and exterior state spaces and interpret the Poincaré polynomial as their graded state count.

**Exercise 12.24** (Affine Hecke algebras and Baxterization). The extended affine braid group of type  $GL_r$  is generated by  $b_1, \dots, b_{r-1}$  and commuting invertible elements  $X_1, \dots, X_r$ , subject to the braid relations and

$$b_i X_i b_i = X_{i+1}, \quad b_i X_j = X_j b_i \quad (j \neq i, i+1).$$

The extended affine Hecke algebra  $\mathcal{H}_v^{\text{aff}}(GL_r)$  is generated by  $T_1, \dots, T_{r-1}$  and commuting invertible elements  $X_1, \dots, X_r$ , subject to the finite Hecke relations and

$$T_i X_i T_i = X_{i+1}, \quad T_i X_j = X_j T_i \quad (j \neq i, i+1).$$

Equivalently, it is the quotient of this group algebra by the displayed quadratic Hecke relations.

For an invertible spectral parameter  $z$ , define

$$\check{R}_i(z) = z T_i - z^{-1} T_i^{-1}.$$

Its eigenvalues on the  $v$  and  $-v^{-1}$  eigenspaces of  $T_i$  are

$$zv - z^{-1}v^{-1}, \quad -zv^{-1} + z^{-1}v.$$

Use the braid and Hecke relations for  $T_i$  to prove

$$\check{R}_i(z) \check{R}_{i+1}(zw) \check{R}_i(w) = \check{R}_{i+1}(w) \check{R}_i(zw) \check{R}_{i+1}(z).$$

Prove that  $\check{R}_i(1)$  is a scalar multiple of the identity and compute  $\check{R}_i(z) \check{R}_i(z^{-1})$ . For  $|z| = 1$ , prove that  $\check{R}_i(z)^* = \check{R}_i(z^{-1})$  and normalize the operators to be unitary wherever the scalar product is nonzero. Prove that their two eigenvalues depend on  $z$  and hence that the spectral family itself does not satisfy one parameter-independent Hecke polynomial.

## 12.4 XXZ Spectroscopy

**Exercise 12.25** (Six-vertex transfer matrices and the XXZ spectrum). On the fundamental module  $V_1$ , put

$$\check{R}(z) = zT - z^{-1}T^{-1}, \quad R(z) = \tau \circ \check{R}(z),$$

where  $T = vP_2 - v^{-1}P_0$  and  $\tau$  is the flip. For an auxiliary copy  $V_a \cong V_1$  and a periodic row of length  $N$ , define

$$\mathsf{T}_N(z) = \text{Tr}_{V_a}(R_{aN}(z) \cdots R_{a1}(z)) \quad \text{on } V_1^{\otimes N},$$

and for an  $N \times M$  periodic lattice define

$$Z_{N,M}^{6V}(z) = \text{Tr}_{V_1^{\otimes N}} \mathbb{T}_N(z)^M.$$

Use invariance of  $P_2$  and  $P_0$  under the Cartan action to prove

$$[R(z), K \otimes K] = 0.$$

Relative to the distinguished weight lines of  $V_1$ , deduce that exactly the six arrow configurations preserving total weight can have nonzero local weights, and compute those weights from  $P_2$  and  $P_0$ . With  $v = e^{-i\eta}$  and  $z = e^{-iu}$ , determine the scalar normalization and diagonal gauge that identify these weights with the additive six-vertex matrix  $R(u)$ , and verify all six entries rather than treating the two parametrizations as independent models. Prove the spectral Yang–Baxter equation for  $R$  from the Baxterization identity. Apply it to an auxiliary pair and take their intrinsic traces to prove

$$[\mathbb{T}_N(z), \mathbb{T}_N(w)] = 0.$$

Show that  $Z_{N,M}^{6V}(z)$  is the sum of products of the six local weights over arrow configurations on the periodic square lattice.

Use  $\check{R}(1) = (v - v^{-1})I$  and compute the logarithmic derivative at the regular point. Prove that

$$\hat{H}_N = J \left[ (v - v^{-1}) \mathbb{T}'_N(1) \mathbb{T}_N(1)^{-1} - N(v + v^{-1})I \right]$$

is the periodic XXZ Hamiltonian, with  $\Delta = (v + v^{-1})/2$  and with the telescoping boundary term equal to zero. On the sector with one reversed spin, apply the distinguished Fourier modes and derive

$$E(k) = 4J(\cos k - \Delta), \quad k = \frac{2\pi m}{N}, \quad 0 \leq m < N.$$

Deduce the finite-size line spacings and the band edges. State the energy calibration, periodic-boundary, polarization-sector, and spectral-resolution assumptions required when these values are compared with an XXZ spin-chain simulator, and determine the scalar gauge and real spectral-parameter regime under which the same six weights are nonnegative Boltzmann weights.

**Exercise 12.26** (Coordinate Bethe ansatz and the Baxter equation). In the sector of the periodic XXZ chain with  $M$  reversed spins, label a configuration by

$$1 \leq x_1 < \cdots < x_M \leq N.$$

For momenta  $k_1, \dots, k_M$ , put

$$\psi(x_1, \dots, x_M) = \sum_{\pi \in S_M} A_\pi \exp \left( i \sum_{a=1}^M k_{\pi(a)} x_a \right).$$

Apply the XXZ Hamiltonian first when no two reversed spins are adjacent and then at a collision. Prove

$$E = 4J \sum_{a=1}^M (\cos k_a - \Delta)$$

and derive the two-body exchange factor

$$S(k, k') = - \frac{1 + e^{i(k+k')} - 2\Delta e^{ik}}{1 + e^{i(k+k')} - 2\Delta e^{ik'}}.$$

Use factorization of all coefficient ratios and periodicity to obtain the Bethe equations

$$e^{ik_a N} = \prod_{b \neq a} S(k_a, k_b), \quad 1 \leq a \leq M.$$

Put  $v = e^{-\eta}$ , so  $\Delta = \cosh \eta$ , and reparametrize the momenta by rapidities  $u_j$  through

$$e^{ik_j} = \frac{\sinh(u_j + \eta/2)}{\sinh(u_j - \eta/2)}.$$

Use the additive normalization of the same commuting transfer family, obtained from the multiplicative parameter by  $z = e^{-u}$  followed by a nonzero scalar normalization, and write its eigenvalue as  $\Lambda(u)$ . Define

$$Q(u) = \prod_{j=1}^M \sinh(u - u_j), \quad a(u) = \sinh(u + \eta/2)^N, \quad d(u) = \sinh(u - \eta/2)^N.$$

Derive the algebraic relations of the six-vertex monodromy operator from the Yang–Baxter equation, and use them [45] to prove that a transfer eigenvalue  $\Lambda(u)$  satisfies Baxter’s equation

$$\Lambda(u)Q(u) = a(u)Q(u - \eta) + d(u)Q(u + \eta).$$

Prove that cancellation of the apparent poles at the zeros of  $Q$  gives

$$\left( \frac{\sinh(u_j + \eta/2)}{\sinh(u_j - \eta/2)} \right)^N = \prod_{\ell \neq j} \frac{\sinh(u_j - u_\ell + \eta)}{\sinh(u_j - u_\ell - \eta)},$$

and identify these equations with the momentum form above. Solve the  $M = 1$  sector and one chosen  $M = 2$  finite chain, and state the periodicity, anisotropy, completeness, root-selection, finite-size, and spectral resolution assumptions under which many-body line measurements test the Bethe roots and Baxter polynomial.

**Exercise 12.27** ( $q$ KZ transport). Let  $v$  be the quantum-group parameter and let  $\kappa \neq 0$  be an independent shift parameter. Let  $D \in \text{End}(V)$  be invertible and satisfy  $[D \otimes D, \check{R}(u)] = 0$ . Let  $D_1$  act on the transported tensor factor and define

$$\mathcal{K}_1(z_1, \dots, z_r) = D_1 \check{R}_{1r}(z_1/z_r) \cdots \check{R}_{12}(z_1/z_2).$$

The first  $q$ KZ equation is

$$\Psi(z_2, \dots, z_r, \kappa z_1) = \mathcal{K}_1(z_1, \dots, z_r) \Psi(z_1, \dots, z_r),$$

and the remaining equations are obtained by cyclic conjugation.

Use Baxterization to prove consistency of the adjacent exchange rules and the  $q$ KZ difference equations. Identify the constant exchanges and winding operators with the generators of  $\mathcal{H}_v^{\text{aff}}(GL_r)$ , keeping  $v$  and  $\kappa$  independent. In the physical regime assume

$$\check{R}(u)^* = \check{R}(u)^{-1} = \check{R}(u^{-1}), \quad D^* D = I.$$

Put

$$\sigma_\kappa(z_1, \dots, z_r) = (z_2, \dots, z_r, \kappa z_1)$$

and define the full-cycle transport

$$\mathcal{M}_\kappa(z) = \mathcal{K}_1(\sigma_\kappa^{r-1}z) \cdots \mathcal{K}_1(\sigma_\kappa z) \mathcal{K}_1(z).$$

Multiply the  $r$  cyclic equations to prove

$$\Psi(\kappa z_1, \dots, \kappa z_r) = \mathcal{M}_\kappa(z) \Psi(z_1, \dots, z_r),$$

and prove that  $\mathcal{M}_\kappa(z)$  is unitary when every exchange parameter encountered along this path lies in the displayed unitary regime. Prepare and detect three resolved XXZ spin sectors and prove that the two factorizations of a three-spin exchange give the same output probabilities. Express this equality as the  $Q$ -Racah recoupling identity. If the prepared state lies in the  $e^{i\phi}$ -eigenspace of  $\mathcal{M}_\kappa(z)$ , derive the interference phase  $\phi$  accumulated in one simultaneous  $\kappa$ -shift cycle. In a balanced reference interferometer with calibrated phase  $\phi_{\text{ref}}$ , derive the output probabilities

$$P_\pm = \frac{1}{2} (1 \pm \cos(\phi + \phi_{\text{ref}})).$$

State the twist calibration, spectral-parameter path, unitarity, coherence, sector preparation, and phase-resolution assumptions separately from the recoupling probabilities and phase prediction.

## 13 Double Affine Hecke Algebras

### 13.1 Double-Affine Braid Groups

**Exercise 13.1** (Root systems, affine Weyl groups, and alcoves). The DAHA Hecke parameter below is  $t^{1/2}$ . Under comparison with the quantum-group Hecke action, use the convention  $t^{1/2} = v$ ; otherwise  $q$  and  $t$  are independent parameters. Let  $E$  be a finite-dimensional real Euclidean space. A finite *root system* is a finite spanning set  $R \subset E \setminus \{0\}$  such that, with

$$\alpha^\vee = \frac{2\alpha}{\langle \alpha, \alpha \rangle}, \quad s_\alpha(x) = x - \langle x, \alpha^\vee \rangle \alpha,$$

one has  $s_\alpha(R) = R$  and  $\langle \beta, \alpha^\vee \rangle \in \mathbb{Z}$  for all  $\alpha, \beta \in R$ . Define

$$W = \langle s_\alpha : \alpha \in R \rangle, \quad Q = \text{span}_{\mathbb{Z}}(R), \quad Q^\vee = \text{span}_{\mathbb{Z}}\{\alpha^\vee : \alpha \in R\},$$

$$P = \{\lambda \in E : \langle \lambda, \alpha^\vee \rangle \in \mathbb{Z} \text{ for every } \alpha \in R\}.$$

A *positive system* is a subset  $R_+ \subset R$  such that  $R = R_+ \sqcup (-R_+)$  and the sum of two members of  $R_+$ , when it is a root, again lies in  $R_+$ . Its indecomposable members are the *simple roots*; their nonnegative span is the positive root cone, the closed dominant chamber is

$$\overline{C}_+ = \{x \in E : \langle x, \alpha^\vee \rangle \geq 0 \text{ for every simple } \alpha\},$$

and

$$P_+ = \{\lambda \in P : \langle \lambda, \alpha^\vee \rangle \geq 0 \text{ for every simple } \alpha\}$$

is the dominant cone. Prove that  $W$  is generated by the simple reflections and acts simply transitively on the chambers.

For type  $A_{r-1}$  take

$$E = \{x \in \mathbb{R}^r : \sum_i x_i = 0\}, \quad R = \{e_i - e_j : i \neq j\},$$

and prove  $W \cong S_r$ . For the nonreduced system  $BC_r$ , identify the reflection hyperplanes associated with

$$\pm e_i, \quad \pm 2e_i, \quad \pm e_i \pm e_j$$

and prove that its Weyl group is the signed permutation group.

Define the affine reflections in  $\langle x, \alpha^\vee \rangle = m$ ,  $m \in \mathbb{Z}$ , the affine Weyl group  $W_{\text{aff}} = Q^\vee \rtimes W$ , and its fundamental alcove. Show in type  $A_{r-1}$  that the affine braid group is generated by wall crossings of the alcoves and that imposing quadratic Hecke relations yields the affine Hecke algebra. Identify the finite and affine Weyl symmetries that occur in the type- $A$  and  $BC_r$  polynomial models below.

**Exercise 13.2** (Double-affine braid groups and Hecke quotients). For  $r \geq 2$ , the *double-affine braid group*  $\tilde{B}_r$  of type  $GL_r$  is generated by  $T_i$  for  $1 \leq i < r$ , commuting invertible elements  $X_j$ , commuting invertible elements  $Y_j$ , and a central invertible element  $\mathbf{q}$ , subject to the braid relations and

$$\begin{aligned} T_i X_i T_i &= X_{i+1}, & T_i X_j &= X_j T_i \quad (j \neq i, i+1), \\ T_i^{-1} Y_i T_i^{-1} &= Y_{i+1}, & T_i Y_j &= Y_j T_i \quad (j \neq i, i+1), \\ Y_1 X_1 \cdots X_r &= \mathbf{q} X_1 \cdots X_r Y_1, & X_1^{-1} Y_2 &= Y_2 X_1^{-1} T_1^{-2}. \end{aligned}$$

Let  $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ . Interpret  $T_i$  as an adjacent braid and  $X_j, Y_j$  as motions of the  $j$ th point around the two cycles of  $T^2$ . Derive the displayed relations from these motions and identify  $\tilde{B}_r$  with the central extension of

$$\pi_1(\text{Conf}_r(T^2)/S_r)$$

whose total cycle commutator is  $\mathbf{q}$ . Identify the subgroup generated by  $T_i, X_j$  with the extended affine braid group of  $\mathbb{Z}^r \rtimes S_r$ , and prove that imposing

$$(T_i - v)(T_i + v^{-1}) = 0$$

on this subgroup gives the extended affine Hecke algebra. Prove that deleting either the  $X$ -family or the  $Y$ -family leaves an affine braid group. For nonzero  $q, t$ , define the *double affine Hecke algebra* of type  $GL_r$  by

$$\mathbf{H}_{q,t}(GL_r) = \mathbb{C}(q, t)[\tilde{B}_r] / \left( \mathbf{q} - q, (T_i - t^{1/2})(T_i + t^{-1/2}) \right).$$

Identify its two extended affine Hecke subalgebras and its finite Hecke subalgebra.

## 13.2 Polynomial Representations

**Exercise 13.3** (Rank-one DAHA polynomial representation). For nonzero  $k_0, k_1, u_0, u_1$  and a choice of  $q^{1/2}$ , the rank-one DAHA  $\mathbf{H}_q(C_1^\vee, C_1)$  is generated by  $T_0^{\pm 1}, T_1^{\pm 1}, (T_0^\vee)^{\pm 1}, (T_1^\vee)^{\pm 1}$  subject to

$$(T_i - k_i)(T_i + k_i^{-1}) = 0, \quad (T_i^\vee - u_i)(T_i^\vee + u_i^{-1}) = 0 \quad (i = 0, 1),$$

$$T_1^\vee T_1 T_0 T_0^\vee = q^{-1/2}.$$

On  $\mathcal{L} = \mathbb{C}[z^{\pm 1}]$  put

$$\begin{aligned} (s_1 f)(z) &= f(z^{-1}), & (s_0 f)(z) &= f(qz^{-1}), \\ (T_1 f)(z) &= k_1 f(z^{-1}) + \frac{(k_1 - k_1^{-1}) + (u_1 - u_1^{-1})z}{1 - z^2} (f(z) - f(z^{-1})), \\ (T_0 f)(z) &= k_0 f(qz^{-1}) + \frac{(k_0 - k_0^{-1}) + (u_0 - u_0^{-1})q^{1/2}z^{-1}}{1 - qz^{-2}} (f(z) - f(qz^{-1})). \end{aligned}$$

Let  $X$  act by multiplication by  $z$ . The Askey–Wilson parameters in this convention are

$$a = k_1 u_1, \quad b = -k_1 u_1^{-1}, \quad c = q^{1/2} k_0 u_0, \quad d = -q^{1/2} k_0 u_0^{-1}.$$

Put

$$T_1^\vee = X^{-1} T_1^{-1}, \quad T_0^\vee = q^{-1/2} T_0^{-1} X.$$

Verify that the displayed reflection–difference operators preserve  $\mathcal{L}$ , that  $T_0, T_1, T_0^\vee, T_1^\vee$  satisfy the four quadratic Hecke relations and the product relation, and that they give the rank-one DAHA polynomial representation with the stated parameter map.

### 13.3 Elliptic Hypergeometric Functions

**Exercise 13.4** (Elliptic gamma functions and the deformation hierarchy). For  $|p|, |q| < 1$  and  $z \in \mathbb{C}^\times$ , define

$$(z; p)_\infty = \prod_{j=0}^{\infty} (1 - z p^j), \quad \theta(z; p) = (z; p)_\infty (p/z; p)_\infty,$$

and use the abbreviations

$$\theta(z_1, \dots, z_s; p) = \prod_{j=1}^s \theta(z_j; p),$$

$$(a; q, p)_m = \prod_{j=0}^{m-1} \theta(a q^j; p),$$

$$(a_1, \dots, a_s; q, p)_m = \prod_{i=1}^s (a_i; q, p)_m,$$

and

$$\Gamma(z; p, q) = \prod_{j, k \geq 0} \frac{1 - p^{j+1} q^{k+1} / z}{1 - p^j q^k z}.$$

Prove

$$\theta(pz; p) = -z^{-1} \theta(z; p), \quad \theta(z; 0) = 1 - z,$$

and that the zeros of  $\theta(\cdot; p)$  are the points  $p^n$ ,  $n \in \mathbb{Z}$ , with multiplicity one when  $p \neq 0$ . Prove

$$\Gamma(qz; p, q) = \theta(z; p) \Gamma(z; p, q),$$

$$\Gamma(pz; p, q) = \theta(z; q) \Gamma(z; p, q), \quad \Gamma(z; p, q) \Gamma(pq/z; p, q) = 1.$$

Prove

$$(a; q, 0)_m = (a; q)_m, \quad \Gamma(z; 0, q) = \frac{1}{(z; q)_\infty}.$$

**Exercise 13.5** (The Frenkel–Turaev sum). For  $a^2 q^{n+1} = bcde$ , define

$$\begin{aligned} {}_{10}V_9(a; b, c, d, e, q^{-n}; q, p) &= \sum_{m=0}^n \frac{\theta(a q^{2m}; p)}{\theta(a; p)} \\ &\quad \times \frac{(a, b, c, d, e, q^{-n}; q, p)_m}{(q, aq/b, aq/c, aq/d, aq/e, aq^{n+1}; q, p)_m} q^m. \end{aligned}$$



(a) Prove the multiplicative theta addition formula

$$\begin{aligned} \theta(xz, x/z, yw, y/w; p) - \theta(xw, x/w, yz, y/z; p) \\ = \frac{y}{z} \theta(xy, x/y, zw, z/w; p) \end{aligned}$$

by comparing quasi-periodicity and zeros.

(b) Use it to telescope the difference between the  $n$ th and  $(n-1)$ st terminating sums and prove

$${}_{10}V_9(a; b, c, d, e, q^{-n}; q, p) = \frac{(aq, aq/(bc), aq/(bd), aq/(cd); q, p)_n}{(aq/b, aq/c, aq/d, aq/(bcd); q, p)_n}.$$

(c) Take  $p \rightarrow 0$  and identify the terminating very-well-poised basic hypergeometric summation.

**Exercise 13.6** (Elliptic face weights and vertex–face intertwiners). Fix logarithms of  $p, q$  with  $|p|, |q| < 1$  and put

$$[x]_{p,q} = q^{(1-x)/2} \frac{\theta(q^x; p)}{\theta(q; p)}.$$

For integer heights  $i, j, k, l$  satisfying

$$|i - j| = |j - k| = |k - l| = |l - i| = 1,$$

define  $W(i, j, k, l \mid x)$  by

$$\begin{aligned} W(l \pm 1, l \pm 2, l \pm 1, l \mid x) &= [x + 1]_{p,q}, \\ W(l \mp 1, l, l \pm 1, l \mid x) &= \frac{[l + 1 \pm 1]_{p,q} [x]_{p,q}}{[l + 1]_{p,q}}, \\ W(l \pm 1, l, l \pm 1, l \mid x) &= \frac{[l + 1 \mp x]_{p,q}}{[l + 1]_{p,q}}, \end{aligned}$$

whenever the denominators are nonzero, and set all other weights equal to zero. On  $V = \mathbb{C}e_+ \oplus \mathbb{C}e_-$  let  $he_{\pm} = \pm e_{\pm}$  and define  $R(x, \lambda) \in \text{End}(V \otimes V)$  at  $\lambda = l + 1$  by

$$\langle e_{j-k} \otimes e_{k-l}, R(x, l+1)(e_{i-l} \otimes e_{j-i}) \rangle = W(i, j, k, l \mid x),$$

using the orthonormal distinguished weight lines and extending meromorphically in  $\lambda$ .

Let  $R^{\text{vert}}(u) \in \text{End}(V \otimes V)$  be a spectral exchange operator. For each admissible oriented height edge  $|a - b| = 1$ , let  $\psi(a, b \mid u) \in V$  be meromorphic in  $u$ . The pair  $(R^{\text{vert}}, \psi)$  satisfies the vertex–face relation for  $W$  when

$$\begin{aligned} R^{\text{vert}}(u - v)(\psi(a, b \mid u) \otimes \psi(b, c \mid v)) \\ = \sum_{\substack{b' \\ |a-b'|=|b'-c|=1}} W(a, b, c, b' \mid u - v) \psi(a, b' \mid v) \otimes \psi(b', c \mid u). \end{aligned}$$

The intertwiners are nondegenerate when the maps from admissible height-path spaces induced by their tensor products are injective for generic spectral parameters.

(a) Prove  $[-x]_{p,q} = -[x]_{p,q}$  and rewrite the theta addition identity as the corresponding four-term identity for the brackets  $[x]_{p,q}$ .

(b) Verify that  $R(x, \lambda)$  has weight zero and prove the dynamical Yang–Baxter equation

$$\begin{aligned} R_{12}(x, \lambda + h^{(3)})R_{13}(x + y, \lambda)R_{23}(y, \lambda + h^{(1)}) \\ = R_{23}(y, \lambda)R_{13}(x + y, \lambda + h^{(2)})R_{12}(x, \lambda) \end{aligned}$$

by reducing every nontrivial weight component to that bracket identity. Translate the two sides into the two sums over the admissible internal height of a three-face patch. Show directly that the shifts by  $h^{(1)}, h^{(2)}, h^{(3)}$  are the changes of the adjacent face height and that  $\lambda$ , unlike  $x$  and  $y$ , is therefore the dynamical parameter.

(c) For periodic admissible height paths, form the row-to-row transfer maps by multiplying the displayed face weights and summing over the intermediate path. Use the face Yang–Baxter relation to prove commutation of transfer maps with distinct spectral parameters. For fixed boundary heights of a three-face patch with two admissible internal heights, compute their normalized probabilities from the two products of  $W$  and prove that a star–triangle move preserves the boundary partition function. State the real parameter and gauge conditions under which the normalized weights are positive statistical-mechanical probabilities.

**Exercise 13.7** (Vertex–face transfer correspondence). Assume nondegenerate vertex–face intertwiners. Apply the ordinary spectral Yang–Baxter equation for  $R^{\text{vert}}$  to

$$\psi(a, b \mid u) \otimes \psi(b, c \mid v) \otimes \psi(c, d \mid w)$$

and use the vertex–face relation in the two orders. Prove that equality of the resulting coefficients is the face Yang–Baxter relation and hence the dynamical Yang–Baxter equation. Conversely, use nondegeneracy to prove the ordinary Yang–Baxter equation on the image of the height-path map from the face relation.

For inhomogeneities  $z_1, \dots, z_N$ , tensor the edge intertwiners around a periodic row to obtain the map  $\Phi_u$  from periodic height paths to the corresponding vertex sector. Prove the row-transfer intertwining identity

$$\mathsf{T}_N^{\text{vert}}(u)\Phi_u = \Phi_u\mathsf{T}_N^{\text{face}}(u)$$

with the boundary twist dictated by the net height change. Deduce equality of the matched transfer spectra and periodic-strip partition functions. In the trigonometric degeneration  $p \rightarrow 0$ , identify the vertex transfer operator with the six-vertex family and the face transfer operator with the trigonometric solid-on-solid model. State the boundary-sector, normalization, injectivity, and face-gauge conditions required when their finite-size spectra are compared.

**Exercise 13.8** (Terminating elliptic  $6j$ -symbols). For  $bcdefg = a^3q^{n+2}$ , define

$$\begin{aligned} {}_{12}V_{11}(a; b, c, d, e, f, g, q^{-n}; q, p) \\ = \sum_{m=0}^n \frac{\theta(aq^{2m}; p)}{\theta(a; p)} \frac{(a, b, c, d, e, f, g, q^{-n}; q, p)_m}{(q, aq/b, aq/c, aq/d, aq/e, aq/f, aq/g, aq^{n+1}; q, p)_m} q^m. \end{aligned}$$

Let

$$\lambda = \frac{a^2q}{bcd}, \quad g = \frac{\lambda aq^{n+1}}{ef},$$

and assume no displayed denominator vanishes. Expand the series on the right, interchange the

resulting finite sums, and apply the Frenkel–Turaev  ${}_{10}V_9$  summation to prove

$$\begin{aligned} & {}_{12}V_{11}(a; b, c, d, e, f, g, q^{-n}; q, p) \\ &= \frac{(aq, aq/(ef), \lambda q/e, \lambda q/f; q, p)_n}{(aq/e, aq/f, \lambda q/(ef), \lambda q; q, p)_n} \\ & \quad \times {}_{12}V_{11}\left(\lambda; \frac{\lambda b}{a}, \frac{\lambda c}{a}, \frac{\lambda d}{a}, e, f, g, q^{-n}; q, p\right). \end{aligned}$$

Fuse  $n$  consecutive elementary face weights by summing over admissible internal height paths. Factor the endpoint theta products and show that the remaining path sum has the displayed  ${}_{12}V_{11}$  term ratio. Identify the two sides of the transformation with the two fusion orders and hence with the two recouplings of the four external height paths. Deduce the tetrahedral change of channel for the resulting elliptic  $6j$  coefficients and recover the face Yang–Baxter relation for fused weights.

Choose a numerator parameter so that the transformed series reduces to its zeroth term and recover the  ${}_{10}V_9$  summation. Take  $p \rightarrow 0$  to obtain Bailey’s terminating very-well-poised  ${}_{10}\phi_9$  transformation, and then take the spectral degeneration to the  $Q$ -Racah quantum  $6j$  coefficients. For a finite fused face strip, express the relative Boltzmann probabilities of two resolved intermediate heights as ratios of normalized fused face weights. Separately, in a unitary recoupling specialization, express channel probabilities as squared normalized elliptic  $6j$  coefficients. State the positivity, face-gauge, boundary-height, unitarity, and spectral-resolution assumptions required for the two interpretations.

**Exercise 13.9** (Elliptic beta kernels and integral evaluation). Write

$$\Gamma(tz^{\pm 1}; p, q) = \Gamma(tz; p, q)\Gamma(t/z; p, q), \quad \kappa_{p,q} = \frac{(p; p)_{\infty}(q; q)_{\infty}}{2}.$$

For  $|t_j| < 1$  with  $\prod_{j=1}^6 t_j = pq$ , define

$$\mathcal{B}(t_1, \dots, t_6) = \kappa_{p,q} \int_{\mathbb{T}} \frac{\prod_{j=1}^6 \Gamma(t_j z^{\pm 1}; p, q)}{\Gamma(z^{\pm 2}; p, q)} \frac{dz}{2\pi i z}.$$

Use the elliptic-gamma difference equations to derive the parameter-shift recurrences of  $\mathcal{B}$ . Evaluate the residue degeneration in which two parameters multiply to  $pq$  and normalize the product side by the same degeneration. Prove the elliptic-shift uniqueness lemma [61]: a meromorphic function of the balanced parameters that satisfies the two independent  $p$ - and  $q$ -shift recurrences and has the specified residue normalization is uniquely determined. Apply the lemma and analytic continuation to prove

$$\mathcal{B}(t_1, \dots, t_6) = \prod_{1 \leq i < j \leq 6} \Gamma(t_i t_j; p, q).$$

Take  $p \rightarrow 0$  and identify the resulting basic hypergeometric beta integral.

**Exercise 13.10** (Elliptic star–triangle operators and the Askey–Wilson degeneration). For  $|t| < 1$  define

$$(\mathcal{M}_t f)(w) = \frac{\kappa_{p,q}}{\Gamma(t^2; p, q)} \int_{\mathbb{T}} \frac{\Gamma(tw^{\pm 1}z^{\pm 1}; p, q)}{\Gamma(z^{\pm 2}; p, q)} f(z) \frac{dz}{2\pi i z},$$

and

$$\mathcal{D}_s(y, w) = \Gamma\left((pq)^{1/2} s^{-1} y^{\pm 1} w^{\pm 1}; p, q\right).$$

- (a) Apply the elliptic beta integral to the internal integration variable and prove the operator identity

$$\mathcal{M}_s \mathcal{D}_{st}(y, \cdot) \mathcal{M}_t = \mathcal{D}_t(y, \cdot) \mathcal{M}_{st} \mathcal{D}_s(y, \cdot)$$

whenever the contours separate reciprocal pole sequences and the parameters satisfy the corresponding modulus bounds. Interpret the kernels as edge weights and  $\mathcal{D}$  as a site weight of a continuous spin model. Prove that the two local triangular partition functions are equal and that the row transfer operators obtained by the two factorizations commute. For a finite periodic strip, deduce equality of the transfer spectra and partition functions under a star–triangle move; state the positivity regime and contour assumptions under which these are statistical-mechanical observables.

- (b) Put

$$(t_1, t_2, t_3, t_4, t_5, t_6) = \left( a, b, c, d, s, \frac{pq}{abcds} \right).$$

Assume initially that the parameters have moduli for which the unit circle separates the reciprocal pole sequences. In the elliptic beta integral, first take  $p \rightarrow 0$  and then  $s \rightarrow 0$ . Use

$$\Gamma(z; 0, q) = \frac{1}{(z; q)_\infty}, \quad \lim_{p \rightarrow 0} \Gamma(px; p, q) = (q/x; q)_\infty$$

to derive the Askey–Wilson integral

$$\frac{(q; q)_\infty}{2} \int_{\mathbb{T}} w_{AW}(z) \frac{dz}{2\pi iz} = \frac{(abcd; q)_\infty}{(ab, ac, ad, bc, bd, cd; q)_\infty},$$

where

$$w_{AW}(z) = \frac{(z^2, z^{-2}; q)_\infty}{(az, a/z, bz, b/z, cz, c/z, dz, d/z; q)_\infty}.$$

Justify both limits by uniform convergence away from the pole sequences and then extend the identity meromorphically in  $a, b, c, d$ . Identify  $w_{AW}$  as the trigonometric internal-spin measure left by the elliptic star–triangle normalization.

**Exercise 13.11** (Spherical DAHA and the Askey–Wilson operator). Assume  $k_1 + k_1^{-1} \neq 0$ . For the finite Hecke generator  $T_1$ , define

$$e = \frac{T_1 + k_1^{-1}}{k_1 + k_1^{-1}}.$$

For the associated Askey–Wilson parameters  $a, b, c, d$ , with  $0 < q < 1$ , define

$$A(z) = \frac{(1 - az)(1 - bz)(1 - cz)(1 - dz)}{(1 - z^2)(1 - qz^2)}$$

and, on  $f(z) = f(z^{-1})$ ,

$$(\mathcal{D}_{AW} f)(z) = A(z)(f(qz) - f(z)) + A(z^{-1})(f(q^{-1}z) - f(z)).$$

Prove

$$e\mathcal{L} = \mathcal{L}^{z \mapsto z^{-1}} = \mathbb{C}[z + z^{-1}],$$

and prove that the spherical action contains multiplication by  $z + z^{-1}$  and  $\mathcal{D}_{AW}$ . For  $a, b, c, d \in (-1, 1)$ , use the  $q$ -shift ratio of the limiting weight  $w_{AW}$  to prove that  $\mathcal{D}_{AW}$  is symmetric for

$$\langle f, g \rangle_{AW} = \int_{|z|=1} f(z) \overline{g(z)} w_{AW}(z) \frac{dz}{2\pi iz}.$$

Conclude that the  $p \rightarrow 0$  degeneration of the elliptic internal-spin measure is the spectral measure for the spherical rank-one DAHA operator.

**Exercise 13.12** (Askey–Wilson spectral theorem). For  $x = (z + z^{-1})/2$ , define the Askey–Wilson polynomial by

$$p_n(x; a, b, c, d \mid q) = a^{-n}(ab, ac, ad; q)_n \\ \times {}_4\phi_3\left(\begin{matrix} q^{-n}, abcdq^{n-1}, az, az^{-1} \\ ab, ac, ad \end{matrix}; q, q\right).$$

Prove

$$\mathcal{D}_{AW}p_n = (q^{-n} - 1)(1 - abcdq^{n-1})p_n.$$

For  $a, b, c, d \in (-1, 1)$ , deduce from symmetry of  $\mathcal{D}_{AW}$  the orthogonality and three-term recurrence of the Askey–Wilson polynomials.

**Exercise 13.13** (Wilson polynomials and degeneration). For parameters with no denominator Pochhammer symbol vanishing, define

$${}_pF_q\left(\begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_q \end{matrix}; z\right) = \sum_{m=0}^{\infty} \frac{(a_1)_m \cdots (a_p)_m}{(b_1)_m \cdots (b_q)_m} \frac{z^m}{m!},$$

with the terminating interpretation when a numerator parameter is a nonpositive integer. For positive  $a, b, c, d$ , define

$$W_n(x^2; a, b, c, d) = (a+b)_n(a+c)_n(a+d)_n \\ \times {}_4F_3\left(\begin{matrix} -n, n+a+b+c+d-1, a+ix, a-ix \\ a+b, a+c, a+d \end{matrix}; 1\right).$$

Set  $q = e^{-\varepsilon}$  and scale the Askey–Wilson parameters and spectral variable by

$$(a, b, c, d, z) = (q^A, q^B, q^C, q^D, q^{ix}).$$

After the scalar normalization by  $(1-q)^{-3n}$ , prove that the Askey–Wilson polynomial tends to  $W_n(x^2; A, B, C, D)$  as  $\varepsilon \downarrow 0$ . Derive the Wilson difference equation, weight, orthogonality, and three-term recurrence as limits of the corresponding Askey–Wilson identities.

**Exercise 13.14** (Type- $A$  polynomial representations). Let

$$\mathcal{P}_r = \mathbb{C}(q, t^{1/2})[x_1^{\pm 1}, \dots, x_r^{\pm 1}],$$

let  $X_j$  act by multiplication by  $x_j$ , and let  $s_i$  exchange  $x_i$  and  $x_{i+1}$ . Define

$$T_i = t^{1/2}s_i + (t^{1/2} - t^{-1/2})\frac{s_i - 1}{x_i/x_{i+1} - 1}, \quad 1 \leq i < r,$$

and

$$(\omega f)(x_1, \dots, x_r) = f(qx_r, x_1, \dots, x_{r-1}).$$

Define the Cherednik operators by

$$Y_i = T_i \cdots T_{r-1} \omega T_1^{-1} \cdots T_{i-1}^{-1}.$$

For generic independent parameters  $q, t$ , call the operator algebra generated by  $X_j^{\pm 1}$ ,  $T_i$ , and  $\omega^{\pm 1}$  the type- $A$  polynomial operator model.

Prove that the displayed operators preserve  $\mathcal{P}_r$  and satisfy the braid and Hecke relations, and prove that the  $Y_i$  commute. Prove

$$\omega X_1 \omega^{-1} = q X_r, \quad \omega X_i \omega^{-1} = X_{i-1} \quad (2 \leq i \leq r),$$

and identify the two affine-Hecke subalgebras generated respectively by  $T_i, X_j$  and by  $T_i, Y_j$ . Prove that this representation is obtained from the double-affine braid-group algebra by imposing

$$(T_i - t^{1/2})(T_i + t^{-1/2}) = 0$$

and specializing its central parameter to  $q$ . Conclude that the polynomial operator model is a realization of  $H_{q,t}(GL_r)$ .

**Exercise 13.15** (Nonsymmetric Macdonald polynomials and Cherednik intertwiners). For a composition  $\alpha \in \mathbb{Z}^r$ , let  $\alpha^+$  be its dominant rearrangement. Order compositions first by dominance of  $\alpha^+$  and, within a Weyl orbit, by the Bruhat order of the shortest permutation carrying  $\alpha^+$  to  $\alpha$ . Write  $x^\alpha$  for the corresponding Laurent monomial.

Use the triangular action of the commuting Cherednik operators  $Y_i$  to define  $\xi_i(\alpha)$  as the coefficient of  $x^\alpha$  in  $Y_i x^\alpha$ . For generic  $q, t$ , prove that the joint tuples  $\xi(\alpha) = (\xi_1(\alpha), \dots, \xi_r(\alpha))$  are distinct. Construct the unique nonsymmetric Macdonald polynomial

$$E_\alpha = x^\alpha + \sum_{\beta \prec \alpha} c_{\alpha\beta} x^\beta, \quad Y_i E_\alpha = \xi_i(\alpha) E_\alpha.$$

On a joint eigenspace on which the denominator is invertible, define the Cherednik intertwiner

$$\Phi_i = T_i + (t^{1/2} - t^{-1/2}) \frac{Y_i/Y_{i+1}}{1 - Y_i/Y_{i+1}}.$$

Use the affine-Hecke relations to prove

$$\Phi_i Y_i = Y_{i+1} \Phi_i, \quad \Phi_i Y_{i+1} = Y_i \Phi_i, \quad \Phi_i Y_j = Y_j \Phi_i \quad (j \neq i, i+1),$$

and deduce that a nonzero  $\Phi_i E_\alpha$  is proportional to  $E_{s_i \alpha}$ . Include the affine intertwiner obtained from  $\omega$  and show that intertwiners generate every  $E_\alpha$  from a dominant seed.

Define the finite-Hecke symmetrizer  $e$  to be the normalized idempotent satisfying

$$T_i e = t^{1/2} e = e T_i \quad (1 \leq i < r).$$

Prove that  $e E_\alpha$  depends only on  $\alpha^+$  and, after monic normalization, equals the symmetric Macdonald polynomial  $P_{\alpha^+}$ . Deduce that the symmetric spectral theory is the Weyl-invariant part of a joint nonsymmetric  $Y$ -spectrum, and state how symmetry-resolved frequency measurements distinguish a full Weyl orbit before symmetrization.

**Exercise 13.16** (Macdonald operators and polynomials). Let

$$(T_{q,x_i} f)(x_1, \dots, x_r) = f(x_1, \dots, q x_i, \dots, x_r).$$

The first type- $A$  Macdonald operator is

$$D_{q,t} = \sum_{i=1}^r \left( \prod_{j \neq i} \frac{t x_i - x_j}{x_i - x_j} \right) T_{q,x_i}.$$

Use the displayed polynomial representation and the affine braid relations to prove that symmetric Laurent polynomials in the  $Y_i$  preserve  $\mathcal{P}_r^{S_r}$  and identify  $t^{(r-1)/2}(Y_1 + \dots + Y_r)$  with  $D_{q,t}$ .

**Exercise 13.17** (Macdonald spectral theory). For generic  $q, t$ , the *Macdonald polynomial*  $P_\lambda(x; q, t)$  is the symmetric polynomial triangular in the monomial symmetric functions and satisfying

$$D_{q,t}P_\lambda = \left( \sum_{i=1}^r q^{\lambda_i} t^{r-i} \right) P_\lambda, \quad \lambda_1 \geq \cdots \geq \lambda_r \geq 0.$$

For  $0 < q, t < 1$ , define

$$\Delta_{q,t}(x) = \prod_{i \neq j} \frac{(x_i/x_j; q)_\infty}{(tx_i/x_j; q)_\infty},$$

and

$$\langle f, g \rangle_{q,t} = \frac{1}{r!} \int_{(S^1)^r} f(x) \overline{g(x)} \Delta_{q,t}(x) \prod_{j=1}^r \frac{dx_j}{2\pi i x_j}.$$

Order monomial symmetric functions by dominance and prove that  $D_{q,t}$  is triangular with the displayed diagonal entries. For generic  $q, t$ , deduce existence and uniqueness of  $P_\lambda$ . Prove directly from the torus integral that  $D_{q,t}$  is symmetric when  $0 < q, t < 1$ , and deduce orthogonality and joint diagonalization for symmetric Laurent polynomials in the commuting  $Y_i$ .

**Exercise 13.18** (DAHA Fourier duality). For  $\rho = (r-1, r-2, \dots, 0)$  and a partition  $\lambda$ , put

$$\xi_\lambda = (q^{\lambda_1} t^{r-1}, q^{\lambda_2} t^{r-2}, \dots, q^{\lambda_r}).$$

Prove the polynomial Fourier-duality theorem for the displayed  $GL_r$  convention [18]: for generic  $q, t$ , the polynomial representation and its contragredient spectral representation are joined by a nondegenerate transform that exchanges the commuting multiplication and Cherednik operator algebras, with the parameter and inverse conventions determined by the contragredient pairing. In the spherical monic normalization, its kernel is normalized to equal 1 at  $(t^\rho, t^\rho)$  and is symmetric in its two spectral labels. Apply this symmetry to prove the evaluation identity

$$\frac{P_\lambda(\xi_\mu; q, t)}{P_\lambda(t^\rho; q, t)} = \frac{P_\mu(\xi_\lambda; q, t)}{P_\mu(t^\rho; q, t)}.$$

Derive the duality between multiplication recurrences in  $\lambda$  and the commuting Macdonald difference equations in the spectral variable.

**Exercise 13.19** (Macdonald pairings and the Cauchy identity). Let  $\Lambda_{\text{sym}} = \mathbb{C}[p_1, p_2, \dots]$  be graded by  $\deg p_r = r$ . For a partition  $\mu$ , let  $m_r(\mu)$  be the number of its parts equal to  $r$ , and define

$$z_\mu = \prod_{r \geq 1} r^{m_r(\mu)} m_r(\mu)!, \quad p_\mu = \prod_j p_{\mu_j}, \quad p_r(x) = \sum_i x_i^r.$$

Define the symmetric-function Macdonald pairing by

$$\langle p_\mu, p_\nu \rangle_{q,t}^{\text{sym}} = \delta_{\mu\nu} z_\mu \prod_j \frac{1 - q^{\mu_j}}{1 - t^{\mu_j}}.$$

For a cell  $s = (i, j)$  in the diagram of  $\lambda$ , define its arm and leg by

$$a_\lambda(s) = \lambda_i - j, \quad \ell_\lambda(s) = \lambda'_j - i,$$

and put

$$b_\lambda(q, t) = \prod_{s \in \lambda} \frac{1 - q^{a_\lambda(s)} t^{\ell_\lambda(s)+1}}{1 - q^{a_\lambda(s)+1} t^{\ell_\lambda(s)}}.$$

Write the individual factor as

$$\mathbf{b}_\lambda(s) = \frac{1 - q^{a_\lambda(s)} t^{\ell_\lambda(s)+1}}{1 - q^{a_\lambda(s)+1} t^{\ell_\lambda(s)}}.$$

If  $\lambda/\mu$  is a vertical strip, let  $C_{\lambda/\mu}$  and  $R_{\lambda/\mu}$  be the cells of  $\mu$  in columns and rows meeting the strip, respectively, and define

$$\psi'_{\lambda/\mu} = \prod_{s \in C_{\lambda/\mu} \setminus R_{\lambda/\mu}} \frac{\mathbf{b}_\lambda(s)}{\mathbf{b}_\mu(s)}.$$

Prove that the finite-variable Macdonald polynomials are stable under adjoining a zero variable and hence determine elements  $P_\lambda$  of  $\Lambda_{\text{sym}}$ . Put  $Q_\lambda = b_\lambda(q, t)P_\lambda$ . Use the adjointness of the DAHA difference operators and triangularity to prove

$$\langle P_\lambda, P_\mu \rangle_{q,t}^{\text{sym}} = \delta_{\lambda\mu} b_\lambda(q, t)^{-1}.$$

Deduce that  $P_\lambda$  and  $Q_\lambda$  are dual. Expand the reproducing kernel in the power sums and use the  $q$ -binomial identity to prove

$$\sum_\lambda P_\lambda(a; q, t) Q_\lambda(b; q, t) = \prod_{i,j} \frac{(ta_i b_j; q)_\infty}{(a_i b_j; q)_\infty}$$

for finite alphabets with  $|a_i b_j| < 1$ . Prove the first Macdonald Pieri identity

$$(x_1 + \cdots + x_r) P_\mu = \sum_{\substack{\lambda \supset \mu \\ |\lambda| - |\mu| = 1}} \psi'_{\lambda/\mu} P_\lambda$$

in the displayed monic normalization, and derive its adjoint from the Macdonald pairing.

**Exercise 13.20** (Finite Macdonald measures). Let  $0 \leq q, t < 1$ , let  $a = (a_1, \dots, a_N)$  and  $b = (b_1, \dots, b_M)$  be nonnegative with  $a_i b_j < 1$ , and let  $P_\lambda, Q_\lambda$  be the Macdonald functions. Define

$$\Pi(a; b) = \sum_\lambda P_\lambda(a; q, t) Q_\lambda(b; q, t), \quad \mathbb{P}_{q,t}(\lambda) = \frac{P_\lambda(a; q, t) Q_\lambda(b; q, t)}{\Pi(a; b)}.$$

Apply the first Macdonald difference operator in the  $a$  variables to the Cauchy identity. For distinct  $a_i$ , prove

$$\mathbb{E}_{q,t} \left[ \sum_{i=1}^N q^{\lambda_i} t^{N-i} \right] = \sum_{i=1}^N \prod_{j \neq i} \frac{ta_i - a_j}{a_i - a_j} \prod_{k=1}^M \frac{1 - a_i b_k}{1 - ta_i b_k},$$

and extend the identity to coincident parameters by continuity. Verify the specialization  $q = t$  directly and the  $q$ -Whittaker specialization  $t = 0$ . For samples from the finite measure, identify the displayed expectation as a directly estimable statistic and state the positivity, truncation, and sampling assumptions required for comparison with the formula.



**Exercise 13.21** (The Calogero–Sutherland limit). Write  $x_i = e^{z_i}$ , set  $q = e^{-\epsilon}$  and  $t = e^{-g\epsilon}$ , and expand  $D_{q,t}$  through order  $\epsilon^2$ . After subtracting its scalar value on 1 and removing the center-of-mass drift, obtain

$$\mathcal{L}_g = \sum_i \partial_{z_i}^2 + g \sum_{i < j} \coth\left(\frac{z_i - z_j}{2}\right) (\partial_{z_i} - \partial_{z_j}).$$

On the circle, put  $z_i = i\theta_i$  and conjugate by

$$\Psi_0(\theta) = \prod_{i < j} \left| 2 \sin \frac{\theta_i - \theta_j}{2} \right|^g.$$

Recover, up to its ground-state energy, the Sutherland Hamiltonian

$$H_g = - \sum_i \partial_{\theta_i}^2 + \frac{g(g-1)}{2} \sum_{i < j} \frac{1}{\sin^2((\theta_i - \theta_j)/2)}.$$

Compute the eigenfunctions of degrees zero and one and, for  $r = 2$ , degree two. Identify their Sutherland eigenvalues.

### 13.4 Integrable Particle Spectra

**Exercise 13.22** (The Sutherland model and Jack polynomials). For  $N$  identical particles of mass  $m$  on a ring of radius  $R$  and  $g \geq 1$ , let

$$\text{Conf}_N(S^1) = \{(\theta_1, \dots, \theta_N) : \theta_i \neq \theta_j \text{ for } i \neq j\}.$$

On  $C_c^\infty(\text{Conf}_N(S^1))^{S_N}$  define the nonnegative quadratic form

$$q_g[\psi] = \frac{\hbar^2}{2mR^2} \int_{(S^1)^N} \left[ \sum_{j=1}^N |\partial_{\theta_j} \psi|^2 + \frac{g(g-1)}{2} \sum_{i < j} \frac{|\psi|^2}{\sin^2((\theta_i - \theta_j)/2)} \right] d\theta.$$

Prove [57] that  $q_g$  is closable for  $g \geq 1$ . Define  $\hat{H}_g$  to be the unique self-adjoint operator associated with its closure in  $L^2((S^1)^N)^{S_N}$ ; this is the Friedrichs realization of the displayed differential expression. Define

$$\Psi_0(\theta) = \prod_{i < j} |e^{i\theta_i} - e^{i\theta_j}|^g.$$

For partitions of equal size, write  $\mu < \lambda$  in dominance order when

$$\sum_{i=1}^k \mu_i \leq \sum_{i=1}^k \lambda_i$$

for every  $k \geq 1$ , with missing parts interpreted as zero, and with at least one inequality strict. For a partition  $\lambda$  of length at most  $N$ , let  $m_\lambda$  be the sum of the distinct permutations of  $z_1^{\lambda_1} \cdots z_N^{\lambda_N}$ .

The Jack polynomial  $P_\lambda^{(1/g)}$  is the symmetric polynomial normalized by

$$P_\lambda^{(1/g)} = m_\lambda + \sum_{\mu < \lambda} c_{\lambda\mu}(g) m_\mu$$

and by satisfying

$$\left[ \sum_i (z_i \partial_{z_i})^2 + g \sum_{i < j} \frac{z_i + z_j}{z_i - z_j} (z_i \partial_{z_i} - z_j \partial_{z_j}) \right] P_\lambda^{(1/g)} = \left[ \sum_i (\lambda_i^2 + g(N+1-2i)\lambda_i) \right] P_\lambda^{(1/g)};$$

here  $<$  is dominance order.

Conjugate the quantum Hamiltonian by  $\Psi_0$ , identify its commuting quantum integrals, and prove that

$$\Psi_\lambda = \Psi_0 P_\lambda^{(1/g)}(e^{i\theta_1}, \dots, e^{i\theta_N})$$

is a joint eigenfunction with

$$E_\lambda = \frac{\hbar^2}{2mR^2} \left[ \frac{g^2 N(N^2 - 1)}{12} + \sum_{j=1}^N (\lambda_j^2 + g(N+1-2j)\lambda_j) \right].$$

**Exercise 13.23** (Jack transition spectrum). Let  $\rho_r = \sum_{j=1}^N e^{ir\theta_j}$  be a density mode. Derive the Jack Pieri identity in orthonormal normalization for  $\rho_1 P_\lambda^{(1/g)}$  and show that only nonincreasing tuples obtained from  $\lambda$  by increasing one entry by one occur. For a weak periodic drive proportional to  $\rho_1 e^{-i\Omega t} + \rho_{-1} e^{i\Omega t}$ , deduce the allowed resonance frequencies

$$\Omega_{\mu\lambda} = \frac{|E_\mu - E_\lambda|}{\hbar}$$

and express their relative intensities as squared normalized Jack Pieri coefficients. State how spectroscopy of particles confined to a ring determines  $g$  from the line shifts and tests the Jack selection rule from the missing lines.

**Exercise 13.24** (Hypergeometric degeneration). If no denominator parameter  $b_j$  is a nonpositive integer, define

$${}_rF_s \left( \begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; z \right) = \sum_{m=0}^{\infty} \frac{(a_1)_m \cdots (a_r)_m}{(b_1)_m \cdots (b_s)_m} \frac{z^m}{m!}.$$

The series is finite when a numerator parameter is a nonpositive integer.

Set  $q = e^{-\varepsilon}$ . Prove

$$\lim_{\varepsilon \rightarrow 0^+} \frac{(q^a; q)_m}{(1-q)^m} = (a)_m, \quad \lim_{q \rightarrow 1^-} (1-q)^{1-a} \frac{(q; q)_\infty}{(q^a; q)_\infty} = \Gamma(a).$$

Deduce

$$\lim_{q \rightarrow 1^-} {}_r\phi_{r-1} \left( \begin{matrix} q^{a_1}, \dots, q^{a_r} \\ q^{b_1}, \dots, q^{b_{r-1}} \end{matrix}; q, z \right) = {}_rF_{r-1} \left( \begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_{r-1} \end{matrix}; z \right).$$

Derive Gauss's summation from the  $q$ -Gauss summation. Prove the corresponding degeneration of the basic hypergeometric  $q$ -difference equation to the Gauss equation.

**Exercise 13.25** ( $BC_r$  weights and the Koornwinder operator). Write  $f(z^{\pm m}) = f(z^m)f(z^{-m})$  and

$$f(z_i^{\pm 1} z_j^{\pm 1}) = \prod_{\epsilon, \delta = \pm 1} f(z_i^\epsilon z_j^\delta).$$

For  $0 < q, t < 1$  and  $a_1, a_2, a_3, a_4 \in (-1, 1)$ , define

$$\Delta_{BC_r}(z) = \prod_{i=1}^r \frac{(z_i^{\pm 2}; q)_\infty}{\prod_{s=1}^4 (a_s z_i^{\pm 1}; q)_\infty} \times \prod_{1 \leq i < j \leq r} \frac{(z_i^{\pm 1} z_j^{\pm 1}; q)_\infty}{(t z_i^{\pm 1} z_j^{\pm 1}; q)_\infty}.$$

Use the normalized torus measure

$$\frac{1}{2^r r!} \prod_{i=1}^r \frac{dz_i}{2\pi i z_i}.$$

For  $z = (z_1, \dots, z_r)$  put

$$A_i(z) = \frac{\prod_{s=1}^4 (1 - a_s z_i)}{(1 - z_i^2)(1 - q z_i^2)} \prod_{j \neq i} \frac{(1 - t z_i z_j)(1 - t z_i / z_j)}{(1 - z_i z_j)(1 - z_i / z_j)}.$$

On Laurent polynomials invariant under permutations and inversions define

$$\mathcal{D}_K = \sum_{i=1}^r \left[ A_i(z)(T_{q, z_i} - 1) + A_i(z^{-1})(T_{q, z_i}^{-1} - 1) \right].$$

Prove that  $\Delta_{BC_r}$  is invariant under permutations and independent inversions of the  $z_i$ . Identify its factors with the roots  $\pm e_i$ ,  $\pm 2e_i$ , and  $\pm e_i \pm e_j$ . For  $r = 1$ , recover the Askey–Wilson weight and its order-two Weyl normalization, and identify  $\mathcal{D}_K$  with  $\mathcal{D}_{AW}$ .

**Exercise 13.26** (Koornwinder spectral theory and  $BC_r$  frequencies). The Koornwinder polynomial  $K_\lambda$  is the Weyl-invariant Laurent polynomial triangular in orbit sums and satisfying

$$\mathcal{D}_K K_\lambda = \left( \sum_{i=1}^r \left[ a_1 a_2 a_3 a_4 q^{-1} t^{2r-i-1} (q^{\lambda_i} - 1) + t^{i-1} (q^{-\lambda_i} - 1) \right] \right) K_\lambda,$$

with the scalar eigenvalue fixed by the displayed convention.

Prove Koornwinder's symmetry theorem [44]:  $\mathcal{D}_K$  is symmetric for the  $\Delta_{BC_r}$  inner product. Deduce that its triangular eigenfunctions are mutually orthogonal when the displayed eigenvalues are distinct.

Assume  $r \geq 2$ ,  $M \geq 2$ ,  $0 < q, t < 1$ ,  $a_1, a_2, a_3, a_4 \in (-1, 1)$ , and the displayed eigenvalues are distinct on

$$\Lambda_M = \{\lambda : \ell(\lambda) \leq r, |\lambda| \leq M\}.$$

Let  $\mathcal{H}_M$  be the span of the normalized Koornwinder polynomials  $K_\lambda$ ,  $\lambda \in \Lambda_M$ , and define the finite programmable  $q$ -difference Hamiltonian

$$\hat{H}_M = \varepsilon \mathcal{D}_K|_{\mathcal{H}_M}, \quad \varepsilon > 0.$$

Prove that  $\mathcal{H}_M$  is invariant and that  $\hat{H}_M$  is self-adjoint with spectral values

$$E_\lambda = \varepsilon \sum_{i=1}^r \left[ a_1 a_2 a_3 a_4 q^{-1} t^{2r-i-1} (q^{\lambda_i} - 1) + t^{i-1} (q^{-\lambda_i} - 1) \right].$$

For prepared normalized eigenstates  $K_\lambda$  and  $K_\mu$ , derive the spectral spacing

$$\Omega_{\mu\lambda} = \frac{|E_\mu - E_\lambda|}{\hbar}$$

and compute all pairwise gaps among the four levels indexed by  $0, (1), (2), (1, 1)$ . State the positivity, finite-truncation, state-preparation, resolved-partition, energy-calibration, and nondegeneracy assumptions under which the observed lines test the Koornwinder eigenvalue formula.

**Exercise 13.27** (Askey–Wilson spectrum). Fix  $M \geq 4$  and parameters  $a, b, c, d \in (-1, 1)$  for which the Askey–Wilson weight is positive and the first  $M + 1$  eigenvalues are distinct. Let  $\mathcal{H}_M$  be the span of the normalized Askey–Wilson polynomials  $p_0, \dots, p_M$  with their positive Askey–Wilson inner product. Use the self-adjoint Hamiltonian  $\hat{H}_M = \varepsilon \mathcal{D}_{AW}|_{\mathcal{H}_M}$ , where  $\varepsilon > 0$  is a calibrated energy scale. Prove that its spectral values are

$$E_n = \varepsilon(q^{-n} - 1)(1 - abcdq^{n-1}).$$

Write the three-term recurrence in the form

$$(z + z^{-1})p_n = A_n p_{n+1} + B_n p_n + C_n p_{n-1}$$

and compute the corresponding normalized matrix elements. Deduce that a periodic perturbation proportional to the compression of  $z + z^{-1}$  to  $\mathcal{H}_M$  produces only neighboring transitions, with resonance frequencies  $|E_{n+1} - E_n|/\hbar$  and intensities determined by  $A_n$  and  $C_{n+1}$ . Assuming  $\varepsilon$  is known and at least three nondegenerate lines are resolved, derive the equations satisfied by  $q$  and the product  $abcd$ . Compute the Jacobian of two independent gap ratios and state the nonvanishing and parameter-range hypotheses under which these two quantities are locally identifiable.

Form the data map from  $(a, b, c, d)$  to four selected squared normalized neighboring matrix elements. Prove that it is invariant under permutation of the parameters. At a chosen parameter point, compute its differential in the four elementary symmetric functions and state the full-rank hypothesis under which the measured strengths locally determine the unordered set  $\{a, b, c, d\}$ . When this rank condition fails, identify the remaining fiber instead of asserting identifiability; state also that squared strengths cannot distinguish any additional discrete ambiguity in that fiber. State that these are predictions for the engineered simulator.

**Exercise 13.28** (Macdonald–Ruijsenaars transition spectrum). Let  $0 < q, t < 1$ , fix  $M \geq 0$ , and define

$$\mathcal{H}_{q,t}^{(M)} = \text{span}\{P_\lambda(x; q, t) : |\lambda| \leq M\}$$

with the positive Macdonald inner product. Put

$$d_0 = \sum_{i=1}^r t^{r-i}, \quad \hat{H}_{q,t}^{(M)} = \varepsilon(d_0 - D_{q,t})|_{\mathcal{H}_{q,t}^{(M)}}, \quad \varepsilon > 0.$$

Identify  $D_{q,t}$  with the first symmetric function of the commuting DAHA winding operators and derive its compatibility with the Hecke exchange relations. Prove that  $\mathcal{H}_{q,t}^{(M)}$  is invariant, that  $\hat{H}_{q,t}^{(M)}$  is self-adjoint, and that the normalized Macdonald states have energies

$$E_\lambda = \varepsilon \sum_{i=1}^r t^{r-i} (1 - q^{\lambda_i}).$$

Model a finite programmable  $q$ -difference simulator by this operator. Use the displayed Macdonald Pieri identity for  $\rho_1 = x_1 + \cdots + x_r$  and its adjoint to determine every nonzero matrix element of the compression of  $\rho_1 + \rho_1^*$  to  $\mathcal{H}_{q,t}^{(M)}$ . For a periodic drive proportional to this compressed observable, derive the resonance frequencies

$$\Omega_{\mu\lambda} = |E_\mu - E_\lambda|/\hbar$$

and the relative intensities as squared normalized Macdonald Pieri coefficients. Explain that recovery of  $q, t$  requires known  $\varepsilon$  and  $r$ , resolved partition labels, sufficiently many nondegenerate transitions, and a fixed convention for any discrete parameter ambiguity, and that the prediction concerns the engineered simulator.

**Exercise 13.29** (Macdonald–Jack degeneration). Set  $q = e^{-\delta}$  and  $t = e^{-g\delta}$ ; after the scalar and center-of-mass normalizations in the Calogero–Sutherland limit exercise, prove that

$$\lim_{\delta \downarrow 0} P_\lambda(x; e^{-\delta}, e^{-g\delta}) = P_\lambda^{(1/g)}(x)$$

in the monic normalization. Prove that the rescaled spectral gaps and normalized Macdonald Pieri coefficients converge to the Sutherland energies and Jack Pieri coefficients. State how measured line locations and strengths test this degeneration. Explain that recovery of the deformation parameter requires resolved partition labels and sufficiently many nondegenerate transitions.

**Exercise 13.30** (Elliptic Ruijsenaars spectral tori). Let  $0 < p < 1$  and choose  $\tau \in i\mathbb{R}_{>0}$  with  $p = e^{2\pi i\tau}$ .

- (a) Prove that multiplication by  $p$  acts freely and properly discontinuously on  $\mathbb{C}^\times$  and that the exponential map induces a biholomorphism

$$E_p := \mathbb{C}^\times / p^\mathbb{Z} \cong \mathbb{C} / (\mathbb{Z} + \tau\mathbb{Z}).$$

Call this compact complex torus the *multiplicative elliptic torus*. For fixed  $t \in \mathbb{C}^\times$ , compute

$$\frac{\theta(tpz; p)}{\theta(pz; p)} = t^{-1} \frac{\theta(tz; p)}{\theta(z; p)}$$

and interpret this automorphy as a meromorphic section on  $E_p$ .

- (b) Put

$$m(\tau) = \frac{\vartheta_2(\tau)^4}{\vartheta_3(\tau)^4}.$$

Prove the analytic Legendre uniformization theorem [29, 26]: the torus  $E_p$  is biholomorphic to the smooth compactification of

$$C_m : \quad y^2 = x(x-1)(x-m(\tau)).$$

Prove from the theta series that

$$m(\tau) = 16p^{1/2} + O(p) \quad (p \downarrow 0),$$

so the elliptic torus degenerates toward the double-root cubic  $y^2 = x^2(x-1)$ .

- (c) For type  $A_{r-1}$  and  $0 < q, t < 1$ , define

$$D_{p,q,t}^{\text{ell}} = \sum_{i=1}^r \left( \prod_{j \neq i} \frac{\theta(tx_i/x_j; p)}{\theta(x_i/x_j; p)} \right) T_{q,x_i}.$$

Determine the automorphy of each coefficient under multiplication of a ratio  $x_i/x_j$  by  $p$ . Prove directly from  $\theta(z; 0) = 1 - z$  that

$$D_{p,q,t}^{\text{ell}} \longrightarrow D_{q,t} \quad (p \rightarrow 0)$$

locally uniformly away from the polar divisors. After taking this trigonometric degeneration, set  $q = e^{-\varepsilon}$  and  $t = e^{-g\varepsilon}$ , derive the Heckman–Opdam differential operator from the second-order term as  $\varepsilon \rightarrow 0$ , and take its rational scaling limit to obtain the rational Calogero–Moser operator.

(d) Fix a finite multiplicative grid

$$\mathcal{G}_M = \{(q^{n_1} x_1^0, \dots, q^{n_r} x_r^0) : -M \leq n_i \leq M\}$$

whose ratios avoid every displayed polar divisor for  $0 \leq p \leq p_0$ . Let  $C_p^{(M)}$  be the compression of  $D_{p,q,t}^{\text{ell}}$  to functions on  $\mathcal{G}_M$ , discarding a shifted value when it leaves the grid, and define the self-adjoint programmable Hamiltonian

$$\hat{H}_p^{(M)} = \frac{\epsilon}{2} \left( C_p^{(M)} + (C_p^{(M)})^* \right), \quad \epsilon > 0.$$

Prove entrywise and operator-norm convergence

$$\hat{H}_p^{(M)} \longrightarrow \hat{H}_0^{(M)} \quad (p \downarrow 0),$$

and hence convergence of every ordered spectral value and transition frequency. At a parameter where a selected nondegenerate gap has nonzero derivative with respect to  $p$ , prove that its measured value locally determines  $p$  and therefore the Legendre modulus  $m(\tau)$ . Predict the  $p$ -dependent hopping amplitudes, ordered gaps, and their trigonometric limits. State the finite-grid, boundary-compression, coupling-calibration, pole-avoidance, state-preparation, and spectral-resolution assumptions under which the measured hopping amplitudes and resonance gaps test the theta-function deformation and its nodal degeneration.

## 14 Algebraic Curves

### 14.1 Legendre Involution

**Exercise 14.1** (Affine schemes and the Legendre family). Let  $m$  be an indeterminate and form the finitely generated  $\mathbb{C}[m]$ -algebra

$$A = \mathbb{C}[m, x, y] / (y^2 - x(x-1)(x-m)).$$

For a commutative ring  $R$ , an ideal  $\mathfrak{p} \subsetneq R$  is *prime* when  $ab \in \mathfrak{p}$  implies  $a \in \mathfrak{p}$  or  $b \in \mathfrak{p}$ . Define

$$\text{Spec } R = \{\mathfrak{p} : \mathfrak{p} \text{ is prime}\}, \quad V(I) = \{\mathfrak{p} : I \subseteq \mathfrak{p}\}, \quad D(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}.$$

Prove that the sets  $V(I)$  are the closed sets of a topology and that the principal open sets  $D(f)$  are closed under finite intersections and form a basis for the topology.

For  $f \in R$ , define the localization  $R_f$  by the universal property that  $f$  becomes invertible and every homomorphism from  $R$  in which  $f$  is invertible factors uniquely through  $R_f$ . A *presheaf* of rings on

a space  $X$  assigns a ring  $\mathcal{F}(U)$  to each open set and compatible restriction maps. It is a *sheaf* when compatible local sections on an open cover glue uniquely. Set

$$\mathcal{O}_{\mathrm{Spec} R}(D(f)) = R_f$$

with the localization restriction maps. Prove the sheaf gluing condition on principal-open covers, extend this prescription uniquely to a sheaf on  $\mathrm{Spec} R$ , and prove

$$R \cong \Gamma(\mathrm{Spec} R, \mathcal{O}_{\mathrm{Spec} R}).$$

For a sheaf of rings  $\mathcal{F}$  and  $p \in X$ , define its stalk by

$$\mathcal{F}_p = \varinjlim_{p \in U} \mathcal{F}(U).$$

A commutative ring is *local* when it has a unique maximal ideal. A *locally ringed space* is a space  $X$  with a sheaf of commutative rings  $\mathcal{O}_X$  whose stalks are local rings. Prove

$$\mathcal{O}_{\mathrm{Spec} R, \mathfrak{p}} \cong R_{\mathfrak{p}}, \quad \mathfrak{m}_{\mathfrak{p}} = \mathfrak{p}R_{\mathfrak{p}}.$$

For a continuous map  $f : X \rightarrow Y$ , set  $(f_*\mathcal{O}_X)(U) = \mathcal{O}_X(f^{-1}(U))$ . A morphism of locally ringed spaces

$$(f, f^\#) : (X, \mathcal{O}_X) \longrightarrow (Y, \mathcal{O}_Y)$$

consists of  $f$  and a morphism of sheaves  $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$  whose induced homomorphisms on stalks are local. Here a morphism of sheaves of rings is a family of ring homomorphisms on open sets commuting with restriction maps, and a homomorphism of local rings is *local* when the inverse image of the maximal ideal is the maximal ideal.

An *affine scheme* is a locally ringed space isomorphic to  $(\mathrm{Spec} R, \mathcal{O}_{\mathrm{Spec} R})$ . A *scheme* is a locally ringed space admitting an open cover by affine schemes, and a *scheme morphism* is a morphism of locally ringed spaces. A *complex scheme* is a scheme equipped with a morphism to  $\mathrm{Spec} \mathbb{C}$ , and its morphisms are required to commute with these structure morphisms. It is *of finite type* when it admits a finite affine open cover by spectra of finitely generated commutative  $\mathbb{C}$ -algebras.

For a homomorphism  $\varphi : R \rightarrow S$ , prove that inverse image of prime ideals and localization induce a scheme morphism

$$\varphi^* : \mathrm{Spec} S \longrightarrow \mathrm{Spec} R$$

and establish the affine anti-equivalence

$$\mathrm{Hom}_{\mathrm{Sch}}(\mathrm{Spec} S, \mathrm{Spec} R) \cong \mathrm{Hom}_{\mathrm{Ring}}(R, S).$$

The inclusion  $\mathbb{C}[m] \rightarrow A$  therefore defines the Legendre-family morphism

$$\pi : \mathcal{E}_{\mathrm{aff}} = \mathrm{Spec} A \longrightarrow \mathbb{A}_m^1 = \mathrm{Spec} \mathbb{C}[m].$$

Prove that both schemes are complex and of finite type and that  $\pi$  commutes with their structure morphisms to  $\mathrm{Spec} \mathbb{C}$ . For  $m_0 \in \mathbb{C}$ , define its algebraic fiber by

$$\mathcal{E}_{m_0}^{\mathrm{aff}} = \mathrm{Spec} \left( A \otimes_{\mathbb{C}[m]} \mathbb{C}[m] / (m - m_0) \right)$$

Prove the weak Nullstellensatz [36]: every maximal ideal of  $\mathbb{C}[z_1, \dots, z_n]$  is the kernel of evaluation at a unique point of  $\mathbb{C}^n$ . Identify the closed points of the fiber with

$$y^2 = x(x-1)(x-m_0).$$

Show that replacing an ideal by its radical preserves the closed points but can change the structure sheaf. Compare  $\operatorname{Spec} \mathbb{C}[t]/(t)$  with  $\operatorname{Spec} \mathbb{C}[t]/(t^2)$ , identify the nonzero nilpotent class in the latter ring, and compute its square.

Let  $\iota : A \rightarrow A$  fix  $m, x$  and send  $y$  to  $-y$ . Prove that it is an involution over  $\mathbb{C}[m]$ , that

$$A^\iota = \mathbb{C}[m, x],$$

and that the quotient morphism is the double-cover map to the  $x$ -line. Determine its fixed closed points on every fiber.

**Exercise 14.2** (Local rings, tangent spaces, and singular fibers). For  $p \in X = \operatorname{Spec} R$  corresponding to  $\mathfrak{p}$ , define the residue field, cotangent space, and tangent space by

$$\kappa(p) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}, \quad T_p^*X = \mathfrak{p}R_{\mathfrak{p}}/(\mathfrak{p}R_{\mathfrak{p}})^2, \quad T_pX = \operatorname{Hom}_{\kappa(p)}(T_p^*X, \kappa(p)).$$

Define the Krull dimension of a ring as the supremum of the lengths of strict chains of prime ideals and define the local dimension of  $X$  at  $p$  as  $\dim R_{\mathfrak{p}}$ .

For a  $\mathbb{C}$ -algebra  $R$ , define the module of Kähler differentials  $\Omega_{R/\mathbb{C}}^1$  by a derivation  $d : R \rightarrow \Omega_{R/\mathbb{C}}^1$  universal among  $\mathbb{C}$ -linear derivations from  $R$  to  $R$ -modules. If

$$R = \mathbb{C}[x_1, \dots, x_n]/(f_1, \dots, f_r),$$

derive the conormal sequence and prove at a closed point  $p$  that

$$T_pX = \{v \in \mathbb{C}^n : df_j(p)(v) = 0 \text{ for } 1 \leq j \leq r\}.$$

Prove the Jacobian criterion over  $\mathbb{C}$  [36]:  $p$  is smooth exactly when  $\dim_{\mathbb{C}} T_pX = \dim \mathcal{O}_{X,p}$ .

Apply the criterion to the fibers  $\mathcal{E}_m^{\text{aff}}$  constructed above. Prove that they are smooth for  $m \notin \{0, 1\}$  and that the fibers at  $m = 0$  and  $m = 1$  have singular points  $(0, 0)$  and  $(1, 0)$ , respectively. Compute the local dimension at every closed point of these fibers and compare it with the tangent-space dimension at the two singular points. Define the tangent cone of a plane curve at a point by the lowest nonzero homogeneous part of a translated local equation and prove that both singular fibers have two distinct tangent directions over  $\mathbb{C}$ .

Determine the smooth and singular points of

$$y^2 - x^3, \quad y^2 - x^2(x + 1), \quad xy,$$

and distinguish the cusp, node, and crossing by their local rings and tangent cones. For the morphism

$$\operatorname{Spec} \mathbb{C}[u, t]/(t^2 - u) \longrightarrow \operatorname{Spec} \mathbb{C}[u],$$

show that every fiber over  $u \neq 0$  consists of two reduced points, whereas the fiber over  $u = 0$  is  $\operatorname{Spec} \mathbb{C}[t]/(t^2)$ , and compute its tangent space and length-two local algebra.

**Exercise 14.3** (Function fields, rational maps, and normalization). Call a curve *integral* when it is irreducible and reduced. For an integral affine curve  $U = \operatorname{Spec} A$ , define its function field by

$$K(U) = \operatorname{Frac}(A).$$

If  $U$  and  $V$  are nonempty affine open subsets of the same integral curve  $C$ , use principal affine neighborhoods and localization to construct a canonical isomorphism  $K(U) \cong K(V)$ . Denote the resulting field by  $K(C)$  and identify it with the local ring at the generic point.



A *rational map*  $C \dashrightarrow D$  is a morphism from a nonempty open subset of  $C$  to  $D$ , with two representatives equivalent when they agree on a nonempty open subset. It is *dominant* when its image is dense. Prove the contravariant correspondence

$$\{\text{dominant rational maps } C \dashrightarrow D\} \longleftrightarrow \{\mathbb{C}\text{-field embeddings } K(D) \hookrightarrow K(C)\}$$

for integral curves of finite type over  $\mathbb{C}$ .

Call an integral scheme *normal* when all of its local rings are integrally closed domains. For  $C = \text{Spec } A$ , let  $\tilde{A}$  be the integral closure of  $A$  in  $K(C)$ . Prove Noether normalization in dimension one and use it to prove [36] that  $\tilde{A}$  is a finite  $A$ -module for a one-dimensional finitely generated  $\mathbb{C}$ -domain. Prove that the affine normalizations glue to a finite birational morphism

$$\nu : \tilde{C} \longrightarrow C$$

and prove its universal property: every dominant morphism from a normal integral curve to  $C$  factors uniquely through  $\nu$ . Show that every smooth curve is normal by proving that its local rings at closed points are one-dimensional regular local rings and hence discrete valuation rings.

For the cusp and node, prove that  $\mathbb{C}[t]$  is the integral closure of the respective coordinate ring in

$$\mathbb{C}[t^2, t^3] \subset \mathbb{C}[t], \quad \mathbb{C}[t^2 - 1, t(t^2 - 1)] \subset \mathbb{C}[t]$$

inside its function field. Deduce the affine normalization maps

$$t \longmapsto (t^2, t^3), \quad t \longmapsto (t^2 - 1, t(t^2 - 1)).$$

Show that the cusp has one point above its singularity, whereas the node has the two points  $t = 1$  and  $t = -1$  above its singularity, and relate this distinction to the tangent cones computed above.

**Exercise 14.4** (Projective curves and the Legendre compactification). Let  $R = \bigoplus_{n \geq 0} R_n$  be a finitely generated graded  $\mathbb{C}$ -algebra generated by  $R_1$ , and let  $R_+ = \bigoplus_{n > 0} R_n$ . Define

$$\text{Proj } R = \{\mathfrak{p} \subset R : \mathfrak{p} \text{ is homogeneous prime and } R_+ \not\subset \mathfrak{p}\}.$$

For homogeneous  $f$  of positive degree, define

$$D_+(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}, \quad \mathcal{O}_{\text{Proj } R}(D_+(f)) = (R_f)_0.$$

Prove that these affine charts glue to a scheme. For a finite-dimensional complex vector space  $V$ , prove

$$\mathbb{P}(V) = \text{Proj}(\text{Sym } V^\vee)$$

and recover its interpretation as the space of complex lines in  $V$ .

For a homogeneous ideal  $I \subset R$ , define the projective subscheme  $\text{Proj}(R/I)$ . Show that homogenization of an affine ideal gives its projective closure and identify the points added at infinity for a parabola and a hyperbola. Call a complex scheme *projective* when it is isomorphic to a closed subscheme of some  $\mathbb{P}(V)$ . Define a *projective algebraic curve* to be a projective complex scheme of finite type whose irreducible components all have Krull dimension one. For a projective plane curve  $C = V(F)$  defined by a homogeneous polynomial, define  $\deg C = \deg F$ .

For  $m \in \mathbb{C} \setminus \{0, 1\}$ , construct the projective closure

$$E_m : Y^2 Z = X(X - Z)(X - mZ) \subset \mathbb{P}^2.$$

Prove that  $E_m$  is a projective algebraic curve of degree three. Use the Jacobian criterion on its affine charts to prove that  $E_m$  is nonsingular and that it has the unique point  $O = [0 : 1 : 0]$  over the line at infinity. Prove that the rational function  $x = X/Z$  extends to a morphism

$$\pi_m : E_m \longrightarrow \mathbb{P}^1, \quad \pi_m([X : Y : Z]) = [X : Z] \quad ([X : Y : Z] \neq O), \quad \pi_m(O) = [1 : 0].$$

**Exercise 14.5** (Intersection multiplicity, resultants, and discriminants). Let  $C = V(F)$  and  $D = V(G)$  be projective plane curves with no common component. At  $p \in C \cap D$ , let  $F_p, G_p \in \mathcal{O}_{\mathbb{P}^2, p}$  be local equations and define their local intersection multiplicity by

$$I_p(C, D) = \dim_{\mathbb{C}} \mathcal{O}_{\mathbb{P}^2, p} / (F_p, G_p).$$

Prove that this length is finite, is invariant under multiplication of  $F$  and  $G$  by local units, and equals one exactly for a transverse intersection of smooth branches.

For a two-dimensional vector space  $W$  and binary forms  $F \in \text{Sym}^m W^\vee$  and  $G \in \text{Sym}^n W^\vee$ , define their resultant as the determinant-line element induced by

$$\text{Sym}^{n-1} W^\vee \oplus \text{Sym}^{m-1} W^\vee \longrightarrow \text{Sym}^{m+n-1} W^\vee, \quad (A, B) \longmapsto AF + BG;$$

it lies in the highest exterior power of the target tensored with the dual of the highest exterior power of the source, so its vanishing is intrinsic although a scalar representative depends on compatible volume elements. Prove that it vanishes exactly when  $F$  and  $G$  have a common projective zero. For  $f(t) = a_n t^n + \cdots + a_0$ , normalize the scalar resultant by the monomial volume elements and define

$$\text{Disc}(f) = (-1)^{n(n-1)/2} a_n^{-1} \text{Res}(f, f').$$

Prove that

$$\text{Disc}(t^3 + at + b) = -4a^3 - 27b^2.$$

Compute also

$$\text{Disc}(x(x-1)(x-m)) = m^2(1-m)^2$$

and recover algebraically the two singular Legendre fibers found by the Jacobian criterion.

Prove Bézout's theorem for plane curves [36]:

$$\sum_{p \in C \cap D} I_p(C, D) = \deg C \deg D.$$

Derive the possible intersection patterns of a line with a smooth cubic and prove that tangency is detected by a repeated root and hence by the corresponding discriminant locus.

**Exercise 14.6** (Analytification of projective curves and algebraic-holomorphic comparison). Let  $C$  be a smooth projective complex algebraic curve. On an affine chart  $U = \text{Spec } A$  presented inside  $\mathbb{A}^n$ , give the closed points  $U(\mathbb{C})$  the subspace topology from  $\mathbb{C}^n$  and give them the sheaf of functions locally representable by convergent power series. Prove the holomorphic implicit-function theorem from the contraction mapping theorem, and combine it with the Jacobian criterion to prove that these charts glue to a Riemann surface  $C^{\text{an}}$ . Prove that projectivity makes  $C^{\text{an}}$  compact and that a morphism of smooth projective complex curves induces a holomorphic map of their analytifications.

For a rational function on  $C$ , prove that its restriction to closed points is meromorphic on  $C^{\text{an}}$  and gives an embedding

$$K(C) \longrightarrow \mathcal{M}(C^{\text{an}}).$$

Prove the dimension-one algebraic–analytic comparison and GAGA theorems [60]: this map is an isomorphism, analytification gives an equivalence between algebraic and holomorphic line bundles, and the induced maps on their spaces of sections are isomorphisms. Deduce that algebraic divisors, meromorphic sections, and degrees agree with their analytic counterparts.

Apply the construction to the projective Legendre curve  $E_m$ . Verify on its affine charts that the algebraic differential  $dx/y$  analytifies to a holomorphic one-form and that the algebraic morphism  $E_m \rightarrow \mathbb{P}^1$  becomes the holomorphic branched cover used below.

## 14.2 Branched Covers

**Exercise 14.7** (Branched covers and the Riemann–Hurwitz formula). For a compact connected oriented surface  $S$ , define its genus  $g(S)$  by

$$\chi(S) = 2 - 2g(S).$$

Let  $f : C \rightarrow D$  be a nonconstant holomorphic map between compact connected Riemann surfaces. At  $p \in C$ , choose local parameters  $z$  at  $p$  and  $w$  at  $f(p)$ . Define the *ramification index*  $e_p \geq 1$  by

$$w \circ f = z^{e_p} u(z), \quad u(0) \neq 0,$$

and define the ramification divisor

$$R_f = \sum_{p \in C} (e_p - 1)[p].$$

Show that  $e_p$  is independent of the local parameters and that the number of points in a nearby regular fiber, counted with multiplicity, is a constant  $d$ , the degree of  $f$ .

Triangulate  $D$  so that every branch value is a vertex, lift the triangulation to  $C$ , and compare vertices, edges, and faces to prove the Riemann–Hurwitz formula

$$2g(C) - 2 = d(2g(D) - 2) + \deg R_f.$$

For a square-free polynomial  $q \in \mathbb{C}[x]$  of degree  $2g + 1$  or  $2g + 2$ , let  $C_q$  be the nonsingular projective curve obtained from  $y^2 = q(x)$ . Show that the projection  $x : C_q \rightarrow \mathbb{P}^1$  has degree two, determine its branch points including the point or points over infinity, and deduce that  $C_q$  has genus  $g$ .

**Exercise 14.8** (Legendre ramification and holomorphic differentials). For  $m \in \mathbb{C} \setminus \{0, 1\}$ , let  $\pi_m : E_m \rightarrow \mathbb{P}^1$  be the projective Legendre curve and morphism constructed above. Prove that it has degree two and is ramified exactly over  $0, 1, m, \infty$ . Use Riemann–Hurwitz to obtain

$$\chi(E_m) = 2\chi(\mathbb{P}^1) - 4 = 0$$

and hence that  $E_m$  has genus one.

On the affine chart  $Z = 1$ , put

$$\eta_m = \frac{dx}{y}, \quad y^2 = x(x - 1)(x - m).$$

Check at each ramification point and at infinity that  $\eta_m$  extends to a nowhere-vanishing holomorphic one-form on  $E_m$ . For  $0 < k < 1$ , define

$$K(k) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}.$$

For  $0 < m < 1$ , let  $\gamma_m$  be the closed real oval over  $[0, m]$ . With  $m = k^2$ , prove that its orientation may be chosen so that

$$\oint_{\gamma_m} \eta_m = 4K(k).$$

**Exercise 14.9** (Surface homology and the intersection pairing). A *CW complex* is built inductively by attaching closed disks along maps from their boundary spheres to the preceding skeleton. Prove the cellular-homology theorem: its chain groups are the relative groups of consecutive skeleta, its differential is determined by the attaching maps, and its homology agrees naturally with singular homology. Let  $C$  be a closed connected oriented surface of genus  $g$ . Use the polygon with edge word

$$a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}$$

to give  $C$  a CW decomposition with one vertex,  $2g$  one-cells, and one two-cell. Apply the cellular-homology theorem to the singular homology defined earlier and prove

$$H_0(C; \mathbb{Z}) \cong \mathbb{Z}, \quad H_1(C; \mathbb{Z}) \cong \mathbb{Z}^{2g}, \quad H_2(C; \mathbb{Z}) \cong \mathbb{Z}.$$

Represent the one-cell classes by oriented simple closed curves  $\alpha_i, \beta_i$  and prove that their oriented intersection numbers are

$$\alpha_i \cdot \alpha_j = 0, \quad \beta_i \cdot \beta_j = 0, \quad \alpha_i \cdot \beta_j = \delta_{ij}.$$

Prove Poincaré duality for the surface [14], and deduce that this alternating pairing is unimodular. Apply the de Rham theorem and the universal coefficient theorem to identify

$$H_{\text{dR}}^1(C) \cong \text{Hom}(H_1(C; \mathbb{Z}), \mathbb{R})$$

after extension of scalars, with the isomorphism given by periods. For a complex-valued closed one-form, prove that its periods on the  $\alpha_i, \beta_i$  determine its de Rham class. Specialize to  $g = 1$  and obtain the two generators used in the period-lattice construction.

### 14.3 Elliptic Integrals

**Exercise 14.10** (Riemann bilinear identity and period lattices). For a compact genus-one Riemann surface  $C$  and a nowhere-vanishing holomorphic one-form  $\eta$ , define the period homomorphism and period subgroup by

$$\text{per}_\eta : H_1(C; \mathbb{Z}) \longrightarrow \mathbb{C}, \quad [\gamma] \longmapsto \int_\gamma \eta, \quad \Lambda_\eta = \text{im}(\text{per}_\eta).$$

Let  $\alpha, \beta$  be oriented simple closed curves meeting once positively and generating the first homology of  $C$ , and put

$$A = \int_\alpha \eta, \quad B = \int_\beta \eta.$$

Cut  $C$  along  $\alpha \cup \beta$ , choose a primitive of  $\eta$  on the resulting polygon, and apply Stokes' theorem to its boundary to prove the genus-one Riemann bilinear identity

$$\frac{i}{2} \int_C \eta \wedge \bar{\eta} = \text{Im}(\overline{A}B) > 0.$$

Deduce that  $A$  and  $B$  are linearly independent over  $\mathbb{R}$  and that

$$\Lambda_\eta = A\mathbb{Z} + B\mathbb{Z}$$

is a discrete subgroup of rank two. Apply this to  $(E_m, \eta_m)$  and identify the real generator  $A = 4K(\sqrt{m})$  for  $0 < m < 1$ .

**Exercise 14.11** (Abel maps and elliptic uniformization). Let  $\Lambda_{\eta_m}$  be the period lattice obtained above. For a fixed  $p_0 \in E_m$ , define

$$\mathcal{A}_{p_0}(p) = \left[ \int_{p_0}^p \eta_m \right] \in \mathbb{C}/\Lambda_{\eta_m}.$$

Show that this is independent of the path modulo periods, is holomorphic, and has nowhere-vanishing differential. Show that it is therefore a finite covering. Identify its induced map on first homology with  $\text{per}_{\eta_m} : H_1(E_m; \mathbb{Z}) \rightarrow \Lambda_{\eta_m}$  and use the period-pairing calculation to prove that the covering has one sheet. Conclude that it is a biholomorphism

$$E_m \cong \mathbb{C}/\Lambda_{\eta_m}.$$

Under this identification, show that translation on the complex torus defines an intrinsic group law on  $E_m$  with identity  $p_0$ .

**Exercise 14.12** (Picard–Fuchs equations and complete elliptic integrals). Let  $U = \mathbb{C} \setminus \{0, 1\}$ . Along a path in  $U$ , transport a closed cycle continuously through the family  $E_m$  and call the resulting identification of the groups  $H_1(E_m; \mathbb{Z})$  *Gauss–Manin transport*. Transport around a loop defines its *monodromy*.

For a transported cycle  $\gamma_m$ , put

$$\Pi(m) = \oint_{\gamma_m} \eta_m.$$

Differentiate under the integral sign and reduce differentials modulo exact differentials on  $E_m$  to prove the Picard–Fuchs equation

$$m(1-m)\Pi''(m) + (1-2m)\Pi'(m) - \frac{1}{4}\Pi(m) = 0.$$

Identify this with the Gauss hypergeometric equation. For the real oval, use  $\lim_{m \downarrow 0} \Pi(m) = 2\pi$  and the recurrence at  $m = 0$  to derive

$$\Pi(m) = 4K(\sqrt{m}) = 2\pi \sum_{n=0}^{\infty} \left( \frac{(1/2)_n}{n!} \right)^2 m^n = 2\pi {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; m\right), \quad |m| < 1.$$

Prove that the projective cubic has discriminant

$$\Delta(m) = 16m^2(1-m)^2$$

in the displayed normalization and that the family has singular fibers at  $m = 0, 1, \infty$ .

Show that local monodromy at  $m = 1$  fixes one analytic solution and sends a logarithmic companion to itself plus a multiple of the analytic solution. Use the hypergeometric connection formula to obtain

$$K(\sqrt{m}) = \log \frac{4}{\sqrt{1-m}} + O((1-m)|\log(1-m)|) \quad (m \uparrow 1).$$

## 14.4 Pendulum Periods

**Exercise 14.13** (Complete elliptic integrals and pendulum periods). On  $T^*S^1$  consider the pendulum Hamiltonian

$$H(\theta, p) = \frac{p^2}{2M\ell^2} + Mg\ell(1 - \cos \theta).$$

For an oscillation of amplitude  $0 < \theta_0 < \pi$ , put  $m = \sin^2(\theta_0/2)$ . Use energy conservation and the real period above to prove

$$T(\theta_0) = \sqrt{\frac{\ell}{g}} \Pi(m) = 4\sqrt{\frac{\ell}{g}} K\left(\sin \frac{\theta_0}{2}\right).$$

Use the hypergeometric series to derive

$$T(\theta_0) = 2\pi\sqrt{\frac{\ell}{g}} \left(1 + \frac{\theta_0^2}{16} + O(\theta_0^4)\right) \quad (\theta_0 \rightarrow 0),$$

and use the connection formula to derive

$$T(\theta_0) = 4\sqrt{\frac{\ell}{g}} \log \frac{4}{\cos(\theta_0/2)} + O\left(\sqrt{\frac{\ell}{g}} \cos^2 \frac{\theta_0}{2} \left|\log \cos^2 \frac{\theta_0}{2}\right|\right).$$

Predict the increase of period with amplitude, the leading nonlinear frequency shift, and the logarithmic divergence at the separatrix. State the point-mass, rigid-length, planar-motion, negligible-damping, amplitude-calibration, and timing-resolution assumptions under which a period-versus-amplitude measurement tests the hypergeometric period and the location and logarithmic type of the singular fiber.

## 15 Riemann–Roch

### 15.1 Picard Group

**Exercise 15.1** (Divisors, line bundles, and first Chern classes). For a scheme  $X$ , an *algebraic line bundle* is an  $\mathcal{O}_X$ -module locally isomorphic to  $\mathcal{O}_X$ . For  $X = \text{Proj } R$  and  $n \in \mathbb{Z}$ , define the graded shift by  $R(n)_d = R_{n+d}$  and define the twisting sheaf  $\mathcal{O}_X(n)$  on the standard affine charts by

$$\mathcal{O}_X(n)(D_+(f)) = (R(n)_f)_0.$$

For projective space, prove intrinsically for  $n \geq 0$  that

$$\Gamma(\mathbb{P}(V), \mathcal{O}(n)) \cong \text{Sym}^n V^\vee,$$

and that  $\mathcal{O}(-1)$  is the tautological line bundle whose fiber over a line  $\ell \subset V$  is  $\ell$ . For a smooth projective plane curve  $C$ , define  $\mathcal{O}_C(n)$  by restriction. Prove [36, 29] that an algebraic line bundle on a smooth complex projective curve determines a holomorphic line bundle and that this correspondence preserves tensor products and global sections.

On a compact Riemann surface  $C$ , a *divisor* is a finite formal sum

$$D = \sum_{p \in C} n_p [p], \quad n_p \in \mathbb{Z}, \quad \deg D = \sum_p n_p.$$

For a nonzero meromorphic function  $f$ , define  $\operatorname{div}(f) = \sum_p \operatorname{ord}_p(f)[p]$ . Use the argument principle to prove

$$\deg \operatorname{div}(f) = 0.$$

A *holomorphic line bundle* on  $C$  is a complex line bundle with local product maps whose transition functions are nowhere-vanishing holomorphic functions  $g_{ij}$  satisfying  $g_{ij}g_{jk} = g_{ik}$ . For a divisor  $D = \sum_p n_p[p]$ , define  $\mathcal{O}(D)$  to be the sheaf of meromorphic functions  $h$  satisfying  $\operatorname{div}(h) + D \geq 0$  on each open set. Show that this sheaf is the sheaf of holomorphic sections of a holomorphic line bundle. Two divisors are *linearly equivalent* when their difference is principal, namely the divisor of a nonzero meromorphic function. Prove

$$\mathcal{O}(D) \cong \mathcal{O}(D') \iff D \text{ and } D' \text{ are linearly equivalent.}$$

For a nonzero meromorphic section  $s$ , define  $\operatorname{div}(s)$  from its local representatives. Prove the meromorphic-section theorem [29]: every holomorphic line bundle on a compact Riemann surface has a nonzero meromorphic section. Prove that

$$\deg L := \deg \operatorname{div}(s)$$

is independent of  $s$ , that tensor products add degrees, and that  $L \cong \mathcal{O}(\operatorname{div}(s))$ . Use Bézout's theorem to prove

$$\deg \mathcal{O}_C(1) = \deg C$$

for every smooth projective plane curve  $C$ .

For a Hermitian connection  $\nabla$  on  $L$ , define

$$c_1(L) = \left[ \frac{i}{2\pi} F_\nabla \right] \in H_{\text{dR}}^2(C; \mathbb{R}).$$

Show that this class is independent of the Hermitian connection and is natural under pullback. Compare connection forms on overlaps and use Stokes' theorem around the zeros and poles of a meromorphic section to prove

$$\int_C c_1(L) = \deg L.$$

Define  $\operatorname{Pic}(C)$  to be the isomorphism classes of holomorphic line bundles with tensor product. Prove that dual line bundles give inverses, that  $\operatorname{Pic}(C)$  is an abelian group, and that degree is a homomorphism

$$\deg : \operatorname{Pic}(C) \longrightarrow \mathbb{Z}.$$

Write  $\operatorname{Pic}^0(C)$  for its kernel and prove that tensor product gives an action of  $\operatorname{Pic}^0(C)$  on every degree component.

## 15.2 Sheaf Cohomology

**Exercise 15.2** (Čech cohomology and exact sequences). Let  $\mathcal{F}$  be a sheaf of complex vector spaces on a compact Riemann surface  $C$ , and let  $\mathfrak{U} = (U_i)_{i \in I}$  be a finite ordered open cover. Define the alternating Čech cochains and coboundary by

$$\check{C}^q(\mathfrak{U}, \mathcal{F}) = \prod_{i_0 < \dots < i_q} \mathcal{F}(U_{i_0} \cap \dots \cap U_{i_q}),$$

$$(\delta s)_{i_0 \dots i_{q+1}} = \sum_{r=0}^{q+1} (-1)^r s_{i_0 \dots \widehat{i_r} \dots i_{q+1}}|_{U_{i_0} \cap \dots \cap U_{i_{q+1}}}.$$

Prove  $\delta^2 = 0$  and define  $\check{H}^q(\mathfrak{U}, \mathcal{F}) = \ker \delta / \operatorname{im} \delta$ . Show directly from the sheaf gluing axiom that

$$\check{H}^0(\mathfrak{U}, \mathcal{F}) = \mathcal{F}(C).$$

Prove [29] that a finite cover whose nonempty intersections are Stein computes the cohomology  $H^q(C, \mathcal{F})$  of a coherent analytic sheaf, that coherent sheaves on  $C$  have finite-dimensional cohomology and vanish in degrees  $q > 1$ , and that a short exact sequence of coherent sheaves induces a long exact cohomology sequence. For a coherent algebraic sheaf on a smooth projective curve, define its algebraic Čech complex on a finite affine cover, compare it with the analytic Čech complex after analytification, and prove the GAGA cohomology isomorphism.

For a holomorphic line bundle  $L$ , a point  $p \in C$ , and its inclusion  $i_p : \{p\} \rightarrow C$ , prove that evaluation at  $p$  gives an exact sequence

$$0 \longrightarrow L(-p) \longrightarrow L \longrightarrow (i_p)_* L_p \longrightarrow 0.$$

Writing  $h^q(L) = \dim H^q(C, L)$  and  $\chi(L) = h^0(L) - h^1(L)$ , use its long exact sequence to prove

$$\chi(L) = \chi(L(-p)) + 1.$$

Iterate this identity through the positive and negative parts of a divisor and use the meromorphic-section theorem to deduce

$$\chi(L) = \deg L + \chi(\mathcal{O}_C)$$

for every holomorphic line bundle on  $C$ .

**Exercise 15.3** (Serre duality and the Riemann–Roch theorem). Define the canonical line bundle  $K_C$  to be the holomorphic cotangent bundle of a compact connected Riemann surface  $C$ . Let  $g$  be its topological genus and prove [29] the analytic genus identity  $\dim H^0(C, K_C) = g$ . Prove Serre duality [29] in the form of a natural perfect pairing

$$H^1(C, L) \otimes H^0(C, K_C \otimes L^{-1}) \longrightarrow \mathbb{C}, \quad H^1(C, L) \cong H^0(C, K_C \otimes L^{-1})^\vee.$$

Show that every holomorphic function on  $C$  is constant and deduce

$$\chi(\mathcal{O}_C) = 1 - g.$$

Combine this with the Čech calculation above to prove the Riemann–Roch theorem

$$h^0(L) - h^0(K_C \otimes L^{-1}) = \deg L + 1 - g.$$

Apply it to  $L = K_C$  to prove  $\deg K_C = 2g - 2$ . Deduce that

$$h^0(L) = \deg L + 1 - g \quad \text{when } \deg L > 2g - 2,$$

and that a negative-degree line bundle has no nonzero holomorphic sections. Specialize to  $g = 1$  and prove that a line bundle of degree  $N > 0$  has  $h^0(L) = N$ .



**Exercise 15.4** (Polarized quantum spaces). Let  $(M, \omega)$  be prequantizable. A *complex polarization* is an involutive Lagrangian subbundle  $P \subset T_{\mathbb{C}}M$ . Define its annihilator and canonical line by

$$P^\circ = \{\alpha \in T_{\mathbb{C}}^*M : \alpha|_P = 0\}, \quad K_P = \Lambda^{\text{top}} P^\circ.$$

A *half-form bundle* is a line bundle  $\delta_P$  with  $\delta_P^{\otimes 2} \cong K_P$ . Prove [70] the induced partial connection on  $\delta_P$  along  $P$  and the half-form pairing. Define the corrected polarized quantum space to be the square-integrable sections of  $L \otimes \delta_P$  annihilated by the tensor-product partial connection along  $P$ . For the vertical polarization on  $T^*V$ , defined as the tangent distribution to the fibers of  $T^*V \rightarrow V$ , recover  $L^2(V)$  and the operators

$$\psi(x) \mapsto \ell(x)\psi(x), \quad \psi(x) \mapsto -i\hbar d\psi_x(v)$$

for  $\ell \in V^*$  and  $v \in V$ .

A *Kähler manifold* is a complex manifold  $(M, J)$  with a symplectic form  $\omega$  satisfying

$$\omega(Ju, Jv) = \omega(u, v), \quad g(u, v) = \omega(u, Jv)$$

for a Riemannian metric  $g$ . For a compact Kähler manifold with  $P = T^{0,1}M$ , prove

$$P^\circ = T^{*1,0}M, \quad K_P = K_M := \Lambda^{\text{top}} T^{*1,0}M.$$

Call a Hermitian holomorphic line bundle *positive* when its first Chern form satisfies  $c_1(L, \nabla)(u, Ju) > 0$  for every nonzero tangent vector  $u$ . When the prequantum line bundle is positive and a square root  $K_M^{1/2}$  exists, identify the corrected quantum space with

$$H^0(M, L \otimes K_M^{1/2}),$$

where  $H^0$  denotes holomorphic sections. On an elliptic curve, use the trivialization by  $dz$  to show that the half-form correction does not change the theta-section count.

**Exercise 15.5** (Borel–Weil–Bott cohomology on flag varieties). Let  $V = \mathbb{C}^2$ , let  $G = SL_2(\mathbb{C})$ , and let  $B \subset G$  be the subgroup stabilizing a line in  $V$ . Prove that the orbit map identifies

$$G/B \cong \mathbb{P}(V) \cong \mathbb{P}^1.$$

Equip  $\mathcal{O}(-1)$  with its tautological  $G$ -action and  $\mathcal{O}(n)$  with the induced action. Use the previously established identification of its global sections to prove, as  $G$ -representations,

$$H^0(\mathbb{P}^1, \mathcal{O}(n)) \cong \text{Sym}^n V^\vee \quad (n \geq 0).$$

Compute the transition of  $dz$  between the two standard affine charts and prove  $K_{\mathbb{P}^1} \cong \mathcal{O}(-2)$ . Apply Riemann–Roch and equivariant Serre duality to derive

$$H^q(\mathbb{P}^1, \mathcal{O}(n)) \cong \begin{cases} \text{Sym}^n V^\vee, & q = 0, \ n \geq 0, \\ \text{Sym}^{-n-2} V, & q = 1, \ n \leq -2, \\ 0, & \text{otherwise.} \end{cases}$$

Identify the weight lattice of  $SL_2$  with  $\mathbb{Z}$ , put  $\rho = 1$ , and let the nontrivial Weyl element act by  $s(n) = -n$ . Prove that its shifted action is

$$s \cdot n = s(n + \rho) - \rho = -n - 2,$$

and use it to rewrite the displayed cohomology calculation as the rank-one Borel–Weil–Bott theorem. For a connected simply connected complex semisimple group  $G$ , fix a Borel subgroup  $B$ , a maximal torus  $T \subset B$ , and the corresponding positive roots. Let  $X^*(T)$  be the integral weight lattice, let  $\rho$  be half the sum of the positive roots, and define

$$w \cdot \lambda = w(\lambda + \rho) - \rho.$$

For  $\lambda \in X^*(T)$ , define the equivariant line bundle

$$\mathcal{L}_\lambda = G \times_B \mathbb{C}_{-\lambda} \longrightarrow G/B,$$

using the associated-bundle convention for which dominant weights give  $H^0(G/B, \mathcal{L}_\lambda) \cong V_\lambda^\vee$ . Prove the Borel–Weil–Bott theorem [15]: if  $\lambda + \rho$  is singular then every  $H^q(G/B, \mathcal{L}_\lambda)$  vanishes; if it is regular, there is a unique  $w$  such that  $w \cdot \lambda$  is dominant and

$$H^q(G/B, \mathcal{L}_\lambda) \cong \begin{cases} V_{w \cdot \lambda}^\vee, & q = \ell(w), \\ 0, & q \neq \ell(w). \end{cases}$$

Verify that this theorem specializes to every case in the calculation on  $\mathbb{P}^1$ .

**Exercise 15.6** (Twisted differential operators and infinitesimal characters). Fix  $n \in \mathbb{Z}_{\geq 0}$ . On the affine chart of  $\mathbb{P}^1$  with coordinate  $z$ , represent a section of  $\mathcal{O}(n)$  by a polynomial  $p(z)$  of degree at most  $n$ . Differentiate the lifted Möbius action on  $\mathcal{O}(n)$  and obtain

$$e = \frac{d}{dz}, \quad h = -2z \frac{d}{dz} + n, \quad f = -z^2 \frac{d}{dz} + nz.$$

Verify

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h,$$

and prove directly that the polynomials of degree at most  $n$  form an invariant irreducible module of highest weight  $n$ .

For a holomorphic line bundle  $L$  on a complex manifold  $X$ , define  $\mathcal{D}_L^{-1} = 0$ , define  $\mathcal{D}_L^0 = \mathcal{O}_X$  acting by multiplication, and define inductively the sheaf  $\mathcal{D}_L^r$  of differential operators of order at most  $r \geq 1$  by

$$P \in \mathcal{D}_L^r(U) \iff [P, m_a] \in \mathcal{D}_L^{r-1}(U) \text{ for every } a \in \mathcal{O}_X(U).$$

Put  $\mathcal{D}_L = \bigcup_{r \geq 0} \mathcal{D}_L^r$  and  $\mathcal{D}_n = \mathcal{D}_{\mathcal{O}(n)}$ . Extend the infinitesimal action to an algebra homomorphism

$$U(\mathfrak{sl}_2) \longrightarrow \Gamma(\mathbb{P}^1, \mathcal{D}_n).$$

Define

$$\Omega = \frac{1}{2}h^2 + ef + fe.$$

Use the commutation relations to prove that  $\Omega$  is central and that it acts on  $H^0(\mathbb{P}^1, \mathcal{O}(n))$  by

$$\frac{1}{2}n(n+2).$$

Filter  $U(\mathfrak{sl}_2)$  by word length, prove that the associated graded algebra is  $S(\mathfrak{sl}_2)$ , and deduce the Poincaré–Birkhoff–Witt theorem. Use the resulting ordered monomials to prove the first identity below. Prove the rank-one global differential operator calculation [11], which says that the

infinitesimal-action map is surjective with kernel generated by the displayed central relation, to deduce the second:

$$Z(U(\mathfrak{sl}_2)) = \mathbb{C}[\Omega], \quad \Gamma(\mathbb{P}^1, \mathcal{D}_n) \cong \frac{U(\mathfrak{sl}_2)}{U(\mathfrak{sl}_2)(\Omega - \frac{1}{2}n(n+2))}.$$

For a complex semisimple Lie algebra  $\mathfrak{g}$  with Cartan subalgebra  $\mathfrak{h}$ , define an infinitesimal character to be an algebra map

$$\chi : Z(U(\mathfrak{g})) \longrightarrow \mathbb{C}.$$

Prove the Harish–Chandra isomorphism  $Z(U(\mathfrak{g})) \cong S(\mathfrak{h})^W$ , normalized so that the character  $\chi_\lambda$  is evaluation at the Weyl orbit of  $\lambda + \rho$ . Deduce  $\chi_{w \cdot \lambda} = \chi_\lambda$  and identify the rank-one scalar above with  $\chi_n(\Omega)$ .

**Exercise 15.7** (Beilinson–Bernstein localization). Let  $G$  be a connected simply connected complex semisimple group,  $X = G/B$ , and  $\mathfrak{g} = \text{Lie}(G)$ . For  $\lambda \in \mathfrak{h}^*$  define the central reduction

$$U_\lambda = U(\mathfrak{g})/U(\mathfrak{g}) \ker \chi_\lambda.$$

Assume first that  $\lambda$  is integral and define  $\mathcal{D}_\lambda = \mathcal{D}_{\mathcal{L}_\lambda}$  by the commutator filtration on differential operators acting on the equivariant line bundle  $\mathcal{L}_\lambda$ . Differentiate the  $G$ -action, prove that the map  $U(\mathfrak{g}) \rightarrow \Gamma(X, \mathcal{D}_\lambda)$  annihilates  $\ker \chi_\lambda$  with the same  $\rho$ -normalization as the Harish–Chandra character, and prove [11] the global-sections isomorphism

$$\Gamma(X, \mathcal{D}_\lambda) \cong U_\lambda.$$

Put  $\mathcal{D}_X = \mathcal{D}_{\mathcal{O}_X}$ . Identify  $\mathcal{D}_\lambda$  locally with  $\mathcal{D}_X$  after choosing a nonvanishing local section of  $\mathcal{L}_\lambda$ , compute its gluing by conjugation with the transition functions, and recover  $\mathcal{D}_\lambda$  on  $\mathbb{P}^1$ .

Let  $U_\lambda\text{-mod}_{\text{fg}}$  be the category of finitely generated left  $U_\lambda$ -modules and let  $\text{Coh}(\mathcal{D}_\lambda)$  be the category of left  $\mathcal{D}_\lambda$ -modules locally admitting finite generators and relations over  $\mathcal{D}_\lambda$ . Define

$$\text{Loc}_\lambda(M) = \mathcal{D}_\lambda \otimes_{U_\lambda} M, \quad \Gamma(\mathcal{M}) = \Gamma(X, \mathcal{M}).$$

Assume that  $\lambda$  is integral and dominant, meaning  $\langle \lambda, \alpha_i^\vee \rangle \geq 0$  for every simple coroot; then  $\lambda + \rho$  is regular. Prove the Beilinson–Bernstein localization theorem with this  $\rho$ -normalization: in this range the two functors are exact quasi-inverse equivalences

$$U_\lambda\text{-mod}_{\text{fg}} \simeq \text{Coh}(\mathcal{D}_\lambda).$$

For  $n \geq 0$ , apply the equivalence to  $M = H^0(\mathbb{P}^1, \mathcal{O}(n))$  and prove that the action map gives

$$\mathcal{D}_n \otimes_{U_n} M \cong \mathcal{O}(n)$$

as  $\mathcal{D}_n$ -modules. Deduce that Borel–Weil realizes a particular finite-dimensional module by sections, whereas localization realizes every finitely generated module with the fixed regular infinitesimal character as a twisted system of differential equations on the same flag variety.

### 15.3 Theta Functions

**Exercise 15.8** (Abel's theorem and theta divisors). For  $a, b \in \mathbb{R}$ ,  $z \in \mathbb{C}$ , and  $\tau$  in the upper half-plane, define

$$\vartheta \begin{bmatrix} a \\ b \end{bmatrix} (z \mid \tau) = \sum_{n \in \mathbb{Z}} \exp\left(\pi i(n+a)^2 \tau + 2\pi i(n+a)(z+b)\right).$$

Derive its two automorphy laws by shifting the summation index. Let  $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$  and put

$$\vartheta_1(z \mid \tau) = \vartheta \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} (z \mid \tau).$$

Use the theta product and automorphy laws to prove that  $\vartheta_1$  has one simple zero modulo the period lattice, at the origin.

For a degree-zero divisor  $D = \sum_p n_p [p]$ , define its Abel sum by

$$\mathfrak{a}(D) = \sum_p n_p p \in E_\tau.$$

Show that this is independent of the chosen lifts of the points to  $\mathbb{C}$ . If

$$D = [p_1] + \cdots + [p_N] - [q_1] - \cdots - [q_N]$$

and  $\mathfrak{a}(D) = 0$ , choose lifts with  $\sum p_j - \sum q_j = m + n\tau$  and prove that

$$f_D(z) = e^{-2\pi i n z} \frac{\prod_{j=1}^N \vartheta_1(z - p_j \mid \tau)}{\prod_{j=1}^N \vartheta_1(z - q_j \mid \tau)}$$

is elliptic and satisfies  $\text{div}(f_D) = D$ . Conversely, apply the argument principle to  $f'/f$  and  $zf'/f$  around a period parallelogram to prove that a principal divisor has degree zero and Abel sum zero. Conclude the genus-one Abel theorem

$$D \text{ is principal} \iff \deg D = 0 \text{ and } \mathfrak{a}(D) = 0.$$

Let  $\text{Pic}^d(E_\tau)$  denote the isomorphism classes of holomorphic line bundles of degree  $d$ , with tensor product as the operation on  $\text{Pic}^0(E_\tau)$ . Fix  $0 \in E_\tau$  and prove that

$$E_\tau \longrightarrow \text{Pic}^0(E_\tau), \quad p \longmapsto \mathcal{O}([p] - [0]),$$

is an isomorphism of complex groups. Show that tensoring with  $\mathcal{O}(N[0])$  identifies  $\text{Pic}^0(E_\tau)$  with  $\text{Pic}^N(E_\tau)$ . Identify effective divisors of degree  $N$  with unordered  $N$ -tuples of theta zeros and show that their Abel sum determines the corresponding line bundle.

**Exercise 15.9** (Theta sections and genus-one Riemann–Roch). On  $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ , define the standard degree- $N$  theta line bundle  $L_N$  by representing its holomorphic sections as functions on  $\mathbb{C}$  satisfying

$$f(z+1) = f(z), \quad f(z+\tau) = e^{-\pi i N \tau - 2\pi i N z} f(z).$$

Use these automorphy factors and the argument principle to prove that every nonzero holomorphic section has a zero divisor of degree  $N$ . For  $r \in \mathbb{Z}/N\mathbb{Z}$ , define

$$f_r(z) = \vartheta \begin{bmatrix} r/N \\ 0 \end{bmatrix} (Nz \mid N\tau) = \theta_{N,r}(z \mid \tau)$$

and derive its two automorphy laws directly from the series.

For any line bundle  $L$  of degree  $N > 0$ , use Abel's theorem to write

$$L \cong \mathcal{O}((N-1)[0] + [p])$$

for a unique  $p \in E_\tau$ . Prove from  $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$  that multiplication by  $N$  is surjective, choose  $a \in E_\tau$  with  $-Na = p$ , and identify  $L$  with the pullback of the standard degree- $N$  theta bundle by translation by  $a$ . Translate the distinguished level- $N$  theta sections and use their distinct magnetic-translation characters to construct  $N$  linearly independent sections of  $L$ . Conversely, expand a section on the universal cover in Fourier modes and use its two automorphy laws to show that all coefficients are determined by  $N$  initial values. Deduce

$$\dim H^0(E_\tau, L) = N.$$

Relate translation of theta-function zeros to  $\text{Pic}^N(E_\tau)$ .

## 15.4 Landau Levels

**Exercise 15.10** (Theta-function Landau spectra). Let

$$Q = \mathbb{R}^2 / (L_x\mathbb{Z} \oplus L_y\mathbb{Z}) \cong E_\tau, \quad \tau = i \frac{L_y}{L_x}, \quad A = L_x L_y,$$

and let a particle of charge  $e$  experience a magnetic field through the curvature convention

$$F_\nabla = -\frac{ie}{\hbar} B dx \wedge dy.$$

(a) Suppose  $B > 0$  is constant and put

$$N = \frac{eBA}{2\pi\hbar}.$$

Use prequantization to prove that a magnetic line bundle exists exactly when  $N \in \mathbb{Z}$ . For  $N > 0$ , use the automorphy laws above to identify

$$H^0(Q, L) = \bigoplus_{r \in \mathbb{Z}/N\mathbb{Z}} \mathbb{C} \theta_{N,r}(z | \tau), \quad z = \frac{x}{L_x} + \tau \frac{y}{L_y}.$$

Compute the magnetic-translation eigenvalues of the theta sections and relate their holomorphic-gauge representatives to physical wavefunctions by the common unitary-gauge factor. For translations by  $L_x/N$  and  $L_y/N$ , use holonomy around a fundamental magnetic cell to prove

$$UV = e^{2\pi i/N} VU.$$

Define the covariant Landau Hamiltonian by

$$\hat{H} = \frac{\hbar^2}{2m} \nabla^* \nabla.$$

Construct raising and lowering operators and prove

$$E_n = \hbar \frac{eB}{m} \left( n + \frac{1}{2} \right), \quad \dim \ker(\hat{H} - E_n) = N.$$

- (b) Let  $B$  now be nonuniform, let  $L \rightarrow E_\tau$  be fixed with a unitary connection  $\nabla$  satisfying the displayed curvature convention, and assume

$$N = \frac{e}{2\pi\hbar} \int_{E_\tau} B \, dx \, dy > 0.$$

Use the complex structure to define the positive Pauli Hamiltonian

$$H_+ = \frac{\hbar^2}{2m} \left( \nabla^* \nabla - \frac{e}{\hbar} B \right) = \frac{\hbar^2}{m} \bar{\partial}_\nabla^* \bar{\partial}_\nabla.$$

Prove that its zero modes are the holomorphic sections of  $L$  and use the first-Chern calculation above and genus-one Riemann–Roch to prove

$$\dim \ker H_+ = N.$$

Deduce that this count is unchanged under smooth deformations of the unitary connection on the fixed line bundle.

Deduce the cyclotron-resonance spacing, degeneracy density  $eB/(2\pi\hbar)$  in the constant-field model, magnetic-translation interference phase, and robust Pauli zero-mode count. State separately the flat-torus, flux-integrality, spin-coupling, positivity, noninteracting, and spectral-resolution assumptions under which these measurements test the theta-function and Riemann–Roch predictions.

## 16 Affine Grassmannian

### 16.1 Loop Groups

**Exercise 16.1** (Loop groups, affine Grassmannians, and Schubert varieties). Let  $G$  be a connected complex reductive algebraic group and put

$$\mathcal{O} = \mathbb{C}[[t]], \quad \mathcal{K} = \mathbb{C}((t)), \quad L^+G = G(\mathcal{O}), \quad LG = G(\mathcal{K}).$$

Define the affine Grassmannian as the quotient

$$\mathrm{Gr}_G = LG/L^+G.$$

Call a filtered union  $X = \bigcup_{n \geq 0} X_n$  an *ind-projective variety* when every  $X_n$  is projective and every inclusion  $X_n \hookrightarrow X_{n+1}$  is closed. For a  $\mathbb{C}$ -algebra  $R$ , define the affine-Grassmannian functor to assign the  $G$ -bundles on  $\mathrm{Spec} R[[t]]$  equipped with a trivialization over  $\mathrm{Spec} R((t))$ . Construct its increasing Schubert filtration and prove [52] that it is represented by an ind-projective variety whose complex points are  $LG/L^+G$ .

Fix a maximal torus  $T \subset G$ . A coweight is an algebraic homomorphism  $\lambda : \mathbb{C}^\times \rightarrow T$ ; write  $t^\lambda \in T(\mathcal{K})$  for its value at  $t$ . Define dominant coweights using the positive roots and put

$$\mathrm{Gr}_\lambda = L^+G t^\lambda / L^+G, \quad \overline{\mathrm{Gr}}_\lambda = \overline{\mathrm{Gr}_\lambda}.$$

Prove the Cartan and Schubert decompositions

$$\mathrm{Gr}_G = \coprod_{\lambda \in X_*(T)_+} \mathrm{Gr}_\lambda, \quad \overline{\mathrm{Gr}}_\lambda = \coprod_{\mu \leq \lambda} \mathrm{Gr}_\mu, \quad \dim_{\mathbb{C}} \mathrm{Gr}_\lambda = \langle 2\rho, \lambda \rangle,$$

where  $\mu \leq \lambda$  means that  $\lambda - \mu$  is a nonnegative integral sum of positive coroots.

For  $G = PGL_2(\mathbb{C})$ , identify  $\text{Gr}_G$  with the homothety classes of  $\mathcal{O}$ -lattices in  $\mathcal{K}^2$ . Show that the dominant orbits are indexed by  $m \in \mathbb{Z}_{\geq 0}$ , that  $\dim_{\mathbb{C}} \text{Gr}_m = m$ , and that  $\overline{\text{Gr}}_0$  is a point. Represent the lattices in  $\overline{\text{Gr}}_1$  by

$$t\mathcal{O}^2 \subset L \subset \mathcal{O}^2, \quad \dim_{\mathbb{C}}(\mathcal{O}^2/L) = 1,$$

and prove intrinsically

$$\overline{\text{Gr}}_1 \cong \mathbb{P}^1.$$

If  $\varpi^\vee$  denotes the dominant generator of the coweight lattice, verify that the positive coroot is  $2\varpi^\vee$  and hence

$$\overline{\text{Gr}}_m = \coprod_{0 \leq r \leq \lfloor m/2 \rfloor} \text{Gr}_{m-2r}.$$

## 16.2 Perverse-Sheaf Convolution

**Exercise 16.2** (Holonomic differential equations). Let  $X$  be a smooth complex algebraic variety of complex dimension  $d$ , and put  $\mathcal{D}_X = \mathcal{D}_{\mathcal{O}_X}$  with its order filtration  $F_\bullet \mathcal{D}_X$ . For a coherent left  $\mathcal{D}_X$ -module  $\mathcal{M}$ , call an increasing filtration  $F_\bullet \mathcal{M}$  *good* when it is exhaustive,

$$F_p \mathcal{D}_X F_q \mathcal{M} \subseteq F_{p+q} \mathcal{M},$$

and  $\text{gr}_F \mathcal{M}$  is coherent over  $\text{gr}_F \mathcal{D}_X$ . Prove from the commutator definition of differential-operator order that the principal-symbol map gives an isomorphism

$$\text{gr}_F \mathcal{D}_X \cong \text{Sym}_{\mathcal{O}_X} \mathcal{T}_X$$

of sheaves of graded algebras. Define the characteristic variety by

$$\text{Ch}(\mathcal{M}) = \text{Supp}(\text{gr}_F \mathcal{M}) \subseteq T^*X,$$

and prove that it is independent of the good filtration. Use Bernstein's inequality  $\dim_{\mathbb{C}} \text{Ch}(\mathcal{M}) \geq d$  for nonzero  $\mathcal{M}$ , and call  $\mathcal{M}$  *holonomic* when equality holds.

For a smooth compactification  $j : C \hookrightarrow \overline{C}$  of a smooth curve, put  $D = \overline{C} \setminus C$ . Call an integrable connection  $(\mathcal{E}, \nabla)$  on  $C$  *regular singular* along  $D$  when it has a meromorphic extension  $\mathcal{E}(*D)$  containing a locally free  $\mathcal{O}_{\overline{C}}$ -lattice  $E$  satisfying

$$E \otimes \mathcal{O}_{\overline{C}}(*D) = \mathcal{E}(*D), \quad \nabla E \subseteq E \otimes \Omega_{\overline{C}}^1(\log D).$$

For a smooth curve map  $i : C \rightarrow X$ , call a point noncharacteristic for  $\mathcal{M}$  when the characteristic variety meets the kernel of the cotangent map

$$T^*X|_C \longrightarrow T^*C$$

only in the zero section over that point. Prove the curve criterion for regularity: a holonomic  $\mathcal{D}_X$ -module is regular when every connection obtained by inverse image to a smooth curve on its noncharacteristic locus is regular singular at the omitted points. Denote the derived category of complexes with regular holonomic cohomology by  $D_{\text{rh}}^b(\mathcal{D}_X)$ .

Give  $\mathcal{O}_X$  the filtration

$$F_p \mathcal{O}_X = \begin{cases} 0, & p < 0, \\ \mathcal{O}_X, & p \geq 0. \end{cases}$$

Prove that it is good, identify  $\mathrm{Ch}(\mathcal{O}_X)$  with the zero section of  $T^*X$ , and deduce that  $\mathcal{O}_X$  is holonomic. Pull its connection back to a smooth curve and use the lattice  $\mathcal{O}_{\overline{C}}$  to prove that  $\mathcal{O}_X$  is regular holonomic.

**Exercise 16.3** (Perverse sheaves and the Riemann–Hilbert correspondence). Let  $X$  be a complex algebraic variety with a finite stratification by smooth locally closed subvarieties. A bounded constructible complex on  $X$  is a finite cochain complex of sheaves whose cohomology sheaves have finite-dimensional locally constant restrictions to the strata. Define  $D_c^b(X)$  by inverting the quasi-isomorphisms of such complexes. For the inclusion  $i_S : S \hookrightarrow X$  of a stratum, use the derived stalk  $i_S^*$  and costalk  $i_S^!$ . Call  $\mathcal{F} \in D_c^b(X)$  *perverse* for the middle perversity when

$$H^k(i_S^* \mathcal{F}) = 0 \quad (k > -\dim_{\mathbb{C}} S), \quad H^k(i_S^! \mathcal{F}) = 0 \quad (k < -\dim_{\mathbb{C}} S).$$

Prove the basic perverse-sheaf and intermediate-extension theorems [10]: perverse complexes form an abelian category, Verdier duality preserves it, and for a local system  $\mathcal{L}$  on a smooth dense stratum  $j : S \hookrightarrow \overline{S}$  there is a unique perverse intermediate extension

$$\mathrm{IC}(\overline{S}, \mathcal{L}) = j_{!*}(\mathcal{L}[\dim_{\mathbb{C}} S])$$

having no nonzero subobject or quotient supported on the boundary. Prove that when  $\overline{S}$  is smooth and  $\mathcal{L}$  extends to a local system  $\overline{\mathcal{L}}$  on  $\overline{S}$ , this is  $\overline{\mathcal{L}}[\dim_{\mathbb{C}} S]$ .

Suppose now that  $X$  is smooth of complex dimension  $d$ . The action of vector fields on a left  $\mathcal{D}_X$ -module induces its de Rham differential. Define the shifted de Rham complex by

$$\mathrm{DR}_X(\mathcal{M}) = [\mathcal{M} \rightarrow \Omega_X^1 \otimes \mathcal{M} \rightarrow \cdots \rightarrow \Omega_X^d \otimes \mathcal{M}][d].$$

Prove the Riemann–Hilbert correspondence [35]:

$$\mathrm{DR}_X : D_{\mathrm{rh}}^b(\mathcal{D}_X) \xrightarrow{\sim} D_c^b(X^{\mathrm{an}}),$$

which restricts to an equivalence from regular holonomic modules to perverse sheaves. Prove

$$\mathrm{DR}_X(\mathcal{O}_X) \cong \mathbb{C}_{X^{\mathrm{an}}}[d]$$

by the holomorphic Poincaré lemma, and show that a regular-singular flat bundle is sent to its local system of horizontal sections shifted by  $d$ . Apply the Morita untwisting  $\mathcal{M} \mapsto \mathcal{O}(-n) \otimes \mathcal{M}$  to the  $\mathcal{D}_n$ -module  $\mathcal{O}(n)$  and show that its de Rham complex is  $\mathbb{C}_{\mathbb{P}^1}[1]$ . Conclude that the regular holonomic part of Beilinson–Bernstein localization becomes a category of perverse solution complexes on the flag variety.

**Exercise 16.4** (Schubert intersection complexes). For the Schubert stratification of  $\mathrm{Gr}_G$ , define

$$\mathrm{IC}_{\lambda} = \mathrm{IC}(\overline{\mathrm{Gr}}_{\lambda}, \mathbb{C}),$$

extended by zero to  $\mathrm{Gr}_G$ . Define  $\mathrm{Perv}_{L^+G}(\mathrm{Gr}_G)$  to consist of perverse sheaves supported on a finite Schubert union and equipped with compatible  $L^+G$ -equivariance. Prove in the  $PGL_2$  model that

$$\mathrm{IC}_0 = \mathbb{C}_{\mathrm{pt}}, \quad \mathrm{IC}_1 = \mathbb{C}_{\mathbb{P}^1}[1],$$

and compute the total dimensions of their hypercohomology as 1 and 2.



**Exercise 16.5** (Convolution and the geometric Satake equivalence). Define the twisted product

$$\mathrm{Gr}_G \tilde{\times} \mathrm{Gr}_G = LG \times^{L+G} \mathrm{Gr}_G, \quad (g, x) \sim (gh, h^{-1}x),$$

and the convolution map

$$m : \mathrm{Gr}_G \tilde{\times} \mathrm{Gr}_G \longrightarrow \mathrm{Gr}_G, \quad m([g, x]) = gx.$$

For equivariant constructible complexes  $\mathcal{F}, \mathcal{G}$ , let  $\mathcal{F} \tilde{\boxtimes} \mathcal{G}$  be the complex on the twisted product descended from their exterior product and define

$$\mathcal{F} * \mathcal{G} = Rm_!(\mathcal{F} \tilde{\boxtimes} \mathcal{G}).$$

Prove [52] that convolution preserves  $\mathrm{Perv}_{L+G}(\mathrm{Gr}_G)$ , that  $\mathrm{IC}_0$  is its tensor unit. Over the configuration space of two points of a smooth complex curve, construct the Beilinson–Drinfeld Grassmannian classifying a  $G$ -bundle with trivialization away from those points. Prove that its restriction away from the diagonal is the exterior product of two affine Grassmannians and that its diagonal fiber is the convolution space. Use nearby cycles and exchange of the two points to construct the associativity and parity-corrected commutativity constraints, and prove that they equip  $\mathrm{Perv}_{L+G}(\mathrm{Gr}_G)$  with a symmetric monoidal structure.

For  $G = \mathrm{PGL}_2(\mathbb{C})$  and  $m \geq 1$ , restrict convolution to the proper map

$$m_{1,m} : \overline{\mathrm{Gr}}_1 \tilde{\times} \overline{\mathrm{Gr}}_m \longrightarrow \overline{\mathrm{Gr}}_{m+1}.$$

Interpret its source as chains of homothety classes of lattices  $([L_0], [L_1], [L_2])$ , with  $[L_0] = [\mathcal{O}^2]$ , whose first relative position is 1 and whose second relative position is at most  $m$ . For  $[L_2] \in \mathrm{Gr}_{m+1-2r}$ , prove from the lattice model that

$$m_{1,m}^{-1}([L_2]) \cong \begin{cases} \mathrm{pt}, & r = 0, \\ \mathbb{P}^1, & 1 \leq r \leq \lfloor (m+1)/2 \rfloor. \end{cases}$$

Call a proper map  $f : Y \rightarrow Z$ , stratified so that its fiber dimension is constant on each stratum  $S \subset Z$ , *semismall* when

$$2 \dim f^{-1}(s) \leq \mathrm{codim}_Z S \quad (s \in S).$$

Use  $\mathrm{codim} \mathrm{Gr}_{m+1-2r} = 2r$  to prove that  $m_{1,m}$  is semismall and that only  $\mathrm{Gr}_{m+1}$  and  $\mathrm{Gr}_{m-1}$  attain equality. Prove the semismall decomposition theorem [10]: the direct image of the intersection complex is the sum of the intersection complexes of the equality strata, with multiplicity local systems given by the top homology of their fibers. Identify  $\mathrm{IC}_1 \tilde{\boxtimes} \mathrm{IC}_m$  with the intersection complex of the twisted Schubert product. Deduce from the point and  $\mathbb{P}^1$  fibers, and from equivariance of their top homology, that

$$\mathrm{IC}_1 * \mathrm{IC}_m \cong \mathrm{IC}_{m+1} \oplus \mathrm{IC}_{m-1} \quad (m \geq 1).$$

Define the Langlands dual group  $G^\vee$  to be the connected complex reductive group whose roots, coroots, weights, and coweights are the coroots, roots, coweights, and weights of  $G$ , respectively. Construct the geometric Satake equivalence

$$(\mathrm{Perv}_{L+G}(\mathrm{Gr}_G), *) \simeq^{\otimes} \mathrm{Rep}_{\mathbb{C}}(G^\vee),$$

under which total hypercohomology, with the standard parity-corrected commutativity constraint, is the fiber functor and

$$\mathrm{IC}_\lambda \longleftrightarrow V_\lambda$$

for the irreducible representation of highest weight  $\lambda$ , and prove that it is an equivalence of symmetric monoidal categories. Deduce the ring isomorphism

$$K_0(\mathrm{Perv}_{L+G}(\mathrm{Gr}_G)) \cong R(G^\vee)$$

carrying convolution to tensor product and Schubert intersection complexes to irreducible characters.

### 16.3 Satake Characters

**Exercise 16.6** (Satake characters). Take  $G = PGL_2(\mathbb{C})$ , so  $G^\vee = SL_2(\mathbb{C})$ , and write  $\mathrm{IC}_m$  for the Schubert intersection complex indexed by  $m \geq 0$ . Verify that Langlands duality sends  $\varpi^\vee$  to the fundamental weight of  $SL_2$ . Under geometric Satake identify

$$\mathrm{IC}_m \longleftrightarrow V_m = \mathrm{Sym}^m(\mathbb{C}^2).$$

Define the monic Chebyshev polynomial  $C_m(y) = U_m(y/2)$ . Use the geometric recurrence above to prove inductively in the Grothendieck ring that

$$[\mathrm{IC}_m] = C_m([\mathrm{IC}_1]).$$

For a maximal-torus element  $u_\theta \in SU(2)$ , use the established character formula

$$\chi_m(u_\theta) = U_m\left(\cos \frac{\theta}{2}\right)$$

and the trigonometric definition of the Chebyshev polynomials of the second kind to prove

$$U_m(x)U_n(x) = \sum_{r=0}^{\min(m,n)} U_{m+n-2r}(x).$$

Apply this identity to the classes  $C_m([\mathrm{IC}_1])$ . Prove that  $\mathrm{Rep}_{\mathbb{C}}(SL_2)$  is semisimple and transport this property through the Satake equivalence to deduce

$$\mathrm{IC}_m * \mathrm{IC}_n \cong \bigoplus_{r=0}^{\min(m,n)} \mathrm{IC}_{m+n-2r}.$$

Under the Satake equivalence recover simultaneously

$$V_m \otimes V_n \cong \bigoplus_{r=0}^{\min(m,n)} V_{m+n-2r}.$$

### 16.4 't Hooft-Line Fusion

**Exercise 16.7** ('t Hooft-line fusion). Let  $T_m$  denote the purely magnetic 't Hooft line of dominant magnetic coweight  $m$  in the topologically twisted  $\mathcal{N} = 4$  super-Yang–Mills theory with compact gauge group  $PSU(2)$ . Prove from the moduli of singular monopoles [34] that the protected purely

magnetic line-operator product is the convolution product in the Satake category. Predict the operator-product selection rule

$$T_m T_n = \sum_{r=0}^{\min(m,n)} T_{m+n-2r},$$

so the allowed magnetic-charge channels are  $m+n, m+n-2, \dots, |m-n|$ , each with multiplicity one. Explain how the lower channels encode monopole bubbling and how protected line-operator correlation functions distinguish the predicted multiplicities. State separately the topological-twist, BPS protection, global gauge-group, purely magnetic, operator-separation, and S-duality assumptions, and note that the selection rule is a protected quantum-field-theory prediction rather than a direct laboratory cross section.

## 17 Finite-Gap Spectral Curves

### 17.1 Bloch Monodromy

**Exercise 17.1** (Floquet operators and Bloch spectral curves). Let

$$L = -\frac{d^2}{du^2} + V(u)$$

with real smooth potential of period  $P$ . For  $E \in \mathbb{C}$ , let  $\mathcal{S}_E$  be the two-dimensional solution space of  $L\psi = E\psi$ , and define the monodromy automorphism

$$M(E) : \mathcal{S}_E \longrightarrow \mathcal{S}_E, \quad \psi(u) \longmapsto \psi(u+P).$$

Use constancy of the Wronskian to prove  $\det M(E) = 1$ . Define

$$\Delta(E) = \text{tr } M(E)$$

and call  $\rho$  a Bloch multiplier when  $M(E)\psi = \rho\psi$  for a nonzero solution. Prove

$$\rho + \rho^{-1} = \Delta(E)$$

Let  $H^2([0, P])$  be the space of  $L^2$  functions whose first two distributional derivatives belong to  $L^2$ . Prove from the fundamental theorem of calculus and density of smooth functions that endpoint evaluation of a function and its first derivative extends continuously to  $H^2([0, P])$  [38]. For  $\rho \in \mathbb{C}^\times$ , define the twisted Sobolev domain

$$H_\rho^2 = \{\psi \in H^2([0, P]) : \psi(P) = \rho\psi(0), \psi'(P) = \rho\psi'(0)\}.$$

Regard

$$\mathcal{L}_{\rho,E} : H_\rho^2 \longrightarrow L^2([0, P]), \quad \mathcal{L}_{\rho,E}\psi = (L - E)\psi,$$

as a bounded operator. For a bounded operator  $T : H_0 \rightarrow H_1$ , define its cokernel by

$$\text{coker } T = H_1 / \text{im } T.$$

Use variation of parameters to prove that the image of  $\mathcal{L}_{\rho,E}$  is closed and that its kernel and cokernel are finite-dimensional. Prove

$$\ker \mathcal{L}_{\rho,E} \cong \ker(M(E) - \rho I)$$

and that  $\mathcal{L}_{\rho,E}$  is invertible exactly when  $M(E) - \rho I$  is invertible. Deduce that the noninvertibility locus of the family  $\mathcal{L}_{\rho,E}$  is the Bloch spectral curve

$$\mathcal{B}_L : \quad \rho^2 - \Delta(E)\rho + 1 = 0.$$

For  $E \in \mathbb{R}$  and  $|\rho| = 1$ , use Green's identity to prove that  $\mathcal{L}_{\rho,E}$  is self-adjoint as an operator on  $L^2([0, P])$  with domain  $H_\rho^2$ , and deduce

$$\dim \ker \mathcal{L}_{\rho,E} - \dim \operatorname{coker} \mathcal{L}_{\rho,E} = 0.$$

Show that its branch points occur at the periodic and antiperiodic energies  $\Delta(E) = \pm 2$ , and that a real energy supports a propagating Bloch wave exactly when  $|\Delta(E)| \leq 2$ .

## 17.2 Hyperelliptic Curves

**Exercise 17.2** (Lax pairs and isospectral KdV flow). Let  $u(x, t)$  be a smooth real periodic function and define

$$L(t) = -\partial_x^2 - u(x, t), \quad P(t) = -4\partial_x^3 - 6u(x, t)\partial_x - 3u_x(x, t)$$

on smooth periodic functions. Compute the commutator and prove that

$$\dot{L} = [P, L]$$

is equivalent to

$$u_t + 6uu_x + u_{xxx} = 0.$$

Prove that  $P$  is skew-adjoint on the periodic domain. If  $U(t)$  solves  $\dot{U} = PU$ ,  $U(0) = I$ , prove

$$L(t) = U(t)L(0)U(t)^{-1}$$

and deduce preservation of the periodic and antiperiodic spectra.

For the equation  $L(t)\psi = E\psi$  on the line, evolve solutions by  $\psi_t = P\psi$ . Prove that the fundamental solution over one spatial period changes by conjugation and hence that the Floquet discriminant, Bloch multipliers, and Bloch spectral curve are independent of  $t$ .

**Exercise 17.3** (Hyperelliptic curves and holomorphic differentials). Let

$$C : \quad y^2 = \prod_{r=1}^{2g+1} (x - e_r), \quad e_r \neq e_s \quad (r \neq s),$$

denote the normalization of its projective plane closure. Use Riemann–Hurwitz to prove that  $C$  has genus  $g$ . Near infinity use

$$X = x^{-1}, \quad Y = yx^{-(g+1)}, \quad Y^2 = X \prod_{r=1}^{2g+1} (1 - e_r X)$$

and the Jacobian criterion, together with the affine chart, to verify that the normalization is smooth and has one point over infinity. Analyze the finite branch points and the point over infinity to show that the distinguished differentials

$$\eta_j = \frac{x^{j-1}dx}{y}, \quad 1 \leq j \leq g,$$

are holomorphic. Compare their vanishing orders at infinity and use  $\dim H^0(C, K_C) = g$  to show that they exhaust the holomorphic differentials.

Choose a symplectic marking  $(a_1, \dots, a_g, b_1, \dots, b_g)$  of  $H_1(C; \mathbb{Z})$ , define the period maps

$$A(\eta) = \left( \int_{a_1} \eta, \dots, \int_{a_g} \eta \right), \quad B(\eta) = \left( \int_{b_1} \eta, \dots, \int_{b_g} \eta \right),$$

and prove the genus- $g$  Riemann bilinear identities

$$\int_C \alpha \wedge \beta = \sum_{j=1}^g \left( \int_{a_j} \alpha \int_{b_j} \beta - \int_{b_j} \alpha \int_{a_j} \beta \right)$$

for closed complex one-forms. Prove that  $A$  is an isomorphism. Let  $\omega_j$  be the uniquely normalized differentials satisfying  $\int_{a_i} \omega_j = \delta_{ij}$  and define

$$\Omega_{ij} = \int_{b_i} \omega_j.$$

Use the bilinear identity to prove

$$\Omega^T = \Omega, \quad \text{Im } \Omega > 0.$$

### 17.3 Baker–Akhiezer Functions

**Exercise 17.4** (Jacobians, Abel–Jacobi maps, and Riemann theta functions). For a compact Riemann surface  $C$ , define its Jacobian intrinsically by

$$\text{Jac}(C) = H^0(C, K_C)^\vee / \left\{ \eta \mapsto \int_\gamma \eta : \gamma \in H_1(C; \mathbb{Z}) \right\}.$$

Using the symplectic marking and normalized periods above, identify it with

$$\mathbb{C}^g / (\mathbb{Z}^g + \Omega \mathbb{Z}^g).$$

Define the Riemann theta function by the normally convergent series

$$\vartheta(z \mid \Omega) = \sum_{n \in \mathbb{Z}^g} \exp(\pi i n^T \Omega n + 2\pi i n^T z).$$

Prove its holomorphy and automorphy laws

$$\begin{aligned} \vartheta(z + m \mid \Omega) &= \vartheta(z \mid \Omega), \\ \vartheta(z + \Omega n \mid \Omega) &= e^{-\pi i n^T \Omega n - 2\pi i n^T z} \vartheta(z \mid \Omega). \end{aligned}$$

Fix  $p_0 \in C$  and define the Abel–Jacobi map by

$$\mathcal{A}_{p_0}(p) = \left[ \eta \mapsto \int_{p_0}^p \eta \right] \in \text{Jac}(C).$$

Prove that it is well defined and extend it additively to divisors. Prove the Abel–Jacobi theorem [29]: a degree-zero divisor is principal exactly when its Abel–Jacobi image vanishes, and every point of the Jacobian is represented by a degree-zero divisor. Deduce

$$\text{Pic}^0(C) \cong \text{Jac}(C).$$

Prove Riemann’s vanishing theorem [26] to identify a translate of the zero locus of  $\vartheta$  with the Abel–Jacobi images of effective divisors of degree  $g-1$ . Recover the genus-one Abel and theta-divisor statements above when  $g=1$ .

**Exercise 17.5** (Jacobi elliptic functions and elliptic uniformization). For  $0 < k < 1$ , define  $\text{sn}(u; k)$  near  $u = 0$  by

$$u = \int_0^{\text{sn}(u; k)} \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}$$

and use Abel uniformization of the smooth elliptic curve  $y^2 = (1-x^2)(1-k^2x^2)$  to extend this local inverse uniquely to a single-valued meromorphic doubly periodic function. Define  $\text{cn}(u; k)$  and  $\text{dn}(u; k)$  by

$$\text{cn}^2 = 1 - \text{sn}^2, \quad \text{dn}^2 = 1 - k^2 \text{sn}^2, \quad \text{cn}(0; k) = \text{dn}(0; k) = 1.$$

Derive

$$\text{sn}' = \text{cn} \text{dn}, \quad \text{cn}' = -\text{sn} \text{dn}, \quad \text{dn}' = -k^2 \text{sn} \text{cn}.$$

Put  $m = k^2$  and define

$$x(u) = m \text{sn}^2(u; k), \quad y(u) = m \text{sn}(u; k) \text{cn}(u; k) \text{dn}(u; k).$$

Prove

$$y(u)^2 = x(u)(x(u) - 1)(x(u) - m), \quad \frac{dx}{y} = 2 du.$$

Identify the period lattice of  $(x(u), y(u))$  and recover the uniformization of the Legendre curve constructed earlier.

**Exercise 17.6** (Lamé theta functions and the spectral curve). Let  $K = K(k)$ ,  $K' = K(\sqrt{1-k^2})$ , and  $\tau = iK'/K$ . Using the theta functions with characteristics, put

$$H(u) = \vartheta \left[ \begin{smallmatrix} \frac{1}{2} \\ \frac{1}{2} \end{smallmatrix} \right] \left( \frac{u}{2K} \middle| \tau \right), \quad \Theta(u) = \vartheta \left[ \begin{smallmatrix} 0 \\ \frac{1}{2} \end{smallmatrix} \right] \left( \frac{u}{2K} \middle| \tau \right), \quad Z(u) = \frac{d}{du} \log \Theta(u).$$

For the Lamé operator

$$L_k = -\frac{d^2}{du^2} + 2k^2 \text{sn}^2(u; k),$$

use the theta addition identities to verify the Hermite solutions

$$\psi_{\pm}(u, a) = \frac{H(u \pm a)}{\Theta(u)} \exp(\mp u Z(a)),$$

$$L_k \psi_{\pm}(\cdot, a) = E(a) \psi_{\pm}(\cdot, a), \quad E(a) = 1 + k^2 - k^2 \text{sn}^2(a; k).$$

From the theta automorphy laws, derive their Bloch multipliers over the potential period  $2K$ :

$$\rho_{\pm}(a) = -\exp(\mp 2K Z(a)).$$

Define

$$\mu(a) = \frac{i}{2} \frac{dE(a)}{da}.$$

Use the Jacobi differential identities to prove

$$\mu(a)^2 = (E(a) - k^2)(E(a) - 1)(E(a) - 1 - k^2).$$

Thus the parameter  $a$  uniformizes the Lamé spectral curve

$$\Sigma_k: \quad \mu^2 = (E - k^2)(E - 1)(E - 1 - k^2).$$

**Exercise 17.7** (Baker–Akhiezer functions and finite-gap reconstruction). Let

$$\Sigma : \quad \mu^2 = \prod_{j=0}^{2g} (E - E_j), \quad E_0 < E_1 < \cdots < E_{2g},$$

be a nonsingular real hyperelliptic curve with marked point  $\infty$ , and let  $D$  be an effective divisor of degree  $g$ . Let  $d\Omega_1$  be the unique meromorphic differential with zero  $a$ -periods, no pole away from  $\infty$ , and principal part  $d(i\sqrt{E})$  at  $\infty$ . Put

$$\Omega_1(p) = \int^p d\Omega_1, \quad U_j = \frac{1}{2\pi i} \int_{b_j} d\Omega_1,$$

take the Abel map  $\mathcal{A}$  to be based at  $\infty$ , and let  $Z \in \text{Jac}(\Sigma)$  be the theta translate determined by  $D$  through Riemann’s vanishing theorem.

For  $x \in \mathbb{R}$ , let  $F_x$  be the flat holomorphic line bundle whose holonomy around a closed curve  $\gamma$  is

$$\text{hol}_{F_x}(\gamma) = \exp \left( -x \int_{\gamma} d\Omega_1 \right).$$

Call  $(D, Z)$  *admissible* when, for every  $x \in \mathbb{R}$ ,

$$H^0(\Sigma, K_{\Sigma}(-D) \otimes F_x^{-1}) = 0, \quad \vartheta(Ux + Z \mid \Omega) \neq 0,$$

and  $\frac{d^2}{dx^2} \log \vartheta(Ux + Z \mid \Omega)$  is real. Since  $\deg F_x = 0$ , use Riemann–Roch to prove

$$\dim H^0(\Sigma, \mathcal{O}(D) \otimes F_x) = 1.$$

For admissible  $(D, Z)$ , define

$$\psi(x, p) = \frac{\vartheta(\mathcal{A}(p) + Ux + Z \mid \Omega) \vartheta(Z \mid \Omega)}{\vartheta(\mathcal{A}(p) + Z \mid \Omega) \vartheta(Ux + Z \mid \Omega)} e^{x\Omega_1(p)}.$$

Use the theta automorphy laws to prove that the apparent monodromy of the quotient cancels that of the exponential, so  $\psi$  is a function on  $\Sigma \setminus \{\infty\}$  with poles prescribed by  $D$ . Prove that  $\psi(x, p)e^{-x\Omega_1(p)}$  is the unique section of  $\mathcal{O}(D) \otimes F_x$  normalized to have value 1 at  $\infty$  in the local trivialization used in the theta formula. Deduce from the Riemann–Roch dimension count that  $\psi$  is the unique normalized Baker–Akhiezer function with these divisor, monodromy, and asymptotic data.

**Exercise 17.8** (Finite-gap reconstruction and KdV linearization). For admissible spectral data  $(\Sigma, D, Z)$  and its normalized Baker–Akhiezer function  $\psi$ , prove the finite-gap reconstruction theorem [12]: there is a constant  $c$  determined by the expansion at infinity such that

$$u(x) = c + 2 \frac{d^2}{dx^2} \log \vartheta(Ux + Z \mid \Omega)$$

is real and smooth for admissible  $(D, Z)$ , and

$$\left( -\frac{d^2}{dx^2} - u(x) \right) \psi(x, p) = E(p) \psi(x, p).$$

Prove as part of the same theorem the real finite-gap Floquet characterization: the multiplier has unit modulus precisely over

$$[E_0, E_1] \cup [E_2, E_3] \cup \cdots \cup [E_{2g-2}, E_{2g-1}] \cup [E_{2g}, \infty).$$

Deduce from the two points over a regular  $E$  and their reciprocal Bloch multipliers that this set is the spectrum and that its complementary intervals are the open gaps. Let  $d\Omega_3$  be the unique second-kind differential with zero  $a$ -periods, no pole away from  $\infty$ , and principal part  $d(4iE^{3/2})$  there in the branch compatible with  $d\Omega_1$ , and put

$$V_j = \frac{1}{2\pi i} \int_{b_j} d\Omega_3.$$

Prove the finite-gap linearization theorem [12]: the time-dependent Baker–Akhiezer function is obtained by adjoining the exponential of  $t \int^P d\Omega_3$ , its moving pole divisor satisfies

$$\mathcal{A}(D(t)) = \mathcal{A}(D(0)) + Vt,$$

and the reconstructed KdV potential is

$$u(x, t) = c + 2 \frac{d^2}{dx^2} \log \vartheta(Ux + Vt + Z \mid \Omega).$$

Deduce that KdV flow is translation on the Jacobian and preserves the spectral curve. Use the theta transformation law under a change of symplectic marking to prove that  $\Omega, U, V, Z$  change while the normalized Baker–Akhiezer function and reconstructed potential do not. Specialize to  $g = 1$  and recover, up to translation and an additive energy constant, the Lamé potential and its Hermite theta solutions.

## 17.4 Finite-Gap Transport

**Exercise 17.9** (Lamé band edges and attenuation prediction). Use the half-period values of the Jacobi functions to recover the three band-edge solutions

$$L_k \operatorname{dn} = k^2 \operatorname{dn}, \quad L_k \operatorname{cn} = \operatorname{cn}, \quad L_k \operatorname{sn} = (1 + k^2) \operatorname{sn}.$$

Determine which are periodic and antiperiodic over  $2K$ . Use the theta-function Bloch multipliers to prove

$$\operatorname{spec}(L_k) = [k^2, 1] \cup [1 + k^2, \infty)$$

and identify  $(1, 1 + k^2)$  as the unique open gap.

For an energy in the gap, choose  $a$  with  $E = E(a)$  and prove that a Bloch wave changes in magnitude by  $e^{-2K\kappa(E)}$  per cell, where

$$\kappa(E) = \frac{1}{2K} |\log |\rho_{\pm}(a)|| = |\operatorname{Re} Z(a)|.$$

Predict the three band-edge energies and show that, up to interface factors, the transmitted amplitude through  $M$  cells is suppressed by  $e^{-2KM\kappa(E)}$  and the transmitted intensity by  $e^{-4KM\kappa(E)}$ . Show that the gap closes as  $k \rightarrow 0$ , and state the periodic-medium, coherence, finite-sample, and energy-resolution assumptions under which a transmission experiment tests the spectral curve.



**Exercise 17.10** (Cnoidal KdV waves and finite-gap transport). For the normalized Korteweg–de Vries equation

$$u_t + 6uu_x + u_{xxx} = 0,$$

put  $\xi = x - vt$  and

$$u(x, t) = a + 2k^2\kappa^2 \operatorname{cn}^2(\kappa\xi; k), \quad 0 < k < 1.$$

Derive

$$\frac{d^2}{dz^2} \operatorname{cn}^2(z; k) = 2(1 - k^2) + 4(2k^2 - 1) \operatorname{cn}^2(z; k) - 6k^2 \operatorname{cn}^4(z; k)$$

and prove that the displayed wave solves KdV exactly when

$$v = 6a + 4\kappa^2(2k^2 - 1).$$

Deduce its amplitude, wavelength, and speed:

$$A = 2k^2\kappa^2, \quad \lambda = \frac{2K(k)}{\kappa}, \quad v = 6a + 4\kappa^2(2k^2 - 1).$$

Define the complete elliptic integral of the second kind and the elliptic nome by

$$E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \theta} d\theta, \quad q_J = e^{-\pi K'(k)/K(k)}.$$

If  $\tau = iK'(k)/K(k)$ , record that the modular nome is  $q_{\text{mod}} = e^{2\pi i\tau} = q_J^2$ . Prove the Fourier expansion

$$\operatorname{cn}^2(z; k) = \frac{E(k) - (1 - k^2)K(k)}{k^2 K(k)} + \frac{2\pi^2}{k^2 K(k)^2} \sum_{n \geq 1} \frac{nq_J^n}{1 - q_J^{2n}} \cos \frac{n\pi z}{K(k)}.$$

Specialize the finite-gap reconstruction to genus one, put  $V = -u$  in the Floquet operator at the beginning of the section, and show that it gives the cnoidal wave above and that  $k \uparrow 1$  gives the solitary wave

$$u(x, t) = a + 2\kappa^2 \operatorname{sech}^2(\kappa(x - (6a + 4\kappa^2)t)).$$

Predict the wavelength, phase speed, and harmonic-amplitude ratios of a weakly dispersive one-dimensional wave train from  $K$ ,  $E$ , and the nome. For the dimensional equation

$$\eta_t + c_0\eta_x + \alpha\eta\eta_x + \beta\eta_{xxx} = 0, \quad \alpha\beta \neq 0,$$

choose a length  $L$  and prove that

$$X = \frac{x - c_0 t}{L}, \quad T = \frac{\beta t}{L^3}, \quad u = \frac{\alpha L^2}{6\beta} \eta$$

gives the normalized KdV equation. Deduce the dimensional predictions

$$A_{\text{phys}} = \frac{12\beta k^2 \kappa^2}{\alpha L^2}, \quad \lambda_{\text{phys}} = \frac{2LK(k)}{\kappa}, \quad c_{\text{phys}} = c_0 + \frac{\beta}{L^2}(6a + 4\kappa^2(2k^2 - 1)).$$

State the KdV scaling, weak-nonlinearity, long-wavelength, negligible-loss, one-dimensional, and finite-window assumptions under which a wave-tank or transmission-line experiment tests these predictions.

## 18 Index Theory

### 18.1 Lattice Translations

**Exercise 18.1** (Fredholm operators and Bloch families). For a real smooth  $P$ -periodic potential on the line, let the lattice  $\mathbb{Z}$  act on distributional solutions by

$$(\tau_n \psi)(x) = \psi(x + nP).$$

Its unitary characters have the form  $\chi_\rho(n) = \rho^n$  with  $|\rho| = 1$ . Prove that the  $\chi_\rho$ -equivariant solutions are exactly the functions determined by their restriction to  $[0, P]$  with boundary conditions

$$\psi(P) = \rho\psi(0), \quad \psi'(P) = \rho\psi'(0),$$

so the twisted domains below are the lattice-translation sectors.

For the twisted periodic Schrödinger operators

$$\mathcal{L}_{\rho,E} : H_\rho^2 \longrightarrow L^2([0, P])$$

with  $|\rho| = 1$  and  $E \in \mathbb{R}$ , the preceding closed-range and kernel–cokernel calculations show that the image is closed and the kernel and cokernel are finite-dimensional. A bounded operator  $T : H_0 \rightarrow H_1$  is *Fredholm* when it has these three properties. Define

$$\text{ind}_{\text{an}}(T) = \dim \ker T - \dim \text{coker } T.$$

Use its self-adjoint realization on  $L^2([0, P])$  to prove

$$\text{ind}_{\text{an}}(\mathcal{L}_{\rho,E}) = 0, \quad \ker \mathcal{L}_{\rho,E} \neq 0 \iff \rho^2 - \Delta(E)\rho + 1 = 0.$$

Let  $H_0$  and  $H_1$  be complex Hilbert spaces. Write  $\mathcal{B}(H_0, H_1)$  for the bounded operators and  $\mathcal{K}(H_0, H_1)$  for the compact operators. Prove directly from the compactness of the closed-unit-ball image that composition on either side by a bounded operator preserves compactness and that  $\mathcal{K}(H_0, H_1)$  is operator-norm closed. For  $T \in \mathcal{K}(H_0, H_1)$ , apply the compact spectral theorem to  $T^*T$  and use its spectral projections to construct finite-rank operators converging to  $T$  in norm. Deduce that  $\mathcal{K}(H_0, H_1)$  is exactly the operator-norm closure of the finite-rank operators and that  $\mathcal{K}(H) = \mathcal{K}(H, H)$  is a closed two-sided ideal in  $\mathcal{B}(H) = \mathcal{B}(H, H)$ . Define the *Calkin algebra* by

$$\mathcal{Q}(H) = \mathcal{B}(H)/\mathcal{K}(H).$$

Prove Atkinson’s theorem [19]:  $T$  is Fredholm exactly when there is a bounded  $S : H_1 \rightarrow H_0$  for which

$$ST - I_{H_0} \in \mathcal{K}(H_0), \quad TS - I_{H_1} \in \mathcal{K}(H_1).$$

For  $H_0 = H_1$ , identify this condition with invertibility of the image of  $T$  in the Calkin algebra. Prove that compact perturbations preserve the Fredholm property and the index, that

$$\text{ind}_{\text{an}}(T^*) = -\text{ind}_{\text{an}}(T), \quad \text{ind}_{\text{an}}(T_2 T_1) = \text{ind}_{\text{an}}(T_2) + \text{ind}_{\text{an}}(T_1),$$

and that the index is constant along a norm-continuous path of Fredholm operators. On  $\ell^2(\mathbb{N})$ , let  $S$  be the unilateral shift  $Se_n = e_{n+1}$ . Construct a parametrix modulo finite-rank operators and compute

$$\text{ind}_{\text{an}}(S) = -1, \quad \text{ind}_{\text{an}}(S^*) = 1.$$

## 18.2 K-Theory Classification

**Exercise 18.2** (Elliptic operators and analytic index). Let  $E, F \rightarrow M$  be smooth complex vector bundles over a closed smooth manifold. For  $f \in C^\infty(M)$  let  $M_f$  denote multiplication by  $f$ . A linear map

$$D : \Gamma^\infty(E) \longrightarrow \Gamma^\infty(F)$$

is a *differential operator of order at most  $m$*  when every  $(m+1)$ -fold iterated commutator with multiplication operators vanishes. For an operator of order  $m$ ,  $x \in M$ ,  $\xi \in T_x^*M$ , and  $v \in E_x$ , choose  $f$  with  $df_x = \xi$  and a section  $s$  with  $s(x) = v$ , and define

$$\sigma_m(D)(x, \xi)v = \frac{1}{m!} \text{ad}_{M_f}^m(D)s(x), \quad \text{ad}_{M_f}(D) = [D, M_f].$$

Prove that this is independent of  $f$  and  $s$ , depends homogeneously of degree  $m$  on  $\xi$ , and defines the *principal symbol*

$$\sigma_m(D) : \pi^*E \longrightarrow \pi^*F \quad \text{over } T^*M.$$

Call  $D$  *elliptic* when  $\sigma_m(D)(x, \xi)$  is invertible for every  $\xi \neq 0$ .

Choose Hermitian and Riemannian metrics and a compatible connection on  $E$ . Using the spectral calculus for the nonnegative connection Laplacian  $\Delta_E = \nabla^* \nabla$ , define  $H^s(M; E)$  as the completion of smooth sections for the norm

$$\|u\|_{H^s} = \|(I + \Delta_E)^{s/2} u\|_{L^2}.$$

Prove [38] that different choices give equivalent norms. Develop coordinate-invariant symbol quantization and prove elliptic regularity and the parametrix theorem [38]: for every real  $s$ , an elliptic  $D$  extends to

$$D_s : H^s(M; E) \longrightarrow H^{s-m}(M; F)$$

and admits a bounded parametrix whose two remainders are compact. Deduce from Atkinson's theorem that  $D_s$  is Fredholm, that its index is independent of  $s$ , and use symbol quantization to prove that it depends only on the homotopy class of the elliptic principal symbol. Define this integer to be  $\text{ind}_{\text{an}}(D)$  and prove

$$\text{ind}_{\text{an}}(D^*) = -\text{ind}_{\text{an}}(D).$$

For a holomorphic line bundle  $L$  on a compact Riemann surface  $C$ , show that

$$\bar{\partial}_L : \Omega^{0,0}(C; L) \longrightarrow \Omega^{0,1}(C; L)$$

is elliptic. Prove [29, 38] the Dolbeault identification of its kernel and cokernel with  $H^0(C, L)$  and  $H^1(C, L)$ . Recover from Riemann–Roch the analytic-index formula

$$\text{ind}_{\text{an}}(\bar{\partial}_L) = \deg L + 1 - g(C).$$

**Exercise 18.3** (Homotopy, suspension, and clutching). A *based space* is a space  $X$  with a distinguished point  $x_0$ . Define based maps, based homotopies, and the set  $[X, Y]_*$  of based homotopy classes. Define the reduced suspension and based loop space by

$$\Sigma X = X \times [0, 1] / (X \times \{0, 1\} \cup \{x_0\} \times [0, 1]),$$

$$\Omega Y = \{\gamma : [0, 1] \rightarrow Y : \gamma(0) = \gamma(1) = y_0\}.$$

Construct the suspension–loop adjunction

$$[\Sigma X, Y]_* \cong [X, \Omega Y]_*.$$

For  $n \geq 1$ , define

$$\pi_n(Y) = [S^n, Y]_*, \quad S^n = \Sigma S^{n-1}.$$

Use the pinch map  $S^n \rightarrow S^n \vee S^n$  to define multiplication and loop reversal to define inverses. Prove that  $\pi_n(Y)$  is a group, abelian for  $n \geq 2$ , and that a based map induces a homomorphism functorially. Use the suspension–loop adjunction to prove

$$\pi_n(Y) \cong \pi_{n-1}(\Omega Y) \quad (n \geq 2),$$

and define the suspension homomorphism  $\pi_n(Y) \rightarrow \pi_{n+1}(\Sigma Y)$ .

Let  $V$  be a finite-dimensional complex vector space and let  $g : X \rightarrow \text{Aut}(V)$  be a based map. Glue the trivial bundles on the two reduced cones on  $X$  by  $g$  along their common copy of  $X$  and denote the resulting bundle on  $\Sigma X$  by  $E_g$ . Prove that based homotopic clutching maps give isomorphic bundles and that direct sum of clutching maps corresponds to direct sum of bundles. Call bundles  $E$  and  $F$  *stably isomorphic* when

$$E \oplus \mathbb{C}^r \cong F \oplus \mathbb{C}^s$$

for some  $r, s$ . Prove the clutching theorem [4]: for compact Hausdorff  $X$ , stable bundles on  $\Sigma X$  are classified by stable homotopy classes of clutching maps.

Take  $X = S^1$ ,  $V = \mathbb{C}$ , and let  $g(z)$  be multiplication by  $z$ . Identify  $\Sigma S^1$  with  $S^2$ , identify  $E_g$  with the Hopf line bundle after choosing the orientation compatible with the clutching convention, and use the circle-cocycle calculation to prove

$$\int_{S^2} c_1(E_g) = 1.$$

**Exercise 18.4** (Connections, curvature, and Chern–Weil forms). Let  $E \rightarrow M$  be a smooth complex vector bundle of rank  $r$ . Define its direct sum, tensor product, dual, and pullback by the corresponding fiberwise operations. Call a connection *complex* when  $\nabla_X(is) = i\nabla_X s$ . For a complex connection, extend  $\nabla$  to  $E$ -valued forms by the graded Leibniz rule and write

$$F_\nabla = \nabla^2 \in \Omega^2(M; \text{End } E)$$

for its curvature.

Prove the Bianchi identity  $\nabla F_\nabla = 0$ . Using trace and determinant, define

$$\text{ch}(E, \nabla) = \text{Tr} \exp\left(\frac{iF_\nabla}{2\pi}\right), \quad c(E, \nabla) = \det\left(I + \frac{iF_\nabla}{2\pi}\right) = 1 + c_1(E, \nabla) + \cdots + c_r(E, \nabla).$$

Use Bianchi and cyclicity of trace to prove that the homogeneous components of  $\text{ch}(E, \nabla)$  are closed. For a path  $\nabla_t$  of connections, derive

$$\frac{d}{dt} \text{Tr} \exp\left(\frac{iF_{\nabla_t}}{2\pi}\right) = d \text{Tr} \left( \frac{i\dot{\nabla}_t}{2\pi} \exp\left(\frac{iF_{\nabla_t}}{2\pi}\right) \right).$$

Prove [14] the corresponding Chern–Weil transgression formula for the determinant invariant polynomials. Deduce that the de Rham classes  $\text{ch}(E)$  and  $c_j(E)$  are independent of the connection, natural under pullback, and satisfy

$$c(E \oplus F) = c(E)c(F), \quad \text{ch}(E \oplus F) = \text{ch}(E) + \text{ch}(F), \quad \text{ch}(E \otimes F) = \text{ch}(E) \text{ch}(F).$$

Prove also that

$$\mathrm{ch}_q(E^\vee) = (-1)^q \mathrm{ch}_q(E).$$

Recover the first Chern form of a line bundle used above.

**Exercise 18.5** (Spectral projections and Berry bundles). Let  $W$  be a finite-dimensional Hermitian vector space, let  $B$  be a smooth manifold, and let  $H : B \rightarrow \mathrm{End}(W)$  be a smooth family of self-adjoint operators. Suppose a fixed contour  $\Gamma$  encloses an isolated spectral cluster of constant multiplicity. Define

$$P_b = \frac{1}{2\pi i} \oint_{\Gamma} (z - H(b))^{-1} dz, \quad \mathcal{E} = \coprod_{b \in B} \mathrm{im} P_b.$$

Prove that  $P_b$  is a smooth family of orthogonal projections and that  $\mathcal{E} \rightarrow B$  is a smooth Hermitian vector bundle. Construct its local product maps from the restrictions of nearby projections to a fixed fiber, without choosing a frame. Define the Berry connection by

$$\nabla^B s = P ds$$

for sections satisfying  $Ps = s$ , and prove

$$F_B = P(dP \wedge dP)P, \quad c_1(\mathcal{E}) = \left[ \frac{i}{2\pi} \mathrm{Tr} P(dP \wedge dP) \right].$$

**Exercise 18.6** (Two-level Berry degree and gap protection). For the Pauli endomorphisms and the distinguished two-band Hamiltonian on the Brillouin torus  $B = T^2$ ,

$$H(k) = d_1(k)\sigma_1 + d_2(k)\sigma_2 + d_3(k)\sigma_3, \quad d(k) \neq 0,$$

let  $\hat{d} = d/|d|$  and let  $\mathcal{E}_-$  be the negative-energy line bundle. From  $P_- = (I - \hat{d} \cdot \sigma)/2$ , derive

$$\int_{T^2} c_1(\mathcal{E}_-) = \frac{1}{4\pi} \int_{T^2} \hat{d} \cdot (\partial_{k_1} \hat{d} \times \partial_{k_2} \hat{d}) dk_1 dk_2 = \deg(\hat{d}).$$

Show that this integer cannot change under a smooth deformation that keeps the spectral gap open.

**Exercise 18.7** (Grothendieck  $K$ -theory and exterior products). For a compact Hausdorff space  $X$ , let  $\mathrm{Vect}_{\mathbb{C}}(X)$  be the commutative monoid of isomorphism classes of finite-rank complex vector bundles under direct sum. Define  $K^0(X)$  as its Grothendieck group: elements are formal differences  $[E] - [F]$ , with

$$[E] - [F] = [E'] - [F']$$

when there is a bundle  $G$  for which

$$E \oplus F' \oplus G \cong E' \oplus F \oplus G.$$

Use tensor product to make  $K^0(X)$  a ring. For based spaces define the smash product

$$X \wedge Y = X \times Y / (X \vee Y), \quad X \vee Y = X \times \{y_0\} \cup \{x_0\} \times Y,$$

and define the exterior product in  $K^0(X \times Y)$  by

$$a \boxtimes b = \mathrm{pr}_X^* a \cdot \mathrm{pr}_Y^* b.$$

Define

$$\widetilde{K}^0(X) = \ker(K^0(X) \longrightarrow K^0(\{x_0\})).$$

For a closed based subspace  $A \subset Z$ , define

$$K^0(Z, A) = \widetilde{K}^0(Z/A).$$

For  $a = [E] - [E'] \in \widetilde{K}^0(X)$ , stabilize and choose an isomorphism  $E_{x_0} \cong E'_{x_0}$ , and do the same for a representative of  $b \in \widetilde{K}^0(Y)$ . Show that  $a \boxtimes b$ , together with the induced stable trivialization on  $X \vee Y$ , determines a relative class in

$$K^0(X \times Y, X \vee Y) \cong \widetilde{K}^0(X \wedge Y),$$

independent of the stabilizations and isomorphisms. Deduce the bilinear reduced exterior product

$$\widetilde{K}^0(X) \otimes \widetilde{K}^0(Y) \longrightarrow \widetilde{K}^0(X \wedge Y).$$

**Exercise 18.8** (Bott periodicity and topological  $K$ -groups). Define

$$\widetilde{K}^{-n}(X) = \widetilde{K}^0(\Sigma^n X) \quad (n \geq 0), \quad K^{-n}(X) = \widetilde{K}^{-n}(X_+), \quad K^1(X) = K^{-1}(X).$$

Here  $X_+ = X \sqcup \{*\}$  for an unbased compact space  $X$ .

Let  $BU(r)$  be the classifying space characterized by the natural correspondence between rank- $r$  complex bundles on a compact space and homotopy classes of maps to  $BU(r)$ . Let  $BU$  and  $U$  denote the stable classifying and unitary spaces, and write  $[X, Y]$  for unbased homotopy classes. Prove [4] the stable classification

$$K^0(X) \cong [X, \mathbb{Z} \times BU], \quad K^1(X) \cong [X, U],$$

and complex Bott periodicity

$$\Omega(\mathbb{Z} \times BU) \simeq U, \quad \Omega U \simeq \mathbb{Z} \times BU.$$

Let

$$\beta = [E_g] - [\mathbb{C}] \in \widetilde{K}^0(S^2)$$

be the Bott class of the Hopf bundle. Prove as part of Bott periodicity that the displayed loop-space equivalence is represented in reduced  $K$ -theory by exterior product with  $\beta$ , and deduce

$$\widetilde{K}^0(\Sigma^n X) \cong \widetilde{K}^0(\Sigma^n X \wedge S^2) \cong \widetilde{K}^0(\Sigma^{n+2} X) \quad (n \geq 0).$$

Use this isomorphism to extend reduced and unreduced  $K$ -theory two-periodically to all degrees. Deduce

$$\widetilde{K}^0(S^{2r}) \cong \mathbb{Z}, \quad \widetilde{K}^0(S^{2r+1}) = 0, \quad K^1(S^{2r}) = 0, \quad K^1(S^{2r+1}) \cong \mathbb{Z}.$$

For a CW complex, call an inclusion  $A \hookrightarrow X$  a *cofibration* when it has the homotopy-extension property. Construct the boundary map and prove the six-term exact sequence in reduced  $K$ -theory [4] for a cofibration and the standard cell attachment  $S^1 \vee S^1 \rightarrow T^2 \rightarrow S^2$  to compute

$$K^0(T^2) \cong \mathbb{Z}^2, \quad K^1(T^2) \cong \mathbb{Z}^2.$$

**Exercise 18.9** (Chern characters and gapped band topology). For a complex vector bundle  $E \rightarrow X$ , prove the projective-bundle cohomology formula by induction over a trivializing cover and the Mayer–Vietoris sequence. Iterate it to construct the full flag bundle  $p : \text{Fl}(E) \rightarrow X$ , prove that  $p^*$  is injective on rational cohomology, and prove that  $p^*E$  has a filtration with line-bundle quotients.

Deduce the splitting principle [14]. For compact smooth  $X$ , prove integrality of the first Chern class of a line bundle from its circle cocycle, and use the splitting principle to prove that the Chern–Weil classes  $c_j(E)$  are the real images of integral cohomology classes. Use their rational polynomial relation with the Chern character and the Chern–Weil identities to prove that

$$\text{ch}([E] - [F]) = \text{ch}(E) - \text{ch}(F)$$

defines a natural ring homomorphism

$$\text{ch} : K^0(X) \longrightarrow H^{\text{even}}(X; \mathbb{Q}).$$

Prove [4] that these classes define the same natural transformation for finite CW complexes and that it becomes an isomorphism after tensoring with  $\mathbb{Q}$ .

For a continuous finite-rank spectral projection  $P(x)$  of a gapped Hamiltonian over a compact parameter space  $X$ , form its image bundle and class in  $K^0(X)$ . Prove that this class is unchanged by a gap-preserving homotopy and identify, for  $X = T^2$ , its rank and first-Chern components with the two integers in the displayed computation.

**Exercise 18.10** (Real  $K$ -theory and Clifford periodicity). For a compact Hausdorff space  $X$ , define  $KO^0(X)$  as the Grothendieck group of finite-rank real vector bundles. For a based compact space put

$$\widetilde{KO}^0(X) = \ker(KO^0(X) \longrightarrow KO^0(\{x_0\})), \quad \widetilde{KO}^{-n}(X) = \widetilde{KO}^0(\Sigma^n X),$$

and for an unbased compact space put

$$KO^{-n}(X) = \widetilde{KO}^{-n}(X_+) \quad (n \geq 0).$$

Let  $Cl_n$  be the real  $*$ -algebra generated by skew-adjoint elements  $e_1, \dots, e_n$  satisfying

$$e_i e_j + e_j e_i = -2\delta_{ij}.$$

Use this negative Riemannian Clifford convention throughout the geometric sections. Under complexification it is related to the Lorentzian gamma convention  $\gamma(v)\gamma(w) + \gamma(w)\gamma(v) = 2g(v, w)$  by the Wick rotation  $c(v) = i\gamma(v)$  on a positive-definite subspace. Use the universal property to prove

$$Cl_{m+n} \cong Cl_m \hat{\otimes} Cl_n.$$

Prove the real Clifford Morita periodicity  $Cl_{n+8} \simeq_{\text{Morita}} Cl_n$ , the Atiyah–Bott–Shapiro construction, and real Bott periodicity [4]:

$$KO^{-n-8}(X) \cong KO^{-n}(X),$$

with coefficient groups

|                      |              |                |                |   |              |   |   |    |
|----------------------|--------------|----------------|----------------|---|--------------|---|---|----|
| $n \bmod 8$          | 0            | 1              | 2              | 3 | 4            | 5 | 6 | 7  |
| $KO^{-n}(\text{pt})$ | $\mathbb{Z}$ | $\mathbb{Z}/2$ | $\mathbb{Z}/2$ | 0 | $\mathbb{Z}$ | 0 | 0 | 0. |

Identify the two  $\mathbb{Z}/2$  classes with the obstructions to extending the corresponding irreducible Clifford-module structures by one generator.

**Exercise 18.11** (The tenfold way and Majorana localization). A finite-range translation-invariant free-fermion Hamiltonian is *stably equivalent* to another when one may adjoin spectrally flat filled and empty bands and join the flattened Hamiltonians through a symmetry-preserving gapped

path. Prove the free-fermion tenfold-way theorem [42]. An antiunitary time-reversal operator  $T$ , an antiunitary particle–hole operator  $C$ , and a unitary chiral operator  $S$  satisfy

$$TH(k)T^{-1} = H(-k), \quad CH(k)C^{-1} = -H(-k), \quad SH(k)S^{-1} = -H(k).$$

Record absence of a symmetry by 0 and its square, when present, by +1 or −1. The ten symmetry classes and their periodic labels are

| class  | $T^2$ | $C^2$ | $S$ | $s$ | theory |
|--------|-------|-------|-----|-----|--------|
| $A$    | 0     | 0     | 0   | 0   | $K$    |
| $AIII$ | 0     | 0     | 1   | 1   | $K$    |
| $AI$   | +1    | 0     | 0   | 0   | $KO$   |
| $BDI$  | +1    | +1    | 1   | 1   | $KO$   |
| $D$    | 0     | +1    | 0   | 2   | $KO$   |
| $DIII$ | −1    | +1    | 1   | 3   | $KO$   |
| $AII$  | −1    | 0     | 0   | 4   | $KO$   |
| $CII$  | −1    | −1    | 1   | 5   | $KO$   |
| $C$    | 0     | −1    | 0   | 6   | $KO$   |
| $CI$   | +1    | −1    | 1   | 7   | $KO$   |

where  $s$  is read modulo two in the complex rows and modulo eight in the real rows. When both  $T$  and  $C$  occur, prove that their product supplies the displayed chiral symmetry after a phase normalization. Prove from the tenfold-way theorem that, after fixing the Hamiltonian on the boundary of a Brillouin cell so that its strong momentum-space quotient is  $S^d$ , the strong stable classes are  $K^{d-s}(\text{pt})$  in the complex rows and  $KO^{d-s}(\text{pt})$  in the real rows. Use complex and real Bott periodicity to compute the classification group for every class in dimensions 0, 1, 2, 3. In particular recover the integer class-A invariant in two dimensions, the  $\mathbb{Z}/2$  class-D invariant in one dimension, and the  $\mathbb{Z}/2$  class-AII invariant in two and three dimensions.

For a one-dimensional class-D Bogoliubov–de Gennes Hamiltonian, let  $A(0)$  and  $A(\pi)$  be its real invertible skew forms on the particle–hole fixed fibers. Define the Pfaffian intrinsically by

$$\frac{1}{n!} A(k)^{\wedge n} = \text{Pf}(A(k)) \quad \text{in } \det(V_k)^*, \quad \dim_{\mathbb{R}} V_k = 2n.$$

Use the Real bundle over a half Brillouin circle to transport the two determinant lines and compare the two nonzero Pfaffians. Prove that the resulting relative sign, written in any common oriented trivialization as

$$\nu = \text{sgn}(\text{Pf } A(0) \text{ Pf } A(\pi)) \in \{\pm 1\}$$

is independent of that trivialization, invariant under stabilization and gapped class-D homotopy, and can change only when the bulk gap closes. Let  $\tau_y, \tau_z$  be the distinguished Pauli endomorphisms of the particle–hole space. For the Kitaev-chain symbol

$$H(k) = (-\mu - 2t \cos k) \tau_z + 2\Delta \sin k \tau_y$$

derive

$$\nu = \text{sgn}(\mu^2 - 4t^2).$$

At  $\Delta = t > 0$ , solve the open-chain zero-mode recursion and prove that a Majorana end mode has amplitude proportional to  $(-\mu/(2t))^{j-1}$  exactly when  $|\mu| < 2t$ . Derive its localization length in lattice spacings,

$$\xi^{-1} = \log \frac{2t}{|\mu|},$$



and predict the bulk transition at  $|\mu| = 2t$  and the end-mode decay on the topological side. State the quadratic, translation-invariant-bulk, particle-hole, open-boundary, gap, localization-length, and parity-resolution assumptions under which the Pfaffian invariant predicts a zero-bias end mode.

**Exercise 18.12** (Spectral flow and Toeplitz index). Let  $A_t$ ,  $0 \leq t \leq 1$ , be a norm-continuous path of bounded self-adjoint Fredholm operators with invertible endpoints. Choose a partition  $0 = t_0 < \dots < t_N = 1$  and numbers  $\varepsilon_j > 0$  so that the finite-rank spectral projections

$$\mathbf{1}_{[-\varepsilon_j, \varepsilon_j]}(A_t)$$

vary continuously for  $t \in [t_{j-1}, t_j]$ . Define the *spectral flow* by

$$\text{sf}(A_t) = \sum_{j=1}^N \left( \text{rank } \mathbf{1}_{[0, \varepsilon_j]}(A_{t_j}) - \text{rank } \mathbf{1}_{[0, \varepsilon_j]}(A_{t_{j-1}}) \right).$$

Prove independence of the choices, additivity under concatenation, and invariance under endpoint-fixed homotopies through such paths.

Let  $W$  be a finite-dimensional Hermitian vector space, let  $u : S^1 \rightarrow U(W)$  be smooth, and define

$$\text{wind}(\det u) = \frac{1}{2\pi i} \int_{S^1} (\det u)^{-1} d(\det u).$$

Let  $H^2(S^1; W) \subset L^2(S^1; W)$  be the closed span of the nonnegative Fourier modes, let  $P_+$  be its orthogonal projection, and define the Toeplitz operator

$$T_u = P_+ M_u|_{H^2(S^1; W)}.$$

Approximate  $u$  and  $u^{-1}$  uniformly by trigonometric polynomials, prove that  $[P_+, M_u]$  is compact, and deduce from  $T_u T_{u^{-1}} = I + \text{compact}$  that  $T_u$  is Fredholm; prove that homotopies of  $u$  produce norm-continuous Fredholm paths. For  $W = \mathbb{C} \oplus W'$  compute the index when  $u(z) = z^n \oplus I_{W'}$  from the unilateral shift. Prove by induction from  $SU(r-1) \rightarrow SU(r) \rightarrow S^{2r-1}$  that  $SU(r)$  is simply connected for  $r \geq 2$ , and use the determinant fibration

$$SU(r) \longrightarrow U(r) \xrightarrow{\det} U(1)$$

to prove that determinant induces  $\pi_1(U(r)) \cong \mathbb{Z}$ . Combine this with index homotopy invariance and stabilization to prove

$$\text{ind}_{\text{an}}(T_u) = -\text{wind}(\det u).$$

Define the odd  $K$ -theory pairing on the circle by

$$\langle [u], [S^1] \rangle = \text{wind}(\det u) = -\text{ind}_{\text{an}}(T_u).$$

For a loop  $A_t$  of self-adjoint Fredholm operators based at an invertible operator, define its suspended operator on periodic paths by

$$\mathcal{D}_A = \frac{d}{dt} + A_t.$$

Prove the spectral-flow suspension theorem [8]

$$\text{ind}_{\text{an}}(\mathcal{D}_A) = \text{sf}(A_t)$$

and identify both integers with the corresponding  $K^1$  pairing.

**Exercise 18.13** (Thom classes and the topological index). For a locally compact space  $Y$ , define compactly supported  $K$ -theory by

$$K_c^j(Y) = \widetilde{K}^j(Y^+), \quad j \in \mathbb{Z}/2,$$

where  $Y^+$  is the one-point compactification. For a complex vector bundle  $E \rightarrow X$ , define its Thom space by

$$\text{Th}(E) = D(E)/S(E),$$

where  $D(E)$  and  $S(E)$  are its unit disk and sphere bundles for any Hermitian metric. Define a *Thom class*  $u_E \in K_c^0(E)$  to be a class whose restriction to each fiber is the Bott generator in  $K_c^0(\mathbb{C}^{\text{rank } E})$ . Use Bott periodicity to construct  $u_E$  and prove the  $K$ -theoretic Thom isomorphism [4]

$$K^j(X) \longrightarrow K_c^j(E), \quad a \longmapsto \pi^* a \cdot u_E.$$

For an elliptic operator  $D : \Gamma(E_0) \rightarrow \Gamma(E_1)$  on a closed manifold  $M$ , its principal symbol is an isomorphism away from the zero section. Define its *symbol class*

$$[\sigma(D)] \in K_c^0(T^*M)$$

as the stable-homotopy class of the triple  $(\pi^*E_0, \pi^*E_1, \sigma(D))$ , where in general a compactly supported class is represented by two bundles together with an isomorphism outside a compact set. Verify compatibility of this description with direct sum, homotopy, and stabilization.

Prove [14] the Whitney embedding and tubular-neighborhood theorems and choose an embedding  $M \hookrightarrow \mathbb{R}^N$ . Show that the Euclidean orthogonal splitting induces an embedding

$$T^*M \hookrightarrow T^*\mathbb{R}^N \cong \mathbb{R}^{2N}$$

whose normal bundle is the pullback of  $\nu \oplus \nu$ , where  $\nu$  is the normal bundle of  $M$ , and give  $\nu \oplus \nu$  the complex structure  $J(v, w) = (-w, v)$ . Use its Thom class, extension by zero, and Bott periodicity to define

$$\text{ind}_{\text{top}} : K_c^0(T^*M) \longrightarrow K_c^0(\mathbb{R}^{2N}) \cong \mathbb{Z}.$$

Prove [4] isotopy invariance and stabilization of embeddings to prove that this map is independent of all choices.

**Exercise 18.14** (Atiyah–Singer index theorem). Apply the splitting principle to a complex vector bundle  $E$ . If the first Chern classes of the resulting line bundles are  $x_j$ , call the  $x_j$  the *formal Chern roots* of  $E$ , so  $c(E) = \prod_j (1 + x_j)$ . Define

$$\text{Td}(E) = \prod_j \frac{x_j}{1 - e^{-x_j}},$$

and for an oriented real bundle  $V$  whose complexification has formal roots  $\{\pm x_j\}$  define

$$\widehat{A}(V) = \prod_j \frac{x_j/2}{\sinh(x_j/2)}.$$

Use the splitting principle to interpret these symmetric power series as characteristic classes. Use the  $K$ -theoretic Chern character constructed above and prove the multiplicativity of the two displayed classes under direct sum.

A *spin structure* on an oriented Riemannian  $n$ -manifold is a lift of its oriented orthonormal frame bundle through  $\text{Spin}(n) \rightarrow \text{SO}(n)$ . Let  $S = S^+ \oplus S^-$  be the associated complex spinor bundle in even dimension, characterized as a module over the complex Clifford bundle generated by  $v$  with  $v^2 = -\|v\|^2$ . For a Hermitian bundle  $E$  with connection, define the twisted chiral Dirac operator

$$\mathcal{D}_E^+ : \Gamma(S^+ \otimes E) \longrightarrow \Gamma(S^- \otimes E)$$

by composing the tensor-product connection with Clifford multiplication. Compute its principal symbol and prove ellipticity.

Define

$$\text{Spin}^c(n) = (\text{Spin}(n) \times U(1)) / \{(1, 1), (-1, -1)\}.$$

A *spin<sup>c</sup> structure* is a lift of the oriented frame bundle through  $\text{Spin}^c(n) \rightarrow \text{SO}(n)$ . Construct from the homomorphism  $U(n) \rightarrow \text{Spin}^c(2n)$  the canonical spin<sup>c</sup> structure of a Hermitian complex manifold [47], and identify its spinor bundle  $\Lambda^{0,*}T^*M$ . Define its chiral Dolbeault–Dirac operator for a holomorphic Hermitian bundle  $E$  with its Chern connection

$$\mathcal{D}_{\bar{\partial}, E}^+ = \sqrt{2}(\bar{\partial}_E + \bar{\partial}_E^*) : \Omega^{0, \text{even}}(M; E) \longrightarrow \Omega^{0, \text{odd}}(M; E)$$

and compute its principal symbol.

Prove the Atiyah–Singer index theorem [5]:

$$\text{ind}_{\text{an}}(D) = \text{ind}_{\text{top}}[\sigma(D)]$$

for every elliptic differential operator on a closed manifold. Derive its cohomological consequences

$$\text{ind}_{\text{an}}(\mathcal{D}_{\bar{\partial}, E}^+) = \int_M \text{ch}(E) \text{Td}(T^{1,0}M)$$

for a compact complex manifold and

$$\text{ind}_{\text{an}}(\mathcal{D}_E^+) = \int_M \hat{A}(TM) \text{ch}(E)$$

for a closed spin manifold.

For a compact Riemann surface  $C$ , use  $T^{1,0}C \cong K_C^\vee$  and the Riemann–Roch computation of  $\deg K_C$ , and identify the chiral Dolbeault–Dirac index with that of  $\bar{\partial}_L : \Omega^{0,0}(C; L) \rightarrow \Omega^{0,1}(C; L)$  to derive

$$\text{ind}_{\text{an}}(\bar{\partial}_L) = \int_C \left( c_1(L) + \frac{1}{2}c_1(T^{1,0}C) \right) = \deg L + 1 - g.$$

For a closed spin surface and a line bundle  $L$  of degree  $N$ , derive

$$\text{ind}_{\text{an}}(\mathcal{D}_L^+) = N$$

and identify this integer as the difference of the two chiral zero-mode counts.

### 18.3 Monopole Harmonics

**Exercise 18.15** (Monopole harmonics and spherical Landau levels). Let  $S_R^2$  be the round sphere of radius  $R$ , let  $N \in \mathbb{Z}$ , and let  $L_N \rightarrow S_R^2$  be a Hermitian line bundle with unitary connection of constant curvature

$$F_\nabla = -\frac{iN}{2R^2} dA, \quad \frac{i}{2\pi} \int_{S_R^2} F_\nabla = N.$$

For a particle of charge  $e$ , impose the convention  $F_{\nabla} = -ieB dA/\hbar$  and deduce the magnetic-flux quantization

$$\Phi_B = \int_{S_R^2} B dA = \frac{Nh}{e}.$$

Put  $q = |N|/2$ . Identify sections of  $L_N$  intrinsically with functions  $F : SU(2) \rightarrow \mathbb{C}$  satisfying

$$F(gh) = \chi_N(h)^{-1} F(g)$$

under the identification  $S^2 \cong SU(2)/U(1)$ , where

$$h_{\theta} = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}, \quad \chi_N(h_{\theta}) = e^{iN\theta}.$$

Use Peter–Weyl theory to recover the monopole-harmonic decomposition and show that

$$\ell = q + n, \quad n = 0, 1, 2, \dots, \quad \dim \mathcal{H}_{\ell} = 2\ell + 1 = |N| + 2n + 1.$$

In the distinguished polar chart for  $SU(2)/U(1)$ , derive the Jacobi polynomial expression for these eigensections from the Wigner matrix coefficients; use it to determine their nodal profiles without redefining the global sections. For the scalar magnetic Hamiltonian

$$H_N = \frac{\hbar^2}{2m} \nabla^* \nabla,$$

derive

$$E_n = \frac{\hbar^2}{2mR^2} \left( (q+n)(q+n+1) - q^2 \right), \quad E_{n+1} - E_n = \frac{\hbar^2}{mR^2} (n+q+1).$$

For  $N \geq 0$ , equip  $S^2 \cong \mathbb{P}^1$  with its complex structure, identify  $L_N$  with  $\mathcal{O}(N)$ , and define the spin<sup>c</sup> Dolbeault–Pauli Hamiltonian

$$H_+ = \frac{\hbar^2}{m} \bar{\partial}_{\nabla}^* \bar{\partial}_{\nabla}.$$

Use Riemann–Roch and the Dolbeault index theorem to prove

$$\dim \ker H_+ = N + 1,$$

identify these zero modes with the lowest monopole-harmonic multiplet, and predict the complete level degeneracies and spacings above. State the uniform-field, round-sphere, fixed-charge, spin<sup>c</sup> coupling, noninteracting, and spectral-resolution assumptions under which spherical Landau-level spectroscopy tests the Jacobi-polynomial and index predictions.

## 18.4 Quantized Hall Transport

**Exercise 18.16** (Family index and quantized Hall transport). Let  $B$  be compact and let  $T_b : H_0 \rightarrow H_1$  be a norm-continuous family of Fredholm operators. Prove [6] that there is a finite-dimensional subspace  $V \subset H_1$  for which

$$H_0 \oplus V \longrightarrow H_1, \quad (u, v) \longmapsto T_b u + v$$

is surjective for every  $b \in B$ . Prove that its kernels form a vector bundle  $\mathcal{K}_T \rightarrow B$  and define the *family index* by

$$\text{Ind}(T) = [\mathcal{K}_T] - [B \times V] \in K^0(B).$$

Prove independence of  $V$ , homotopy invariance, and additivity under direct sum. Show that the construction is local on  $B$  and therefore extends to continuous Fredholm morphisms between Hilbert bundles.

Let  $E_\tau$  carry a degree- $N > 0$  magnetic line bundle  $L_N$ , and let

$$\widehat{E}_\tau = \text{Hom}(H_1(E_\tau; \mathbb{Z}), U(1))$$

be the torus of boundary-condition twists. Let  $\mathcal{P} \rightarrow E_\tau \times \widehat{E}_\tau$  be the normalized Poincaré line bundle, whose restriction to  $E_\tau \times \{\chi\}$  is the flat line bundle  $F_\chi$ . Orient  $\widehat{E}_\tau$  so that

$$\frac{1}{2} \int_{E_\tau} c_1(\mathcal{P})^2$$

is its positive integral generator. Define the lowest-Landau-level bundle and filled-level line by

$$\mathcal{H}_N|_\chi = H^0(E_\tau, L_N \otimes F_\chi), \quad \mathcal{D}_N = \bigwedge^N \mathcal{H}_N.$$

Use the theta automorphy laws to realize the fibers by level- $N$  theta functions and Riemann–Roch to prove that  $\mathcal{H}_N$  has rank  $N$ . Show that the Sobolev completions of  $\bar{\partial}_{L_N \otimes F_\chi}$  form a Fredholm family and that Riemann–Roch identifies its family index with  $[\mathcal{H}_N]$ .

Prove the families Atiyah–Singer formula [6]: for a holomorphic submersion  $p : X \rightarrow B$  with compact fibers and a holomorphic bundle  $\mathcal{L} \rightarrow X$ ,

$$\text{ch Ind}(\bar{\partial}_{\mathcal{L}}) = \int_{X/B} \text{ch}(\mathcal{L}) \text{Td}(T^{1,0}(X/B)),$$

where the right side is integration along the compact complex fibers. Apply it to  $X = E_\tau \times \widehat{E}_\tau$ ,  $B = \widehat{E}_\tau$ , and  $\mathcal{L} = \text{pr}_1^* L_N \otimes \mathcal{P}$ . Use the triviality of the elliptic-curve tangent bundle to prove

$$\text{ch}(\mathcal{H}_N) = \int_{E_\tau} \exp(c_1(L_N) + c_1(\mathcal{P})).$$

Extract its degree-zero and degree-two parts to prove

$$\text{rank } \mathcal{H}_N = N, \quad \int_{\widehat{E}_\tau} c_1(\mathcal{D}_N) = 1.$$

For the Fermi projection  $P(\phi_1, \phi_2)$  over the boundary-twist torus, prove [64] the gapped noninteracting Kubo–Chern formula and the adiabatic theorem,

$$\sigma_{xy} = \frac{e^2}{h} \frac{i}{2\pi} \int_{\widehat{E}_\tau} \text{Tr}(P dP \wedge dP) = \frac{e^2}{h} \int_{\widehat{E}_\tau} c_1(\mathcal{D}_N).$$

Deduce for one completely filled Landau level

$$\sigma_{xy} = \frac{e^2}{h}, \quad Q_{\text{pump}} = e$$

for a unit winding of a boundary twist. Express the Berry connection of  $\mathcal{D}_N$  through the normalized theta sections and identify the same integer from their automorphy. Identify the pumped charge with the spectral-flow and  $K^1$ -pairing integer. State separately the spectral-gap, adiabatic, noninteracting, filled-level, mobility, contact, temperature, and finite-size assumptions under which Hall-voltage and charge-pump measurements test these predictions.

## 19 Cobordism

### 19.1 $SO$ , Spin, and $U$ Structures

**Exercise 19.1** (Oriented, spin, and complex bordism rings). For an oriented manifold  $M$ , write  $-M$  for the same manifold with the opposite orientation. An *oriented bordism* from a closed oriented  $n$ -manifold  $M_0$  to a closed oriented  $n$ -manifold  $M_1$  is a compact oriented  $(n+1)$ -manifold  $W$  with an orientation-preserving identification

$$\partial W \cong (-M_0) \sqcup M_1,$$

product neighborhoods of its boundary components, and the boundary orientation determined by the outward-normal-first convention. Call  $M_0$  and  $M_1$  *oriented bordant* when such a  $W$  exists.

Prove that oriented bordism is an equivalence relation. Let  $\Omega_n^{SO}$  be the set of its equivalence classes. Prove that disjoint union makes  $\Omega_n^{SO}$  an abelian group with zero the empty manifold and inverse  $[M] \mapsto [-M]$ . Use the product orientation to make

$$\Omega_*^{SO} = \bigoplus_{n \geq 0} \Omega_n^{SO}$$

a graded-commutative ring with unit the positively oriented point. Prove

$$\Omega_0^{SO} \cong \mathbb{Z}, \quad \Omega_1^{SO} = 0, \quad \Omega_2^{SO} = 0$$

by signed point count and disks. Prove the classification of closed oriented surfaces from polygonal decompositions and construct oriented handlebodies to complete the calculation in degree two.

For a spin manifold with boundary, use the outward unit normal to induce a spin structure on its boundary. Define *spin bordism* by requiring the boundary identification to preserve these structures, and construct the graded ring  $\Omega_*^{\text{Spin}}$ . A *stable complex structure* on  $M$  is a complex-vector-bundle structure on  $TM \oplus \mathbb{R}^r$  for some  $r$ , considered modulo addition of trivial complex line bundles and homotopy. Define stable complex bordism and its ring  $\Omega_*^U$ . Prove that products inherit the corresponding structures and construct the forgetful homomorphisms

$$\Omega_*^{\text{Spin}} \longrightarrow \Omega_*^{SO}, \quad \Omega_*^U \longrightarrow \Omega_*^{SO}.$$

### 19.2 Pontryagin–Thom Transform

**Exercise 19.2** (Pontryagin–Thom collapse and Thom spectra). For based spaces  $E_k$ , a sequence of based structure maps  $\Sigma E_k \rightarrow E_{k+1}$  is a *spectrum*; its stable homotopy groups are

$$\pi_n(E) = \varinjlim_k \pi_{n+k}(E_k),$$

with transition maps induced by suspension and the structure maps. A sequence with structure maps  $\Sigma^2 E_k \rightarrow E_{k+1}$  is indexed in even degrees and has stable groups  $\varinjlim_k \pi_{n+2k}(E_k)$ .

For  $G = SO, \text{Spin}$ , let  $BG(k)$  be the classifying space for rank- $k$  real vector bundles with  $G$ -structure and let  $\gamma_k^G \rightarrow BG(k)$  be the universal bundle. Put

$$MG(k) = \text{Th}(\gamma_k^G), \quad \pi_n(MG) = \varinjlim_k \pi_{n+k}(MG(k)),$$

where stabilization of the universal bundles induces the structure maps. For complex rank- $k$  bundles, put

$$MU(k) = \text{Th}(\gamma_k^U), \quad \pi_n(MU) = \varinjlim_k \pi_{n+2k}(MU(k)).$$

Call the resulting sequences of Thom spaces the spectra  $MSO$ ,  $M\text{Spin}$ , and  $MU$ . Construct their structure maps from stabilization by a trivial real line or complex line and verify the displayed formulas for their stable homotopy groups.

Choose an embedding  $M^n \hookrightarrow \mathbb{R}^{n+k}$  of a closed  $G$ -manifold, give its normal bundle the stable  $G$ -structure complementary to that on  $TM$ , and let its classifying map pull back  $\gamma_k^G$  to the normal bundle. Collapse the complement of a tubular neighborhood to construct

$$S^{n+k} \longrightarrow \text{Th}(\nu_M) \longrightarrow MG(k).$$

Prove that isotopy, stabilization, and a  $G$ -bordism do not change the resulting stable homotopy class. Prove the transversality and regular-value construction in the Pontryagin–Thom theorem [63] to prove that collapse induces ring isomorphisms

$$\Omega_n^{SO} \cong \pi_n(MSO), \quad \Omega_n^{\text{Spin}} \cong \pi_n(M\text{Spin}), \quad \Omega_n^U \cong \pi_n(MU).$$

A *stable framing* of  $M$  is a trivialization of its stable normal bundle. Let  $\Omega_n^{\text{fr}}$  be framed bordism and define the  $n$ th stable stem by

$$\pi_n^S = \varinjlim_k [S^{n+k}, S^k]_*.$$

Show that a framing turns the collapse map into a map  $S^{n+k} \rightarrow S^k$  and recover

$$\Omega_n^{\text{fr}} \cong \pi_n^S.$$

### 19.3 Bordism Genera

**Exercise 19.3** (Pontryagin numbers and the signature genus). For a real vector bundle  $V$ , define its Pontryagin classes by

$$p_j(V) = (-1)^j c_{2j}(V \otimes_{\mathbb{R}} \mathbb{C}) \in H_{\text{dR}}^{4j}(M).$$

For a closed oriented  $4k$ -manifold and a partition  $I = (i_1, \dots, i_r)$  of  $k$ , define the Pontryagin number

$$p_I[M] = \int_M p_{i_1}(TM) \wedge \cdots \wedge p_{i_r}(TM).$$

If  $M = \partial W$  as an oriented manifold, use

$$TW|_M \cong TM \oplus \underline{\mathbb{R}}$$

and choose a connection that is a product near the boundary. Use Stokes' theorem for its Chern–Weil representatives to prove  $p_I[M] = 0$ . Deduce that every Pontryagin number defines a homomorphism  $\Omega_{4k}^{SO} \rightarrow \mathbb{Q}$ . Prove Thom's rational bordism theorem [63]: two oriented manifolds determine the same class in  $\Omega_*^{SO} \otimes \mathbb{Q}$  exactly when all their Pontryagin numbers agree.

For a closed oriented  $4k$ -manifold, prove Poincaré duality [14] to prove that

$$([\alpha], [\beta]) \longmapsto \int_M \alpha \wedge \beta$$

is a nondegenerate symmetric form on  $H_{\text{dR}}^{2k}(M)$ , and define  $\text{sign}(M)$  to be its signature. If the formal Pontryagin roots satisfy  $p(TM) = \prod_j (1 + x_j^2)$ , define

$$L(TM) = \prod_j \frac{x_j}{\tanh x_j}.$$

Prove the Hirzebruch signature theorem [37]

$$\text{sign}(M) = \int_M L(TM)$$

and deduce that the signature is a ring homomorphism  $\Omega_*^{SO} \rightarrow \mathbb{Z}$ . In dimension four derive

$$\text{sign}(M) = \frac{1}{3} \int_M p_1(TM).$$

For  $h = c_1(\mathcal{O}(1))$  on  $\mathbb{CP}^2$ , prove the Euler sequence [36]

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}(1)^{\oplus 3} \longrightarrow T\mathbb{CP}^2 \longrightarrow 0$$

and prove

$$c(T\mathbb{CP}^2) = (1 + h)^3, \quad \int_{\mathbb{CP}^2} p_1 = 3, \quad \text{sign}(\mathbb{CP}^2) = 1,$$

and conclude that  $[\mathbb{CP}^2]$  has infinite order in  $\Omega_4^{SO}$ .

**Exercise 19.4** (Dirac and Dolbeault bordism genera). For a commutative ring  $R$ , a *genus* is a ring homomorphism from a bordism ring to  $R$ . Given a formal power series  $Q(x) \in R[[x]]$  with  $Q(0) = 1$ , use the splitting principle to construct the multiplicative characteristic class  $\prod_j Q(x_j)$  from the appropriate formal roots. Prove that its top-degree characteristic number is additive under disjoint union and multiplicative under products, and determine the condition under which stabilization leaves it unchanged.

For a closed spin  $4k$ -manifold, define

$$\hat{A}[M] = \int_M \hat{A}(TM).$$

Construct the graded tensor-product spinor bundle on a product, prove the product formula for its Dirac operator, and use the Atiyah–Singer theorem to prove

$$\hat{A}[M] = \text{ind}_{\text{an}}(\mathcal{D}_M^+) \in \mathbb{Z}$$

and that  $\hat{A}$  is a genus  $\Omega_*^{\text{Spin}} \rightarrow \mathbb{Z}$ . Prove its bordism invariance directly from the stable tangent bundle of a boundary and Stokes' theorem.

For a closed stable complex manifold  $X$ , define

$$\text{Td}[X] = \int_X \text{Td}(TX).$$

Prove the canonical  $\text{spin}^c$  Dirac operator associated to a stable complex structure [47], and apply the index theorem to prove that  $\text{Td}[X]$  is integral and defines a genus  $\Omega_*^U \rightarrow \mathbb{Z}$ . When  $X$  is a compact complex manifold, identify it with

$$\text{Td}[X] = \sum_q (-1)^q \dim H^q(X, \mathcal{O}_X).$$



From the degree-zero and degree-two terms of the Todd class, prove

$$\mathrm{Td}[\mathrm{pt}] = 1, \quad \mathrm{Td}[C] = 1 - g$$

for a compact Riemann surface  $C$  of genus  $g$ .

**Exercise 19.5** (Atiyah–Patodi–Singer index and spectral eta functions). Let  $W$  be a compact even-dimensional spin Riemannian manifold whose metric is a product near its boundary  $M$ . Let  $E \rightarrow W$  be a Hermitian bundle with unitary connection of product type there. Write  $\mathcal{D}_{W,E}^+$  for the chiral Dirac operator and  $A_{M,E}$  for its self-adjoint tangential boundary operator. For  $\mathrm{Re} s$  sufficiently large, define the eta function

$$\eta_{A_{M,E}}(s) = \sum_{\lambda \neq 0} \mathrm{sign}(\lambda) |\lambda|^{-s},$$

where the spectral values are repeated with multiplicity. Construct the small-time heat-kernel expansion and apply its Mellin transform to prove that this function has a meromorphic continuation regular at  $s = 0$  [7].

Let  $P_{\geq 0}(A_{M,E})$  be the spectral projection onto the nonnegative spectrum and impose the Atiyah–Patodi–Singer boundary condition

$$P_{\geq 0}(A_{M,E})(\psi|_M) = 0.$$

Prove the Atiyah–Patodi–Singer index theorem [7]

$$\mathrm{ind}_{\mathrm{APS}}(\mathcal{D}_{W,E}^+) = \int_W \hat{A}(TW) \mathrm{ch}(E) - \frac{\eta_{A_{M,E}}(0) + h_{M,E}}{2}, \quad h_{M,E} = \dim \ker A_{M,E}.$$

Define the reduced eta invariant by

$$\bar{\eta}(A) = \frac{\eta_A(0) + \dim \ker A}{2} \quad \text{in } \mathbb{R}/\mathbb{Z}.$$

Deduce

$$\bar{\eta}(A_{M,E}) = \int_W \hat{A}(TW) \mathrm{ch}(E) \pmod{\mathbb{Z}}.$$

Prove

$$\bar{\eta}(-A) = -\bar{\eta}(A) \pmod{\mathbb{Z}}$$

from spectral symmetry under orientation reversal. When two product-type bordisms are glued along oppositely oriented copies of the same boundary, use additivity of the characteristic integrals to show that their reduced eta corrections cancel modulo  $\mathbb{Z}$ . Recover the closed Atiyah–Singer formula when the boundary is empty.

**Exercise 19.6** (Hurwitz zeta functions and flux spectral asymmetry). For  $0 < a < 1$ , let

$$A_a = -i \frac{d}{d\theta} + a$$

act on periodic functions on  $\mathbb{R}/2\pi\mathbb{Z}$ . Prove that its spectral values are  $n + a$ ,  $n \in \mathbb{Z}$ . Define the Hurwitz zeta function for  $\mathrm{Re} s > 1$  by

$$\zeta(s, a) = \sum_{n \geq 0} (n + a)^{-s}.$$

Separate the positive and negative spectral values to prove

$$\eta_{A_a}(s) = \zeta(s, a) - \zeta(s, 1 - a).$$

Apply Euler–Maclaurin to the first  $N$  terms and the tail to obtain, first for  $\operatorname{Re} s > 1$  and then by continuation near  $s = 0$ ,

$$\zeta(s, a) = \sum_{n=0}^{N-1} (n+a)^{-s} + \frac{(N+a)^{1-s}}{s-1} + \frac{1}{2}(N+a)^{-s} + sR_N(s, a),$$

where  $R_N$  is holomorphic near  $s = 0$ . Set  $s = 0$  to prove  $\zeta(0, a) = \frac{1}{2} - a$  and derive

$$\eta_{A_a}(0) = 1 - 2a, \quad \bar{\eta}(A_a) = \frac{1}{2} - a \pmod{\mathbb{Z}}.$$

For a single chiral massless charged Dirac channel on a coherent ring of circumference  $\ell$  threaded by flux  $\Phi$ , put

$$a = \frac{\Phi}{\Phi_0} \pmod{\mathbb{Z}}, \quad \Phi_0 = \frac{h}{e}, \quad E_n = \frac{2\pi\hbar v}{\ell}(n+a).$$

Define the eta-regularized vacuum charge by

$$Q_{\text{vac}} = -\frac{e}{2}\eta_{A_a}(0).$$

Predict

$$\frac{Q_{\text{vac}}}{e} = a - \frac{1}{2} \pmod{\mathbb{Z}}$$

and prove that insertion of one flux quantum produces unit spectral flow. State the single-channel, linear-dispersion, coherent-ring, charge regularization, flux-calibration, temperature, and level-resolution assumptions under which charge sensing and flux-resolved spectroscopy test the prediction.

**Exercise 19.7** (Jacobi theta functions and the K3 elliptic genus). For a closed complex manifold  $X$  of complex dimension  $d$  with formal Chern roots  $x_1, \dots, x_d$ , define its two-variable elliptic genus by

$$\operatorname{Ell}_X(\tau, z) = \int_X \prod_{j=1}^d x_j \frac{\vartheta_1\left(\frac{x_j}{2\pi i} - z \mid \tau\right)}{\vartheta_1\left(\frac{x_j}{2\pi i} \mid \tau\right)}, \quad \operatorname{Im} \tau > 0.$$

Prove [69] that this characteristic series defines a stable complex bordism genus with values in holomorphic functions of  $(\tau, z)$ .

Define

$$\vartheta_2(z \mid \tau) = \vartheta \begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix} (z \mid \tau), \quad \vartheta_3(z \mid \tau) = \vartheta \begin{bmatrix} 0 \\ 0 \end{bmatrix} (z \mid \tau), \quad \vartheta_4(z \mid \tau) = \vartheta \begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix} (z \mid \tau).$$

A *K3 surface* is a compact connected complex surface  $X$  with  $K_X \cong \mathcal{O}_X$  and  $H^1(X, \mathcal{O}_X) = 0$ . Use Serre duality and the Dolbeault index formula to prove

$$\operatorname{Td}[X] = 2, \quad \int_X c_2(TX) = 24.$$

Prove the required theta addition identities from the theta series and their divisor and automorphy laws [24]. Use them with the Chern-root formula and the K3 conditions  $c_1 = 0$ ,  $c_2[X] = 24$  to derive the K3 identity

$$\text{Ell}_{K3}(\tau, z) = 8 \left[ \left( \frac{\vartheta_2(z | \tau)}{\vartheta_2(0 | \tau)} \right)^2 + \left( \frac{\vartheta_3(z | \tau)}{\vartheta_3(0 | \tau)} \right)^2 + \left( \frac{\vartheta_4(z | \tau)}{\vartheta_4(0 | \tau)} \right)^2 \right].$$

Put  $q = e^{2\pi i \tau}$  and  $y = e^{2\pi i z}$ . Derive

$$\text{Ell}_{K3}(\tau, z) = 2y + 20 + 2y^{-1} + O(q), \quad \text{Ell}_{K3}(\tau, 0) = 24.$$

An  $\mathcal{N} = (4, 4)$  supersymmetric sigma model has commuting left- and right-moving small  $\mathcal{N} = 4$  superconformal algebras, each with four supercurrents and an  $SU(2)$  current algebra;  $L_0, \bar{L}_0$  are their energy operators,  $J_0$  is a chosen left  $SU(2)$  Cartan charge, and  $F_L, F_R$  are the fermion parities. For such a model with K3 target, prove [69, 24] the Ramond-sector identity

$$\text{Ell}_{K3}(\tau, z) = \text{Tr}_{\mathcal{H}_{RR}} \left( (-1)^{F_L + F_R} y^{J_0} q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - \bar{c}/24} \right)$$

and prove that right-moving supersymmetry cancels every non-ground-state pair, making the trace independent of  $\bar{q}$ . Predict 24 protected Ramond ground states with charge-resolved multiplicities 2, 20, 2 at charges 1, 0,  $-1$ . State the exact-supersymmetry, Ramond-boundary, charge-normalization, unbroken-supersymmetry, continuum-scaling, finite-volume, temperature, and symmetry-resolved spectroscopy assumptions under which a programmable quantum system tests these protected counts.

**Exercise 19.8** (Complex orientations and elliptic formal groups). A *multiplicative generalized cohomology theory* is a homotopy-invariant contravariant functor on pairs with products, suspension isomorphisms, and long exact sequences. A *ring spectrum* is a spectrum  $E$  with a unit and multiplication  $E \wedge E \rightarrow E$  that are associative and unital up to coherent homotopy; it represents such a theory. Ordinary integral cohomology and complex  $K$ -theory are examples. Let  $\mathcal{L} \rightarrow \mathbb{CP}^\infty$  be the tautological line bundle. A *complex orientation* of  $E$  is a class

$$x_E \in \tilde{E}^2(\mathbb{CP}^\infty)$$

whose restriction to  $\mathbb{CP}^1$  is the suspended unit. For a complex line bundle  $L \rightarrow X$  classified by  $f_L : X \rightarrow \mathbb{CP}^\infty$ , put  $c_1^E(L) = f_L^* x_E$ .

Prove the projective-bundle theorem for a complex-oriented theory [56],

$$E^*(\mathbb{CP}^\infty) \cong E^*[[x_E]].$$

The tensor product of line bundles induces a map  $m : \mathbb{CP}^\infty \times \mathbb{CP}^\infty \rightarrow \mathbb{CP}^\infty$ . Define  $F_E(x, y)$  by

$$m^* x_E = F_E(x_E \otimes 1, 1 \otimes x_E).$$

Use the unit, symmetry, and associativity of tensor product to prove

$$F_E(x, 0) = x, \quad F_E(x, y) = F_E(y, x), \quad F_E(F_E(x, y), z) = F_E(x, F_E(y, z)),$$

so  $F_E$  is a one-dimensional commutative formal group law. Derive

$$F_{H\mathbb{Z}}(x, y) = x + y.$$

For periodic complex  $K$ -theory with Bott element  $\beta \in K^{-2}(\text{pt})$ , take  $x_K = \beta^{-1}(1 - [\mathcal{L}])$  and derive

$$F_K(x, y) = x + y - \beta xy.$$

Prove Quillen's complex-cobordism theorem [56]:  $MU$  is the universal complex-oriented ring spectrum and its coefficient ring carries the universal one-dimensional commutative formal group law. Prove that a chosen complex orientation of a ring spectrum  $E$  induces a map  $MU \rightarrow E$  and specializes the universal formal group law to  $F_E$ .

Let  $C$  be a smooth elliptic curve over a commutative ring  $R$  with identity section. Complete its group law at the identity and, after choosing a local parameter, obtain a formal group law  $F_C$ . Call  $E$  *even-periodic* when its odd coefficient groups vanish and it has an invertible class of degree two inducing two-periodicity. Call an even-periodic complex-oriented theory  $E$  an *elliptic cohomology theory associated to  $C$*  when  $E^0 \cong R$  and its formal group is identified with the formal completion of  $C$ . Prove that changing the local parameter gives an isomorphic formal group law and hence does not change the coordinate-free formal group. Prove [3] that topological modular forms assemble these local elliptic cohomology theories over the moduli of elliptic curves, retaining the descent and torsion information absent from a single modular-form expansion.

**Exercise 19.9** (String bordism, the Witten genus, and topological modular forms). Write  $K(\mathbb{Z}, 4)$  for an Eilenberg–Mac Lane space representing integral degree-four cohomology. For a based map  $f : X \rightarrow Y$ , define its homotopy fiber to be the space of pairs  $(x, \gamma)$  in which  $\gamma$  is a path from  $f(x)$  to the base point of  $Y$ . Let  $B\text{String}$  be the homotopy fiber of the universal class

$$\lambda : B\text{Spin} \longrightarrow K(\mathbb{Z}, 4),$$

and define  $M\text{String}$  from the pulled-back universal stable bundle by the Thom-spectrum construction above. A *string structure* on a spin manifold  $M$  is a lift of the stable tangent classifying map to  $B\text{String}$ ; its obstruction is

$$\lambda(TM) = \frac{p_1(TM)}{2} \in H^4(M; \mathbb{Z}).$$

Let  $M$  be a closed string manifold of dimension  $4m$ , and write the formal roots of  $TM \otimes \mathbb{C}$  as  $\{\pm 2\pi i x_j\}$ . Define its Witten genus by

$$\varphi_W(M; \tau) = \int_M \prod_j x_j \frac{\vartheta'_1(0 \mid \tau)}{\vartheta_1(x_j \mid \tau)}.$$

Use the theta transformation laws and the vanishing of  $\lambda$  to prove that the integrand's anomalous exponential factor cancels and that  $\varphi_W(M; \tau)$  is a modular form of weight  $2m$ . Expand the theta product and prove that its constant term is  $\hat{A}[M]$ , that the genus is multiplicative, and that it vanishes on string boundaries.

Let  $\text{tmf}$  denote the connective spectrum of topological modular forms. Prove the sigma orientation and its rational coefficient theorem [3, 2]: there is a ring map of spectra

$$M\text{String} \longrightarrow \text{tmf}$$

whose modular-form realization is the Witten genus, and

$$\pi_{2k} \text{tmf} \otimes \mathbb{Q} \cong M_k \otimes \mathbb{Q}$$

for  $k \geq 0$  in the even connective grading. Explain why the tmf class refines the rational modular form by retaining integral and torsion information that the  $q$ -series alone cannot detect.

For a 24-dimensional string manifold, deduce

$$\varphi_W(M) = aE_4^3 + b\Delta$$

for rational  $a, b$  subject to the tmf integrality refinement, and express the first Fourier coefficients in terms of  $a$  and  $b$ . Interpret the Witten genus as the supersymmetric index of a two-dimensional sigma model and state the string-condition, anomaly-cancellation, supersymmetry, compactness, modular-normalization, and finite-energy assumptions under which its signed excitation counts may be compared with a physical partition function. Distinguish this index from an unsigned state degeneracy and from the two-variable K3 elliptic genus above.

## 19.4 SPT Phases

**Exercise 19.10** (Bordism characters and invertible topological phases). Let  $G$  be a finite internal symmetry group and let  $H = SO$  or  $\text{Spin}$ . An  $H$ -bordism over  $BG$  from  $(M_0, A_0)$  to  $(M_1, A_1)$  is an  $H$ -bordism  $W$  together with a map  $A : W \rightarrow BG$  whose restrictions are  $A_0$  and  $A_1$ . Let  $\Omega_d^H(BG)$  be the resulting bordism group of closed  $d$ -manifolds equipped with maps to  $BG$ . Prove that disjoint union defines its addition and that forgetting the map to  $BG$  gives a homomorphism

$$\Omega_d^H(BG) \longrightarrow \Omega_d^H.$$

Define a  $U(1)$ -valued bordism character to be a homomorphism

$$Z : \Omega_d^H(BG) \longrightarrow U(1).$$

Write its value on  $(M, A)$  as  $Z(M, A)$ . Prove

$$Z(\emptyset) = 1, \quad Z((-M), A) = \overline{Z(M, A)}, \quad Z(M \sqcup N, A \sqcup B) = Z(M, A)Z(N, B).$$

Define stacking by pointwise multiplication of characters and prove that complex conjugation supplies the stacking inverse. Call a gapped  $G$ -symmetric phase *invertible* when, after adjoining decoupled product degrees of freedom, it has a stacking inverse, and call two such phases equivalent when a uniformly gapped  $G$ -symmetric path joins their Hamiltonians. Prove [30] that, under the reflection-positivity and low-energy topological-field-theory hypotheses, deformation classes of invertible phases are computed from the associated Thom spectrum and their closed-manifold partition functions give bordism characters. Distinguish this full stable-homotopy classification from the information retained by partition functions on closed manifolds.

A *normalized*  $U(1)$ -valued 2-cocycle on  $G$  is a function  $\omega : G \times G \rightarrow U(1)$  satisfying

$$\omega(e, g) = \omega(g, e) = 1, \quad \omega(g, h)\omega(gh, k) = \omega(h, k)\omega(g, hk).$$

For  $\beta : G \rightarrow U(1)$  with  $\beta(e) = 1$ , define

$$(\delta\beta)(g, h) = \frac{\beta(g)\beta(h)}{\beta(gh)},$$

and define  $H^2(G, U(1))$  to be normalized cocycles modulo multiplication by such coboundaries. Prove that operators satisfying

$$U_g U_h = \omega(g, h) U_{gh}, \quad U_e = 1,$$

associate precisely because  $\omega$  is a cocycle and that rephasing  $U_g$  changes  $\omega$  by a coboundary. For finite-dimensional physical and auxiliary Hilbert spaces  $H$  and  $W$ , define the periodic matrix-product state associated to  $A : W \rightarrow H \otimes W$  by contracting the  $W$ -outputs and inputs around a circle. Call  $A$  *injective* when, for some length  $r$ , contraction over  $r$  consecutive sites gives an injective map  $\text{End}(W) \rightarrow H^{\otimes r}$ . Prove the canonical-form theorem for injective matrix-product states, derive the projective virtual symmetry action, and prove [17, 54] that one-dimensional bosonic gapped phases with a unique bulk ground state, unbroken onsite symmetry  $G$ , and finite correlation length are classified, under symmetric gapped-path equivalence, by  $H^2(G, U(1))$ , with boundary degrees of freedom carrying the corresponding projective representation.

For commuting  $g, h \in G$ , define the twisted torus response

$$Z_\omega(T^2; g, h) = \frac{\omega(g, h)}{\omega(h, g)}.$$

Prove that it depends only on the cohomology class of  $\omega$ . For  $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ , written additively as pairs  $(r, s)$ , prove that

$$\omega((r, s), (r', s')) = (-1)^{sr'}$$

is a normalized cocycle and that the two generating holonomies give

$$Z_\omega(T^2; (1, 0), (0, 1)) = -1.$$

Interpret the phase  $\pi$  as the quantized interference response of the nontrivial bosonic symmetry-protected phase, stating separately the onsite-symmetry, gap, unique-bulk-ground-state, flat-background-field, coherence, and phase-resolution assumptions.

**Exercise 19.11** (Spin bordism and the Majorana chain). For a closed spin surface  $(\Sigma, \sigma)$ , prove [33] that  $\sigma$  determines a quadratic refinement

$$q_\sigma : H_1(\Sigma; \mathbb{Z}_2) \longrightarrow \mathbb{Z}_2, \quad q_\sigma(x + y) = q_\sigma(x) + q_\sigma(y) + x \cdot y,$$

of the mod-2 intersection pairing. Define its Arf invariant intrinsically by the Gauss sum

$$(-1)^{\text{Arf}(\sigma)} = 2^{-g} \sum_{x \in H_1(\Sigma; \mathbb{Z}_2)} (-1)^{q_\sigma(x)},$$

where  $g$  is the genus. Prove [30]

$$\Omega_2^{\text{Spin}} \cong \mathbb{Z}_2$$

with Arf as generator of the character group. Prove that

$$Z_{\text{Arf}}(\Sigma, \sigma) = (-1)^{\text{Arf}(\sigma)}$$

is multiplicative under disjoint union, is unchanged by spin bordism, and squares to the trivial response under stacking.

For fermionic annihilation operators  $c_1, \dots, c_N$  satisfying

$$\{c_j, c_k\} = 0, \quad \{c_j, c_k^*\} = \delta_{jk},$$

define the Majorana operators

$$\gamma_{j,A} = c_j + c_j^*, \quad \gamma_{j,B} = -i(c_j - c_j^*).$$

Derive

$$\gamma_{j,\alpha}^* = \gamma_{j,\alpha}, \quad \{\gamma_{j,\alpha}, \gamma_{k,\beta}\} = 2\delta_{jk}\delta_{\alpha\beta}.$$

On an open chain define the distinguished Kitaev Hamiltonian from [41]

$$H_{\text{top}} = -J \sum_{j=1}^{N-1} i\gamma_{j,B}\gamma_{j+1,A}, \quad J > 0.$$

Prove that its  $N - 1$  bond operators commute, square to the identity, and leave  $\gamma_{1,A}$  and  $\gamma_{N,B}$  unpaired. Deduce the spectral values

$$E_m = -J(N - 1) + 2Jm, \quad 0 \leq m \leq N - 1,$$

with multiplicity  $2\binom{N-1}{m}$ , a bulk gap  $2J$ , and two ground states of opposite total fermion parity. Compare with

$$H_{\text{triv}} = -J \sum_{j=1}^N i\gamma_{j,A}\gamma_{j,B},$$

which has a unique ground state. Stack two copies of  $H_{\text{top}}$  and construct parity-even bilinear couplings between the two Majoranas at each end that remove the edge degeneracy without closing the bulk gap. Relate this  $\mathbb{Z}_2$  stacking law to  $Z_{\text{Arf}}$ .

Assume a uniformly gapped parity-preserving local deformation remains in the nontrivial phase and admits parity-odd edge operators whose commutators with the Hamiltonian are  $O(e^{-N/\xi})$  for some localization length  $\xi$ . Deduce an  $O(e^{-N/\xi})$  splitting and predict an exponentially split pair of parity-resolved subgap levels and, for a parity-conserving weak link with phase difference  $\phi$ , use the low-energy coupling

$$H_{\text{link}}(\phi) = it \cos(\phi/2) \gamma_L \gamma_R$$

to derive the two branches

$$E_{\pm}(\phi) = \pm t \cos(\phi/2).$$

Show that a  $2\pi$  advance carries a fixed-parity state to the opposite energy branch, so that each fixed-parity branch has  $4\pi$  spectral flow. State separately the quadratic-model, open-boundary, bulk-gap, fermion-parity, quasiparticle-poisoning, coherence-length, temperature, and spectral-resolution assumptions under which tunneling spectroscopy and Josephson measurements test these predictions.

**Exercise 19.12** (Projective characters and the cluster chain). Let  $N \geq 4$  be even and put

$$K_j = Z_{j-1}X_jZ_{j+1}, \quad H_{\text{cl}} = -J \sum_{j=2}^{N-1} K_j, \quad J > 0,$$

on an open chain of  $N$  qubits, where  $X_j$  and  $Z_j$  are the Pauli operators on the  $j$ th tensor factor. Define the onsite  $G = \mathbb{Z}_2 \times \mathbb{Z}_2$  symmetry operators

$$S_e = \prod_{j \text{ even}} X_j, \quad S_o = \prod_{j \text{ odd}} X_j.$$

Prove that the  $K_j$  are independent commuting involutions and that both symmetries commute with every  $K_j$ . Derive the complete spectrum

$$E_m = -J(N - 2) + 2Jm, \quad 0 \leq m \leq N - 2,$$

with multiplicity  $4\binom{N-2}{m}$ , including a four-dimensional ground space and gap  $2J$ .

On the ground space prove

$$S_e = Z_1 (Z_{N-1} X_N), \quad S_o = (X_1 Z_2) Z_N.$$

Show that the two left factors anticommute, as do the two right factors, although  $S_e$  and  $S_o$  commute. With

$$U_{(r,s)} = Z^r X^s,$$

derive

$$U_{(r,s)} U_{(r',s')} = (-1)^{sr'} U_{(r+r',s+s')}$$

and prove that each edge realizes the nontrivial projective class whose twisted torus response is  $-1$ . Show that a symmetry-preserving operator supported at one edge acts as a scalar on its irreducible projective doublet. Deduce that the projective multipliers at the two ends cancel in the linear global symmetry action and identify this cancellation as anomaly inflow from the bulk response.

For the irreducible edge representation, define its projective character by

$$\psi_\omega(r, s) = \text{Tr } U_{(r,s)}.$$

Compute

$$\psi_\omega(0, 0) = 2, \quad \psi_\omega(r, s) = 0 \quad \text{for } (r, s) \neq (0, 0),$$

and verify the projective-character orthogonality normalization

$$\frac{1}{4} \sum_{r,s \in \mathbb{Z}_2} |\psi_\omega(r, s)|^2 = 1.$$

Define the four group characters

$$\chi_{p,q}(r, s) = (-1)^{pr+qs}, \quad (p, q), (r, s) \in \mathbb{Z}_2^2.$$

Prove their orthogonality and completeness identities

$$\frac{1}{4} \sum_{r,s \in \mathbb{Z}_2} \chi_{p,q}(r, s) \overline{\chi_{p',q'}(r, s)} = \delta_{p,p'} \delta_{q,q'},$$

$$\frac{1}{4} \sum_{p,q \in \mathbb{Z}_2} \chi_{p,q}(r, s) \overline{\chi_{p,q}(r', s')} = \delta_{r,r'} \delta_{s,s'}.$$

Use the character projectors

$$P_{p,q} = \frac{1}{4} \sum_{r,s \in \mathbb{Z}_2} \chi_{p,q}(r, s) S_e^r S_o^s$$

to prove that every energy  $E_m$  occurs with multiplicity  $\binom{N-2}{m}$  in each symmetry sector. If an operator  $O$  transforms by  $\chi_{u,v}$ , derive the selection rule

$$P_{p,q} O P_{p',q'} = 0 \quad \text{unless} \quad (p, q) = (p' + u, q' + v).$$

For  $2 \leq r \leq s \leq N-1$  with  $r \equiv s \pmod{2}$ , prove the exact string-correlation identity

$$\left\langle Z_{r-1} \left( \prod_{k=0}^{(s-r)/2} X_{r+2k} \right) Z_{s+1} \right\rangle = 1$$



in every ground state. Predict four symmetry-resolved ground levels, the gap  $2J$ , the character selection rule, unit string correlation, and the edge-pulse commutator phase  $\pi$ . State separately the commuting-stabilizer, open-boundary, onsite-symmetry, coupling-calibration, state-preparation, temperature, coherence, symmetry-resolved-readout, and finite-size assumptions under which qubit-array spectroscopy, string measurements, and edge interferometry test these predictions; for small symmetry-preserving perturbations, distinguish exact quantization of the thermodynamic projective class from finite-size edge splitting and nonuniversal deformation of the string correlator.

## 20 Fusion Categories

### 20.1 Temperley–Lieb Categories

**Exercise 20.1** (Temperley–Lieb categories and Jones–Wenzl projectors). For  $d \in \mathbb{C}$ , the *Temperley–Lieb category*  $\mathrm{TL}_d$  has objects  $n \in \mathbb{Z}_{\geq 0}$ . The morphism space  $\mathrm{Hom}_{\mathrm{TL}_d}(m, n)$  is the complex span of planar nonintersecting pairings of  $m$  lower and  $n$  upper boundary points. Composition is vertical stacking, with every resulting closed loop removed and replaced by the scalar  $d$ ; tensor product is horizontal juxtaposition and has tensor unit 0. For  $d \in \mathbb{R}$ , reflection across a horizontal line defines  $f^*$ . The cup and cap diagrams give coevaluation and evaluation morphisms.

The *Temperley–Lieb algebra* is  $TL_n(d) = \mathrm{End}_{\mathrm{TL}_d}(n)$ . It is generated by  $U_1, \dots, U_{n-1}$  with

$$U_i^2 = dU_i, \quad U_i U_{i\pm 1} U_i = U_i, \quad U_i U_j = U_j U_i \quad (|i - j| > 1).$$

Its diagram trace is normalized by

$$\mathrm{tr}_0(1) = 1, \quad \mathrm{tr}_{n+1}(x) = d \mathrm{tr}_n(x), \quad \mathrm{tr}_{n+1}(xU_n) = \mathrm{tr}_n(x),$$

where  $x \in TL_n(d)$  is embedded in  $TL_{n+1}(d)$ .

Put

$$\zeta = e^{\pi i/(k+2)}, \quad d = \zeta + \zeta^{-1} = [2]_\zeta.$$

With  $f_0 = f_1 = 1$ , define recursively in  $TL_{n+1}(d)$

$$f_{n+1} = f_n - \frac{[n]_\zeta}{[n+1]_\zeta} f_n U_n f_n \quad (1 \leq n \leq k).$$

(a) Use the Temperley–Lieb relations to prove inductively

$$f_n^2 = f_n, \quad f_n^* = f_n, \quad U_i f_n = f_n U_i = 0 \quad (1 \leq i < n), \quad \mathrm{tr}(f_n) = [n+1]_\zeta.$$

(b) Deduce  $\mathrm{tr}(f_{k+1}) = [k+2]_\zeta = 0$ .

### 20.2 Fusion Semisimplification

**Exercise 20.2** (Temperley–Lieb semisimplification and the positive quotient). At  $d = \zeta + \zeta^{-1}$ , a morphism  $f : m \rightarrow n$  in  $\mathrm{TL}_d$  is negligible when

$$\mathrm{tr}_m(gf) = 0 \quad \text{for every } g : n \rightarrow m.$$

Let  $\mathcal{N}(m, n)$  be the space of negligible morphisms. A tensor ideal  $\mathcal{I}$  is a collection of morphism subspaces closed under composition and tensoring. The quotient has morphisms

$$\mathrm{Hom}_{\mathcal{C}/\mathcal{I}}(X, Y) = \mathrm{Hom}_{\mathcal{C}}(X, Y)/\mathcal{I}(X, Y).$$

The  $C^*$ -idempotent completion has objects  $(X, p)$  with  $p = p^2 = p^*$  and morphisms  $q \operatorname{Hom}(X, Y)p$ . Let  $\mathcal{C}_k$  be the idempotent completion of  $\operatorname{TL}_d/\mathcal{N}$ , and let  $V_a$  be the image of  $f_a$ .

Prove the root-of-unity Markov-form theorem [66]: the forms

$$\langle f, g \rangle = \operatorname{tr}_m(f^*g), \quad f, g : m \rightarrow n,$$

are positive semidefinite, their radicals form the  $*$ -closed tensor ideal  $\mathcal{N}$  generated by  $f_{k+1}$ , and every negligible idempotent lies in that ideal. Deduce that composition and tensor product descend to positive-definite quotient morphism spaces, that the quotient removes every trace-zero summand, and that its  $C^*$ -idempotent completion has faithful positive categorical trace.

**Exercise 20.3** (Truncated fusion). Derive

$$V_a \otimes V_b \cong \bigoplus_{\substack{|a-b| \leq c \leq \min(a+b, 2k-a-b) \\ a+b+c \equiv 0 \pmod{2}}} V_c$$

and

$$d_a = [a+1]_\zeta = \frac{\sin((a+1)\pi/(k+2))}{\sin(\pi/(k+2))}.$$

**Exercise 20.4** (Unitary fusion structure). A nonzero object is simple when its endomorphism algebra is the scalar multiples of its identity. A  $C^*$ -category is semisimple when every object is a finite orthogonal direct sum of simple objects. A unitary fusion category is a rigid semisimple  $C^*$ -tensor category with simple tensor unit, finite-dimensional morphism spaces, and finitely many simple objects up to unitary isomorphism.

Use left composition on the positive quotient spaces and diagram reflection to equip  $\mathcal{C}_k$  with  $C^*$ -norms. Use the cup and cap diagrams to construct standard conjugates. Prove from truncated fusion that every simple object is unitarily isomorphic to exactly one of

$$V_0, V_1, \dots, V_k,$$

and deduce that  $\mathcal{C}_k$  is a unitary fusion category. Prove that its standard categorical trace agrees with the quotient diagram trace and that the recoupling maps between simple-summand decompositions are unitary and satisfy the pentagon identity. With  $\zeta^{1/2} = e^{\pi i/(2(k+2))}$ , prove that

$$c_i = \zeta^{1/2} \operatorname{id} - \zeta^{-1/2} U_i$$

is unitary, satisfies the braid relations, preserves  $\mathcal{N}$ , and descends to a unitary braiding on  $\mathcal{C}_k$ .

**Exercise 20.5** (Fusion rings and Frobenius–Perron dimensions). Let  $\mathcal{C}$  be a fusion category with simple objects  $X_0 = \mathbf{1}, X_1, \dots, X_r$ . Define its Grothendieck ring to be the free abelian group on the classes  $[X_a]$  with multiplication

$$[X_a][X_b] = \sum_c N_{ab}^c [X_c].$$

For an object  $X$ , let  $N_X$  be left multiplication by  $[X]$  on  $K_0(\mathcal{C}) \otimes \mathbb{R}$ . Prove the Frobenius–Perron dimension theorem [25]: there is a unique ring homomorphism

$$\operatorname{FPdim} : K_0(\mathcal{C}) \longrightarrow \mathbb{R}$$

that is positive on every nonzero object, and prove

$$\operatorname{FPdim}(X_a) = \rho(N_{X_a}).$$

Define

$$\text{FPdim}(\mathcal{C}) = \sum_a \text{FPdim}(X_a)^2.$$

For a unitary fusion category, use positivity of the standard categorical trace and uniqueness of the positive dimension character to prove

$$\dim_{\mathcal{C}}(X) = \text{FPdim}(X).$$

Apply this to  $\mathcal{C}_k$ : prove that  $(d_a)_{a=0}^k$  is the unique positive common dimension character of the fusion matrices and hence that  $\text{FPdim}(V_a) = d_a$ . Use sine orthogonality to compute

$$\text{FPdim}(\mathcal{C}_k) = \frac{k+2}{2 \sin^2(\pi/(k+2))}.$$

Prove that a tensor equivalence preserves the fusion ring and all these dimensions.

**Exercise 20.6** (Root-of-unity quantum groups and fusion semisimplification). Let  $\zeta = e^{\pi i/(k+2)}$ . The divided-power form of  $U_{\zeta}(\mathfrak{sl}_2)$  is the specialization of the integral quantum group generated by  $K^{\pm 1}$  and the divided powers  $E^{(n)}, F^{(n)}$ . Call a finite-dimensional type-1 module *tilting* when it admits both a filtration by Weyl modules and a filtration by dual Weyl modules, and let  $\text{Tilt}_{\zeta}$  be their tensor category. A morphism is *negligible* when every quantum trace obtained by closing it vanishes. Prove that negligible morphisms form a tensor ideal and that the quantum dimension of the tilting module of highest weight  $k+1$  is

$$[k+2]_{\zeta} = 0.$$

Prove the root-of-unity semisimplification theorem [66]: the idempotent completion of the negligible quotient of  $\text{Tilt}_{\zeta}$  is a unitary braided fusion category and there is a braided unitary equivalence, writing  $(-)^{\natural}$  for this completion,

$$(\text{Tilt}_{\zeta}/\mathcal{N})^{\natural} \simeq \mathcal{C}_k$$

sending the fundamental quantum-group module to  $V_1$  and the simple highest-weight modules  $0, \dots, k$  to  $V_0, \dots, V_k$ . Prove that this equivalence transports the truncated fusion rule and categorical dimensions, and verify directly that every summand generated by the weight- $k+1$  object becomes negligible.

Compare this equivalence with the affine  $SU(2)_k$  character system: prove that its modular  $S$ -matrix diagonalizes the same fusion operators and that

$$\frac{S_{a0}}{S_{00}} = \dim_{\mathcal{C}_k}(V_a).$$

Conclude that WZW characters, root-of-unity quantum-group modules, and Temperley–Lieb diagrams give three realizations of the same fusion algebra, with the negligible quotient supplying the required passage from the nonsemisimple quantum-group category to the unitary fusion category.

**Exercise 20.7** (Morita equivalence and Drinfeld centers). Let  $\mathcal{C}$  and  $\mathcal{D}$  be unitary fusion categories. A  $\mathcal{C}$ – $\mathcal{D}$  *bimodule category* is a finite semisimple  $C^*$ -category  $\mathcal{M}$  with commuting unitary left  $\mathcal{C}$ - and right  $\mathcal{D}$ -actions and unitary associativity and unit constraints satisfying the pentagon and triangle identities. For a  $\mathcal{C}$ – $\mathcal{D}$  bimodule  $\mathcal{M}$  and a  $\mathcal{D}$ – $\mathcal{E}$  bimodule  $\mathcal{N}$ , define their *relative tensor product* by the universal property

$$\text{Fun}_{\mathcal{C}\text{--}\mathcal{E}}(\mathcal{M} \boxtimes_{\mathcal{D}} \mathcal{N}, \mathcal{A}) \cong \text{Fun}_{\mathcal{D}\text{--bal}}(\mathcal{M} \times \mathcal{N}, \mathcal{A})$$

for every  $\mathcal{C}$ – $\mathcal{E}$  module category  $\mathcal{A}$ , where the right side consists of bilinear module functors equipped with coherent  $\mathcal{D}$ –balancing isomorphisms. Prove that relative tensor products of finite semisimple module categories exist and are associative up to their canonical equivalence [25]. Call  $\mathcal{C}$  and  $\mathcal{D}$  *Morita equivalent* when there is a  $\mathcal{C}$ – $\mathcal{D}$  bimodule  $\mathcal{M}$  with a  $\mathcal{D}$ – $\mathcal{C}$  bimodule  $\mathcal{M}^\vee$  and equivalences

$$\mathcal{M} \boxtimes_{\mathcal{D}} \mathcal{M}^\vee \simeq \mathcal{C}, \quad \mathcal{M}^\vee \boxtimes_{\mathcal{C}} \mathcal{M} \simeq \mathcal{D}$$

of regular bimodule categories. Use the universal property and associativity of relative tensor products to prove that Morita equivalence is an equivalence relation.

The *Drinfeld center*  $Z(\mathcal{C})$  has objects  $(X, \gamma)$ , where  $X \in \mathcal{C}$  and  $\gamma_Y : X \otimes Y \rightarrow Y \otimes X$  is a unitary natural isomorphism satisfying

$$\gamma_1 = \text{id}_X, \quad \gamma_{Y \otimes Z} = (\text{id}_Y \otimes \gamma_Z)(\gamma_Y \otimes \text{id}_Z),$$

with associators understood; its morphisms intertwine the half-braidings. Construct the tensor product and braiding in  $Z(\mathcal{C})$ . In a unitary ribbon fusion category, call an object  $X$  *transparent* when

$$c_{Y,X} c_{X,Y} = \text{id}_{X \otimes Y} \quad \text{for every object } Y.$$

The full fusion subcategory of transparent objects is the *Müger center*; call the ribbon fusion category *modular* when this center is equivalent to  $\text{Hilb}_{\text{fd}}$ . Construct the braiding and ribbon structure on  $Z(\mathcal{C})$  and prove [25, 66] that it is a unitary modular tensor category and that

$$\mathcal{C} \simeq_{\text{Morita}} \mathcal{D} \iff Z(\mathcal{C}) \simeq Z(\mathcal{D})$$

by a braided unitary equivalence. Deduce that the center is an invariant of the bulk rather than of a chosen boundary presentation.

**Exercise 20.8** (Fully dualizable fusion categories and topological phases). Define the symmetric monoidal Morita 3-category  $\text{Fus}$  to have unitary fusion categories as objects, finite semisimple bimodule categories as 1-morphisms, module functors as 2-morphisms, and natural transformations as 3-morphisms. Define the Deligne tensor product  $\mathcal{C} \boxtimes \mathcal{D}$  by the universal property that biexact bilinear functors  $\mathcal{C} \times \mathcal{D} \rightarrow \mathcal{A}$  are the same as exact linear functors  $\mathcal{C} \boxtimes \mathcal{D} \rightarrow \mathcal{A}$ , and use it as the monoidal product. An object in a symmetric monoidal 3-category is *fully dualizable* when it has a tensor dual and its evaluation, coevaluation, and successive adjunction data admit left and right adjoints through level three. For a unitary fusion category, construct these data from its opposite category, regular bimodule categories, and adjoint bimodule functors, and prove full dualizability in  $\text{Fus}$  [23].

Let  $\text{Bord}_3^{\text{fr}}$  be the symmetric monoidal 3-category of compact framed manifolds and bordisms with corners through dimension three. A *fully extended framed 3-dimensional topological field theory* valued in  $\text{Fus}$  is a symmetric monoidal 3-functor

$$\mathcal{Z} : \text{Bord}_3^{\text{fr}} \longrightarrow \text{Fus}.$$

Prove the cobordism hypothesis [49]: evaluation on the positively framed point identifies the 3-groupoid of such theories with the 3-groupoid of fully dualizable objects of  $\text{Fus}$ . Construct the canonical spherical structure of a unitary fusion category, prove its independence from auxiliary choices, and use it to construct the oriented Turaev–Viro theory. Prove that its bulk line operators form  $Z(\mathcal{C})$  [66, 1].

Let  $\text{Pic}(\text{Fus})$  be the sub-3-groupoid of tensor-invertible objects and invertible morphisms. Prove that the cobordism hypothesis restricts on  $\text{Pic}(\text{Fus})$  to the classification of invertible theories, whereas

all fully dualizable objects classify the non-invertible theories as well. Define the global dimension, with  $\text{Irr}(\mathcal{C})$  the set of unitary isomorphism classes of simple objects, by

$$\text{Dim}(\mathcal{C}) = \sum_{[X] \in \text{Irr}(\mathcal{C})} d_X^2.$$

Use the equality of categorical and Frobenius–Perron dimensions to show

$$\text{Dim}(\mathcal{C}) = \text{FPdim}(\mathcal{C}).$$

Prove [25]

$$\text{Dim } Z(\mathcal{C}) = \text{Dim}(\mathcal{C})^2, \quad \text{Dim}(\mathcal{C} \boxtimes \mathcal{D}) = \text{Dim}(\mathcal{C}) \text{Dim}(\mathcal{D}),$$

and prove that a unitary fusion category is tensor-invertible in  $\mathbf{Fus}$  only when it is Morita equivalent to  $\mathbf{Hilb}_{\text{fd}}$ , the unitary fusion category of finite-dimensional Hilbert spaces. For  $\mathcal{C}_k$ , use the distinct simple objects  $V_0, V_1$  and positivity of their categorical dimensions to prove  $\text{Dim}(\mathcal{C}_k) > 1$  for  $k \geq 1$ , and conclude that its associated state-sum phase is non-invertible. State explicitly that this classifies the fully extended topological low-energy theories and does not by itself prove a classification of all microscopic gapped Hamiltonians.

### 20.3 Modular $S$ -Matrix

**Exercise 20.9** (Categorical Fourier transform, modular data, and Verlinde diagonalization). Put

$$\mathcal{D} = \text{Dim}(\mathcal{C}_k)^{1/2}, \quad \theta_a = \exp\left(\frac{\pi i a(a+2)}{2(k+2)}\right).$$

Prove [66] that these phases, together with the braiding on  $\mathcal{C}_k$ , define its unitary ribbon structure. For simple objects define the normalized Hopf-link matrix intrinsically by

$$S_{ab} = \frac{1}{\mathcal{D}} \text{tr}_{V_a \otimes V_b} (c_{V_b, V_a} c_{V_a, V_b}).$$

Evaluate the closed Temperley–Lieb diagram to derive

$$S_{ab} = \sqrt{\frac{2}{k+2}} \sin\left(\frac{(a+1)(b+1)\pi}{k+2}\right).$$

Let  $N_{ab}^c$  be the multiplicity of  $V_c$  in  $V_a \otimes V_b$ .

Use sine orthogonality to prove that  $S$  is real, symmetric, and unitary. Prove [66] the equivalence between invertibility of the Hopf-link matrix and triviality of the Müger center, and conclude that  $\mathcal{C}_k$  is modular. For each  $a$ , let  $(N_a)_{bc} = N_{ab}^c$ . Use the truncated Chebyshev recursion to prove

$$N_a = S \text{diag}\left(\frac{S_{ax}}{S_{0x}}\right)_{x=0}^k S^{-1}$$

and hence

$$N_{ab}^c = \sum_{x=0}^k \frac{S_{ax} S_{bx} \overline{S_{cx}}}{S_{0x}}.$$

For functions  $f, g$  on the simple-object labels, define fusion convolution and the categorical Fourier transform by

$$(f *_{\text{fus}} g)(c) = \sum_{a,b=0}^k f(a)g(b)N_{ab}^c, \quad \widehat{f}(x) = \sum_{a=0}^k f(a) \frac{S_{ax}}{S_{0x}}.$$

Prove the convolution, inversion, and Plancherel identities

$$\widehat{f *_{\text{fus}} g}(x) = \widehat{f}(x) \widehat{g}(x), \quad f(a) = \sum_{x=0}^k S_{ax} S_{0x} \widehat{f}(x),$$

$$\sum_{a=0}^k |f(a)|^2 = \sum_{x=0}^k |S_{0x}|^2 |\widehat{f}(x)|^2.$$

Deduce that the number of fusion channels from  $a_1, \dots, a_n$  to  $c$  is  $(N_{a_1} \cdots N_{a_n})_{0c}$  and express it as a finite sum of products of sine functions. Let  $\mathcal{C}_k^{\text{rev}}$  have the inverse braiding. Prove [25] the canonical braided equivalence

$$\mathcal{C}_k \boxtimes \mathcal{C}_k^{\text{rev}} \simeq Z(\mathcal{C}_k)$$

and deduce that the simple bulk charges are indexed by pairs  $(x, y)$  with  $0 \leq x, y \leq k$ .

## 20.4 Anyonic Observables

**Exercise 20.10** (Chern–Simons theory, quantum invariants, and chiral transport). Let  $M$  be a closed oriented three-manifold and let  $A$  be a connection on the trivial principal  $SU(2)$ -bundle, regarded as an  $\mathfrak{su}(2)$ -valued one-form. Normalize the invariant trace by

$$\frac{1}{8\pi^2} \int_W \text{Tr}(F_B \wedge F_B) \in \mathbb{Z}$$

for every  $SU(2)$ -connection  $B$  on a closed oriented four-manifold  $W$ , and define, for  $k \in \mathbb{Z}_{>0}$ ,

$$S_k(A) = \frac{k}{4\pi} \int_M \text{Tr} \left( A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right) \mod 2\pi\mathbb{Z}.$$

Compute its change under a gauge transformation and prove that  $e^{iS_k(A)}$  is gauge invariant exactly because  $k$  is integral. For a framed oriented link with components labelled by  $V_{a_1}, \dots, V_{a_n} \in \mathcal{C}_k$ , define the Wilson-loop observable formally by the product of the corresponding holonomy traces.

Prove the Chern–Simons/Reshetikhin–Turaev correspondence [66]: the quantum expectation of framed Wilson links is the  $SU(2)_k$  Reshetikhin–Turaev invariant, and canonical quantization on a surface  $\Sigma$  gives its modular-functor Hilbert space. Deduce

$$\dim \mathcal{H}_k(\Sigma_g) = \sum_{a=0}^k S_{0a}^{2-2g},$$

the torus dimension  $k+1$ , the Hopf-link amplitude  $S_{ab}/S_{00}$ , and the framed-unknot quantum dimension  $S_{a0}/S_{00}$ .

Prove that the torus mapping-class generators act projectively by the matrices  $S$  and

$$T_{ab} = \delta_{ab} e^{-2\pi i c_- / 24} \theta_a, \quad c_- = \frac{3k}{k+2} \pmod{8}.$$

Use the Gauss-sum identity

$$\frac{1}{\sqrt{\text{Dim}(\mathcal{C}_k)}} \sum_{a=0}^k d_a^2 \theta_a = e^{2\pi i c_- / 8}$$

to recover the chiral central charge modulo eight. Derive the low-temperature thermal Hall prediction

$$\frac{\kappa_{xy}}{T} = \frac{\pi^2 k_B^2}{3h} c_-.$$

Contrast this chiral theory with the doubled center  $Z(\mathcal{C}_k)$ , whose opposite chiralities cancel. State the bulk-gap, edge-equilibration, framing, contact, temperature, finite-size, and interferometric-calibration assumptions under which thermal transport, ground-state degeneracy, and linked-anyon amplitudes test these formulas.

**Exercise 20.11** (Factorization homology and modular surface state spaces). Put  $\mathcal{A} = Z(\mathcal{C}_k)$ . Let  $\text{Disk}(\Sigma)$  have finite disjoint unions of oriented disks embedded in  $\Sigma$  as objects and isotopies followed by inclusions as morphisms; send an  $r$ -disk object to  $\mathcal{A}^{\boxtimes r}$ . Define  $\int_{\Sigma} \mathcal{A}$  by the universal property that compatible families of functors from this diagram to any target category factor uniquely through it. Prove that this symmetric-monoidal colimit exists, and prove from disk isotopy and the ribbon relations [9, 1] that this construction determines an isotopy-invariant symmetric monoidal assignment

$$\Sigma \mapsto \int_{\Sigma} \mathcal{A},$$

called factorization homology and characterized by disk normalization, disjoint union, and collar excision. The first two properties are

$$\int_{\mathbb{R}^2} \mathcal{A} \simeq \mathcal{A}, \quad \int_{\Sigma \sqcup \Sigma'} \mathcal{A} \simeq \left( \int_{\Sigma} \mathcal{A} \right) \boxtimes \left( \int_{\Sigma'} \mathcal{A} \right).$$

For a collar gluing  $\Sigma = \Sigma_- \cup_{N \times \mathbb{R}} \Sigma_+$ , excision is the equivalence

$$\int_{\Sigma} \mathcal{A} \simeq \left( \int_{\Sigma_-} \mathcal{A} \right) \int_{N \times \mathbb{R}} \mathcal{A} \boxtimes \left( \int_{\Sigma_+} \mathcal{A} \right),$$

where the collar embeddings give the two surface values their right and left module structures over  $\int_{N \times \mathbb{R}} \mathcal{A}$ . On a closed oriented surface, construct from a handle decomposition a distinguished finite-dimensional Hilbert space  $\mathcal{H}_{\mathcal{A}}(\Sigma)$  canonically isomorphic to the Reshetikhin–Turaev surface state space, and prove invariance under handle moves. Apply collar excision to a pair-of-pants decomposition and use Verlinde diagonalization to prove

$$\dim \mathcal{H}_{Z(\mathcal{C}_k)}(\Sigma_g) = \left( \sum_{x=0}^k S_{0x}^{2-2g} \right)^2.$$

Insert the sine formula for  $S_{0x}$ , recover dimension 1 on the sphere and  $(k+1)^2$  on the torus, and determine the growth with  $g$  for a fixed chosen  $k$ .

Choose oriented meridian and longitude classes on the torus and use the intrinsic orthogonal decomposition into simple bulk-charge sectors  $(x, y)$  to prove that the anomaly-free doubled mapping-class operators have kernels

$$(U_S)_{(x,y),(x',y')} = S_{xx'} \overline{S_{yy'}}, \quad (U_T)_{(x,y),(x',y')} = \delta_{xx'} \delta_{yy'} \theta_x \overline{\theta_y}.$$

Deduce the transition probability  $|S_{xx'}|^2 |S_{yy'}|^2$  under an adiabatic modular  $S$  operation and the Dehn-twist Berry phase  $\arg \theta_x - \arg \theta_y$  in sector  $(x, y)$ . State the gap, adiabaticity, ground-space preparation, mapping-class calibration, finite-size, and phase-resolution assumptions under which these numbers are measurable.

For a stratified collar labelled by a  $\mathcal{C}$ – $\mathcal{D}$  bimodule category  $\mathcal{M}$ , construct the corresponding two-sided bar resolution and prove stratified excision [1]. Deduce that gluing interfaces composes their labels by relative tensor product. Prove that a Morita-invertible  $\mathcal{M}$  gives a transparent wall by applying

$$\mathcal{M} \boxtimes_{\mathcal{D}} \mathcal{M}^{\text{op}} \simeq \mathcal{C}, \quad \mathcal{M}^{\text{op}} \boxtimes_{\mathcal{C}} \mathcal{M} \simeq \mathcal{D}.$$

**Exercise 20.12** (Levin–Wen string-net Hamiltonians). Let  $\Gamma$  be a trivalent cellulation of a closed oriented surface and let  $\mathcal{C}_k$  be the unitary fusion category with simple objects  $V_0, \dots, V_k$ , fusion multiplicities  $N_{ab}^c$ , standard categorical trace, and quantum dimensions  $d_a$ . Write  $a^\vee$  for the unique label satisfying  $\bar{V}_a \cong V_{a^\vee}$ . Put

$$\mathcal{H}_\Gamma = \ell^2\left(\{0, \dots, k\}^{E(\Gamma)}\right), \quad \mathcal{D}^2 = \text{Dim}(\mathcal{C}_k).$$

For each vertex  $v$ , let  $Q_v$  be the projection onto edge labelings admissible under the fusion rules. For a face  $p$ , let  $B_p^a$  insert a coevaluation–evaluation loop labelled  $V_a$  inside  $p$  and resolve it into the boundary edges by the associator and the quantum recoupling maps; set it equal to zero off the subspace on which the vertices incident to  $p$  are admissible. Define

$$B_p = \frac{1}{\mathcal{D}^2} \sum_{a=0}^k d_a B_p^a$$

and, for  $\Delta_v, \Delta_p > 0$ ,

$$\hat{H}_\Gamma = \Delta_v \sum_v (I - Q_v) + \Delta_p \sum_p (I - B_p).$$

Use the snake identities, the standard categorical trace, and the quantum recoupling pentagon to prove

$$(B_p^a)^* = B_p^{a^\vee}, \quad B_p^a B_p^b = \sum_c N_{ab}^c B_p^c.$$

Use

$$d_a d_b = \sum_c N_{ab}^c d_c$$

to prove

$$Q_v^2 = Q_v = Q_v^*, \quad B_p^2 = B_p = B_p^*, \quad [Q_v, B_p] = [B_p, B_{p'}] = 0.$$

Deduce that  $\hat{H}_\Gamma$  is positive and frustration-free and that its ground space is the simultaneous  $+1$  space of all  $Q_v$  and  $B_p$ . Prove that recoupling across an edge gives a unitary identification of the ground spaces before and after the corresponding local change of cellulation. Prove [48, 1] that these identifications assemble to a canonical unitary isomorphism

$$\ker \hat{H}_\Gamma \cong \mathcal{H}_{Z(\mathcal{C}_k)}(\Sigma)$$

compatible with collar gluing and the mapping-class action.

For the fusion category  $\text{Vec}_{\mathbb{Z}/2}$  with trivial associator, identify the labels with  $\mathbb{Z}/2\mathbb{Z}$ , let  $Z_e$  act on a label  $a_e$  by  $(-1)^{a_e}$ , and let  $X_e$  reverse that label. Construct the corresponding string-net Hamiltonian and show that

$$Q_v = \frac{1}{2} \left( I + \prod_{e \ni v} Z_e \right), \quad B_p = \frac{1}{2} \left( I + \prod_{e \subset \partial p} X_e \right).$$



On a torus, count the two independent relations among the vertex and face constraints, prove that the ground-state degeneracy is 4, and verify agreement with the factorization-homology genus formula. Identify the two pairs of noncontractible string operators and their mutual sign, and state the boundary conditions and gap-preserving perturbation regime under which the fourfold low-energy multiplicity is a measurable prediction of a qubit realization.

For  $\mathcal{C}_1 = SU(2)_1$ , use its associator and ribbon twist  $\theta_1 = i$  to show that the same fusion rule does not give the displayed toric-code plaquette operator. Use  $Z(\mathcal{C}_1) \simeq \mathcal{C}_1 \boxtimes \mathcal{C}_1^{\text{rev}}$  to derive the four torus twists

$$1, \quad i, \quad -i, \quad 1,$$

and contrast them with the toric-code twists  $1, 1, 1, -1$ . Deduce that the equal ground-state degeneracies do not identify the two phases and that a Dehn-twist Berry-phase measurement distinguishes doubled semion from toric-code order.

**Exercise 20.13** (String-net loop operators and spectroscopy). On a torus let  $W_a^{\alpha,+}$  and  $W_a^{\alpha,-}$  insert a noncontractible  $V_a$  loop along a cycle  $\alpha$ , using respectively the braiding and inverse braiding of  $\mathcal{C}_k$  when the loop is resolved through the string net. Define  $W_a^{\beta,+}$  and  $W_a^{\beta,-}$  for a cycle  $\beta$  with intersection number one with  $\alpha$ .

Decompose the string-net ground space into its one-dimensional simple bulk-charge sectors indexed by pairs  $(x, y)$  with  $0 \leq x, y \leq k$ , and recover

$$\dim \ker \hat{H}_\Gamma = (k+1)^2.$$

Prove the loop fusion relations

$$W_a^{\alpha,\pm} W_b^{\alpha,\pm} = \sum_c N_{ab}^c W_c^{\alpha,\pm}$$

and use Verlinde diagonalization to show that their eigenvalues in the sector  $(x, y)$  are

$$W_a^{\alpha,+} : \frac{S_{ax}}{S_{0x}}, \quad W_a^{\alpha,-} : \frac{\overline{S_{ay}}}{S_{0y}}.$$

For a weak calibrated loop perturbation

$$\hat{V} = - \sum_a \left( \epsilon_a^+ W_a^{\alpha,+} + \epsilon_a^- W_a^{\alpha,-} + \text{adjoints} \right),$$

derive the first-order splitting

$$\delta E_{x,y} = -2 \operatorname{Re} \sum_a \left( \epsilon_a^+ \frac{S_{ax}}{S_{0x}} + \epsilon_a^- \frac{\overline{S_{ay}}}{S_{0y}} \right).$$

Insert the sine formula for  $S$  and list the distinct splittings for a chosen  $k$ . State the topological-gap, system-size, loop-support, calibration, and sector-resolution assumptions under which these spectral lines test the doubled  $SU(2)_k$  string-net model.

**Exercise 20.14** (Anyon interference and fusion counting). Use the ribbon twist  $\theta_a$  and the Hopf-link matrix already obtained. On every simple fusion channel  $V_c \subseteq V_a \otimes V_b$ , prove the balancing identity

$$c_{V_b, V_a} c_{V_a, V_b} |_{V_c} = \frac{\theta_c}{\theta_a \theta_b} \operatorname{id}_{V_c}.$$

For simple charges  $a, b$ , prove that the normalized monodromy amplitude is

$$M_{ab} = \frac{S_{ab}S_{00}}{S_{0a}S_{0b}}.$$

Identify

$$\frac{S_{ab}}{S_{0b}} = U_a \left( \cos \frac{(b+1)\pi}{k+2} \right)$$

with the specialized  $SU(2)$  Weyl character, where  $U_a$  is the Chebyshev family. In a two-path anyon interferometer with path intensities  $I_1, I_2$ , derive

$$I_b(\phi) = I_1 + I_2 + 2\sqrt{I_1 I_2} \Re(e^{i\phi} M_{ab}).$$

Prove that a calibrated full framing twist of the probe path replaces  $M_{ab}$  by  $\theta_a M_{ab}$  and shifts the interference phase by  $\arg \theta_a$ . Insert the sine formula and predict the visibility and phase for every enclosed charge. For a collection of prepared charges, use the Verlinde formula to predict the exact number of fusion outcomes in each total-charge sector. Identify indistinguishable charges for one probe and prove that the full family of probes distinguishes them. State the coherence, topological-gap, charge-preparation, and readout assumptions separately from the predicted fringes and integer state counts.

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